Recurrences, Absolute Differences, and Collatz Orbits

Algebra, Prime Patterns, and Zero Distributions

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Abstract

We bring together recurrence-order arguments for binomial-power sums, extremal finite absolute-difference triangles, explicitly constructed Collatz trajectories, and the algebra and geometry of their orbit polynomials. The collection includes proposed solutions to precisely stated problems from the literature, proofs of auxiliary classification results, and explicit boundaries between theorems, conditional statements, and numerical evidence. For terminating Collatz polynomials we study the zero accumulation annulus, prime-dependent arithmetic separation, logarithmic independence, and Euler–Mascheroni boundary-distance limits. A further extension determines maximal Kummer kernels for an odd-degree family and its entire power-of-two radical tower, while a deterministic inverse-orbit construction realizes prescribed rotationally invariant zero distributions. All proofs, finite-certificate procedures, figures, and references needed by the stated arguments are contained in this source. No universal Collatz termination, prime Gilbreath conjecture, twin-prime conjecture, Mersenne-prime infinitude, or unresolved constant's transcendence is asserted.

Status. This is an AI-assisted research manuscript, not an externally peer-reviewed publication. Statements presented as proved are supported by the arguments in this document; historical priority and acceptance by the mathematical community have not been established. A supplied large computational census is not reclassified as an independently rerun proof. Such observations are marked separately wherever no complete certificate has been retained.

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Reading guide and scope of the consolidation

Part I develops recurrence and finite-difference methods and prime-pattern constructions. Part II applies them to specific questions from the literature, retaining the exact scope of partial and negative answers. Part III assembles the zero geometry and arithmetic results. Part IV develops full Galois groups and limiting zero distributions and contains the principal further extensions. The parts use a common bibliography and globally distinct references; their local definitions are retained when the meaning of a symbol changes.

The unaccelerated Collatz map is denoted C, the shortcut map T, and the odd-to-odd map S in Part I. Part II sometimes writes c for the shortcut map and uses its own local trajectory notation. Throughout, an orbit polynomial ends at the *first* occurrence of 1. Statements about a full polynomial apply only when that starting value is known to terminate. This restriction is not an assumption that the Collatz conjecture holds. The symbol P_a in the inverse-Mersenne family of Part I denotes an integer, whereas $P_N(z)$ denotes an orbit polynomial elsewhere.

The recurrence order of a Franel sequence refers to rational-function coefficients and shift width, not the degree of the polynomial coefficients. Finite Gilbreath extension questions are distinct from Gilbreath's conjecture for the actual primes. Prime-chain existence varies an affine parameter; it does not guarantee a new prime above a prescribed fixed giant seed. Zero accumulation as a set is distinct from weak convergence of full root counting measures. These distinctions are used throughout the proofs.

For the large finite searches inherited from the problem-oriented source, the present edition records whether a displayed claim is an analytic theorem, an explicit finite certificate, or reported exploratory evidence. The latter does not establish a universal conjecture. The main analytic statements do not rely on the exploratory searches.

Further results established in this edition. The main additions beyond the supplied sources are the following.

1. Theorem IV.3.1 removes the former order-of-2 restriction from the full radical-tower calculation: it holds for every odd $k \geq 3$ with $5 \nmid k$. For prime $k \geq 23$ with $11 \nmid k+1$, the full Collatz-polynomial group is $(C_4)^k \rtimes (S_k \times C_2)$.
2. Theorem IV.6.2 realizes every rotationally invariant probability measure on $2\,3^{-2/7} \leq |z| \leq 2$, improving the earlier endpoint $2\,3^{-1/4}$. The uniform two-inverse-step exponent cost 7 is optimal, although the resulting radial endpoint need not be.
3. Theorem II.4.35 gives infinitely many odd composite starts whose first prime is any prescribed prime other than 3, with explicit arrival times and a logarithmic counting bound.
4. Corollary I.10.7 completes the even-exponent Mersenne-shaped divisor classification. Theorem III.8.2 extends the fixed-prefix prime-only obstruction to o-minimal rules, including fixed expressions using real exponentials and logarithms.

These are explicit extensions and auxiliary classifications. They are not presented as resolutions of the famous conjectures excluded above. The additional G_2 shift-filter rigidity result appears in Theorem II.5.14.

Part I

Recurrences, finite differences, and prime orbit families

I.1 Framework and conventions

The starting point is the binomial-power sequence

$$A_s(n) = \sum_{k=0}^{n} \binom{n}{k}^s, \qquad s \geq 1,$$

and the proposed formula

$$\operatorname{ord}_{\mathbb{Q}(n)} A_s = \left\lfloor \frac{s+1}{2} \right\rfloor.$$

Here recurrence order means the number of shifts in a minimal linear recurrence with rational-function coefficients, not the polynomial degrees of those coefficients. Connecting this question to absolute differences does not identify Franel sequences with the sequence of primes. Each connection below has its own hypotheses and proof.

We use three versions of the Collatz map:

$$C(x) = \begin{cases} x/2, & x \text{ even}, \\ 3x+1, & x \text{ odd}, \end{cases} \qquad T(x) = \begin{cases} x/2, & x \text{ even}, \\ (3x+1)/2, & x \text{ odd}, \end{cases}$$

and the map on positive odd integers

$$S(x) = \frac{3x+1}{2^{v_2(3x+1)}}.$$

A path of L shortcut steps, containing R odd branches, uses $L+R$ standard C-steps. A path of t odd-to-odd steps whose halving exponents sum to K uses $K+t$ standard steps. Infinite orbits retain the cycles $1 \mapsto 4 \mapsto 2 \mapsto 1$ for C and $1 \mapsto 2 \mapsto 1$ for T. Orbit polynomials, in contrast, stop at the first occurrence of 1.

For a sequence a, write $(Ea)_n = a_{n+1}$, $(\Delta a)_n = a_{n+1} - a_n$, and $(Da)_n = |a_{n+1} - a_n|$. A sequence is P-recursive if it satisfies a nonzero linear recurrence with polynomial coefficients for all sufficiently large indices. Unless explicitly restricted, an extension of a finite Gilbreath sequence is an integer; it need not exceed the last entry.

Object	Result treated here	Principal method
Franel moments	Minimal order and stability under sampling and filtering	Critical-value singularities
Finite Gilbreath triangles	First and second extension-width extrema	Pascal bounds and deficits
Collatz polynomials	Exact zero accumulation set	Finite parity words and analytic limits
Orbit observations	Recurrence rigidity	Finite-state determinism and integrality
Prime orbit families	Coalescence and finite prime descents	BFTB and Maynard theorems
Fixed inverse families	Divisibility and large-factor blocks	Congruences and gcd identities
Giant-prime certification	Exact conditions and limited obstructions	Digit sums, Pocklington and Hasse bounds

The arguments are presented as research proofs, with their hypotheses exposed for independent checking. Finite computations certify the displayed examples; they do not establish the universal statements in place of the proofs.

I.1.1 Additional results in this revision

Beyond making all verification internal to the manuscript, this revision strengthens six parts of the investigation.

1. Arithmetic sampling and any nonzero polynomial shift filter preserve Franel recurrence order (Theorem I.2.5).
2. The Collatz recurrence obstruction extends to nonconstant algebraic branches, nonzero-index scaling observations, and logarithmic observations (Theorem I.4.3).
3. Prime descents may follow any prescribed periodic odd-step word with net contraction, subject to an explicit congruence compatibility condition (Theorem I.7.3).
4. Every prescribed-divisor index set of the fixed inverse family is classified by an exact periodic congruence, with its density (Theorem I.10.2).
5. Blocks with many distinct large prime factors occur throughout an arithmetic progression while retaining their original pairwise gcd matrix (Theorem I.10.5).
6. For a prime inverse term P_a, the existence of an elliptic curve whose order is divisible by $2^r - 1$ has an exact threshold, and all admissible traces are classified (Theorem I.11.6).

These statements have proofs below. The elliptic result uses Waterhouse's classical classification; the prime-chain results use established sieve theorems. Claims of historical novelty require a separate literature review.

I.2 Franel sequences and absolute-difference triangles

I.2.1 Recurrence order and visible critical values

For integers $s \geq 1$, put

$$A_s(n) = \sum_{k=0}^{n} \binom{n}{k}^s, \qquad m_s = \left\lfloor \frac{s+1}{2} \right\rfloor.$$

The parameter s selects a sequence; n indexes its terms. By the *recurrence order* of a sequence we mean the least shift width r of a nontrivial relation, valid for all sufficiently large n,

$$\sum_{j=0}^{r} p_j(n) a(n+j) = 0, \qquad p_j \in \mathbb{Q}(n), \quad p_0 p_r \neq 0.$$

This is different from the degrees of the coefficient polynomials. Stoll proves an upper bound m_s. Straub–Zudilin prove the corresponding lower bound for telescoping recurrences based on the original binomial summand [1, 2]. The argument below proposes the unrestricted lower bound using a real moment representation. Its proof is included in full; it is not being attributed to the creative-telescoping theorem.

Lemma I.2.1 (Singularities bounded by shift width)**.** *Suppose that $F(z) = \sum_{n\geq 0} a(n) z^n$ has positive radius of convergence and its coefficients satisfy a rational-coefficient recurrence of shift width r. The union of the finite nonzero nonremovable singularities of branches obtained by analytic continuation of its germ at zero has at most r points.*

Proof. Clear denominators in the recurrence and write $\theta = z\,d/dz$. Multiplication by z^{n+r} and summation give

$$\sum_{j=0}^{r} z^{r-j} p_j(\theta - j) F(z) = R(z),$$

where R is a polynomial accounting for the finitely many initial terms. Let $D = \max_j \deg p_j$ and $c_j = [n^D]p_j(n)$. The coefficient of the highest derivative is $z^D H(z)$, where

$$H(z) = \sum_{j=0}^{r} c_j z^{r-j} \neq 0, \qquad \deg H \leq r.$$

Every point with $zH(z) \neq 0$ is an ordinary point of this linear differential equation, so every continued branch extends holomorphically there. For $D = 0$ the same conclusion follows directly from $HF = R$. □

Lemma I.2.2 (A signed Morse cusp remains visible)**.** *Let Φ be a real analytic function on a compact real analytic manifold of dimension $d \geq 1$, endowed with a positive analytic density μ. Suppose the critical points above a nonzero value v are finite, nondegenerate, and all maxima or all minima. Let w be a real analytic function of one variable which is not identically zero near v. Then*

$$F_w(z) = \sum_{n \geq 0} \left(\int \Phi^n w(\Phi)\, d\mu \right) z^n$$

has a continued branch with a nonremovable singularity at $z = 1/v$. The same conclusion holds for an analytic amplitude on the manifold whose value has a common nonzero sign at these critical points.

Proof. Use analytic Morse coordinates to write $\Phi = v + \epsilon|y|^2$ near each relevant critical point, with a common $\epsilon \in \{1, -1\}$. For the unweighted pushforward measure $\nu = \Phi_* \mu$, integration over spheres shows that its local density is

$$\rho(v + t) = r(t) + \mathbf{1}_{\{\epsilon t > 0\}} |t|^{d/2-1} h(|t|), \qquad r \in C^\infty, \quad h \text{ analytic}, \quad h(0) > 0.$$

Indeed the contribution at radius $\sqrt{u}$ is $\frac{1}{2} u^{d/2-1} \int_{\mathbb{S}^{d-1}} g(\sqrt{u}\omega)\, d\omega$; the odd terms of the analytic density g integrate to zero. Smooth cutoffs supported away from the critical points contribute only r. All critical-point contributions have positive leading coefficients on the same side, so they cannot cancel.

Write $w(v + t) = t^e a(t)$, where $e \geq 0$ is finite and $a(0) \neq 0$. The weighted density has a one-sided term with exponent $d/2 - 1 + e$ and nonzero leading coefficient. If this exponent is a nonnegative integer, its corresponding derivative jumps; otherwise the appropriate finite derivative fails to be continuous. Thus the density is not C^∞ at v. For an amplitude on the manifold with a common nonzero sign, the same spherical averaging gives a common nonzero leading coefficient, and the argument is unchanged.

For the resulting finite signed measure ν_w, its Cauchy transform

$$G(w_0) = \int \frac{d\nu_w(u)}{w_0 - u}$$

is analytic in the upper half-plane and agrees near infinity with $w_0^{-1} F_w(w_0^{-1})$. The distributions $-\pi^{-1} \operatorname{Im} G(x + iy)$ converge to ν_w as $y \downarrow 0$. If G extended holomorphically across a disk centered at v, that boundary density would be real analytic on the disk's real diameter. This contradicts the one-sided cusp. The inversion $z = 1/w_0$ proves the assertion for a branch connected to the original power series. □

Theorem I.2.3 (Franel minimality and finite filters)**.** *For every $s \geq 1$,*

$$\operatorname{ord}_{\mathbb{Q}(n)} A_s = m_s.$$

For $s \geq 2$, every nonzero $q \in \mathbb{Q}[z]$ also satisfies

$$\operatorname{ord}_{\mathbb{Q}(n)}(q(E)A_s) = m_s, \qquad (Ea)(n) = a(n+1).$$

Proof. For $s \geq 2$, set $d = s-1$, $X = x_1 \cdots x_d$, and

$$P_s = \left(\prod_{i=1}^{d}(1+x_i)\right)(1+X^{-1}).$$

Expansion gives $A_s(n) = \operatorname{CT} P_s^n$. On the unit torus $x_i = e^{i\theta_i}$ this Laurent polynomial is the real function

$$\Phi_s(\theta) = 2^s \prod_{i=1}^{d} \cos(\theta_i/2)\cos\big((\theta_1 + \cdots + \theta_d)/2\big).$$

The constant term is therefore its moment under Haar probability measure. On a nonzero fiber all the factors are nonzero. The critical point equations are

$$\frac{x_i}{1+x_i} - \frac{1}{1+X} = 0 \qquad (1 \leq i \leq d).$$

They imply $x_i X = 1$ for all i, hence $x_1 = \cdots = x_d = \omega$, $\omega^s = 1$, and $\omega \neq -1$. Consequently the distinct nonzero critical values are exactly

$$v_j = (-1)^j\big(2\cos(\pi j/s)\big)^s, \qquad 0 \leq j < s/2.$$

There are m_s such values and their absolute values strictly decrease. At the representative with $\omega = e^{2\pi i j/s}$ the Hessian in angle coordinates is

$$-(-1)^j\big(2\cos(\pi j/s)\big)^{s-2}(I_d + J_d),$$

where J_d is the all-ones matrix. The eigenvalues of $I_d + J_d$ are 1 and s, so the Hessian is definite. The conjugate point has the same critical value and the same type of extremum. Lemmas I.2.1 and I.2.2 give the lower bound m_s, while Stoll's theorem gives the upper bound.

Applying $q(E)$ replaces the moment measure by $q(u)d\nu(u)$. At each v_j, a polynomial $q \neq 0$ can vanish only to finite order. Lemma I.2.2 proves that every one of the m_s singularities survives, even when $q(v_j) = 0$. This gives the lower bound for the filtered sequence. Conversely, all shifts of A_s belong to the m_s-dimensional vector space over $\mathbb{Q}(n)$ generated by m_s consecutive shifts. Any $m_s + 1$ shifts of $q(E)A_s$ are therefore dependent, giving the upper bound. For $s = 1$, $A_1(n) = 2^n$ has order one. The restriction $s \geq 2$ in the filter claim is essential, since $(E-2)A_1 = 0$. □

Corollary I.2.4 (Every fixed absolute-difference row)**.** *Let $Da(n) = |a(n+1) - a(n)|$ and $\Delta = E - 1$. For all $s \geq 1$ and $n, k \geq 0$,*

$$D^k A_s(n) = \Delta^k A_s(n) > 0.$$

For fixed k, this sequence has recurrence order m_s.

Proof. For $s \geq 2$, the coefficients of P_s are nonnegative and its constant term is 2. Thus $P_s - 1$ has nonnegative coefficients and constant term 1, and

$$\Delta^k A_s(n) = \operatorname{CT}\big(P_s^n (P_s - 1)^k\big) \geq 2^n > 0.$$

Every ordinary difference is positive, so every absolute value can be removed. Apply Theorem I.2.3 with $q(z) = (z-1)^k$. The case $s = 1$ is immediate from $A_1(n) = 2^n$. □

I.2.2 An additional strengthening: arithmetic subsequences

The same mechanism answers a further question: can retaining only one arithmetic progression of indices make the recurrence shorter?

Theorem I.2.5 (Sampling does not reduce the order)**.** *Fix integers $s \geq 2$, $a \geq 1$, and $b \geq 0$. Put $B(n) = A_s(an + b)$. For every nonzero $q \in \mathbb{Q}[z]$,*

$$\boxed{\operatorname{ord}_{\mathbb{Q}(n)}(q(E)B) = m_s.}$$

Proof. Use the preceding torus representation and put $\Psi = \Phi_s^a$. The sequence in question is the moment sequence

$$(q(E)B)(n) = \int \Psi^n \, \Phi_s^b q(\Psi) \, d\mu.$$

On nonzero fibers, $d\Psi = a\Phi_s^{a-1} d\Phi_s$, so its critical points are exactly the nonzero critical points already classified. Their critical values are $\lambda_j = v_j^a$. These remain pairwise distinct even when a is even, since the $|v_j|$ are pairwise distinct. At a critical point,

$$\operatorname{Hess} \Psi = a v_j^{a-1} \operatorname{Hess} \Phi_s.$$

It is definite, and all points in a given critical fiber have the same type. If a is even, all these nonzero critical points are maxima.

The analytic amplitude Φ_s^b equals $v_j^b \neq 0$ at every critical point above λ_j, with a common sign. If q vanishes to order e_j at λ_j, local spherical averaging gives a one-sided density term with exponent

$$d/2 - 1 + e_j$$

and nonzero leading coefficient. Regular parts of the fiber contribute only a smooth density and cannot cancel this cusp. This is precisely the signed-amplitude argument in Lemma I.2.2; positivity of the global weighted measure is not required. Thus the generating function has nonremovable singularities at all m_s points $1/\lambda_j$. Lemma I.2.1 gives the lower bound.

For the upper bound, write a recurrence of order m_s for A_s as a first-order rational matrix system on m_s consecutive shifts. Advancing by a steps multiplies its rational transition matrices, and evaluation at $an + b$ yields another system of the same dimension over $\mathbb{Q}(n)$. The sampled sequence and all its shifts are rational linear functionals of this system. The same holds after applying $q(E)$; its $m_s + 1$ consecutive shifts are dependent. This proves the upper bound. □

In particular, taking every second, third, or any fixed ath Franel term, and then taking any fixed ordinary difference, preserves the minimal recurrence order. This is an extension of the proof in this manuscript, not a claim of historical priority.

A fixed row $D^k A_s(n)$ with variable n is different from the left edge $D^k A_s(0)$ with variable k. For example, the latter begins $1, 1, 7, 31, 175, \ldots$ when $s = 3$. Moreover, recurrence order is unchanged by modifying finitely many terms, so order information alone cannot exclude an arbitrarily long finite block of zeros or of values in $\{0, d\}$. These observations prevent an automatic deduction of the prime Gilbreath conjecture from Franel minimality.

I.2.3 Maximum extension width for finite Gilbreath sequences

For a strictly increasing integer sequence $S = (s_1, \ldots, s_n)$ with $(s_1, s_2) = (2, 3)$ define

$$t_{0,j} = s_j, \qquad t_{i+1,j} = |t_{i,j+1} - t_{i,j}|.$$

Write $S \in \mathcal{G}_n$ if $t_{i,1} = 1$ for every $1 \le i < n$. Let K_S be the set of integers k for which (S, k) has the same left-edge property. The extension need not satisfy $k > s_n$; this is Muney's extension-set convention [3].

Lemma I.2.6 (A Pascal bound)**.** *Every nonnegative absolute-difference triangle satisfies*

$$t_{i,j} \le \sum_{h=0}^{j-1} \binom{j-1}{h} t_{i+h,1} \qquad (i \ge 1)$$

whenever the entries exist. In particular, $S \in \mathcal{G}_n$ implies $t_{i,j} \le 2^{j-1}$.

Proof. Iterate $t_{i,j+1} \le t_{i,j} + t_{i+1,j}$ in the column index and use Pascal's identity. The stated special case is the sum of a binomial row. □

Theorem I.2.7 (Maximum width and its unique maximizer)**.** *For every $n \ge 2$,*

$$\max_{S \in \mathcal{G}_n} |K_S| = 2^{n-1} + 1.$$

The unique maximizing sequence is $U_n = (2, 3, 5, 9, \ldots, 2^{n-1} + 1)$, and $K_{U_n} = \{1, 3, \ldots, 2^n + 1\}$.

Proof. Put $e_i = t_{i,n-i}$ and $A = \sum_{i=1}^{n-1} e_i$. After appending k, the new right boundary is

$$r_0 = |k - s_n|, \qquad r_i = |r_{i-1} - e_i|.$$

Validity is equivalent to $r_{n-1} = 1$. If $r_0 > A + 1$, every fold is subtraction and the last value is $r_0 - A > 1$. Thus $r_0 \le A + 1$.

In every positive row the first column is 1, while all later columns are even. The second-column assertion follows from $|t_{i,2} - 1| = 1$; the congruence $t_{i,j+1} \equiv t_{i,j} + t_{i+1,j} \pmod 2$ propagates it. Consequently A is odd, and $r_{n-1} \equiv r_0 + A \pmod 2$ forces r_0 even. The candidate interval contains $A + 2$ integers of the required parity. Hence

$$|K_S| \le A + 2 \le 2 + \sum_{i=1}^{n-1} 2^{n-i-1} = 2^{n-1} + 1.$$

Equality implies, in particular, $t_{1,n-1} = 2^{n-2}$. Equality in the Pascal bound then forces equality in both predecessor bounds. Repeating gives $t_{1,j} = 2^{j-1}$ for all j, and strict increase of S yields $s_j = 2^{j-1} + 1$ for $j \ge 2$.

For U_n, every positive row has $t_{i,j} = 2^{j-1}$, and the folding boundary is $(2^{n-2}, 2^{n-3}, \ldots, 1)$. All even distances $0 \le r_0 \le 2^{n-1}$ fold to 1: after the first fold the even distances range from 0 to 2^{n-2}, and iteration ends by folding 0 or 2 about 1. Thus all candidates are valid. □

This proves the finite maximum-width assertion posed as [3, Conjecture 30], for every $n \ge 2$. The proof is a uniform triangle inequality argument, rather than an inference from enumeration. It does not assume or prove that every initial segment of the primes belongs to $\mathcal{G}_n$.

Corollary I.2.8 (Increasing extensions and the length-one case)**.** *For $n \ge 2$, the maximum number of valid extensions with $k > s_n$ is 2^{n-2}, uniquely attained by U_n. If length one is separately allowed, $S = (2)$ has $K_S = \{1, 3\}$ and one increasing extension.*

Proof. K_S is symmetric about s_n. For U_n the positive distances number 2^{n-2}. If $S \ne U_n$, equality in the bound for A is impossible; since A is odd, $|K_S| \le 2^{n-1} - 1$, which gives at most $2^{n-2} - 1$ positive distances by symmetry. For length one, solve $|k - 2| = 1$ directly. □

For example, $U_4 = (2, 3, 5, 9)$ has the nine extensions $1, 3, 5, 7, 9, 11, 13, 15, 17$. The probabilistic models and deterministic obstacles studied by Chase–Hunter–Tao concern a different route toward the original prime problem [4].

I.2.4 The second-largest extension width

Theorem I.2.9 (Second maximum and uniqueness)**.** *For every $n \geq 4$,*

$$\max_{S \in \mathcal{G}_n \setminus \{U_n\}} |K_S| = 2^{n-1} - 2n + 5.$$

The unique maximizer $W_n = (w_1, \ldots, w_n)$ is given by

$$w_1 = 2, \qquad w_j = 2^{j-1} + 1 - (j-2)(j-3) \quad (2 \leq j \leq n).$$

Thus W_n is the length-n prefix of $2, 3, 5, 7, 11, 21, 45, 99, \ldots$.

Proof. Define the deficits and local slacks

$$\delta_{i,j} = 2^{j-1} - t_{i,j}, \qquad \sigma_{i,j} = \delta_{i,j+1} - \delta_{i,j} - \delta_{i+1,j} = t_{i,j} + t_{i+1,j} - t_{i,j+1}$$

for $i, j \geq 1$ with $i + j + 1 \leq n$. All are nonnegative and $\delta_{i,1} = 0$. The total antidiagonal deficit satisfies

$$D := \sum_{i=1}^{n-1} \delta_{i,n-i} = \sum_{i,j:\, i+j+1 \leq n} \binom{n-j-1}{i-1} \sigma_{i,j}. \tag{I.2.1}$$

To verify the identity, follow a contribution $\sigma_{i,j}$ from $\delta_{i,j+1}$ through $\delta_{i,j+1} = \delta_{i,j} + \delta_{i+1,j} + \sigma_{i,j}$. There are $\binom{n-i-j-1+u}{u}$ paths to antidiagonal row $i-u$, where $0 \leq u \leq i-1$. Their sum is $\binom{n-j-1}{i-1}$ by the hockey-stick identity.

If $S \neq U_n$, at least one slack is nonzero. Choose its earliest column j. All earlier deficits vanish, so $t_{i,j} = t_{i+1,j} = 2^{j-1}$. The absolute-difference relation forces $t_{i,j+1} \in \{0, 2^j\}$; a nonzero slack is exactly 2^j. It cannot occur at $i = 1$, since that would give a zero first-row gap and violate strict increase. Therefore $2 \leq i \leq n - j - 1$, and

$$D \geq 2^j \binom{n-j-1}{i-1} \geq 2^j (n-j-1) \geq 2(n-2).$$

For the middle inequality, the smallest nonendpoint entry of a binomial row with index N is N. For the last, put $N = n - j - 1 \geq 2$ and use $2^{j-1} N \geq N + j - 1$. It is strict for $j \geq 2$, since $(2^{j-1} - 1)N - j + 1 \geq 2^j - j - 1 > 0$. Since $A = 2^{n-1} - 1 - D$, the width bound becomes $|K_S| \leq 2^{n-1} - 2n + 5$.

If equality holds, every weight in (I.2.1) is positive, so there is exactly one nonzero slack. It equals 2, has column 1, and lies in row 2 or $n-2$, because $\binom{n-2}{i-1} = n - 2$. For $n = 4$ these positions coincide.

If $\sigma_{2,1} = 2$, the deficit recurrence gives

$$\begin{aligned} t_{1,1} &= 1, \quad t_{1,j} = 2^{j-1} - 2(j-2) \quad (j \geq 2), \\ t_{2,1} &= 1, \quad t_{2,j} = 2^{j-1} - 2 \quad (j \geq 2), \\ & \qquad t_{i,j} = 2^{j-1} \quad (i \geq 3). \end{aligned}$$

These entries directly satisfy the absolute-difference rule, and all first-row gaps are positive. Summing them gives exactly W_n. Its antidiagonal is

$$e_1 = 2^{n-2} - 2(n-3), \quad e_2 = 2^{n-3} - 2, \quad e_i = 2^{n-i-1} \quad (3 \leq i \leq n-1).$$

For $i \leq n-2$ it satisfies $e_i \leq 1 + \sum_{j>i} e_j$. Starting with the inverse images $\{0, 2\}$ of the final equation $|d - 1| = 1$, suppose that the remaining inverse-image set consists of all even d between 0 and R. If the next fold has even center $e_i \leq R$, its nonnegative inverse images fill all even

values from 0 to $R + e_i$: values below e_i come from $e_i - d$, and those above from $e_i + d$. Reverse induction proves that every candidate distance for W_n is valid. Hence $|K_{W_n}| = A + 2$.

If instead $\sigma_{n-2,1} = 2$, the sequence is U_{n-1} followed by $2^{n-1} - 1$; call it V_n. For $n \geq 5$ its antidiagonal is $e_i = 2^{n-i-1} - 2$ for $i \leq n - 2$, followed by $e_{n-1} = 1$. Starting at distance zero, all folds after the first are subtractions, and the final value is

$$e_1 - \sum_{i=2}^{n-1} e_i = 2n - 7 \geq 3.$$

Thus the candidate $k = s_n$ is invalid and $|K_{V_n}| < A + 2$. For $n = 4$, $V_4 = W_4$ has already been covered. This proves uniqueness. □

I.2.5 A complete finite verification procedure inside the manuscript

The following exact set recursion reproduces every finite extension calculation used here, without any supplementary program. Given S, first form its difference triangle and its antidiagonal $e_1, \ldots, e_{n-1}$. Set $R_{n-1} = \{1\}$, and for $i = n - 1, n - 2, \ldots, 1$ set

$$R_{i-1} = \{e_i + y : y \in R_i\} \cup \{e_i - y : y \in R_i,\ y \leq e_i\}.$$

Repeated elements are removed because these are sets. Then

$$K_S = \{s_n - d, s_n + d : d \in R_0\}.$$

Correctness follows at each step from solving $|x - e_i| = y$ with $x \geq 0$. To enumerate $\mathcal{G}_n$, start at $\mathcal{G}_2 = \{(2, 3)\}$ and, for each sequence already constructed, append every $k \in K_S$ with $k > s_n$. Every admissible prefix appears exactly once. The bound in Theorem I.2.7 makes every stage finite. Computing the triangle of each output also verifies (I.2.1) using only integer additions, subtractions, absolute values, and binomial coefficients.

n	$\lvert\mathcal{G}_n\rvert$	Maximum width	Second maximum
2	1	3	—
3	1	5	—
4	2	9	5
5	6	17	11
6	27	33	25
7	180	65	55
8	1786	129	117

The table records finite checks of the preceding recursion; the proofs of the two extremal theorems do not rely on its entries. In particular, neither extremal theorem determines whether the next actual prime is one of the valid extensions.

I.3 The exact accumulation set of Collatz polynomial zeros

Throughout this section, $T(x) = x/2$ for even x and $T(x) = (3x + 1)/2$ for odd x. For a positive integer N that reaches 1, put

$$\tau(N) = \min\{j \geq 0 : T^j(N) = 1\}, \qquad P_N(z) = \sum_{j=0}^{\tau(N)} T^j(N) z^j.$$

Let $\mathcal{Z}$ be the union of the zero sets of these polynomials, and let $\mathcal{Z}'$ be its derived set: limits of sequences of distinct points of $\mathcal{Z}$. Since $P_1 = 1$, allowing $N = 1$ does not change either set. No assertion that every positive integer reaches 1 is needed.

Hohertz's questions [5, Section 5] include an explicit lower bound depending only on N and the existence of zeros approaching 1. The theorem below supplies such a bound and proves the latter existence statement. His accompanying reference to a parabola does not specify a precise global inequality; we make no claim to settle every interpretation of that additional wording.

Theorem I.3.1 (Exact derived set and a bound depending on the starting value)**.** *One has*

$$\mathcal{Z}' = \{z \in \mathbb{C} : 2/3 \le |z| \le 2\}, \tag{I.3.1}$$

$$P_N(\xi) = 0 \quad \Longrightarrow \quad |\xi| \ge R_N := \frac{4N}{5N + 8 + \sqrt{N^2 + 56N + 64}}, \tag{I.3.2}$$

where $N \ge 2$ and $\tau(N) < \infty$. In particular, $R_N \to 2/3$. More precisely, for $0 \le r < 2/3$, the existence of a zero with $|\xi| \le r$ implies

$$N + 1 \le \frac{1 + 3r/2}{(1 - 3r/2)(1 - r)}. \tag{I.3.3}$$

Consequently, every compact subset of $|z| < 2/3$ contains only finitely many distinct Collatz zeros.

I.3.1 Realizing a finite parity word by a terminating orbit

For a parity word $v = (\varepsilon_0, \dots, \varepsilon_{L-1})$ of length L, write $R = \sum \varepsilon_j$. Formal composition of its branches gives

$$T_v(x) = \frac{3^R x + b_v}{2^L}, \qquad b_v = \sum_{\varepsilon_j = 1} 2^j 3^{\sum_{i>j} \varepsilon_i}. \tag{I.3.4}$$

If $R \ge 1$, the last nonzero summand shows that $3 \nmid b_v$. The element 2 generates $(\mathbb{Z}/3^R\mathbb{Z})^\times$: indeed, $v_3(4^u - 1) = 1 + v_3(u)$ gives its order $2 \cdot 3^{R-1}$. Hence the nonnegative integers t satisfying

$$2^{L+t} \equiv b_v \pmod{3^R}$$

form an infinite arithmetic progression. Define

$$N_t = \frac{2^{L+t} - b_v}{3^R}.$$

For an integer x, the condition $3^R x + b_v \equiv 0 \pmod{2^L}$ is equivalent to the first L actual parity bits being v. To see this, reduce modulo 2 to obtain $x \equiv \varepsilon_0 \pmod 2$, apply this actual first branch, and use induction on the remaining word. Therefore $T^L(N_t) = 2^t$. Every prefix composition has positive slope, so all the first L values are greater than 1 for sufficiently large t. The subsequent values are powers of 2, and $\tau(N_t) = L + t$. If $R = 0$, the same conclusion follows simply by starting at 2^{L+t}.

I.3.2 Repeated words and circles of accumulation

Fix a word w of length k and weight $1 \le r \le k$, and put

$$\alpha_j = \frac{3^{\varepsilon_0 + \cdots + \varepsilon_{j-1}}}{2^j}, \quad q = \frac{3^r}{2^k}, \quad B(z) = \sum_{j=0}^{k-1} \alpha_j z^j, \quad Q(z) = q z^k,$$

$$E(z) = \sum_{\varepsilon_j = 1} \alpha_j z^{j+1}, \qquad A(z) = (1 - z/2)B(z) = 1 - Q(z) + E(z).$$

Here E is not the zero polynomial. Apply the preceding construction to w repeated m times, followed by a power-of-two tail, and denote the starting value by $N_{m,t}$. With m fixed first and then $t \to \infty$ through its permitted progression, one has locally uniformly on $|z| < 2$

$$\frac{P_{N_{m,t}}(z)}{N_{m,t}} \longrightarrow f_m(z) := B(z) \sum_{a=0}^{m-1} Q(z)^a + \frac{Q(z)^m}{1 - z/2}. \tag{I.3.5}$$

The finitely many affine correction terms vanish after division by $N_{m,t}$, while the power-of-two tail converges to its geometric series. Thus

$$H_m := (1 - z/2) f_m = 1 + E \sum_{a=0}^{m-1} Q^a, \qquad (1 - Q) H_m = A - Q^m E.$$

On the circle $|z| = \rho := 2/3^{r/k}$, choose a point z_0 with $A(z_0)E(z_0)(1 - Q(z_0)) \neq 0$. Only finitely many points are excluded. On a sufficiently small neighborhood choose an analytic logarithm ℓ of A/E, and write $Q(z_0) = e^{i\theta}$. Choose integers h_m such that $\delta_m = 2\pi h_m - m\theta \in [-\pi, \pi]$. For all sufficiently large m, the equation

$$u = \ell\big(z_0 e^{u/(km)}\big) + i\delta_m$$

has a bounded solution u_m. Indeed, on the circle of radius 1 centered at $\ell(z_0) + i\delta_m$, the difference $\ell(z_0 e^{u/(km)}) - \ell(z_0)$ is uniformly smaller than 1; Rouché's theorem applies to the resulting equation in u. Consequently,

$$z_m = z_0 e^{u_m/(km)} \longrightarrow z_0, \qquad Q(z_m)^m = A(z_m)/E(z_m).$$

Since $1 - Q$ does not vanish nearby, $H_m(z_m) = 0$. For each sufficiently large m, another application of Rouché's theorem to (I.3.5), on a small isolating circle, produces zeros of actual $P_{N_{m,t}}$ arbitrarily close to z_m. The corresponding starting values can be arbitrarily large.

It follows that $\overline{\mathcal{Z}}$ contains the circles of radii $2/3^{r/k}$ for all rational $r/k \in (0, 1]$. These radii are dense in $[2/3, 2]$, so the entire closed annulus belongs to $\overline{\mathcal{Z}}$. Every neighborhood of every point of this annulus contains an open subset of its interior. Such an open subset cannot contain only finitely many points of $\mathcal{Z}$, since a smaller open subset avoiding those points would contradict density. Thus every point of the annulus is in fact in $\mathcal{Z}'$.

I.3.3 Local finiteness inside the annulus and an explicit bound

Write $a_j = T^j(N)$ and $\tau = \tau(N)$, and introduce

$$\widetilde{P}_N(z) = P_N(z) + \sum_{j=0}^{\tau} z^j, \qquad b_j = \frac{a_j + 1}{N + 1}(2/3)^j.$$

On an odd branch, $a_{j+1} + 1 = \frac{3}{2}(a_j + 1)$; on an even branch the corresponding inequality holds. Hence $1 = b_0 \geq b_1 \geq \cdots \geq b_\tau > 0$. Set $b_{\tau+1} = 0$ and $d_j = b_{j-1} - b_j$, so that $d_j \geq 0$ and $\sum_{j=1}^{\tau+1} d_j = 1$. With $w = 3z/2$,

$$(1 - w)\frac{\widetilde{P}_N(z)}{N + 1} = 1 - \sum_{j=1}^{\tau+1} d_j w^j.$$

For $|z| \leq r < 2/3$, this gives

$$|\widetilde{P}_N(z)| \geq (N + 1)\frac{1 - 3r/2}{1 + 3r/2}, \qquad |\widetilde{P}_N(z) - P_N(z)| \leq \frac{1}{1 - r}.$$

If $P_N(z) = 0$, their combination proves (I.3.3). Setting $r = |z|$ and rearranging yields

$$3(N+1)r^2 - (5N+8)r + 2N \le 0.$$

The smaller root is (I.3.2); its rationalized form is the one stated there. Zeros with $|z| \ge 2/3$ satisfy the same lower bound automatically.

For each $r < 2/3$, only finitely many starting integers satisfy (I.3.3), and each terminating polynomial has only finitely many zeros. There are therefore no derived points in $|z| < 2/3$. Finally, $a_j/a_{j+1} \le 2$; the Eneström–Kakeya theorem places every zero in $|z| \le 2$. This proves Theorem I.3.1. □

The lower bound depends on N alone, although it need not improve the best orbit-dependent bound for every individual N. The derived set must not be confused with the closure: isolated zeros may occur in $|z| < 2/3$.

Corollary I.3.2 (Arbitrarily narrow approaches to an interior point)**.** *Let $2/3 < |\zeta| < 2$. If U_j are nonempty open subsets of the annulus with $\sup_{z\in U_j} |z-\zeta| \to 0$, then there are distinct Collatz zeros $\xi_j \in U_j$, after passing to a subsequence if necessary. In particular, for every $p > 0$, zeros approach 1 through the cusp*

$$0 < |\operatorname{Im} z|, \qquad |\operatorname{Re} z - 1| < |\operatorname{Im} z|^p.$$

No nonempty open subset of $2/3 < |z| < 2$ is free of Collatz zeros.

Proof. Each nonempty open subset of the annulus contains infinitely many distinct zeros, as proved above. Select a zero avoiding the finitely many earlier selections. For the cusp, also impose $0 < |z-1| < 1/j$; the resulting open set is nonempty for every sufficiently large j. □

This corollary gives a precise interpretation of arbitrary thin approaches to 1. It also rules out any proposed universal zero-free parabolic region whose interior meets the open annulus.

I.4 Recurrence rigidity of Collatz orbits and absolute-difference rows

Here $a_n = T^n(a_0)$ is an infinite orbit: after reaching 1 it continues around $1 \leftrightarrow 2$. A sequence is *P-recursive* if it satisfies a nonzero polynomial-coefficient linear recurrence for all sufficiently large indices. The coefficient field may be $\mathbb{C}$. An orbit called nonperiodic below means one that is not eventually periodic.

Lemma I.4.1 (Rigidity of power-size observations)**.** *Let a_n be a nonperiodic positive integer Collatz orbit. If $u_n = c\,a_n^d(1+o(1))$, where $c \ne 0$ and $d \in \mathbb{R}\setminus\{0\}$, then u_n is not P-recursive. More generally, the same conclusion holds whenever, for every fixed j, uniformly over the finitely many length-j parity words,*

$$\frac{u_{n+j}}{u_n} = \left(2^{-j}3^{\varepsilon_n+\cdots+\varepsilon_{n+j-1}}\right)^d + o(1). \tag{I.4.1}$$

Proof. A repeated orbit value forces eventual periodicity. Thus the values of a nonperiodic orbit are distinct positive integers and $a_n \to \infty$. For a fixed parity window w_n of length m,

$$a_{n+j} = c_j(w_n)a_n + b_j(w_n), \qquad c_j(w) = 2^{-j}3^{e_0+\cdots+e_{j-1}}.$$

The possible pairs (b_j, c_j) form a finite set. The power asymptotic therefore implies (I.4.1). Suppose $\sum_{j=0}^{m} p_j(n)u_{n+j} = 0$ is a nonzero recurrence. Set $D = \max \deg p_j$, $q_j = [n^D]p_j$, and $r = \max\{j : q_j \neq 0\}$. After division by $n^D u_n$, we obtain

$$\sum_{j=0}^{r} q_j c_j (w_n)^d \longrightarrow 0.$$

The sum takes values in a finite set, so it is exactly zero for all sufficiently large n. If $r = 0$, this contradicts $q_0 \neq 0$. Otherwise, after fixing the first $r-1$ bits, changing the last bit changes the sum by

$$q_r c_{r-1}(w)^d((3/2)^d - (1/2)^d) \neq 0.$$

Every $(r-1)$-bit state therefore admits at most one next bit. The actual parity tail follows a deterministic finite-state path and must eventually be periodic.

It remains to pass from periodic parity to an actual periodic orbit. Let its period have length p and contain h odd branches, with composition $x \mapsto (3^h x + B)/2^p$. Write $x_\ell = a_{N+\ell p}$ and $D_0 = 2^p - 3^h \neq 0$. Then

$$D_0 x_\ell - B = \frac{3^{h\ell}}{2^{p\ell}}(D_0 x_0 - B).$$

Since the left side is an integer and the numerator multiplier is odd, $2^{p\ell} \mid D_0 x_0 - B$ for every ℓ. Thus $D_0 x_0 - B = 0$, so $x_1 = x_0$. This contradicts nonperiodicity. □

Theorem I.4.2 (Orbits, fixed absolute-difference rows, and rational observations)**.** *Define $\mathcal{D}u_n = |u_{n+1} - u_n|$. For fixed $k \geq 0$, put $u_n = \mathcal{D}^k a_n$. Then*

$$u_n = 2^{-k} a_n + e_{k,n}, \qquad |e_{k,n}| \leq \frac{2^k - 2^{-k}}{3}. \tag{I.4.2}$$

The following are equivalent: eventual periodicity of a_n; P-recursiveness of u_n; D-finiteness of $\sum u_n z^n$; and rationality of that generating function. Moreover, for a nonconstant $F \in \mathbb{C}(x)$ for which $F(u_n)$ is eventually defined,

$$F(u_n) \text{ is P-recursive} \iff a_n \text{ is eventually periodic.}$$

Proof. Exactly $\mathcal{D}a_n = (a_n + \varepsilon_n)/2$. The reverse triangle inequality gives

$$|\mathcal{D}v_n - \mathcal{D}w_n| \leq |v_{n+1} - w_{n+1}| + |v_n - w_n|.$$

Thus the error bounds satisfy $C_0 = 0$ and $C_{k+1} = 2C_k + 2^{-k-1}$, giving (I.4.2). A fixed difference row, or a defined pointwise function of an eventually periodic sequence, is eventually periodic. Its generating function is rational, hence D-finite; D-finiteness is equivalent to P-recursiveness of the coefficient sequence. Conversely, a nonperiodic orbit tends to infinity and $u_n \sim 2^{-k} a_n$, so Lemma I.4.1 applies.

At infinity, a nonconstant rational function either satisfies $F(x) \sim cx^d$ with $c \neq 0$ and a nonzero integer d, or satisfies $F(x) - L \sim cx^{-h}$ for a constant L and an integer $h \geq 1$. Apply the same lemma to $F(u_n)$ or to $F(u_n) - L$. Subtracting a constant preserves P-recursiveness. □

I.4.1 Further extension: algebraic, scaling, and logarithmic observations

The obstruction is not restricted to rational functions. The next theorem also covers observations whose growth is too slow to be comparable to a nonzero power of the orbit.

Theorem I.4.3 (Scaling observations and logarithms)**.** *Fix $k \geq 0$ and put $u_n = \mathcal{D}^k a_n$. Each of the following eventually defined observations is P-recursive if and only if the orbit a_n is eventually periodic.*

1. *$F(u_n)$, where, after possibly subtracting a constant, F is eventually nonzero and has the uniform affine scaling property*

$$\frac{F(cx+b)}{F(x)} \longrightarrow c^d \quad (x \to \infty), \qquad d \in \mathbb{R} \setminus \{0\},$$

   for every $c > 0$, uniformly for b in any fixed bounded interval.
2. *$F(u_n)$ for any fixed nonconstant algebraic branch F near positive infinity, provided the same single-valued observation is defined at every sufficiently late u_n.*
3. *$\log u_n$, whenever $u_n > 0$ eventually. More generally, the same holds for $c \log u_n + L + o(1)$, with $c \neq 0$, if this is a fixed pointwise observation that preserves periodic tails.*

In part 1, examples include $F(x) = x^d(\log x)^\beta$ for real $d \neq 0$ and β, on their positive real domains.

Proof. For a nonperiodic orbit, (I.4.2) implies $u_n = 2^{-k}a_n + O(1)$ and $u_{n+j} = c_j(w_n)u_n + O(1)$ for every fixed j. There are finitely many possible slopes, and the additive errors are uniformly bounded. The scaling hypothesis gives (I.4.1), so Lemma I.4.1 applies. Subtracting a constant at the outset does not affect the argument.

A nonconstant algebraic branch has a convergent Puiseux expansion at infinity. Its first nonconstant term, after subtracting a finite limit when needed, is cx^d with a nonzero rational d and $c \neq 0$. The convergent expansion gives the uniform bounded-translation scaling property in part 1. This proves part 2.

For part 3, it suffices to treat $v_n = c \log u_n + L + o(1)$ on a nonperiodic orbit. Since $\log u_n = \log a_n - k \log 2 + o(1)$, its ordinary first difference satisfies

$$v_{n+1} - v_n = c(\varepsilon_n \log 3 - \log 2) + o(1).$$

If v_n were P-recursive, so would be this difference. Let $q_0, \ldots, q_r$, with $q_r \neq 0$, be the highest-degree coefficients of a nonzero recurrence for that difference. Dividing by the corresponding power of n gives

$$\sum_{j=0}^{r} q_j c(\varepsilon_{n+j} \log 3 - \log 2) \longrightarrow 0.$$

Again this expression takes only finitely many values and is eventually zero. The last bit is uniquely determined by the preceding r bits, because changing it changes the sum by $q_r c \log 3 \neq 0$. For $r = 0$ the condition fixes a single bit, or has no solution; either outcome yields eventual constant parity or a contradiction. Thus parity is eventually periodic in every possible case. The divisibility argument in Lemma I.4.1 then forces an actual periodic orbit, a contradiction. The converse follows because a fixed defined pointwise observation of a periodic tail is itself periodic. □

The qualification on fixed observations in part 3 is essential: an arbitrary index-dependent $o(1)$ perturbation of a periodic sequence need not be P-recursive. For example, the theorem applies without that ambiguity to $\log(1+u_n)$ and to a fixed function $F(x) = c \log x + L + o(1)$ as $x \to \infty$, whenever it is defined on the eventual orbit values.

There is also a linear-filter formulation. If E is the shift and $L = \sum \ell_j(n)E^j \neq 0$ has rational-function coefficients, then $L(a)$ is P-recursive if and only if a is eventually periodic. Indeed, if $ML(a) = 0$ for a nonzero recurrence operator M, then $ML \neq 0$ because the Ore polynomial ring $\mathbb{C}(n)[E]$ has no zero divisors. Thus a itself is P-recursive. The converse is a standard closure property of P-recursive sequences.

The depth k in all these results is fixed. They do not apply to the growing-depth diagonal $\mathcal{D}^n a_0$. They neither exclude further positive cycles nor show that every orbit is P-recursive, and therefore do not prove the Collatz or prime Gilbreath conjecture.

I.4.2 A complete first-order classification for the gap between two orbits

Theorem I.4.4 (Gaps between two orbits)**.** *Let $a_n = T^n(x)$ and $b_n = T^n(y)$ with $x, y \geq 1$, and let either $g_n = b_n - a_n$ or $g_n = |b_n - a_n|$. Then g_n satisfies an eventual first-order homogeneous recurrence with rational-function coefficients if and only if one of the following holds:*

1. *The two orbits meet at the same time and subsequently agree.*
2. *Eventually $\{a_n, b_n\} = \{1, 2\}$.*

In the second case the signed gap alternates between 1 and -1, whereas the absolute gap is constantly 1.

Proof. Suppose the orbits do not meet. Then g_n never vanishes, so the recurrence gives $g_{n+1}/g_n = R(n) \in \mathbb{C}(n)$ eventually. If $b_n - a_n$ is even, the two values have the same parity and this ratio is either $1/2$ or $3/2$. If this happens infinitely often, one of those two values occurs infinitely often, forcing the rational function R to be identically that constant. Hence $g_{N+j} = c^j g_N/2^j$ with $c = 1$ or 3. Integrality for every j forces $2^j \mid g_N$ for every j, a contradiction. The two states consequently have opposite parity eventually.

At such a time, write u for the odd state and v for the even state. Then

$$a_{n+1}b_{n+1} = a_n b_n \frac{3 + 1/u}{4} \leq a_n b_n.$$

This positive integer product is eventually constant. Equality forces the odd state to be 1 at every sufficiently late time. The pair $\{1, v\}$ is followed by $\{2, v/2\}$, so $v/2 = 1$ and $v = 2$. Both alternatives in the statement visibly give a first-order recurrence, proving the converse. □

The theorem applies to twin-prime starting values, but primality plays no role in its proof. A classification for arbitrary higher-order P-recursive gaps is not established here.

I.5 Synchronous mergers of consecutive primes

Recall that C is the standard Collatz map and T combines an odd step with the first subsequent halving. An odd step of T therefore takes two steps of C.

Theorem I.5.1 (Consecutive primes merging at prescribed times)**.** *For every integer $m \geq 2$ and every $\delta > 0$, there exist integers $s, c \geq 1$ and distinct positive odd integers $h_1 < \cdots < h_m$ such that, for infinitely many positive integers t, the numbers*

$$p_i = 16^s t + h_i \quad (1 \leq i \leq m)$$

are consecutive in the full sequence of primes and satisfy

$$T^{4s}(p_i) = C^{5s}(p_i) = 3^s t + c < \delta p_1.$$

Each shortcut path contains exactly s odd steps, and $h_m - h_1 < 2^{3s}$. If $0 < \delta \leq 1$, one may take $s = O(m + \log(1/\delta))$.

Proof. For a binary word $w = (e_0, \ldots, e_{L-1})$, put

$$u = \sum_j e_j, \qquad b_w = \sum_{j=0}^{L-1} e_j 2^j 3^{e_{j+1}+\cdots+e_{L-1}}.$$

The integers whose first L shortcut parities equal w form precisely the residue class $3^u n + b_w \equiv 0 \pmod{2^L}$, and on this class $T^L(n) = (3^u n + b_w)/2^L$. Composition proves the forward implication.

Conversely, reduction modulo 2 forces the first parity, and removing that step proves the remaining parities inductively. Distinct words give distinct residue classes. If $u \geq 1$, then

$$0 < b_w \leq 2^{L-u}(3^u - 2^u) < 3^u 2^{L-u} :$$

indeed, the positions $j_1 < \cdots < j_u$ of the ones obey $j_i \leq L - u + i - 1$.

Take $L = 4s$ and words starting with 1 and containing exactly s ones. There are $\binom{4s-1}{s-1}$ of them, and

$$F_s := 3^{-s}\binom{4s-1}{s-1} \geq \frac{(4/3)^{4s-1}}{12s}.$$

To check this bound, let $f_u = 3^{-u}\binom{4s-1}{u-1}$ for $1 \leq u \leq 4s$. The ratio $f_{u+1}/f_u = (4s-u)/(3u)$ shows that $f_s = f_{s+1}$ is maximal, whereas $\sum_{u=1}^{4s} f_u = (4/3)^{4s-1}/3$.

Choose s so that $F_s \geq k^2$. Pigeonholing b_w modulo 3^s leaves at least k^2 words in one class. Choose an integer c_0 with $16^s c_0 \equiv b_w \pmod{3^s}$ in this class and set

$$a_w = (16^s c_0 - b_w)/3^s.$$

These are distinct odd integers and $T^{4s}(16^s t + a_w) = 3^s t + c_0$ for sufficiently large t. For each odd prime $p \leq k$, discard a least populated residue class among the current a_w modulo p. At least

$$k^2 \prod_{\substack{p \leq k \\ p \text{ odd prime}}} (1 - 1/p) \geq k^2 \prod_{j=2}^{k} (1 - 1/j) = k$$

offsets remain. Any k of them give an admissible tuple $\{x + a_w\}$: modulo 2 all offsets are odd; for $p \leq k$ a class is missing; and for $p > k$ at most k classes are forbidden. Add a large common multiple of 16^s to make the offsets positive. This preserves admissibility and the merger identity after changing c_0 to a positive integer c. The bound on b_w gives offset diameter less than 2^{3s}.

The Banks–Freiberg–Turnage-Butterbaugh theorem [7], Theorem 1, says that if k is sufficiently large relative to m, $\{x + r_i\}_{i=1}^{k}$ is admissible, and $\gcd(g, \prod_i r_i) = 1$, then a fixed m-element subset of $\{gt + r_i\}$ consists of consecutive primes for infinitely many t. Here $g = 16^s$ and every offset is odd, so its hypotheses hold. Increasing s until $(3/16)^s < \delta$ gives the desired inequality for all sufficiently large parameters in this infinite set. The $4s$ shortcut steps contain s odd steps and thus take $5s$ standard steps. The sufficient choice $k \log k > e^{8m+4}$ in the cited theorem has $\log k = O(m)$. Both $F_s \geq k^2$ and $(3/16)^s < \delta$ consequently hold with the stated bound on s. □

This result concerns different starting primes merging synchronously. It does not assert this for a preassigned list such as $2, 3, 5, 7, 11$; the arrival value varies with the prime block. Requiring the block size m itself to be prime merely specializes the theorem. Every finite m is allowed, but no fixed finite time can send infinitely many integers to one value: a value of C has at most two predecessors, so a fiber of C^N has at most 2^N elements.

I.6 Gap-two starts and a counterexample for fixed prime gaps

Proposition I.6.1 (Arbitrarily delayed first synchronous merger)**.** *For $j \geq 0$, let $a_j = 56$ when j is even and $a_j = 120$ when j is odd, and put*

$$\mathcal{F}_j = \{8^j(a_j + 128k) - 7 : k \geq 0\}.$$

If $n \in \mathcal{F}_j$, the first synchronous meeting of the T-orbits of n and $n + 2$ occurs at time $3j + 7$. These families are disjoint and their union has natural density $1/112$.

Proof. For every positive integer a, direct substitution gives

$$8a-7 \longmapsto 12a-10 \longmapsto 6a-5 \longmapsto 9a-7,$$
$$8a-5 \longmapsto 12a-7 \longmapsto 18a-10 \longmapsto 9a-5.$$

The differences at these four times are $2, 3, 12a-5, 2$, all nonzero. Repeating the block j times sends $(8^j a-7, 8^j a-5)$ to $(9^j a-7, 9^j a-5)$ without meeting. The base family satisfies

$$T^7(128k+49) = T^7(128k+51) = 27k+11,$$

with successive differences $(2, 3, 192k+79, 2, 1, 72k+30, 36k+15, 0)$. Because $9^j a_j \equiv 56 \pmod{128}$, the iterated block enters this base family. Also $v_2(n+7) = 3j+3$ on $\mathcal{F}_j$, proving disjointness. Its individual density is $1/(128 \cdot 8^j)$. The union over $j \geq J$ lies in $8^{J+1} \mid n+7$, so its upper density tends to zero as $J \to \infty$. Thus the density of the union is the sum of the individual densities, namely $1/112$. □

For example, the primes $6449, 6451$ merge at 1361 after seven shortcut steps. Both paths have three odd steps, so the merger also occurs after ten standard steps. Neither this example nor the family formula implies infinitely many twin primes.

Proposition I.6.2 (Counterexample to the even-gap generalization in A319227)**.** *The standard trajectory of* 171 *first reaches* 1 *after* 124 *steps. It contains the four prime pairs of gap* 6

$$(41, 47), \quad (47, 53), \quad (251, 257), \quad (257, 263),$$

and each pair is adjacent in the sorted set of odd values in that trajectory. Without this adjacency condition there are exactly five pairs, the additional pair being $(103, 109)$*.*

Proof. Here is the complete odd subsequence, including its final value:

$$171, 257, 193, 145, 109, 41, 31, 47, 71, 107, 161, 121,$$
$$91, 137, 103, 155, 233, 175, 263, 395, 593, 445, 167, 251,$$
$$377, 283, 425, 319, 479, 719, 1079, 1619, 2429, 911,$$
$$1367, 2051, 3077, 577, 433, 325, 61, 23, 35, 53, 5, 1.$$

For each adjacent pair x, y in this list, $3x+1 = 2^h y$ for a positive integer h. Summing $h+1$ gives 124. Primality of the relevant entries is checked by trial division by $2, 3, 5, 7, 11, 13$. The odd numbers $43, 45, 49, 51, 253, 255, 259, 261$ are absent, proving adjacency of the four listed pairs. Inspection of the full finite list gives precisely the five unrestricted pairs. All input data for these checks appear above. □

The comment in OEIS A319227 [9] proposes an upper bound of two pairs for every fixed even gap. The displayed trajectory disproves that generalization, including the stronger interpretation requiring adjacency in the sorted odd set. It does not disprove the original gap-two conjecture: the trajectory of 171 has just the twin pair $(107, 109)$. The starts $171 \cdot 2^j$ first halve down to 171, so the counterexample extends to infinitely many starting values.

I.7 Prime descents with congruences and periodic odd-step clocks

Here one trajectory visits several primes in order. The starting and ending primes vary together. No conclusion about infinitely many prime ancestors of one fixed giant prime follows.

Theorem I.7.1 (Prime chains with a constant odd-step clock)**.** *Let $h \geq 2$, $A = 2^h$, $D = A - 3$, and let $Q \geq 1$ satisfy $\gcd(D, Q) = 1$. Write $b = D^{-1} \pmod{Q}$. For every $m \geq 2$ and $0 < \delta < 1$, infinitely many prime starts have a trajectory containing distinct selected primes*

$$p_1 \longrightarrow p_2 \longrightarrow \cdots \longrightarrow p_m, \qquad p_{j+1} < \delta p_j, \qquad p_j \equiv b \pmod{Q}.$$

The arrows denote standard Collatz iterations. The selected visit times can be fixed independently of the starting prime within this infinite family. Between successive underlying odd nodes the exact number of halvings is h.

Proof. Choose a positive multiple d of D such that $(3/A)^d < \delta$. Take k sufficiently large for the Maynard–Tao theorem for general admissible linear forms [8, 7], and let $K = d(k-1)$. The Chinese remainder theorem supplies a positive integer u_0 with

$$u_0 \equiv 0 \pmod{2Q}, \qquad 3^K u_0 \equiv -1 \pmod{D}.$$

For $D = 1$ the second condition is vacuous. Define

$$u(n) = u_0 + 2QDn, \qquad R_i(n) = \frac{A^i 3^{K-i} u(n) + 1}{D} \quad (0 \leq i \leq K).$$

Because $A \equiv 3 \pmod{D}$, these are integer-coefficient linear forms. They are positive and odd, and

$$3R_i(n) + 1 = AR_{i-1}(n), \qquad C^{h+1}(R_i(n)) = R_{i-1}(n).$$

The second identity uses oddness of R_{i-1} to obtain the exact 2-adic valuation h. Every R_i is $b \pmod{Q}$.

Consider the k forms with indices $0, d, \ldots, (k-1)d$. They are admissible. Modulo 2 they are odd; modulo a prime divisor of Q they are the nonzero class b. If $p \nmid 2QD$, choose n with $u(n) \equiv 0 \pmod{p}$; then every form equals $D^{-1} \not\equiv 0 \pmod{p}$. Finally, if $p \mid D$, expansion of $A = 3 + D$ gives

$$R_i(n) \equiv R_0(n) + i3^{K-1} u(n) \pmod{p}.$$

All selected indices are multiples of p, so the forms agree with R_0, whose slope $2Q3^K$ is nonzero modulo p. Avoiding its single root suffices. The positive slopes are distinct, and the forms are nonproportional: writing them as $c_i u(n) + 1/D$, their coefficient determinants are nonzero multiples of $c_i - c_j$.

Maynard–Tao therefore gives infinitely many n for which at least m forms are prime. Pigeonholing their indices yields a fixed set $i_1 < \cdots < i_m$ prime simultaneously for infinitely many n. Read these in decreasing index order. The visit times are $(h+1)(i_m - i_j)$, and

$$\frac{R_{i_j}(n)}{R_{i_{j+1}}(n)} \longrightarrow (3/A)^{i_{j+1} - i_j} \leq (3/A)^d < \delta.$$

Discarding finitely many parameters proves the theorem. □

For $h = 2$ we have $D = 1$ and hence $b = 1$ for every Q. One may prescribe an arbitrary finite chain length, an arbitrarily small decrease ratio, fixed visit times, and any number of terminal digits $00\cdots001$ simultaneously. Other choices give, for $Q = 1000$,

h		2	4	5	6	8	9
D		1	13	29	61	253	509
b	(mod 1000)	001	077	069	541	917	389.

These are existence statements obtained from prime-distribution theorems, not formulas certifying any particular giant candidate.

There is a real local obstruction to using every index when $D > 1$. For $p \mid D$, the values R_i vary linearly with i modulo p; a sufficiently long consecutive index set can cover all residue classes. Restricting the selected indices to multiples of D removes this obstruction. Also, if two nodes joined by one odd step and h halvings are both $b \pmod{Q}$, then necessarily $(2^h - 3)b \equiv 1 \pmod{Q}$. This necessity concerns this one block, not all possible Collatz paths.

The constant-clock construction extends to a whole prescribed word of odd-step valuations. This is an additional result of the present revision.

Lemma I.7.2 (Exact realization of an odd-step word)**.** *Let $\ell \geq 1$, $h_1, \ldots, h_\ell \geq 1$, $H_j = h_1 + \cdots + h_j$ with $H_0 = 0$, and define*

$$H = H_\ell, \quad \alpha = 2^H, \quad \beta = 3^\ell, \quad B = \sum_{j=0}^{\ell-1} 3^{\ell-1-j} 2^{H_j}.$$

If positive odd integers x, y satisfy $\beta x + B = \alpha y$, then $S^\ell(x) = y$ and its successive exact halving valuations are $h_1, \ldots, h_\ell$. Consequently $C^{H+\ell}(x) = y$.

Proof. For $\ell = 1$ the identity is exactly $3x + 1 = 2^{h_1} y$ with y odd. For larger ℓ, reduction modulo 2^{h_1} gives $3^{\ell-1}(3x + 1) \equiv 0 \pmod{2^{h_1}}$. Thus $x_1 = (3x + 1)/2^{h_1}$ is a positive integer. Dividing the identity by 2^{h_1} leaves

$$3^{\ell-1} x_1 + B' = 2^{H-h_1} y, \qquad B' = \sum_{j=1}^{\ell-1} 3^{\ell-1-j} 2^{H_j - h_1}.$$

Here B' is odd and $H - h_1 \geq 1$, so x_1 is odd. The first valuation is therefore exactly h_1, and induction applied to this remaining word proves all the rest. Positivity holds at every step. □

Theorem I.7.3 (Prime chains with a prescribed periodic word)**.** *Use the notation of Lemma I.7.2, and assume*

$$D := \alpha - \beta > 0, \qquad g = \gcd(B, D), \qquad E = D/g, \qquad c = B/g.$$

Let $Q \geq 1$ satisfy $\gcd(Q, cE) = 1$, and set $b = cE^{-1} \pmod{Q}$. For every $m \geq 2$ and $0 < \delta < 1$, there exist infinitely many prime starts whose trajectories contain m selected distinct primes, all congruent to $b \pmod{Q}$, with each selected successor less than δ times its predecessor. The selected visit times are fixed. Between selected visits the odd-step valuation sequence is a positive integral number of repetitions of $(h_1, \ldots, h_\ell)$.

Proof. The numbers B, D, c, E are coprime to 6, and $\gcd(c, E) = 1$. Choose a positive multiple d of E such that $(\beta/\alpha)^d < \delta$, take sufficiently large k, and put $K = d(k - 1)$. Choose a positive u_0 satisfying

$$u_0 \equiv 0 \pmod{2Q}, \qquad \beta^K u_0 \equiv -c \pmod{E},$$

and define

$$u(n) = u_0 + 2QEn, \qquad R_i(n) = \frac{\alpha^i \beta^{K-i} u(n) + c}{E} \quad (0 \leq i \leq K).$$

Every R_i is an integer-coefficient positive linear form with odd values. Since $(\alpha - \beta)c/E = B$, their defining identity is

$$\beta R_i + B = \alpha R_{i-1} \quad (i \geq 1).$$

Reducing the fraction B/D has therefore preserved the original odd-step word; it has not changed its affine constant. Lemma I.7.2 proves, not merely formal affine iteration, that each block has exactly the stated positive odd intermediate nodes and the prescribed valuations. All R_i are $b \pmod{Q}$.

Select the indices $0, d, \dots, (k-1)d$. For a prime $p \mid 2Q$, oddness or the unit class b excludes a fixed prime divisor. If $p \nmid 2QE$ and $p \nmid c$, choose $u(n) \equiv 0 \pmod{p}$, giving $R_i \equiv c/E \neq 0$. If $p \mid c$, then $p \nmid 6QE$; choosing $u(n) \neq 0 \pmod{p}$ makes every $\alpha^i \beta^{K-i} u(n)/E$ nonzero. For $p \mid E$, the binomial expansion of $\alpha = \beta + gE$ yields

$$R_i \equiv R_0 + ig\beta^{K-1}u(n) \pmod{p}.$$

All selected indices are divisible by p, and R_0 has slope $2Q\beta^K$ coprime to p. Thus the selected tuple is admissible. Its distinct positive slopes and its common nonzero additive term c/E ensure nonproportionality. Maynard–Tao and the same finite pigeonhole argument used above give a fixed set of m prime indices for infinitely many n. Index gaps are at least d, so limiting successive ratios are at most $(\beta/\alpha)^d < \delta$. The visit-time differences are $(H + \ell)$ times the index gaps. □

Corollary I.7.4 (Alternating clocks and a new terminal-digit class)**.** *For arbitrary finite m and $0 < \delta < 1$, infinitely many prime starts have m selected prime visits, all ending in* 609*, with consecutive selected values decreasing by a factor less than δ and fixed visit times. Along the intervening odd trajectory, the exact halving valuations alternate* $2, 3, 2, 3, \dots$.

Proof. Take $(h_1, h_2) = (2, 3)$: then $H = 5$, $\alpha = 32$, $\beta = 9$, $B = 7$, $D = 23$, and $7 \cdot 23^{-1} \equiv 609 \pmod{1000}$. All hypotheses of Theorem I.7.3 hold. Each full two-odd-step block takes seven standard steps. □

Neither theorem requires the selected primes to be consecutive in the full prime sequence, or the intervening odd nodes to be prime. Neither proves that all these infinitely many paths reach 1. The number of selected primes is an arbitrary finite integer, not an infinite subsequence of one path. No priority claim is made for these deductions from the cited prime-distribution results.

I.8 Pascal–Franel weights and a certified ten-prime trajectory

I.8.1 An array descending every three standard steps

Fix $K \geq 0$ and $n \geq 1$, and define

$$L_i = 2n4^i 3^{K-i} + 1 \quad (0 \leq i \leq K).$$

Then $3L_i + 1 = 4L_{i-1}$ and $C^3(L_i) = L_{i-1}$. Thus $L_K, L_{K-1}, \dots, L_0$ occur as odd nodes on one actual trajectory. The values need not all be prime. For sufficiently large K, however, Maynard–Tao supplies infinitely many n for which any prescribed finite number of them are prime. Admissibility is immediate: for each prime p, choose $n \equiv 0 \pmod{p}$, making all forms $1 \pmod{p}$. The prime positions can be fixed by the finite pigeonhole principle.

Proposition I.8.1 (Pascal difference identities)**.** *For $i + j \leq K$ and $j \geq 1$,*

$$\Delta^j L_i = 2n4^i 3^{K-i-j} > 0, \qquad \Delta^K L_0 = 2n \quad (K \geq 1).$$

Hence ordinary and iterated absolute differences agree on this array. Also

$$\sum_{i=0}^{K} \binom{K}{i} (L_i - 1) = 2n7^K.$$

Proof. Each difference removes a factor 3 because $4 - 3 = 1$; induction proves the first formula. The final identity is the binomial theorem. The case $j = 0$ retains the added constant 1 and is not covered by the positive-order formula. □

Keeping only prime positions destroys the equally spaced index structure. Nevertheless, all entries are 1 (mod $2n$), so every positive-depth absolute difference of the prime subsequence is divisible by $2n$. This is a different input from the entire prime sequence in Gilbreath's conjecture.

Proposition I.8.2 (A weighted Franel sum and the Legendre recurrence)**.** *Define* $V_K = \sum_{i=0}^{K} \binom{K}{i}^2 4^i 3^{K-i}$. *Then*

$$\sum_{i=0}^{K} \binom{K}{i}^2 (L_i - 1) = 2nV_K, \qquad \sum_{K\geq 0} V_K z^K = (1 - 14z + z^2)^{-1/2},$$
$$(K+1)V_{K+1} = 7(2K+1)V_K - KV_{K-1}, \qquad V_0 = 1, \quad V_1 = 7.$$

The first values are $1, 7, 73, 847, 10321, 129367$.

Proof. Binomial expansion gives $V_K = \mathrm{CT}_x(7 + 4x + 3/x)^K$. In formal series,

$$\mathrm{CT}_x \frac{1}{1 - z(7 + 4x + 3/x)} = \frac{1}{1 - 7z} \sum_{j\geq 0} \binom{2j}{j} \frac{12^j z^{2j}}{(1 - 7z)^{2j}} = (1 - 14z + z^2)^{-1/2}.$$

For this generating function G, the identity $(1 - 14z + z^2)G' = (7 - z)G$ gives the recurrence by coefficient comparison. It is also the generating function for the Legendre values $P_K(7)$ [11]. This recurrence for a weighted sum is not a primality test for its constituent nodes. □

I.8.2 Ten distinct certified prime visits within 48 steps

Take $K = 16$ and $n = 1280689410$. Starting at L_{16}, the standard Collatz clock gives the following selected prime visits.

Time	i	Prime L_i	Lucas base g
0	16	11001038264567070721	17
6	14	6188084023818977281	38
12	12	3480797263398174721	31
24	8	1101346009122078721	29
30	6	619507130131169281	19
33	5	464630347598376961	23
36	4	348472760698782721	23
39	3	261354570524087041	31
42	2	196015927893065281	29
48	0	110258959439849221	6

These are deterministic certificates, with all certificate inputs contained in the paper. For every displayed index,

$$L_i - 1 = 2^{2+2i} 3^{17-i} \cdot 5 \cdot 7 \cdot 11^2 \cdot 13 \cdot 3877.$$

Each displayed base g satisfies

$$g^{L_i - 1} \equiv 1 \pmod{L_i}, \qquad \gcd(g^{(L_i-1)/q} - 1, L_i) = 1 \quad (q \in \{2, 3, 5, 7, 11, 13, 3877\}).$$

All the small factors are prime by trial division up to $\sqrt{3877} < 63$. Repeated squaring and the Euclidean algorithm reproduce the congruences and gcds; Appendix I.15 states the exact procedures. For any prime divisor λ of L_i, these conditions force the order of g modulo λ to contain every prime-power factor of $L_i - 1$. Therefore $L_i - 1 \mid \lambda - 1$, whence $\lambda \geq L_i$, proving primality. This is the complete-factorization Lucas criterion also stated in Lemma I.15.1.

Iteration of the explicitly given map from the displayed starting value first reaches 1 after 428 standard steps and has maximum 33003114793701212164. The finite-iteration protocol in Appendix I.15 reproduces both values from the starting integer alone. This calculation concerns one orbit, not the convergence of the infinite families in the existence theorems.

I.8.3 Franel-weighted filters of a fixed inverse array

For $P_a = (4^a M - 1)/3$, with $a \geq 1$ and $M > 1$ odd and $M \equiv 1 \pmod 3$, define

$$F_{n,s}(z) = \sum_{k=0}^{n} \binom{n}{k}^s z^k, \qquad H_{n,s}(a) = \sum_{k=0}^{n} (-1)^k \binom{n}{k}^s P_{a+k} \quad (n \geq 1,\ s \geq 1).$$

Substitution gives

$$3H_{n,s}(a) = 4^a M F_{n,s}(-4) - F_{n,s}(-1).$$

For odd n, symmetry gives $F_{n,s}(-1) = 0$; moreover $F_{n,s}(-4)/3$ is a nonzero integer, since its numerator is $0 \pmod 3$ and $1 \pmod 4$. Hence $|H_{n,s}(a)|$ is composite: it is the product of $M > 1$ and a nonzero integer of absolute value at least 4^a. For $s = 1$ this specializes to

$$\Delta^n P_a = 3^{n-1} 4^a M \quad (n \geq 1).$$

For $s = 3$ and $n = 2m$, Dixon's identity [12] gives

$$F_{2m,3}(-1) = (-1)^m \frac{(3m)!}{m!^3}.$$

If $M > 3$ is prime and $3m < M$, this is nonzero modulo M, so $M \nmid H_{2m,3}(a)$. Thus an even row can remove the forced factor M, but this is not a prime-generating rule. Indeed both the positive and alternating weighted sums are even for $n \geq 1$: the P_{a+k} are odd, and $\sum_k \binom{n}{k}^s$ is even. Pairing k with $n - k$ proves this when n is odd; for even $n \geq 2$, the unpaired central binomial coefficient is itself even.

I.9 Prime filters for inverse images of a fixed large Mersenne number

Throughout this section, let

$$r = 136279841, \qquad M = 2^r - 1, \qquad P_a = \frac{4^a M - 1}{3} \quad (a \geq 1).$$

The primality of M, announced by GIMPS, is an external established input [17]. The divisibility arguments below use only its displayed exponent; they do not repeat the primality computation for M.

Proposition I.9.1 (Necessary conditions on indices, residues, and gaps)**.** *If P_a is prime, then*

$$\gcd(a, 68139920) = 1, \qquad a \not\equiv 2 \pmod 3.$$

In particular, its index modulo 30 *lies in* $\{1, 3, 7, 9, 13, 19, 21, 27\}$*. Every prime term is* 21 *modulo* 80*, and its successor P_{a+1} is composite. If $p = P_a < q = P_b$ are prime terms, then*

$$q \geq 16p + 5.$$

If, in addition, $a \equiv 3 \pmod 6$, then $q \geq 256p + 85$.

Proof. Since $M \equiv 1 \pmod 5$, an even index gives $5 \mid P_a$, whereas an odd index gives $P_a \equiv 1 \pmod 5$. The congruence $M \equiv 4 \pmod 9$ gives $P_a \equiv a + 1 \pmod 3$.

Put $d = \gcd(2a, r - 1)$. Then d is even, and $2^r - 1 \equiv 1$, $2^{2a} \equiv 1 \pmod{2^d - 1}$. Consequently

$$\frac{2^d - 1}{3} \mid P_a.$$

If $d \geq 4$, this divisor is greater than 1 and less than P_a: indeed $d \leq r-1$ and $P_a \geq (4M-1)/3$. Thus a prime term requires $d = 2$, equivalently $\gcd(a, (r-1)/2) = 1$. Here

$$68139920 = 2^4 \cdot 5 \cdot 851749.$$

For $a \geq 2$ we have $P_a \equiv 5 \pmod{16}$. The remaining case $a = 1$ is composite: $2^{35} \equiv 1 \pmod{71}$, $r \equiv 26 \pmod{35}$, and $2^{26} \equiv 19 \pmod{71}$ give $4M - 1 \equiv 0 \pmod{71}$. Combining the congruences modulo 16 and 5 yields $P_a \equiv 21 \pmod{80}$. Prime indices are odd, so $b - a \geq 2$, and

$$P_b = 4^{b-a} P_a + \frac{4^{b-a} - 1}{3}$$

proves the first gap bound. If $a \equiv 3 \pmod{6}$, then $a + 2 \equiv 2 \pmod{3}$ is also excluded; hence $b - a \geq 4$ and the second bound follows. □

For example, $5 \mid a$ implies that both 11 and 31 divide P_a: $M \equiv 1$ modulo each prime, and 4 has order 5 modulo each. A nonzero remainder rules out only the tested divisor, not compositeness. There is also no rule requiring a change in decimal length between prime terms. The two necessary index conditions in the proposition define a periodic set of natural density

$$\frac{1}{2}\frac{4}{5}\frac{851748}{851749}\frac{2}{3} = \frac{4 \cdot 851748}{15 \cdot 851749}.$$

This is the density of a surviving sieve set, not a claimed density of prime terms.

Proposition I.9.2 (An explicit composite subsequence and power residues)**.** *Let* $a_0 = (r-1)/2 = 68139920$ *and* $D = (M+2)/3$. *Then*

$$P_{a_0} = \frac{M-1}{2}\frac{M+2}{3}, \qquad D \mid P_{a_0+tr} \quad (t \geq 0).$$

If $p = P_a$ *is prime, then* M *is a quadratic residue but not a fourth-power residue modulo* p.

Proof. Substitute $4^{a_0} = 2^{r-1} = (M+1)/2$ for the first identity. Since $4^r - 1 = M(M+2) = 3MD$, we have $P_{a+r} = 4^r P_a + MD$, which propagates the divisibility by D. Furthermore $M \equiv 2^{-2a} \pmod{p}$, so it is a quadratic residue. The preceding proposition gives odd a and $p \equiv 5 \pmod{8}$. The supplementary law for the quadratic character of 2 gives $2^{(p-1)/2} \equiv -1 \pmod{p}$, whence

$$M^{(p-1)/4} \equiv (-1)^{-a} = -1 \pmod{p}.$$

This rules out a fourth power. □

Every prime term is also congruent to P_1 modulo $20M$. Indeed, for odd a, $4^a - 4$ is divisible by 60. Thus every entry in every positive-depth absolute-difference row of the subsequence of prime terms is a multiple of $20M$, and cannot equal 1. This statement concerns this selected subsequence and has no bearing on the Gilbreath conjecture for the full increasing sequence of primes.

I.10 Divisibility and large-prime-factor blocks in inverse Collatz families

In this section fix any integer $M \geq 1$ with $M \equiv 1 \pmod{3}$, and put

$$P_a = \frac{4^a M - 1}{3} \quad (a \geq 1), \qquad F_d = \frac{4^d - 1}{3} \quad (d \geq 1).$$

The terms P_a are positive odd integers and satisfy

$$P_{a+d} = 4^d P_a + F_d, \qquad P_{a+1} = 4P_a + 1. \tag{I.10.1}$$

For the standard Collatz map, $C^{2a+1}(P_a) = M$. This asserts that the separate trajectories of all these starting values reach M; it does not assert that consecutive terms P_a, P_{a+1} lie consecutively on one trajectory. The target M need not be odd for this identity with C.

Proposition I.10.1 (Exact common-divisor identity)**.** *For $b > a \geq 1$,*

$$\gcd(P_a, P_b) = \gcd(P_a, F_{b-a}).$$

In particular, a common prime divisor of two distinct terms divides F_{b-a}, a number depending only on their index difference.

Proof. Apply the Euclidean algorithm to (I.10.1). □

I.10.1 Exact divisibility progressions and valuation densities

The following gives a complete answer to the auxiliary question of which indices admit a specified divisor. It is an elementary consequence of multiplicative orders, rather than a primality criterion for the terms.

Theorem I.10.2 (All prescribed-divisor index sets)**.** *Let d be a positive integer. If d is even or $\gcd(d, M) > 1$, no term P_a is divisible by d. Otherwise let*

$$h_d = \operatorname{ord}_{3d}(4).$$

Then $\{a \geq 1 : d \mid P_a\}$ is empty unless M^{-1} belongs to the cyclic subgroup generated by 4 in $(\mathbb{Z}/3d\mathbb{Z})^\times$. When it does belong, there is a unique residue $a_d \pmod{h_d}$ such that

$$d \mid P_a \quad \Longleftrightarrow \quad a \equiv a_d \pmod{h_d}.$$

This index set has natural density $1/h_d$. Moreover, for every integer $e \geq 0$, the set of indices with $v_3(P_a) = e$ has natural density

$$\frac{2}{3^{e+1}}.$$

Proof. Oddness excludes even divisors. If a prime divides both d and M, it cannot divide $4^a M - 1$, so the divisibility is again impossible. In all other cases,

$$d \mid P_a \quad \Longleftrightarrow \quad 4^a M \equiv 1 \pmod{3d}.$$

The powers of 4 form a cyclic group of order h_d, proving both the existence test and the uniqueness of the index class. A residue class modulo h_d has density $1/h_d$.

For $e \geq 1$, the order of 4 modulo 3^{e+1} is exactly 3^e. Indeed, induction or the binomial theorem gives $v_3(4^{3^j} - 1) = j + 1$ for $j \geq 0$. The group of units congruent to 1 modulo 3 has 3^e elements modulo 3^{e+1} and is therefore generated by 4. Since $M \equiv 1 \pmod 3$, a unique index class modulo 3^e satisfies $3^e \mid P_a$. These classes are nested as e increases. Subtracting the densities for e and $e + 1$ gives $3^{-e} - 3^{-e-1} = 2/3^{e+1}$. For $e = 0$, the same conclusion is $1 - 1/3 = 2/3$. □

For finitely many prescribed divisors, the resulting index congruences are simultaneously soluble precisely when their residues agree modulo each pairwise gcd of their periods. When soluble, the combined set is one class modulo the least common multiple of the periods. Exact valuations at finitely many primes can likewise be imposed or excluded by combining the corresponding divisibility classes. This supplies a finite, exact sieve with no unproved probabilistic assumption.

I.10.2 Lifting blocks while preserving their common divisors

Theorem I.10.3 (Successively adjoining coprime new factors)**.** *Fix positive integers A, L, Q, a real number $B \geq 1$, and a finite set of primes $\mathcal{S}$. There are indices*

$$A = A_0 < A_1 < A_2 < \cdots, \qquad A_j \equiv A \pmod{Q},$$

and integers $U_{j,i} > 1$ for $j \geq 1$, $0 \leq i < L$, such that

$$P_{A_j+i} = P_{A+i} \prod_{h=1}^{j} U_{h,i},$$
$$\gcd(U_{j,i}, U_{h,t}) = 1 \qquad \text{if } (j,i) \neq (h,t).$$

Every $U_{j,i}$ is coprime to the initial block product $\prod_{i=0}^{L-1} P_{A+i}$, and all its prime factors exceed B and lie outside $\mathcal{S}$. Consequently, the valuation at every prime in $\mathcal{S}$, and at every prime dividing the initial block, is unchanged at each position. In addition,

$$\gcd(P_{A_j+i}, P_{A_j+t}) = \gcd(P_{A+i}, P_{A+t}) \qquad (i \neq t).$$

Proof. Write the current block as $x_i = P_{A_j+i}$ and choose the odd integer

$$R = \left(\prod_{i=0}^{L-1} x_i\right)\left(\prod_{d=1}^{L-1} F_d\right)\left(\prod_{\substack{p \text{ odd prime} \\ p \leq B \text{ or } p \in \mathcal{S}}} p\right),$$

with empty products interpreted as 1. Set $H = Q\varphi(3R^2) > 0$. Euler's theorem gives $4^H \equiv 1 \pmod{3R^2}$. For $A_{j+1} = A_j + H$ and $y_i = P_{A_{j+1}+i}$, therefore,

$$y_i - x_i = 4^{A_j+i} M \frac{4^H - 1}{3} \equiv 0 \pmod{R^2}.$$

Since $x_i \mid R$, we may write $y_i = x_i U_{j+1,i}$ with $U_{j+1,i} \equiv 1 \pmod{R}$. Also $y_i > x_i$, so $U_{j+1,i} > 1$. Each new factor is thus coprime to every current term and to all designated small or excluded odd primes. All terms and factors are odd, so the assertion at 2 follows too.

If a prime p divided two distinct new factors at the same stage, it would divide the corresponding y_i, y_t. Proposition I.10.1 would give $p \mid F_{|t-i|} \mid R$, contrary to $U_{j+1,i} \equiv 1 \pmod{R}$. Every earlier new factor divides the current block product, so factors from different stages are coprime as well. New factors are coprime to the entire old block and to each other; hence each pairwise gcd and each asserted valuation is preserved. Induction completes the construction. □

Corollary I.10.4 (Arbitrarily long blocks with many large prime factors)**.** *For any $L, k \geq 1$ and $B \geq 1$, infinitely many blocks of L consecutive terms have at least k distinct prime factors greater than B at every position. The k designated prime factors at different positions can be chosen to be distinct from each other. The starting index may be restricted to any prescribed residue class.*

Proof. Take $j \geq k$ in Theorem I.10.3, and select a prime divisor from each $U_{h,i}$ with $1 \leq h \leq k$. Their pairwise coprimality gives all the required distinctions. Choose A, Q to obtain the prescribed class. □

Theorem I.10.5 (Positive-density strengthening with a fixed gcd matrix)**.** *Under the hypotheses of Theorem I.10.3, fix also $k \geq 1$. There is an arithmetic progression of starting indices $n \equiv A \pmod{Q}$ on which every block $(P_n, \ldots, P_{n+L-1})$ has at least k designated distinct prime factors*

greater than B at each position, all distinct across positions and outside $\mathcal{S}$. On this whole progression,

$$\gcd(P_{n+i}, P_{n+t}) = \gcd(P_{A+i}, P_{A+t}) \quad (i \neq t),$$

and the valuations at all primes in $\mathcal{S}$ and in the initial block equal their respective initial valuations. In particular, the set of starting indices with these properties has positive lower density.

Proof. First perform k liftings and put $A_* = A_k$, $x_i = P_{A_*+i}$. Select the k prime factors at each position from the preceding corollary. Now define R by the same product as in the lifting proof, using these x_i, and set $H = Q\varphi(3R^2)$. For every integer $t \geq 0$,

$$P_{A_*+tH+i} \equiv x_i \pmod{R^2}.$$

If $p \mid R$, then $v_p(x_i) \leq v_p(R) < 2v_p(R)$, so this congruence preserves the exact valuation $v_p(x_i)$, including zero. It therefore preserves the selected prime factors and all requested old valuations. A prime common to two positions must divide the relevant $F_{|i-j|}$, which is a factor of R. Its valuation at both positions is preserved, so every pairwise gcd remains unchanged. The progression $A_* + tH$ has density $1/H > 0$ and lies in the required index class. □

All three block statements remain valid for any fixed distinct nonnegative offsets $h_1, \ldots, h_L$ in place of consecutive indices: replace x_i by $P_{A_j+h_i}$ and include $\prod_{i<t} F_{|h_t-h_i|}$ in R. The positive-density result answers the additional question of whether arbitrarily factor-rich blocks can be found throughout an arithmetic progression, with the old gcd pattern intact. It gives no practical upper bound on a small starting index or a short running time.

In particular, infinitely many distinct ordinary primes divide this family, and arbitrarily long composite blocks occur on positive-density sets of indices. Neither conclusion proves that infinitely many *terms* are prime. A prime divisor of an inverse image need not itself be an inverse image of M. These are elementary results and consequences; no claim of historical priority is made.

I.10.3 Complete classification of Mersenne-shaped divisors

Now specialize to $M = 2^r - 1$ with odd $r \geq 3$, so $M \equiv 1 \pmod 3$.

Theorem I.10.6 (Mersenne-shaped divisors)**.** *For every $s \geq 2$ and $a \geq 1$,*

$$(2^s - 1) \mid 3P_a \quad \Longleftrightarrow \quad s \mid (r-1) \text{ and } s \mid 2a.$$

In particular, if s is odd, then

$$(2^s - 1) \mid P_a \quad \Longleftrightarrow \quad s \mid (r-1) \text{ and } s \mid a.$$

Proof. Put $D = 2^s - 1$. Since 2 is invertible modulo D, the first divisibility is equivalent to $2^r - 1 \equiv 2^{-2a} \pmod D$. Let $u = r \bmod s$ and $v = (-2a) \bmod s$, with $0 \leq u, v < s$. Then $2^u - 1 \equiv 2^v \pmod D$. If $s = 2$, oddness of r gives $u = 1$ and $v = 0$, so the assertion holds directly. Suppose henceforth that $s \geq 3$. The difference of these two integers has absolute value less than D, so they must be equal. The equation $2^u - 1 = 2^v$ holds only for $u = 1, v = 0$: the left side is either zero, or odd, forcing $v = 0$ and then $u = 1$. This proves the first assertion. For odd s, $\gcd(2^s - 1, 3) = 1$ and $s \mid 2a$ is equivalent to $s \mid a$. □

Corollary I.10.7 (All Mersenne-shaped divisors of the inverse terms)**.** *Let $r \geq 3$ be odd and $P_a = (4^a(2^r - 1) - 1)/3$, where $a \geq 1$. For every integer $s \geq 2$,*

$$(2^s - 1) \mid P_a \quad \Longleftrightarrow \quad \begin{cases} s \mid (r-1) \text{ and } s \mid a, & s \text{ odd}, \\ s \mid (r-1) \text{ and } a \equiv 1 - r \pmod{3s/2}, & s \text{ even}. \end{cases}$$

Thus the even-exponent case, where division by 3 *matters, also admits an exact classification in the index* a*.*

Proof. The odd case follows from the Mersenne-shaped divisor theorem above. Suppose s is even and put $D = 2^s - 1$, so $3 \mid D$. Divisibility of P_a by D first requires $s \mid (r-1)$ and $s \mid 2a$. Under these conditions write $r = 1 + st$ and $2a = su$, with integers $t \geq 1$ and $u \geq 1$. Since D^2 is divisible by $3D$, binomial expansion gives

$$3P_a = (1+D)^u\big(2(1+D)^t - 1\big) - 1 \equiv (u+2t)D \pmod{3D}.$$

Consequently $D \mid P_a$ if and only if $3 \mid u + 2t$. Because $u + 2t = 2(a + r - 1)/s$, this is equivalent to $a \equiv 1 - r \pmod{3s/2}$. Conversely, that congruence together with $s \mid (r-1)$ already implies $s \mid 2a$, so it supplies all the conditions used in the argument. □

Corollary I.10.8 (Mersenne prime divisors for $r = 136279841$)**.** *For* $M = 2^{136279841} - 1$*, the Mersenne primes that divide at least one term* P_a *are exactly* 3 *and* 31*. More precisely,*

$$3 \mid P_a \iff a \equiv 2 \pmod 3, \qquad 31 \mid P_a \iff 5 \mid a.$$

Proof. If $2^s - 1$ is prime, its exponent s is prime. For $s > 2$, Theorem I.10.6 implies $s \mid 136279840$. The factorization

$$136279840 = 2^5 \cdot 5 \cdot 851749$$

and the primality certificate below leave only the odd possibilities $s = 5$ and $s = 851749$. The latter produces a composite Mersenne number:

$$d = 23629220759 = 2 \cdot 13871 \cdot 851749 + 1, \qquad 2^{851749} \equiv 1 \pmod d.$$

Since $1 < d < 2^{851749} - 1$, this proves compositeness whether or not d itself is prime. The complete small modular certificate is given below. The exponent 5 yields the prime 31, and the divisibility condition is exactly $5 \mid a$. Finally $M \equiv 4 \pmod 9$ gives $P_a \equiv a + 1 \pmod 3$, so the prime 3 occurs precisely in the stated class. Both divisors occur, for example at $a = 2$ and $a = 5$ respectively. □

An in-text certificate for the exponent 851749**.** The following elementary form of the Lucas criterion is sufficient. If the complete factorization of $n - 1$ is known into primes, and a base b satisfies

$$b^{n-1} \equiv 1 \pmod n, \qquad \gcd\big(b^{(n-1)/q} - 1, n\big) = 1 \quad \text{for every prime } q \mid n-1,$$

then n is prime. Indeed, for any prime $\ell \mid n$, the order of b modulo ℓ divides $n - 1$, and each gcd condition forces the full q-part of $n - 1$ into that order. Hence $n - 1 \mid \ell - 1$, so $\ell \geq n$. The table supplies the recursive certificate, starting with the prime 2. In each row $b^{n-1} \bmod n = 1$; the final column lists the residues $b^{(n-1)/q} \bmod n$ in increasing order of the distinct primes q in the factorization. Subtracting 1 from each displayed residue and applying the Euclidean algorithm gives gcd 1 with n.

n	$n-1$	b	Residues for the distinct q
3	2	2	2
5	2^2	2	4
11	$2 \cdot 5$	2	$10, 4$
23	$2 \cdot 11$	5	$22, 2$
257	2^8	3	256
1543	$2 \cdot 3 \cdot 257$	5	$1542, 861, 195$
70979	$2 \cdot 23 \cdot 1543$	2	$70978, 55077, 42048$
851749	$2^2 \cdot 3 \cdot 70979$	10	$851748, 729336, 679554$

An in-text proper-divisor certificate. For $d = 23629220759$, put $b_0 = 2$ and $b_{j+1} = b_j^2 \bmod d$, always using the least nonnegative residue. The following table contains all residues required to verify the displayed divisibility without an auxiliary program.

j	b_j	j	b_j
0	2	10	3279980729
1	4	11	3050265079
2	16	12	13793189074
3	256	13	2291904066
4	65536	14	9489855647
5	4294967296	15	8423096495
6	7418137547	16	21477549615
7	1085980106	17	11471801591
8	22641551493	18	21306683214
9	57186427	19	7577986627

The nonzero binary positions of 851749 are

$$J = \{0, 2, 5, 8, 9, 10, 11, 12, 13, 14, 15, 18, 19\}.$$

Multiplication of the corresponding table entries modulo d gives $\prod_{j\in J} b_j \equiv 1 \pmod{d}$. Together with $851749 = \sum_{j\in J} 2^j$, this is the promised certificate.

Neither the divisor classification nor these small certificates uses the primality of M itself. They use only its exponent and integer identities. Every term in this one-odd-step inverse family is $P_a \equiv 1 \pmod 4$, and so cannot itself equal $2^s - 1$ for $s \geq 2$. This does not exclude Mersenne ancestors deeper in the full inverse Collatz tree. Likewise, "exactly 3 and 31" classifies *Mersenne prime divisors*; the set of ordinary prime divisors is infinite, as proved above.

I.11 Primality-certificate barriers in Mersenne inverse orbits

Throughout this section,

$$S(x) = \frac{3x+1}{2^{v_2(3x+1)}} \qquad (x \text{ odd})$$

is the odd-only Collatz map. Its iteration count differs from that of the standard map C. The notation $M_r = 2^r - 1$ does not assume primality; primality is imposed explicitly whenever it is needed.

I.11.1 The inverse-orbit identity and binary digit sums

Lemma I.11.1 (Exact inverse-orbit identity)**.** *Suppose that N is a positive odd integer and $S^t(N) = M$. Write $N_j = S^j(N)$, $k_j = v_2(3N_{j-1}+1)$, $K_j = k_1 + \cdots + k_j$, $K_0 = 0$, and $K = K_t$. Then*

$$3^t N = 2^K M - B, \qquad B = \sum_{j=0}^{t-1} 3^{t-1-j} 2^{K_j}. \tag{I.11.1}$$

The corresponding number of standard Collatz steps is $L = K + t$.

Proof. Substitute the identities $3N_{j-1} + 1 = 2^{k_j} N_j$ successively. At each additional odd step the previous constant term is multiplied by 3 and 2^{K_j} is added, giving the displayed sum by induction. An odd-only step consists of one operation $x \mapsto 3x + 1$ followed by k_j halvings. □

Lemma I.11.2 (Binary digit sum of a Mersenne multiple)**.** *If $r \geq 2$ and A is a positive multiple of $2^r - 1$, then its binary digit sum satisfies $s_2(A) \geq r$.*

Proof. When $A \geq 2^r$, write $A = q2^r + d$, with $q \geq 1$ and $0 \leq d < 2^r$, and replace A by $q + d$. The new integer is a smaller positive multiple of $2^r - 1$, while $s_2(q + d) \leq s_2(q) + s_2(d) = s_2(A)$. Iteration terminates at the unique positive multiple smaller than 2^r, namely $2^r - 1$, whose binary digit sum is r. □

This is a special case of the theory of sturdy numbers; see Clokie et al. [13]. The elementary proof above is sufficient for every use made here.

Theorem I.11.3 (Odd-step obstruction to reusing a known factor)**.** *If $r \geq 2$, $S^t(N) = M_r$, and $M_r \mid N - 1$, then*

$$r \leq H(t) := \sum_{j=0}^{t-1} s_2(3^j). \tag{I.11.2}$$

For $r = 136279841$, necessarily $t \geq 18545$.

Proof. Reduction of (I.11.1) modulo M_r gives $M_r \mid B + 3^t$. Apply Lemma I.11.2 to this positive integer. Combining its first summand with 3^t yields

$$B + 3^t = 4 \cdot 3^{t-1} + \sum_{j=1}^{t-1} 3^{t-1-j} 2^{K_j}.$$

Subadditivity of the binary digit sum and $s_2(2^a u) = s_2(u)$ show that $s_2(B + 3^t) \leq H(t)$. Exact finite arithmetic gives

$$H(18544) = 136276195 < 136279841, \qquad H(18545) = 136290777.$$

Since H is strictly increasing, the stated threshold follows. □

For complete reproducibility, the finite calculation in this proof is the following integer recurrence. Start with $x_0 = 1$ and $h_0 = 0$, and for $0 \leq j < 18545$ set

$$x_{j+1} = 3x_j, \qquad h_{j+1} = h_j + b(x_j), \quad b(0) = 0, \quad b(2u) = b(u), \quad b(2u + 1) = b(u) + 1.$$

Then h_{18544} and h_{18545} are the two displayed integers. This is a specification of the entire calculation within the manuscript; no auxiliary program or external data are required. The result is a necessary condition, not an existence assertion at the threshold and not a lower bound on the complexity of all primality algorithms.

Proposition I.11.4 (Two-sided obstruction for short standard paths)**.** *Let $r \geq 3$ be odd, $S^t(N) = M_r$, and $t \geq 1$. If the corresponding standard length satisfies $L = K + t \leq r$, then*

$$M_r \nmid N - 1, \qquad M_r \nmid N + 1.$$

Proof. Because $M_r \equiv 1 \pmod 3$, the last exponent k_t is even and at least 2. Thus $K \geq t + 1$. The estimates $K_j \leq K - t + j$ give

$$B \leq 2^{K-t}(3^t - 2^t).$$

With $u = K - t \geq 1$,

$$B + 3^t \leq 2^u 3^t + 3^t - 2^{u+t} < 2^{u+2t} = 2^L.$$

For the last inequality use $2 \cdot 3^t - 2^t \leq 4^t$ and $1 + 2^{-u} < 2$. Since $B + 3^t$ is even, $B + 3^t \leq 2^L - 2 < M_r$. Moreover $B \not\equiv 0 \pmod 3$, so $B \neq 3^t$. Consequently $B + 3^t$ and $B - 3^t$ are nonzero integers of absolute value less than M_r. Reducing (I.11.1) modulo M_r proves both assertions. □

Proposition I.11.5 (One or two odd steps)**.** *If $r \geq 4$, $S^t(N) = M_r$, and $t \in \{1, 2\}$, then $M_r \nmid N - 1$ and $M_r \nmid N + 1$.*

Proof. For $t = 1$, $B+3 = 4$ and $B-3 = -2$. For $t = 2$, $B = 3+2^{k_1}$, so the necessary congruences would be $2^{k_1} \equiv -12$ or $6 \pmod{M_r}$, respectively. As $2^r \equiv 1 \pmod{M_r}$, the least nonnegative residue of 2^{k_1} is one of $1, 2, 4, \ldots, 2^{r-1}$. Neither 6 nor $M_r - 12 = 2^r - 13 \geq 3$, which is 3 modulo 4, is such a residue. □

The restriction $r \geq 4$ is necessary: $S(99) = 149$, $S(149) = 7$, and $7 \mid 99 - 1$.

I.11.2 An exact elliptic-curve threshold for a one-step inverse

The exclusion based on Hasse's bound admits a converse. The following classification is an application of Waterhouse's existence theorem for elliptic curves over finite fields, rather than a new version of that classical theorem.

Theorem I.11.6 (Exact threshold and all admissible traces)**.** *Let $r \geq 5$ be odd, let $M = 2^r - 1$, and suppose $p = P_a = (4^a M - 1)/3$ is prime, where $a \geq 1$. Set $D = (M + 2)/3$ and $m = \lfloor 2\sqrt{p} \rfloor$. Then an elliptic curve $E/\mathbb{F}_p$ with $M \mid \#E(\mathbb{F}_p)$ exists if and only if*

$$a \geq \frac{r-3}{2}. \tag{I.11.3}$$

More precisely, its possible Frobenius traces are exactly

$$\{D + jM : j \in \mathbb{Z},\ |D + jM| \leq m\}.$$

Each such trace is ordinary and specifies one $\mathbb{F}_p$-isogeny class. Thus the exact number of these isogeny classes is

$$\left\lfloor \frac{m-D}{M} \right\rfloor + \left\lfloor \frac{m+D}{M} \right\rfloor + 1. \tag{I.11.4}$$

Primality of M is not required.

Proof. Since

$$p + 1 = \frac{4^a - 1}{3} M + D,$$

the divisibility condition is equivalent to the trace congruence $\tau = p+1-\#E(\mathbb{F}_p) \equiv D \pmod{M}$. We have $0 < D < M/2$ and $p \geq 41$. Hence the closest such trace to zero is D, and Hasse's bound gives the necessary condition $D \leq 2\sqrt{p}$. Every congruent trace in that interval is nonzero and satisfies $|\tau| \leq 2\sqrt{p} < p$, so $\gcd(\tau, p) = 1$. Waterhouse's theorem [18, Theorem 4.1] guarantees an ordinary elliptic curve over $\mathbb{F}_p$ for each integer trace with $|\tau| \leq 2\sqrt{p}$ and $\gcd(\tau, p) = 1$; its trace determines its isogeny class. Thus the condition is also sufficient, and counting the integers j in the displayed interval gives (I.11.4).

It remains to locate the threshold in a. Direct expansion gives

$$D^2 \leq 4p \quad \Longleftrightarrow \quad 12 \cdot 4^a \geq M + 4 + 16/M.$$

For $a \leq (r-5)/2$, the left side is at most $3(M+1)/8$, which is smaller than the right side. At $a = (r-3)/2$ it equals $3(M+1)/2$, which is larger than the right side for $M \geq 31$. It increases with a, proving (I.11.3). □

For $r = 136279841$ the previous exclusion range $a \le 68139918$ is therefore sharp as a conditional existence threshold: every prime term with $a \ge 68139919$ admits such a curve. This statement neither proves that any specified P_a is prime nor constructs a fast certificate for it. For a small explicit instance, $M = 31$, $a = 1$, and $p = 41$ give the curve $y^2 = x^3 + 2x + 17$ over $\mathbb{F}_{41}$ with 31 points. Indeed, its nonsingular cubic has two square roots for exactly the following fifteen abscissae and none for the others:

$$1, 3, 6, 7, 14, 16, 20, 22, 24, 27, 28, 32, 36, 38, 39.$$

The point at infinity contributes the thirty-first point. In contrast, $M \nmid P_a \pm 1$ for every $a \ge 1$, directly from $3(P_a - 1) = 4^a M - 4$ and $3(P_a + 1) = 4^a M + 2$.

I.11.3 Paths descending from a larger Mersenne prime

Theorem I.11.7 (Initial Mersenne orbit and a path-length lower bound)**.** *For odd $s \ge 5$,*

$$S^j(M_s) = 3^j 2^{s-j} - 1 \quad (0 \le j < s), \qquad S^s(M_s) = \frac{3^s - 1}{2},$$

$$S^{s+1}(M_s) = \frac{3^{s+1} - 1}{2^{2+v_2(s+1)}}.$$

The last integer is composite. Consequently, if a path from M_s reaches a different Mersenne prime M_r, its odd-step length is $t \ge s + 2$. If also $s > r$, its standard length satisfies

$$L > (1 + \log_2 3)(s + 2) + (s - r). \tag{I.11.5}$$

In particular, if $r = 136279841$ and M_s is a larger Mersenne prime, then $L \ge 352278291$.

Proof. For $j < s - 1$,

$$3(3^j 2^{s-j} - 1) + 1 = 2(3^{j+1} 2^{s-j-1} - 1),$$

with an odd expression in parentheses. For odd s, $v_2(3^s - 1) = 1$ and $v_2(3^{s+1} - 1) = 2 + v_2(s + 1)$, proving the three formulas. At $1 \le j < s$, the integer $S^j(M_s) + 1$ is divisible by 3, so $S^j(M_s)$ is not Mersenne. At $j = s$, the Mersenne condition would force $3^s + 1 = 2^{r+1}$, impossible because the left side is 4 modulo 8 and greater than 4.

Put $u = (s + 1)/2 \ge 3$. The value $S^{s+1}(M_s)$ is the product of the odd parts of $3^u - 1$ and $3^u + 1$, each greater than 1. For odd u, the first number is 2 modulo 8, while

$$3^u + 1 = 4(3^{u-1} - 3^{u-2} + \cdots - 3 + 1)$$

has an odd factor greater than 1. For even u, the second number is 2 modulo 8. If $3^u - 1$ were a power of 2, then $3^{u/2} - 1$ and $3^{u/2} + 1$ would be powers of 2 differing by 2, forcing $u = 2$, a contradiction. This proves compositeness and $t \ge s + 2$.

By (I.11.1) and $B > 0$, $2^K M_r > 3^t M_s$. Hence

$$L = K + t > (1 + \log_2 3)t + \log_2(M_s/M_r) > (1 + \log_2 3)t + s - r.$$

Distinct odd prime exponents satisfy $s \ge r + 2$, so for the specified r the bound exceeds $(1 + \log_2 3)(136279845) + 2 > 352278290.9290$. □

The decimal inequality can be checked without floating-point assumptions. For rational $0 < x < 1$ put

$$\ell_{30}(x) = 2 \sum_{j=0}^{29} \frac{x^{2j+1}}{2j + 1}, \qquad u_{30}(x) = \ell_{30}(x) + \frac{2x^{61}}{61(1 - x^2)}.$$

The geometric-series integral gives $\ell_{30}(x) < \log((1+x)/(1-x)) < u_{30}(x)$. Using $x = 1/2$ for $\log 3$ and $x = 1/3$ for $\log 2$ yields, by rational arithmetic, lower and upper bounds for the last displayed expression strictly between 352278290 and 352278291.

The initial-orbit formulas are related to the discussion of Ohira–Watanabe [14]. Their approximate total-length law $13.45652s$ is a conjectural heuristic and is not used in this proof. Our inequality bounds the length of an actual Collatz path, not the running time of every possible primality test. The small example $8191 = M_{13} \longrightarrow M_7 = 127$ takes 41 odd-only steps and 112 standard steps. A complete certificate is the halving-exponent word

$$\begin{aligned}(k_1, \ldots, k_{41}) = (&1,1,1,1,1,1,1,1,1,1,1,1,2,2,1,1,1,2,1,3,1,\\ &4,1,2,1,5,5,1,1,1,1,3,1,1,4,3,1,2,2,4,2).\end{aligned}$$

Starting with $n_0 = 8191$, the recurrence $n_j = (3n_{j-1}+1)/2^{k_j}$ gives odd integers at every step and $n_{41} = 127$; the sum of the exponents is 71. Thus exclusion of Mersenne numbers from a one-step inverse family does not exclude them from the full inverse tree.

I.11.4 The candidate $82M + 1$ and the remaining primality condition

Proposition I.11.8 (An exact Pocklington-type residual condition)**.** *Assume the known primality of $M = 2^{136279841} - 1$ and put $Q = 82M + 1$. For each $a \in \{2, 3, 5, 7\}$,*

$$\gcd(a^{82} - 1, Q) = 1, \qquad Q \text{ is prime} \iff a^{82M} \equiv 1 \pmod{Q}.$$

Proof. For the small integer $d = a^{82} - 1$, compute

$$R_a = (82(2^{136279841} \bmod d) - 81) \bmod d.$$

The Euclidean algorithm gives $\gcd(d, R_a) = 1$ for the four bases. For a fully specified verification, modular exponentiation is defined by $\mathcal{P}(b, 0, d) = 1 \bmod d$ and

$$\begin{aligned}\mathcal{P}(b, 2h, d) &= \mathcal{P}(b, h, d)^2 \bmod d,\\ \mathcal{P}(b, 2h+1, d) &= b\mathcal{P}(b, h, d)^2 \bmod d.\end{aligned}$$

Compute $R_a = (82\mathcal{P}(2, 136279841, a^{82}-1)-81) \bmod (a^{82}-1)$ and apply $\gcd(u, v) = \gcd(v, u \bmod v)$ until $v = 0$. For example, $a = 2$ gives

$$d = 48357032784585166988247 03, \qquad R_2 = 687865775, \qquad \gcd(d, R_2) = 1.$$

This specifies all inputs and the exact finite verification, involving only moduli of at most 70 decimal digits.

Suppose the large congruence holds. For every prime divisor ℓ of Q, the order of a modulo ℓ divides $82M$ but does not divide 82. Since M is prime, that order is a multiple of M, whence $\ell \equiv 1 \pmod{M}$ and $\ell \geq M + 1 > \sqrt{Q}$. A composite integer has a prime divisor at most its square root, giving a contradiction. The reverse implication is Fermat's little theorem. □

The four small greatest-common-divisor calculations do not require the full decimal expansion of M. They do not settle the remaining large congruence, which has not been evaluated here; no primality claim for Q is made. For base 2, the remaining condition is equivalently

$$x_0 = 2^{82} \bmod Q, \qquad x_{j+1} = x_j^2 \bmod Q \quad (0 \leq j < r), \qquad x_r = x_0, \quad r = 136279841.$$

This equivalence uses $\gcd(2, Q) = 1$ and $82M = 82(2^r - 1)$. It is not a reduction that removes the r squarings; that large computation has not been performed.

An elliptic-curve variation has its own constraint. If Q is prime and $M \mid \#E(\mathbb{F}_Q)$, the trace satisfies $\tau \equiv 2 \pmod{M}$. For $M > 332$, the inequality $2\sqrt{Q} < M - 2$ forces $\tau = 2$ by Hasse's bound. Consequently $\tau^2 - 4Q = -328M$, so this construction does not automatically supply a fixed small complex-multiplication discriminant. It does not rule out other ECPP constructions. Likewise, for $h \geq 2$, the polynomial $82 \cdot 2^r X^h - 81$ is irreducible over $\mathbb{Q}$: its reciprocal is Eisenstein at 41. Irreducibility of a polynomial does not establish primality of its value at $X = 1$.

A small counterexample to automatic transmission of primality is

$$M = 2^{17} - 1 = 131071 \quad \text{(prime)}, \qquad 82M + 1 = 631 \cdot 17033.$$

The small Mersenne primality assertion also has a short certificate:

$$131071 - 1 = 2 \cdot 3 \cdot 5 \cdot 17 \cdot 257, \qquad 3^{131070} \equiv 1 \pmod{131071}.$$

For the prime factors $q = 2, 3, 5, 17, 257$, the respective residues of $3^{131070/q}$ are

$$131070, \quad 109165, \quad 119782, \quad 32768, \quad 120519.$$

Subtracting 1 from each gives an integer coprime to 131071. Thus 3 has order 131070 modulo every prime divisor of 131071, proving primality by the full-factor order criterion. All residues are reproducible by the displayed modular-exponentiation recurrence. All four small greatest-common-divisor conditions for $a = 2, 3, 5, 7$ still equal 1 in this example; the same modular-exponentiation recurrence, with exponent 17, verifies this assertion.

Mills's theorem provides a real $A > 1$ such that $\lfloor A^{3^n} \rfloor$ is prime for every $n \geq 1$. It supplies no identification of a specified inverse-Collatz candidate P_a with one of those outputs. Effective determination of the constant and of its output integers is a separate issue; see Caldwell–Cheng [15]. Existence of such a constant cannot replace a primality certificate for P_a or Q.

Finally, if M is an odd prime, every prime divisor q of $2^M - 1$ satisfies $\mathrm{ord}_q(2) = M$. The order cannot be 1, and it divides the prime M. Therefore $2M \mid q - 1$ and $q \geq 2M + 1$. This proves existence of a larger prime, but does not extract that factor or certify a specifically supplied integer.

I.12 Certified forward and inverse tables from the prime 184321

I.12.1 The forward orbit

The standard map C first carries 184321 to 1 after exactly 134 steps. Excluding the initial value, the primes congruent to 1 modulo 10 encountered on this orbit are precisely

time	3	6	9	12	27	69	93	115
prime	138241	103681	77761	58321	13841	251	911	61.

Thus there are eight such primes after the starting point, and nine including it. In particular, a request for ten of them in the forward direction has only eight answers after the starting point. The consecutive entries 138241 and 103681 both have six decimal digits: changing the number of digits is not necessary for the next selected term to be prime.

The first inverse example satisfies the exact identity

$$3 \cdot 1006638421 + 1 = 2^{14} \cdot 184321.$$

It therefore reaches 184321 in 15 standard steps. This number of steps is not a uniform spacing for the subsequent inverse chain.

I.12.2 A shortest-distance inverse selection rule

Set $q_0 = 184321$. Given q_{j-1}, consider primes $q > q_{j-1}$ with $q \equiv 1 \pmod{10}$ for which $C^t(q) = q_{j-1}$. First minimize t; among primes at that minimum distance choose the smallest q, denoted q_j. The following is a finite, completely certified calculation.

j	q_j	t_j in $C^{t_j}(q_j) = q_{j-1}$
1	1006638421	15
2	152710776661	15
3	3257829902101	7
4	73218498101521	20
5	11107517341623331	15
6	17973882876558681571	21
7	908901534103510613761	16
8	34470932256370180314481	13
9	1239553918438121337223458 1	24
10	16527385579174951162979441	3

There is a tie at the ninth selection. At minimum distance 24, both

$$1239553918438121337223458 1 \quad \text{and} \quad 44623941063772368140045949 1$$

are eligible primes. The prescribed tie-breaking rule selects the smaller.

The inverse edges used to certify minimality are exactly

$$C^{-1}(y) = \{2y\} \cup \begin{cases} \{(y-1)/3\}, & y \equiv 4 \pmod 6, \\ \varnothing, & \text{otherwise.} \end{cases}$$

Breadth-first search therefore exhausts every candidate at smaller distance. Appendix I.15 gives the full search recurrence, its numerical checkpoints, and 89 recursively complete Lucas certificates. Every one of the 156 rejected eligible candidates has a strong Miller–Rabin *compositeness witness to base* 2. Consequently, no probabilistic primality assumption is used, and no external certificate file is required.

The table asserts only these ten selections. It does not prove that this rule continues indefinitely, or that a larger prime ancestor exists for every prescribed prime root. The distances t_j are allowed to vary.

I.13 Quantifiers, existing theorems, and unresolved arithmetic

I.13.1 Finite prime configurations and different meanings of infinity

The coalescence theorem first fixes a finite integer m and then produces infinitely many configurations of m consecutive primes. The allowed values of m may themselves be restricted to primes, such as $2, 3, 5, 7, \ldots$. This does not say that any preassigned list of primes coalesces synchronously. For example,

$$C^t(2) \neq C^t(5) \qquad (t \geq 0).$$

The values differ at $t = 0, 1, 2$; thereafter both orbits lie on the cycle $1, 4, 2$ but have different phases. Replacing $C(1) = 4$ by an absorbing state at 1 would change this question.

For any fixed t and y, there are at most 2^t positive integers x with $C^t(x) = y$, because every vertex has at most two immediate predecessors. Thus infinitely many distinct primes cannot

merge into one value at one fixed time. Allowing all times gives an infinite inverse tree, but the infinitude of prime vertices in a prescribed inverse tree is not established here.

Similarly, arbitrarily long finite prime descents can occur on different paths. This does not produce one path containing infinitely many distinct primes. Fixed visitation times in an existence theorem mean that a suitable finite timetable can be chosen and then retained for infinitely many starting points. They do not mean that every externally prescribed timetable is possible.

I.13.2 What Tao's results supply

Chase, Hunter and Tao [4] prove a probabilistic analogue of Gilbreath's conjecture in a Cramér random model. Their deterministic analysis, under a prime-gap hypothesis, isolates obstacles involving long zero blocks or shallow $\{0, d\}$-valued blocks. The finite extremal results in this paper do not exclude those obstacles for the actual prime sequence.

Tao's Collatz theorem [10] states that for every function $f(N) \to \infty$,

$$\min_{j \geq 0} C^j(N) \leq f(N)$$

for a set of positive integers of logarithmic density one. The primes have logarithmic density zero, as does a fixed exponentially growing family P_a. A density-one statement for all integers need not hold on either thin set. Nor can one simply intersect the infinite sets furnished by Maynard's theorem with a logarithmic-density-one set and retain infinitude. Such a deduction requires a distribution theorem for the particular parameter family.

I.13.3 Prime exponents and untested Mersenne values

Three larger exponents considered in the motivating investigation were

$$136279901, \qquad 136279933, \qquad 136279967.$$

These three integers are prime; an exhaustive trial-division procedure with explicit bounds is included in the verification appendix. The primality of $2^q - 1$ for any of these exponents is not established here. Neither is a Collatz path from any such value to $M_{136279841}$. They are therefore untested Mersenne search targets with certified prime exponents, not certified Mersenne inverse-orbit vertices.

A prime exponent is necessary for a Mersenne prime but insufficient: $2^{11} - 1 = 23 \cdot 89$. A Mersenne number ending in decimal digit 1 can also be composite, as $2^9 - 1 = 7 \cdot 73$ shows. For $q \geq 2$, that last-digit condition is exactly $q \equiv 1 \pmod 4$; the prime $2^5 - 1 = 31$ also satisfies it. A nonzero remainder in a small modular test only excludes that particular divisor.

I.13.4 An existence proof is not an explicit primality certificate

If $M > 2$ is prime and q is any prime divisor of $2^M - 1$, then

$$\operatorname{ord}_q(2) = M, \qquad M \mid q - 1, \qquad q > M.$$

Indeed, the order divides the prime M and cannot equal 1. This is a short proof that a larger prime exists. It does not determine the integer q or certify a newly displayed record-size number. Calling q the least prime divisor does not perform that missing arithmetic computation.

Mills-type constants encode prime sequences, but evaluating a specific term requires certified precision and control of integer boundaries [15]. Defining a constant so that an untested candidate is one of its purported prime values and then using that definition to infer primality is circular. Transcendence results such as the six-exponentials theorem do not by themselves give a primality criterion for these integer candidates.

I.13.5 Questions left open by the methods

1. For the fixed $M = 2^{136279841} - 1$, does the family $P_a = (4^a M - 1)/3$ contain a prime? Does it contain infinitely many primes? The variable affine families used with Maynard's theorem do not answer either question with M fixed.
2. Can an arithmetic condition on the actual next prime guarantee its membership in the valid-extension set of the preceding prime segment? Counting extension options, even sharply, is insufficient.
3. Which isolated zeros of terminating Collatz polynomials lie in $|z| < 2/3$? Classifying accumulation points does not classify these individual zeros.
4. Can the higher-order P-recursive relations satisfied by the differences of two Collatz orbits be fully classified? The first-order classification proved here does not settle this broader question.
5. Can a complete short primality certificate be obtained for $82M + 1$ or another explicit giant inverse-orbit candidate? The exact criteria and obstructions given here do not evaluate the outstanding giant congruences.

The additional results in this revision answer specified auxiliary questions: stability under new operations, exact local divisibility, and sharper certificate conditions. They do not remove the missing arithmetic assumptions in the five questions above.

I.14 Coverage and interpretation

Topic	Mathematical treatment and limit
Franel order and Gilbreath differences	Exact recurrence-order argument, finite filters, arithmetic sampling, and finite absolute-difference extrema. The actual prime sequence is a separate input.
Small cases and universal statements	The length-one case is stated separately. Induction and equality analysis establish the finite extremal theorems; enumeration only checks small cases.
Collatz zeros and orbit observations	Exact zero accumulation, an initial-value bound, and recurrence rigidity under specified observations.
Twin starts and fixed prime gaps	Explicit delayed coalescence families, density 1/112, the pair $(6449, 6451)$, and a counterexample to a broader fixed-gap claim. No infinitude of twin primes follows.
Prime bundles and a single path	Arbitrary finite bundle sizes, synchronous mergers, and finite descents with timing and congruence conditions. Fixed finite-time inverse sets remain finite.
The starting prime 184321	Eight forward primes ending in digit 1, ten successive reverse choices, prime certificates, and exhaustive depth minimality verification within the paper.
Composite blocks and ratios	Congruence classes, gcd identities, arbitrarily factor-rich blocks, and the displayed variable step counts. No general fixed-ratio or decimal-length primality rule is asserted.
Giant Mersenne inverse families	Prime-index restrictions, exact divisor classification, and necessary primality conditions. One odd reverse step does not itself yield a Mersenne number.
Mills constants and larger primes	A distinction between abstract existence and certification of an explicit integer; no circular constant-based certification.
Certificate reuse	Binary digit-sum and path-length bounds, Pocklington conditions, and an exact elliptic-curve trace analysis.

Topic	Mathematical treatment and limit
Pascal, Franel, and prime chains	Weighted identities, a Legendre recurrence, and a fully certified path visiting ten distinct primes in 48 standard steps.

The finite Gilbreath results answer how many integers can extend a sequence that already has the required left edge. The Collatz zero theorem answers where zeros of the associated finite polynomials can accumulate. Prime-chain results choose their starting points as part of the construction; large-factor block results control divisors of composite integers in a fixed inverse family. These are different conclusions and should not be interchanged.

I.15 Self-contained exact verification and certificates

This appendix contains the finite data and algorithms needed for the numerical assertions in the manuscript. No separate program, certificate file, online factorization service, or primality library is required. Algorithms below use arbitrary-precision integer arithmetic. They specify finite checks, not experimental substitutes for the proofs of the infinite statements. The certificates were checked with exact arithmetic during preparation of this revision.

I.15.1 Elementary arithmetic and primality certificates

For clarity, the two elementary subroutines used below are specified here. The operation $u \bmod m$ returns the least nonnegative remainder.

```
POWMOD(a,e,m):
    b := a mod m; z := 1
    while e > 0:
        if e is odd: z := z*b mod m
        b := b*b mod m
        e := floor(e/2)
    return z

GCD(a,b):
    a := abs(a); b := abs(b)
    while b != 0:
        (a,b) := (b,a mod b)
    return a
```

Repeated squaring proves the first routine correct by preserving $zb^e \equiv a_0^{e_0} \pmod m$; Euclidean division proves the second. For the small integers explicitly identified below, trial division means checking every integer $2 \le d \le \lfloor \sqrt{n} \rfloor$; a number $n \ge 2$ is prime if and only if all these remainders are nonzero.

Lemma I.15.1 (Complete Lucas certificate)**.** *Suppose $n > 2$, the complete factorization $n - 1 = \prod_q q^{e_q}$ into certified primes is given, and an integer g satisfies*

$$g^{n-1} \equiv 1 \pmod n, \qquad \gcd(g^{(n-1)/q} - 1, n) = 1 \quad (q \mid n-1).$$

Then n is prime.

Proof. For a prime divisor ℓ of n, let h be the order of g modulo ℓ. The first congruence gives $h \mid n-1$. Each gcd condition says $h \nmid (n-1)/q$; hence h contains the full q-power dividing $n-1$ for every q, and $h = n-1$. Since $h \mid \ell - 1$, we have $\ell \ge n$. Thus $n = \ell$. □

The ten-prime path has its factors and bases in the main text; its small factor primes $2, 3, 5, 7, 11, 13, 3877$ are verified by trial division. For the inverse chain from 184321, Table 2 gives every prime needed recursively. Its row index defines π_i, and the final column is the complete factorization of $\pi_i - 1$ in earlier rows. Check rows in increasing order, with $\pi_1 = 2$ as the base case. Thus every factor's primality is certified before it is used.

```
CERTIFY-TABLE:
    certify pi[1] = 2
    for i = 2,...,89:
        read n = pi[i], base g, and factors (j,e) in row i
        require 1 <= j < i and e >= 1 for every factor
        require product(pi[j]^e) = n-1
        require POWMOD(g,n-1,n) = 1
        for each distinct factor index j:
            require GCD(POWMOD(g,(n-1)/pi[j],n)-1,n) = 1
        certify n
```

Table 2: Complete recursive Lucas data for the inverse-chain calculation.

i	π_i	g	$\pi_i - 1$
1	2	—	base case
2	3	2	π_1
3	5	2	π_1^2
4	7	3	$\pi_1\pi_2$
5	11	2	$\pi_1\pi_3$
6	13	2	$\pi_1^2\pi_2$
7	17	3	π_1^4
8	19	2	$\pi_1\pi_2^2$
9	23	5	$\pi_1\pi_5$
10	29	2	$\pi_1^2\pi_4$
11	31	3	$\pi_1\pi_2\pi_3$
12	37	2	$\pi_1^2\pi_2^2$
13	41	6	$\pi_1^3\pi_3$
14	43	3	$\pi_1\pi_2\pi_4$
15	47	5	$\pi_1\pi_9$
16	53	2	$\pi_1^2\pi_6$
17	61	2	$\pi_1^2\pi_2\pi_3$
18	67	2	$\pi_1\pi_2\pi_5$
19	73	5	$\pi_1^3\pi_2^2$
20	89	3	$\pi_1^3\pi_5$
21	97	5	$\pi_1^5\pi_2$
22	101	2	$\pi_1^2\pi_3^2$
23	107	2	$\pi_1\pi_{16}$
24	131	2	$\pi_1\pi_3\pi_6$
25	139	2	$\pi_1\pi_2\pi_9$
26	173	2	$\pi_1^2\pi_{14}$
27	179	2	$\pi_1\pi_{20}$
28	181	2	$\pi_1^2\pi_2^2\pi_3$
29	191	19	$\pi_1\pi_3\pi_8$
30	223	3	$\pi_1\pi_2\pi_{12}$
31	263	5	$\pi_1\pi_{24}$
32	277	5	$\pi_1^2\pi_2\pi_9$
33	293	2	$\pi_1^2\pi_{19}$

i	π_i	g	$\pi_i - 1$
34	431	7	$\pi_1\pi_3\pi_{14}$
35	577	5	$\pi_1^6\pi_2^2$
36	613	2	$\pi_1^2\pi_2^2\pi_7$
37	727	5	$\pi_1\pi_2\pi_5^2$
38	863	5	$\pi_1\pi_{34}$
39	977	3	$\pi_1^4\pi_{17}$
40	1049	3	$\pi_1^3\pi_{24}$
41	1381	2	$\pi_1^2\pi_2\pi_3\pi_9$
42	1951	3	$\pi_1\pi_2\pi_3^2\pi_6$
43	2099	2	$\pi_1\pi_{40}$
44	2293	2	$\pi_1^2\pi_2\pi_{29}$
45	2791	6	$\pi_1\pi_2^2\pi_3\pi_{11}$
46	3259	3	$\pi_1\pi_2^2\pi_{28}$
47	3449	3	$\pi_1^3\pi_{34}$
48	3499	2	$\pi_1\pi_2\pi_5\pi_{16}$
49	3691	2	$\pi_1\pi_2^2\pi_3\pi_{13}$
50	4217	3	$\pi_1^3\pi_7\pi_{11}$
51	7993	5	$\pi_1^3\pi_2^3\pi_{12}$
52	8287	3	$\pi_1\pi_2\pi_{41}$
53	10531	3	$\pi_1\pi_2^4\pi_3\pi_6$
54	12401	3	$\pi_1^4\pi_3^2\pi_{11}$
55	19207	5	$\pi_1\pi_2^2\pi_5\pi_{21}$
56	36721	37	$\pi_1^4\pi_2^3\pi_3\pi_7$
57	91463	5	$\pi_1\pi_4\pi_{15}\pi_{25}$
58	103471	3	$\pi_1\pi_2\pi_3\pi_{47}$
59	155209	7	$\pi_1^3\pi_2\pi_{10}\pi_{30}$
60	174469	10	$\pi_1^2\pi_2\pi_4\pi_{11}\pi_{18}$
61	184321	13	$\pi_1^{12}\pi_2^2\pi_3$
62	294869	2	$\pi_1^2\pi_4\pi_{53}$
63	343951	3	$\pi_1\pi_2\pi_3^2\pi_{44}$
64	576211	2	$\pi_1\pi_2\pi_3\pi_{55}$
65	723559	3	$\pi_1\pi_2\pi_5\pi_8\pi_{35}$
66	1014521	3	$\pi_1^3\pi_3\pi_6\pi_{42}$
67	4742417	3	$\pi_1^4\pi_4^2\pi_9\pi_{31}$
68	6935153	3	$\pi_1^4\pi_4\pi_8\pi_{46}$
69	48697009	37	$\pi_1^4\pi_2\pi_{66}$
70	195783047	5	$\pi_1\pi_{22}\pi_{32}\pi_{48}$
71	1006638421	2	$\pi_1^2\pi_2\pi_3\pi_{43}\pi_{51}$
72	2122156819	3	$\pi_1\pi_2^2\pi_7\pi_{68}$
73	3856994711	11	$\pi_1\pi_3\pi_{50}\pi_{57}$
74	5111259247	6	$\pi_1\pi_2^4\pi_{23}\pi_{62}$
75	17945710319	11	$\pi_1\pi_{54}\pi_{65}$
76	20649097397	3	$\pi_1^2\pi_7^2\pi_{11}\pi_{64}$
77	91149254741	2	$\pi_1^2\pi_3\pi_{11}^2\pi_{67}$
78	152710776661	2	$\pi_1^2\pi_2\pi_3\pi_6\pi_{70}$
79	3257829902101	10	$\pi_1^2\pi_2\pi_3^2\pi_{30}\pi_{69}$
80	45612819452287	3	$\pi_1\pi_2^4\pi_{19}\pi_{73}$
81	73218498101521	7	$\pi_1^4\pi_2\pi_3\pi_7\pi_{75}$
82	11107517341623331	3	$\pi_1\pi_2\pi_3\pi_{60}\pi_{72}$
83	116894695091930339	2	$\pi_1\pi_5\pi_{26}\pi_{38}\pi_{58}\pi_{63}$
84	17973882876558681571	3	$\pi_1\pi_2\pi_3\pi_{27}\pi_{56}\pi_{77}$
85	908901534103510613761	11	$\pi_1^8\pi_2^2\pi_3\pi_{12}^2\pi_{45}\pi_{76}$
86	34470932256370180314481	7	$\pi_1^4\pi_2\pi_3\pi_5\pi_{33}\pi_{39}\pi_{80}$
87	12395539184381213372234581	2	$\pi_1^2\pi_2\pi_3\pi_5\pi_6\pi_7\pi_{37}\pi_{83}$
88	16527385579174951162979441	15	$\pi_1^4\pi_3\pi_5\pi_6\pi_7\pi_{37}\pi_{83}$

i	π_i	g	$\pi_i - 1$
89	44623941063772368140045 9491	3	$\pi_1\pi_2\pi_3\pi_{36}\pi_{49}\pi_{52}\pi_{59}\pi_{74}$

I.15.2 Exhaustive shortest-distance verification

For an odd integer $n > 1$, write $n - 1 = 2^s d$ with d odd. Define the following predicate. A value `true` is a proof of compositeness; a value `false` makes no primality assertion.

```
MR2-COMPOSITE(n):
    d := n-1; s := 0
    while d is even: d := d/2; s := s+1
    x := POWMOD(2,d,n)
    if x = 1 or x = n-1: return false
    repeat s-1 times:
        x := x*x mod n
        if x = n-1: return false
    return true
```

If n were prime, repeated squaring from 2^d to $2^{n-1} = 1$ would start at 1 or encounter -1: the only square roots of 1 in a field are ± 1. This proves the claimed implication without a probability estimate.

Let $q_0, \ldots, q_{10}$ and $t_1, \ldots, t_{10}$ have the values in Section I.12. Put

$$E_j = \{q_j\} \quad (j \neq 9), \qquad E_9 = \{q_9, 44623941063772368140045 9491\}.$$

Every member of these sets occurs in Table 2. The following algorithm reconstructs all candidates; it does not require a stored list of the rejected integers.

```
VERIFY-INVERSE-SELECTION(j):
    root := q[j-1]
    frontier := {root}; seen := {root}; rejected := 0
    for h = 1,...,t[j]:
        next := {2*y : y in frontier}
        next := next union {(y-1)/3 : y in frontier, y mod 6 = 4}
        frontier := next minus seen
        seen := seen union frontier
        eligible := {n in frontier : n > root, n mod 10 = 1}
        expected := E[j] if h = t[j], otherwise the empty set
        require expected is a subset of eligible
        for n in expected: require n is certified in the Lucas table
        for n in eligible minus expected:
            require MR2-COMPOSITE(n) = true
            rejected := rejected+1
    require q[j] = minimum(E[j])
    return (cardinality(seen),rejected)
```

The sets `frontier` are exactly the vertices at minimum distance h from the root in the inverse directed graph. Removing previously seen vertices is therefore valid even though the Collatz graph contains a cycle. A previously seen eligible prime would already have been tested at its smaller distance. The returned pairs are as follows.

j	t_j	vertices seen, including the root	rejected eligible candidates
1	15	176	7
2	15	175	7
3	7	21	1
4	20	630	23
5	15	123	6
6	21	742	28
7	16	260	10
8	13	117	6
9	24	1638	68
10	3	5	0

The last column sums to 156. All its candidates fail the strong test to base 2. Together with the Lucas table, the algorithm certifies the prime selections, their minimum distances, and the tie at step nine.

I.15.3 Finite Gilbreath enumeration

For a finite sequence $a = (a_1, \dots, a_n)$, form rows $a_{0,j} = a_j$ and $a_{r+1,j} = |a_{r,j+1} - a_{r,j}|$. The following exact set recursion determines the allowable absolute increments of a new last entry. Start with $D = \{1\}$ and, for $r = n-1, n-2, \dots, 1$, replace it by

$$D \longleftarrow \{a_{r,n-r} + d,\ a_{r,n-r} - d : d \in D\} \cap \mathbb{Z}_{\geq 0}.$$

Call the final set $D(a)$. The full extension width is $2|D(a)| - \mathbf{1}_{0 \in D(a)}$; the increasing extensions are $a \parallel (a_n + d)$ with $d \in D(a)$ and $d > 0$. Starting with the single sequence $(2, 3)$, this recursion produces the following exhaustive counts. The last column excludes the unique maximum.

length n	increasing sequences	maximum width	second maximum
2	1	3	—
3	1	5	—
4	2	9	5
5	6	17	11
6	27	33	25
7	180	65	55
8	1786	129	117

For each sequence in this enumeration, the Pascal-defect identity can be checked by forming $\sigma_{r,j} = a_{r,j} + a_{r+1,j} - a_{r,j+1}$ and verifying

$$\sum_{r=1}^{n-1} (2^{n-r-1} - a_{r,n-r}) = \sum_{r=1}^{n-1} \sum_{j=1}^{n-r-1} \binom{n-j-1}{r-1} \sigma_{r,j}.$$

All indices here are one-based within each row. These finite checks are consistent with, but do not replace, the general proofs in the main text.

I.15.4 Orbits, affine words, and weighted sums

All orbit assertions can be checked by direct iteration of the maps C, T, S defined in the main text, stopping at the stated target or bound. The following specification records the full inputs and expected outputs for the finite audits, so that no hidden sampling convention is needed.

1. Iterate C from 171 to its first 1; the length is 124. In its set of odd prime values, the pairs at gap 6 are exactly $(41,47),(47,53),(103,109),(251,257),(257,263)$. If adjacency in the sorted set of *all odd orbit values* is also required, only $(103,109)$ is removed. The only prime pair at gap 2 is $(107,109)$. Trial division suffices for every value used here.
2. Apply T simultaneously to 6449 and 6451. Their first equality is at step 7, at value 1361. The two starts are prime by trial division. For each $0 \le j \le 8$ and $0 \le k \le 19$, put $n = 8^j(a_j + 128k) - 7$, where $a_j = 56$ for even j and 120 for odd j. The first equality of the two T-orbits from $n, n+2$ is at step $3j+7$.
3. For every binary word $\varepsilon = (\varepsilon_0, \dots, \varepsilon_{L-1})$ with $1 \le L \le 9$, form

$$u = \sum_j \varepsilon_j, \qquad b = \sum_{j=0}^{L-1} \varepsilon_j 2^j 3^{\sum_{k>j} \varepsilon_k}.$$

   Choose the unique $n \in [2^L, 2^{L+1})$ satisfying $3^u n + b \equiv 0 \pmod{2^L}$. Its first L parities are exactly the word, and $T^L(n) = (3^u n + b)/2^L$. For $u > 0$ also check $0 < b \le 2^{L-u}(3^u - 2^u)$. There are 1022 words in this audit.
4. For $1 \le s \le 40$, the $4s$ rational numbers $\binom{4s-1}{u-1}/3^u$ attain their maximum at $u = s$, and their sum is $\frac{1}{3}(\frac{4}{3})^{4s-1}$. Rational cross-multiplication suffices.
5. Put $n = 1280689410$, $K = 16$ and $L_i = 2n4^i3^{K-i} + 1$ for $0 \le i \le K$. The first hitting time of 1 from L_{16} under C is 428, and the largest value is 33003114793701212164. At times $3(K-i)$ the value is L_i. For $1 \le k \le 16$ and $0 \le i \le 16 - k$, check

$$\sum_{j=0}^{k} (-1)^{k-j} \binom{k}{j} L_{i+j} = 2n4^i 3^{16-i-k}.$$

   The ten prime entries use the complete Lucas certificates in the main text.
6. Evaluate $V_k = \sum_{i=0}^{k} \binom{k}{i}^2 4^i 3^{k-i}$ for $0 \le k \le 21$. The first six values are

$$1, 7, 73, 847, 10321, 129367.$$

   For $1 \le k \le 20$ verify $(k+1)V_{k+1} = 7(2k+1)V_k - kV_{k-1}$.
7. Iterate C from 184321 to its first 1, testing every value congruent to 1 modulo 10 by trial division. The length and complete prime list are those in Section I.12.
8. For every $1 \le N \le 1000$, form the terminating T-orbit $a_0 = N, \dots, a_\tau = 1$ and rational numbers $b_j = (a_j + 1)(2/3)^j/(N+1)$. Check $b_0 = 1$ and $b_j \ge b_{j+1}$. Then the list $b_{j-1} - b_j$ $(1 \le j \le \tau)$, followed by b_τ, is nonnegative and sums to 1.

I.15.5 Inverse arithmetic and Mersenne checks

Write $P(M,a) = (4^a M - 1)/3$ for $M \equiv 1 \pmod 3$ and $F(d) = (4^d - 1)/3$. The finite arithmetic audits are specified as follows.

1. For $M = 1, 4, 7, \dots, 49$, $1 \le a \le 8$ and $a < b \le 12$, check $\gcd(P(M,a), P(M,b)) = \gcd(P(M,a), F(b-a))$. For odd $3 \le r \le 25$, $2 \le s \le 21$, and $1 \le a \le 24$, put $M = 2^r - 1$ and check

$$(2^s - 1) \mid 3P(M,a) \quad \Longleftrightarrow \quad s \mid r-1 \text{ and } s \mid 2a.$$

   For odd s, also check $(2^s - 1) \mid P(M,a)$ if and only if $s \mid r-1$ and $s \mid a$. Together these loops contain 6780 parameter choices.
2. For $M = 1$ and $M = 7$, take $a = 1$ and a block of length 2. Let $x_i = P(M, 1+i)$ for $i = 0, 1$, $R = x_0 x_1 F(1)$, and let H be the least positive integer satisfying $4^H \equiv 1 \pmod{3R^2}$. The pairs (R, H) are $(5, 10)$ and $(333, 5994)$, respectively. With $y_i = P(M, 1+H+i)$ and $U_i = y_i/x_i$, verify that the quotients are integers greater than 1, that $U_i \equiv 1 \pmod R$, and that $\gcd(U_0, U_1) = 1$. Their binary lengths are respectively $(21, 21)$ and $(11989, 11989)$.

3. For $M \in \{7, 31, 127\}$, $1 \le a \le 4$, $1 \le \ell \le 8$, $1 \le s \le 4$, define

$$B_{\ell,s}(z) = \sum_{k=0}^{\ell} \binom{\ell}{k}^s z^k, \qquad H = \sum_{k=0}^{\ell} (-1)^k \binom{\ell}{k}^s P(M, a+k).$$

   Check $3H = 4^a M B_{\ell,s}(-4) - B_{\ell,s}(-1)$. For odd ℓ, also check $B_{\ell,s}(-1) = 0$, $M \mid H$, and $3 \mid B_{\ell,s}(-4)$.
4. Trial division certifies the primality of $851749, 136279901, 136279933, 136279967$. For $r = 136279841$, the identity $r-1 = 2^5 \cdot 5 \cdot 851749$ is exact. Set $d = 23629220759 = 2 \cdot 13871 \cdot 851749 + 1$. Repeated squaring gives $2^{851749} \equiv 1 \pmod{d}$ and $1 < d < 2^{851749} - 1$. Only this divisor relation is needed; no primality assertion about d is used.
5. For the same r, compute remainders without constructing 2^r:

$$4(2^r - 1) - 1 \equiv 0 \pmod{213}, \qquad 2^r - 1 \equiv 4 \pmod{9},$$
$$2^r - 1 \equiv 1 \pmod{q} \quad (q = 5, 11, 31).$$

   The first congruence certifies $71 \mid P(2^r - 1, 1)$.

For the digit-sum threshold, use the following exact recurrence, where $\mathrm{popcount}(x)$ counts the ones in the binary expansion of x.

```
x := 1; H := 0
for t = 1,...,18545:
    H := H + popcount(x)
    if t = 18544: require H = 136276195
    if t = 18545: require H = 136290777
    x := 3*x
```

The popcount itself is computed by repeatedly adding $x \bmod 2$ and replacing x by $\lfloor x/2 \rfloor$. Thus the numerical threshold requires only a finite sum of explicitly determined binary digits.

For each $a \in \{2, 3, 5, 7\}$, put $d_a = a^{82} - 1$. Exact Euclidean arithmetic gives

$$\gcd\big(d_a, (82 \text{ POWMOD}(2, r, d_a) - 81) \bmod d_a\big) = 1.$$

The same four gcd tests pass at $r = 17$, although $82(2^{17} - 1) + 1 = 631 \cdot 17033$. Thus these small checks alone do not certify the giant candidate's primality.

For the short odd-orbit examples, direct application of S gives $99 \mapsto 149 \mapsto 7$ and $7 \mid 99 - 1$. Starting from 8191, after 41 odd steps the value is 127, and the sum of the removed powers of 2 is 71, giving 112 standard steps. For each odd $5 \le s \le 59$, the first s odd iterates from $2^s - 1$ obey the initial-orbit formulas proved in the text. For each odd $3 \le r \le 15$ and odd $1 \le N < 10000$, iterate S six times, recording $K_0 = B_0 = 0$ and

$$B_t = 3B_{t-1} + 2^{K_{t-1}}, \qquad K_t = K_{t-1} + v_2(3S^{t-1}(N) + 1).$$

At each stage check $3^t N = 2^{K_t} S^t(N) - B_t$. If $S^t(N) = 2^r - 1$ and $K_t + t \le r$, check that $2^r - 1$ divides neither $N - 1$ nor $N + 1$.

Finally the real-number bound used for the standard-step obstruction has an entirely rational certificate. For $0 < x < 1$ put

$$L(x) = 2 \sum_{j=0}^{29} \frac{x^{2j+1}}{2j+1}, \qquad U(x) = L(x) + \frac{2x^{61}}{61(1-x^2)}.$$

The positive series for $\log((1+x)/(1-x))$ gives strict lower and upper bounds $L(x)$ and $U(x)$. With $r = 136279841$, rational cross-multiplication verifies

$$\begin{aligned} 352278290 &< \left(1 + \frac{L(1/2)}{U(1/3)}\right)(r+4) + 2 \\ &< \left(1 + \frac{U(1/2)}{L(1/3)}\right)(r+4) + 2 < 352278291. \end{aligned}$$

This specifies the rounding certificate without depending on floating-point logarithms. None of the checks in this appendix tests the primality of an uncomputed larger Mersenne number.

Part II

Problems from the literature: exact answers and remaining gaps

II.1 Introduction

II.1.1 What this paper does

The manuscript [21] proves, among other things, the unrestricted minimal-order statement for Franel sums (with an elementary singularity-counting argument first given in [19]), Muney's maximum-width conjecture for finite Gilbreath sequences, the second-largest width, and the exact set of accumulation points of the zeros of Collatz polynomials. Here we take its ten main methods and apply them, alone or in combination, to problems posed by other authors: open questions in two recent papers (Muney on finite Gilbreath sequences; Hohertz, and Hohertz–Kalantari, on Collatz polynomials), a research question of van Hoeij on creative telescoping, and conjectures recorded in the OEIS about Collatz trajectories. We report complete solutions where we found them, counterexamples where a statement is false, and partial results with their exact scope otherwise.

The sections are independent. Each one quotes its problems from the source, then states and proves the results. Labels and notation are local to each section.

II.1.2 The transferred methods

We number the methods of [21] as follows.

N1. *Fold recursion for extension sets*: a candidate k extends a finite Gilbreath sequence if and only if the fold $r_i = |r_{i-1} - e_i|$ along the right antidiagonal ends at 1; reverse-tree description of the extension set.
N2. *Pascal bound and parity* for absolute-difference triangles whose left edge is 1.
N3. *Deficit–slack calculus*: deficits $2^{j-1} - t_{i,j}$ and their expansion in local slacks with binomial (hockey-stick) weights.
N4. *Singularity budget*: a recurrence of shift width r allows at most r nonzero finite singular points on any branch of the generating function.
N5. *Visibility of Morse critical values* of real moment functions, with signed amplitudes, shift filters and arithmetic sampling.
N6. *Realization of parity words* by terminating Collatz orbits (affine compositions, residue classes modulo 2^L and 3^R, power-of-two tails).
N7. *Repeated-word limits and Rouché*, together with monotone normalized coefficients and the Eneström–Kakeya theorem, for zeros of Collatz polynomials.
N8. *Finite-state determinism and 2-adic divisibility* for recurrence rigidity of orbit observations.
N9. *Admissible linear forms along inverse chains* combined with Maynard's theorem and the Banks–Freiberg–Turnage-Butterbaugh theorem.
N10. *Exact arithmetic certification*: order and gcd classifications in exponential families, Lucas–Pocklington certificates, exhaustive inverse-tree search.

Section II.2 uses N1–N3; Section II.3 uses N6, N7 and Galois theory; Section II.4 uses N6, N9 and N10; Section II.5 uses N4 and N5, with a new stationary-phase case of N5. Method N8 did not lead to a problem that we could treat.

II.1.3 Outcomes

Table 3 lists every problem treated. *Closed* means that the statement asked in the source is proved or disproved as posed; *partial* means that only the stated part is proved. Easy closures are marked as such. Of the 23 rows, 8 are closed (11 statements if the four conjectures of row 12 are counted separately), 9 are partial, one is conditional and one is open. Rows 19, 20, 22 and 23 are of a different kind, as indicated.

Table 3: Problems treated and outcomes.

	Problem	Source	Outcome
Finite Gilbreath sequences (Section II.2)			
1	Is $\max_S \#\mathrm{comp}(K_S) = 2^{n-4}$ for $n \geq 5$?	Muney [3, §16, item (4)]	**Closed**: yes, with the unique maximizer V_n; short once the right recursion is written down (Thm. II.2.8)
2	Characterize the S with $\lvert K_S\rvert \leq 9$	Muney [3, §16, item (5)]	*Partial*: characterization through the right antidiagonal; widths 6 and 8 never occur, defective sequences have width ≥ 10; which antidiagonals occur for a given n is open (Thm. II.2.10)
3	Closed form for $\max_S h(S)$	Muney [3, §16, item (3)]	*Partial*: $\max h = 2^{n-1}(1 - o(1))$ with explicit two-sided bounds; no closed form (Thm. II.2.12, Cor. II.2.19)
4	Asymptotics of p_n	Muney [3, §16, item (2)]	*Open*: exact values up to $n = 12$ and a conjecture (Conj. II.2.22)
Collatz polynomials (Section II.3)			
5	Do the Hughes–Nikeghbali assumptions hold for Collatz polynomials?	Hohertz [5, §5, Q5]	**Closed**: no, already for $N = 2^k$ (an easy consequence of the product of the zeros); the normalization $N^{1/n(N)}$ and a rescaled theory (Thm. II.3.20)
6	Explicit closed-form upper bound for $\lvert z_N\rvert$ in terms of N	Hohertz [5, §5, Q2]	*Partial*: explicit bound in terms of N and the orbit length; an explicit family with $2 - \max\lvert z_N\rvert < 9/(\log_2 N)^2$ (Thm. II.3.13, Cor. II.3.15)
7	An algorithm proving that a subset of $\{\lvert z\rvert \leq 2\}$ is free of Collatz zeros	Hohertz [5, §5, Q3]	*Partial*: a decision procedure for the stated classes of sets; source-reported cutoff at $\lvert z\rvert \leq 0.666$, not independently certified here (Thm. II.3.17, Remark II.3.18)
8	Galois groups of P_N for general classes	Hohertz [5, §5, Q6]	*Partial*: structure results and two families; group S_n reported for every $N \leq 10^4$ with $d_N = 1$ except six explicit N, where $P_N = (z+1)H$ and H has group S_{n-1} (Prop. II.3.22, Reported computation II.3.23)
9	Conditions for zeros at real multiples ($\neq 2$) of roots of unity	Hohertz [5, §5, Q7]	*Partial*: sufficient conditions, a Galois obstruction, and a conditional finite-range answer for $N \leq 10^4$ (Prop. II.3.25, Thm. II.3.28, Cor. II.3.29)

	Problem	Source	Outcome
10	Theorems on the zeros that need no computation of the orbit	Hohertz [5, §5, Q8]	*Partial* (open-ended question): $d_N \mid v_2(N)+1$, at least $\log_2 N$ zeros in $1<\lvert z\rvert\le 2$, a counting bound for $d_N>1$ (Thm. II.3.32)
11	Characterize the N with $P_N(-1)=0$	Hohertz–Kalantari [25]	*Partial*: exact criterion; $P_{4N+1}(-1)=P_N(-1)$, sharpening their Lemma 11; rigidity; a source-reported list of 19 such $N\le 2\cdot 10^8$ (Thm. II.3.30)
Collatz trajectories (Section II.4)			
12	Every count $n>10$ of starting values with a common trajectory maximum occurs	Schoenfield (A233293), Hoft (A105730, A087256, A033496) [49, 50, 51, 52]	**Closed**: all four conjectures hold; for every $n\ge 11$ infinitely many k with $\lvert S(4^k)\rvert=n$ (Thm. II.4.28)
13	Every crossing number occurs infinitely often	Xausa, A349325 [42]	**Closed**: proved by realizing sign patterns with parity words (Thm. II.4.17)
14	Every prime except 3 is a first prime from an odd composite start	McEachen, A273216 [40]	**Closed**, elementary (Thm. II.4.14)
15	Steps to 6 equal Collatz steps to 1	Spage, A245691 [53]	**Closed**, elementary conjugacy (Prop. II.4.31)
16	$a(2n)-a(2n-1)=n-1$	Lagneau, A265946 [43]	**Closed**, one-line telescoping (Prop. II.4.20)
17	$a(n)$ exists for every n	Lagneau, A276202 [41]	*Conditional* on the termination of explicit trajectories (Prop. II.4.22)
18	At most two twin-prime pairs on a trajectory	Lagneau, A319227 [9]	*Partial*: reduction to chained pairs and explicitly conditional source-reported censuses (Thm. II.4.5, Cor. II.4.6)
19	At most two prime pairs with gap $2q$, for every q	Lagneau, comment in A319227	Already disproved in [21] (gap 6); further counterexamples with gaps 8 and 18 (Prop. II.4.10, Rem. II.4.12); not counted as new
20	The digital root of the trajectory maximum is 7 or a multiple of 4	Chouraqui, reported in A025586 [54]	Fails as literally read (an observation, not a conjecture); the reading for record maxima is not settled (Prop. II.4.33)
Recurrence orders (Section II.5)			
21	Roots of the characteristic polynomial of the minimal telescoper of $\binom{n}{k}^s$	van Hoeij [55, §7, Question 1]	**Closed** for the observation: proved for all s and corrected for even $s\ge 4$; immediate given [19]; the methodological question is not addressed (Thm. II.5.7)
22	Are the order-d rook-walk recurrences minimal?	Kauers–Zeilberger [56] (empirical; our reading)	*Limitation*: counting distinct singular points cannot reach d for $d=4,5$ (Rem. II.5.11)
23	Minimal order for the G_2-invariant sequence A059710	not posed (Mihailovs' recurrence)	*Additional theorem*: the minimal order is 3 (Thm. II.5.12)

II.1.4 Verification and inherited computations

The analytic arguments and embedded finite certificates are distinguished from exploratory computations reported in the earlier source. No absent script or supplementary archive is a dependency of this edition. Large searches whose per-case certificates are not included are

explicitly marked as reported data or as hypotheses of conditional conclusions. Their ranges were not independently rerun during this consolidation. In contrast, the eighteen finite fan witnesses and the small finite-Gilbreath enumeration have complete integer verification procedures printed below.

II.1.5 Scope

Several closures are modest, and we flag them. Rows 14–16 are elementary; row 1 is short once the right recursion is written down; the negative answer in row 5 follows in a few lines from the product of the zeros (the rescaled theory is the substantive part); and row 21 is immediate once the minimal order of the Franel sums [19] is known. Row 20 refutes an observation in a comment, read literally, and row 19 adds counterexamples to a statement already refuted in [21]. The most substantial results are rows 12 and 13 and, among the partial results, rows 2, 3, 7 and 11. The main conjecture of row 18, Muney's question on p_n and a closed form in row 3 remain open, and nothing here bears on the Collatz conjecture itself. Quotations of arXiv papers and OEIS entries follow the web versions accessed in October 2026. Statements that a problem is open are as of that month, and a missed recent preprint or OEIS update cannot be ruled out.

II.2 Finite Gilbreath sequences: component counts, near-minimal widths and defects

This section treats four of the five open questions listed by Muney in Section 16 of [3]. We use two tools of the manuscript [21] (the Pascal bound with its deficit/slack calculus, and the parity of the difference triangle) together with Muney's reverse-tree description of the valid-extension set. The outcome is as follows.

- Question (4) (maximum number of components) is answered affirmatively, with a unique maximizer (Theorem II.2.8). The argument is short once the right recursion for the number of components is written down.
- Question (5) (widths at most 9) is answered partially, by a characterization through the right antidiagonal (Theorem II.2.10). In particular every defective sequence has width at least 10, and the widths 6 and 8 never occur. Which admissible antidiagonals occur for a given n is left open.
- Question (3) (closed form for the maximal defect) is answered only partially. We prove a deficit–width inequality $D(S)(|K_S| - 1) \geq 2^{n-1}$, which gives $\max h \leq 2^{n-1} - 2^{(n+1)/2}$. We also compute the defect exactly on the two-parameter family of lifted minimal sequences $\Lambda^{n-N}(L_N)$, which contains Muney's family. Together these give $\max h = 2^{n-1}(1 - o(1))$, with $(n+1)/2 \leq \log_2(2^{n-1} - \max h) \leq 0.773\,n + 4$ for $n \geq 18$ (Theorems II.2.12, II.2.17, Corollary II.2.19). Muney's lower bound is thus off by a factor tending to $8/3$. We do not obtain a closed form.
- Question (2) (asymptotics of p_n) is not settled. We extend the exact data to $n = 12$ and record a conjecture (Subsection II.2.6).

Muney's question (1) (Conjecture 30) is proved in [21] and is not treated here.

II.2.1 Setting and the problems

We follow [3]. For an integer sequence $S = (s_1, \ldots, s_n)$ write $t_{0,j} = s_j$ and $t_{i+1,j} = |t_{i,j+1} - t_{i,j}|$; row i has the $n - i$ entries $t_{i,1}, \ldots, t_{i,n-i}$. Let $\mathcal{G}_n$ be the set of strictly increasing S of length n with $(s_1, s_2) = (2, 3)$ and $t_{i,1} = 1$ for $1 \leq i \leq n - 1$; $N_n = |\mathcal{G}_n|$. For $S \in \mathcal{G}_n$:

- K_S is the set of all integers k (not only $k > s_n$) such that $(s_1, \dots, s_n, k)$ has left edge 1 in every row below the top; $|K_S|$ is the *width*;
- $e_i = t_{i,n-i}$ $(1 \le i \le n-1)$ is the right antidiagonal and $A(S) = \sum_i e_i$;
- $C_S = \{k : \ |k - s_n| \le A(S) + 1, \ k \equiv s_n \pmod 2\}$, so $|C_S| = A(S) + 2$ (as $A(S)$ is odd, see (F2) below); $H_S = C_S \setminus K_S$, $h(S) = |H_S|$ is the *defect*;
- $\#\mathrm{comp}(X)$ is the number of maximal runs of common difference 2 of a finite set X of integers of one parity class;
- $p_n = \#\{S \in \mathcal{G}_n : \ h(S) = 0\}/N_n$.

We also use $U_n = (2, 3, 5, 9, \dots, 2^{n-1} + 1)$, $L_n = (2, 3, 5, 7, \dots, 2n - 1)$ and $V_n = (2, 3, 5, 9, \dots, 2^{n-2} + 1, 2^{n-1} - 1)$, that is, U_{n-1} followed by $2^{n-1} - 1$ (Muney's family of Theorem 35 in [3]).

The questions are items (2)–(5) of Section 16 ("Open questions") of [3]. The quotations below are from the arXiv HTML rendering of version 2; elided words are marked [...].

Problem II.2.1 ([3, Sect. 16, item (4)])**.** "Is $\max_{S \in \mathcal{G}_n} \#\mathrm{comp}(K_S) = 2^{n-4}$ for all $n \ge 5$?" "Verified for $n \le 10$."

Problem II.2.2 ([3, Sect. 16, item (5)])**.** "Stability classification near the minimum: characterize $S \in \mathcal{G}_n$ with $|K_S| \le 9$."

Problem II.2.3 ([3, Sect. 16, item (3)])**.** "Closed form for $\max_{S \in \mathcal{G}_n} h(S)$." "The lower bound $3 \cdot 2^{n-4} - 2n + 5$ from Corollary 36 is not tight."

Problem II.2.4 ([3, Sect. 16, item (2)])**.** "Asymptotics of $p_n := \#\{S \in \mathcal{G}_n : h(S) = 0\}/N_n$." "Data through $n = 10$ shows" [...] "Does p_n tend to a limit?" "More generally, what are the asymptotic distributions of $h(S)$" [, $|K_S|$ and $\#\mathrm{comp}(K_S)$] "under uniform sampling on" [$\mathcal{G}_n$].

We checked the questions against the primary source [3]. The probabilistic papers [24, 4] concern the prime problem and its random models. We make no exhaustive novelty claim.

II.2.2 Tools

Throughout, $S \in \mathcal{G}_n$ with $n \ge 3$, and $[\,P\,]$ is 1 if P holds and 0 otherwise.

(F1) The fold and the reverse tree (Muney). For an integer k put $r_0 = |k - s_n|$ and $r_i = |r_{i-1} - e_i|$ $(1 \le i \le n-1)$. Appending k creates the new last entries $t_{i+1,n-i} = r_i$ $(0 \le i \le n-1)$ and changes no other left-edge entry. Hence $k \in K_S$ if and only if $r_{n-1} = 1$ [3, Prop. 2]. Moreover, if $k > s_n$ is valid, then the right antidiagonal of (S, k) is the fold path $(r_0, r_1, \dots, r_{n-1})$. Put $T_{n-1} = \{1\}$ and $T_{i-1} = P_{e_i}(T_i)$ for $i = n-1, \dots, 1$, where

$$P_e(T) = \{e + y : \ y \in T\} \cup \{e - y : \ y \in T, \ y \le e\} = \{x \ge 0 : \ |x - e| \in T\}.$$

Then T_i is the set of values r_i from which the fold ends at 1, and $K_S = \{s_n \pm d : \ d \in T_0\}$ [3, Prop. 18]. Consequently

$$|K_S| = 2|T_0| - [\,0 \in T_0\,], \qquad \#\mathrm{comp}(K_S) = c(T_0), \tag{II.2.1}$$

where, for a finite set T of non-negative even integers, $c(T)$ is the number of maximal runs of step 2 of $T \cup (-T)$.

In bottom indexing we write $a_b = e_{n-b}$ and $\tau_b = T_{n-b}$ $(1 \le b \le n-1)$. Then $\tau_1 = \{1\}$, $\tau_{b+1} = P_{a_b}(\tau_b)$ and $T_0 = \tau_n$. If $(S, s_n + r)$ with $r > 0$ is valid, its bottom-indexed antidiagonal $(a'_1, \dots, a'_n)$ satisfies $a'_n = r$ and $a'_b = |a'_{b+1} - a_b|$ for $1 \le b \le n-1$.

(F2) Parity [21]. For $1 \le i \le n-1$ and $j \ge 2$ the entry $t_{i,j}$ is even. Indeed, for $i \le n-2$ we have $|t_{i,2} - 1| = t_{i+1,1} = 1$, so $t_{i,2} \in \{0, 2\}$; and $t_{i,j+1} \equiv t_{i,j} + t_{i+1,j} \pmod 2$, so evenness propagates along row i once it is known for row $i+1$ (descending induction on i). Hence e_i is even for $i \le n-2$, $e_{n-1} = 1$, $A(S)$ is odd, $T_{n-2} = \{0, 2\}$, and $T_i \subseteq 2\mathbb{Z}_{\ge 0}$ for $i \le n-2$.

(F3) Pascal bound and deficits [21]. Since $t_{i,j+1} = t_{i,j} \pm t_{i+1,j}$, we have $t_{i,j+1} \le t_{i,j} + t_{i+1,j}$ and, with $t_{i,1} = 1$ $(i \ge 1)$, $t_{i,j} \le 2^{j-1}$. Put

$$\delta_{i,j} = 2^{j-1} - t_{i,j} \ge 0, \qquad \sigma_{i,j} = t_{i,j} + t_{i+1,j} - t_{i,j+1} \in \{0,\ 2t_{i+1,j}\},$$

so that $\delta_{i,1} = 0$ and $\delta_{i,j+1} = \delta_{i,j} + \delta_{i+1,j} + \sigma_{i,j}$. In particular deficits do not decrease along a row, and

$$\delta_{i,j+1} \ \ge \ \sum_{l=1}^{j} \delta_{i+1,l} \qquad (i \ge 1). \tag{II.2.2}$$

For the antidiagonal write $\delta_i = \delta_{i,n-i} = 2^{n-i-1} - e_i$, so $\delta_{n-1} = 0$ and δ_i is even for $i \le n-2$. Put $\Delta_i = \sum_{j=i+1}^{n-1} \delta_j$ and $D(S) = \Delta_0 = 2^{n-1} - 1 - A(S)$. Applying the monotonicity along row 1 gives: $\delta_1 = 0$ forces $t_{1,j} = 2^{j-1}$ for all j, that is $S = U_n$. Hence $D(S) = 0$ if and only if $S = U_n$.

Lemma II.2.5 (Maxima and full intervals)**.** *Let $M_i = \max T_i$. Then $M_i = 1 + \sum_{j=i+1}^{n-1} e_j = 2^{n-i-1} - \Delta_i$. Let $0 \le i \le n-2$. If $e_j \le M_j$ for every j with $i < j \le n-2$, then $T_i = 2\mathbb{Z} \cap [0, M_i]$.*

Proof. $\max P_e(T) = e + \max T$, and $M_{n-1} = 1$. This gives the first formula. The second follows from $\sum_{j>i} 2^{n-j-1} = 2^{n-i-1} - 1$. For the interval statement, $T_{n-2} = \{0, 2\} = 2\mathbb{Z} \cap [0, 2]$ and $M_{n-2} = 2$. If $T_j = 2\mathbb{Z} \cap [0, M_j]$ and $e_j \le M_j$ with e_j even, then $e_j + T_j = 2\mathbb{Z} \cap [e_j, e_j + M_j]$ and $\{e_j - y :\ y \in T_j,\ y \le e_j\} = 2\mathbb{Z} \cap [0, e_j]$. Their union is $2\mathbb{Z} \cap [0, M_{j-1}]$. □

We call a level $i \in \{1, \dots, n-2\}$ a *violation* if $e_i > M_i$, that is $\delta_i < \Delta_i$ (equivalently $\delta_i \le \Delta_i - 2$, both sides being even). By Lemma II.2.5 with $i = 0$, if there is no violation then $K_S = C_S$. Muney's Theorem 20 [3] is this statement together with its converse.

Lemma II.2.6 (Size and components under one fold)**.** *Let T be a finite non-empty set of non-negative even integers and $e \ge 0$ even.*

(a) $|P_e(T)| = |T| + |T \cap [0, e]| - [\,0 \in T\,]$. *In particular $|P_e(T)| \ge |T|$, and $|P_e(T)| = 2|T| - [\,0 \in T\,]$ when $e \ge \max T$.*

(b) $c(P_e(T)) \le 2c(T)$, *with equality if and only if $e > \max T$. If $e \le \max T$ then $c(P_e(T)) \le 2c(T) - 1$.*

Proof. (a) The sets $e + T$ and $\{e - y :\ y \in T \cap [0, e]\}$ have $|T|$ and $|T \cap [0, e]|$ elements. A common element satisfies $e + y = e - y'$ with $y, y' \ge 0$, so $y = y' = 0$. Thus they meet exactly in $\{e\}$ when $0 \in T$, and not at all otherwise.

(b) Put $Z = T \cup (-T)$. Then $P_e(T) = (e + Z) \cap [0, \infty)$, and by the symmetry of Z, $-P_e(T) = (-e + Z) \cap (-\infty, 0]$. The set $(e + Z) \cap [0, \infty)$ is a translate of $Z \cap [-e, \infty)$; let ρ be its number of runs. The two mirror halves of $P_e(T) \cup (-P_e(T))$ share a run exactly when $0 \in P_e(T)$, i.e. when $e \in T$. Otherwise they are separated by a gap of length at least 4, since all elements are even. Hence $c(P_e(T)) = 2\rho - [\,e \in T\,]$. Cutting Z by a half-line cannot increase the number of runs, so $\rho \le c(T)$. Equality holds if and only if the leftmost run of Z meets $[-e, \infty)$, i.e. by symmetry if and only if e is at least the smallest element of the last run of T. Since the last run consists of all even numbers between its least element and $\max T$, the conditions $\rho = c(T)$ and $e \notin T$ hold together if and only if $e > \max T$. If $e \le \max T$, then either $e \in T$ or $\rho \le c(T) - 1$, and in both cases $c(P_e(T)) \le 2c(T) - 1$. □

Lemma II.2.7 (Zero rows)**.** *Let $1 \le i \le n-2$ and suppose $e_j = 0$ for all $i < j \le n-2$. Then rows $i+1, \dots, n-2$ equal $(1, 0, \dots, 0)$, and row i equals $(1, x, x, \dots, x)$ with $x \in \{0, 2\}$. In particular $e_i \in \{0, 2\}$.*

Proof. If row $r+1$ is $(1, 0, \dots, 0)$, then row r has $t_{r,1} = 1$, $|t_{r,2} - 1| = 1$ and $t_{r,j+1} = t_{r,j} \pm 0$. So row r is $(1, x, \dots, x)$ with $x \in \{0, 2\}$, and $x = e_r$. Row $n-1$ is (1). Descending induction from $r = n-2$ to $r = i$ gives the claim, using $e_r = 0$ for $r > i$. □

II.2.3 The number of components: answer to Problem II.2.1

Theorem II.2.8. *For every $n \ge 4$ and $S \in \mathcal{G}_n$ we have $\#\mathrm{comp}(K_S) \le 2^{n-4}$. Equality holds if and only if every level $1 \le i \le n-4$ is a violation. For $n \ge 5$ this happens if and only if $S = V_n$. Hence*

$$\max_{S \in \mathcal{G}_n} \#\mathrm{comp}(K_S) = 2^{n-4} \qquad (n \ge 5),$$

with the unique maximizer V_n. This answers Problem II.2.1 affirmatively. For $n = 4$ both elements of $\mathcal{G}_4$ have one component.

Proof. Upper bound. Row $n-2$ is $(1, e_{n-2})$ with $|e_{n-2} - 1| = 1$, so $e_{n-2} \le 2 = M_{n-2}$. Row $n-3$ is $(1, b, e_{n-3})$ with $b \in \{0, 2\}$ and $|e_{n-3} - b| = e_{n-2}$, so $e_{n-3} \le 2 + e_{n-2} = M_{n-3}$. By Lemma II.2.5, T_{n-4} is a full interval $2\mathbb{Z} \cap [0, M_{n-4}]$, so $c(T_{n-4}) = 1$. Applying Lemma II.2.6(b) to the $n-4$ folds $T_{i-1} = P_{e_i}(T_i)$, $i = n-4, \dots, 1$, gives $c(T_0) \le 2^{n-4}$, and (II.2.1) gives the bound. Equality holds if and only if each of these folds doubles c, that is (by Lemma II.2.6(b)) if and only if $e_i > \max T_i = M_i$ for $1 \le i \le n-4$.

V_n attains the bound. The right antidiagonal of V_n is $e_i = 2^{n-i-1} - 2$ ($1 \le i \le n-2$), $e_{n-1} = 1$ [3, Lemma 33], [21]. (Directly: by (F1) the bottom-indexed antidiagonal satisfies $a_{n-1} = 2^{n-2} - 2$ and $a_b = |a_{b+1} - 2^{b-1}|$, which gives $a_b = 2^{b-1} - 2$ for $b \ge 2$ and $a_1 = 1$.) So $\delta_i = 2$ for $i \le n-2$ and $\Delta_i = 2(n-2-i)$. Hence $\delta_i < \Delta_i$ exactly when $i \le n-4$.

Uniqueness ($n \ge 5$). Assume every level $i \le n-4$ is a violation. We show by descending induction on k that rows $k, \dots, n-1$ of S coincide with those of V_n. Recall that row r of V_n ($1 \le r \le n-2$) is $(1, 2, 4, \dots, 2^{n-r-2}, 2^{n-r-1} - 2)$. Equivalently, its deficits are $\delta_{r,j} = 0$ for $j < n-r$ and $\delta_{r,n-r} = 2$.

Base case $k = n-4$. Write rows $n-2, n-3, n-4$ as $(1, a)$, $(1, b, c)$, $(1, d, g, x)$, with $a = e_{n-2}$, $c = e_{n-3}$, $x = e_{n-4}$ and $a, b, d \in \{0, 2\}$. The violation at $n-4$ means $x > M_{n-4} = 2 + c + a$, so $x \ge 4 + c + a$ by parity. On the other hand $x \le g + c \le d + b + c \le 2 + b + c$. Hence $b \ge 2 + a$, which forces $b = 2$, $a = 0$ and equality throughout: $d = 2$, $g = 4$, $x = 4 + c$. From $|c - b| = a = 0$ we get $c = 2$ and $x = 6$. These are the rows $(1, 2, 4, 6)$, $(1, 2, 2)$, $(1, 0)$, (1) of V_n.

Induction step. Let $1 \le k \le n-5$ and assume rows $k+1, \dots, n-1$ agree with V_n. Then $\delta_j = 2$ for $k < j \le n-2$, so $\Delta_k = 2(n-2-k)$. Put $m = n - k \ge 5$. In row $k+1$ we have $\delta_{k+1,j} = 0$ for $j \le m-2$ and $\delta_{k+1,m-1} = 2$. The recursion of (F3) gives $\delta_{k,j} = \sum_{l<j} \sigma_{k,l}$ for $j \le m-1$ and $\delta_k = \delta_{k,m-1} + 2 + \sigma_{k,m-1}$, where $\sigma_{k,l} \in \{0, 2^l\}$ for $l \le m-2$ and $\sigma_{k,m-1} \in \{0, 2^{m-1} - 4\}$, because $\sigma_{k,l} \in \{0, 2t_{k+1,l}\}$.

- If $\sigma_{k,m-1} \ne 0$, then $\delta_k \ge 2^{m-1} - 2$. The violation at k requires $\delta_k \le \Delta_k - 2 = 2m - 6$, which is false for $m \ge 5$.
- Suppose $\sigma_{k,l} \ne 0$ for some $l \le m-2$, and take the least such l, so $\sigma_{k,l} = 2^l$. If $k = 1$, then $t_{1,l+1} = t_{1,l} - t_{2,l} = 2^{l-1} - 2^{l-1} = 0$, a zero gap, contradicting strict increase. If $k \ge 2$, then $\delta_{k,j} \ge 2^l$ for $l < j \le m$. By (II.2.2),

$$\delta_{k-1} = \delta_{k-1,m+1} \;\ge\; \sum_{j=1}^{m} \delta_{k,j} \;\ge\; \delta_k + (m-1-l)\,2^l \;\ge\; \delta_k + 2(m-2).$$

The last step holds because $l \mapsto (m-1-l)2^l$ is non-decreasing on $1 \le l \le m-2$: the ratio of consecutive values is $2(m-2-l)/(m-1-l) \ge 1$ for $l \le m-3$. But the violation at level $k-1$ (note $1 \le k-1 \le n-4$) requires $\delta_{k-1} \le \Delta_{k-1} - 2 = \delta_k + 2m - 6$. This is a contradiction.

Hence every slack in row k vanishes. Then $\delta_{k,j} = 0$ for $j < m$ and $\delta_{k,m} = 2$, so row k agrees with V_n. This completes the induction. Row 1 and $s_1 = 2$ determine S, so $S = V_n$. □

Remark II.2.9. The proof gives a finer bound. By Lemma II.2.6(b), at a non-violating level $c(T_{i-1}) - 1 \le 2(c(T_i) - 1)$, while at a violating level $c(T_{i-1}) - 1 = 2(c(T_i) - 1) + 1$. If i^* is the largest violating level, then $c(T_{i^*}) = 1$ by Lemma II.2.5, and therefore $\#\mathrm{comp}(K_S) \le 1 + \sum_{i \text{ violating}} 2^{i-1} \le 2^{i^*}$, with equality for V_n. The number of components can thus grow also at non-violating levels; for example $S = (2, 3, 5, 9, 17, 19)$ has a single violating level ($i = 2$) and $\#\mathrm{comp}(K_S) = 3$. Exhaustive computation for $n \le 12$ (Subsection II.2.7) confirms the maximum 2^{n-4} and its uniqueness; these finite checks are not used in the proof.

II.2.4 Widths at most nine: partial answer to Problem II.2.2

Theorem II.2.10. *Let $n \ge 3$ and $S \in \mathcal{G}_n$.*

(a) If $h(S) > 0$, then $|K_S| \ge 10$.
(b) $|K_S| \le 9$ if and only if $A(S) \le 7$. In that case $K_S = C_S$ and $|K_S| = A(S) + 2 \in \{5, 7, 9\}$. In particular, no $S \in \mathcal{G}_n$ has width 6 or 8.
(c) $A(S) \le 7$ if and only if $e(S) = e(L_n) + 2\chi_P$ for some set $P \subseteq \{1, \dots, n-2\}$ with $|P| \le 2$. Here $e(L_n) = (2, 0, \dots, 0, 1)$ and χ_P is the indicator vector of P. Explicitly: $e_1 \in \{2, 4\}$, $e_j \in \{0, 2\}$ for $2 \le j \le n-2$, and $(e_1 - 2) + e_2 + \cdots + e_{n-2} \le 4$. Then $|K_S| = 5 + 2|P|$.
(d) $|K_S| = 5$ if and only if $S = L_n$ (this is Muney's Theorem 25 [3]).

Proof. (a) If there is no violation, $h(S) = 0$ by Lemma II.2.5. So let i be the largest violating level. By Lemma II.2.5, $T_i = 2\mathbb{Z} \cap [0, M_i]$. If $e_j = 0$ for all $i < j \le n-2$, then $M_i = 2$, while Lemma II.2.7 gives $e_i \le 2$, so i would not be a violation. Hence some $e_j \ge 2$ with $i < j \le n-2$, and $M_i \ge 4$. As $e_i > M_i$, $T_{i-1} = (e_i + T_i) \cup (e_i - T_i) = 2\mathbb{Z} \cap [e_i - M_i,\, e_i + M_i]$, a set of $M_i + 1 \ge 5$ elements whose least element is at least 2. By Lemma II.2.6(a), $|T_0| \ge |T_{i-1}| \ge 5$. If $|T_0| \ge 6$, then $|K_S| \ge 11$ by (II.2.1). If $|T_0| = 5$, then $|T_{j-1}| = |T_j|$ for all $j < i$. By Lemma II.2.6(a) and induction (using $0 \notin T_j$), $T_j \cap [0, e_j] = \emptyset$. Hence $T_{j-1} = e_j + T_j$ does not contain 0. Thus $0 \notin T_0$ and $|K_S| = 10$.

(b) If $|K_S| \le 9$, then $h(S) = 0$ by (a), so $|K_S| = A + 2$ and $A \le 7$. Conversely, $A \le 7$ gives $|K_S| \le A + 2 \le 9$, hence $h(S) = 0$ by (a) and $|K_S| = A + 2$, which is odd by (F2).

(c) Since $e_1 = s_n - s_{n-1}$ is positive and even, $e_1 \ge 2$. Hence $x := e(S) - e(L_n)$ has non-negative even entries $x_1 = e_1 - 2$ and $x_j = e_j$ $(2 \le j \le n-2)$, and $x_{n-1} = 0$. Its sum is $A - 3 \le 4$. If some entry were ≥ 4, it would be the only non-zero entry. If it is $x_j = e_j = 4$ with $j \ge 2$, then $e_{j'} = 0$ for $j < j' \le n-2$, and Lemma II.2.7 gives $e_j \le 2$, a contradiction. If it is x_1, then $e_2 = \cdots = e_{n-2} = 0$ and Lemma II.2.7 gives $e_1 \le 2$, again a contradiction. So $x = 2\chi_P$ with $|P| \le 2$. The converse and $|K_S| = A + 2 = 5 + 2|P|$ follow from (b).

(d) If $|K_S| = 5$, then $A = 3$ and $e(S) = (2, 0, \dots, 0, 1)$. Lemma II.2.7 with $i = 1$ shows that row 1 is $(1, x, \dots, x)$; strict increase forces $x = 2$, so $S = L_n$. Conversely $|K_{L_n}| = 5$. Both directions are [3, Thm. 25], proved there by the same route; we include the short argument for completeness. □

Remark II.2.11 (What "stability" does and does not mean here). Theorem II.2.10 is a stability statement for the *right antidiagonal*: a width of at most 9 forces $e(S)$ to lie within ℓ^1-distance 4 of $e(L_n)$. It does *not* force S to be close to L_n. For example, $(2, 3, 5, 9, 15, 23, 33, 41, 47, 51, 53) \in \mathcal{G}_{11}$ has antidiagonal $(2, 2, 0, \dots, 0, 1)$ and $|K_S| = 7$. Since $K_S - s_n$ (hence $|K_S|$, $\#\mathrm{comp}(K_S)$ and

$h(S)$) depends on S only through $e(S)$, (b)–(c) characterize the condition $|K_S| \le 9$ completely in terms of $e(S)$. Which of the patterns in (c) actually occur for a given n is a finer reachability question that we leave open. For instance, among $5 \le n \le 12$, width 7 ($|P| = 1$) occurs exactly for $n \in \{5, 7, 8, 9, 11, 12\}$. The exhaustive counts are in Table 4.

n	4	5	6	7	8	9	10	11	12
$\lvert K_S\rvert = 5$	1	1	1	1	1	1	1	1	1
$\lvert K_S\rvert = 7$	0	1	0	4	2	7	0	56	51
$\lvert K_S\rvert = 9$	1	1	7	4	14	29	152	144	170
$\lvert K_S\rvert = 10$ (all defective)	0	1	1	8	9	37	102	968	7995

Table 4: Exhaustive counts (computation; see Subsection II.2.7). Widths 6 and 8 never occur, in accordance with Theorem II.2.10.

II.2.5 The maximal defect: partial answer to Problem II.2.3

Since $h(S) = A(S) + 2 - |K_S| = 2^{n-1} + 1 - D(S) - |K_S|$, maximizing the defect means minimizing $D(S) + |K_S|$. We give a two-sided estimate that determines the leading term 2^{n-1} and bounds the error term from both sides. We do not obtain a closed form.

Theorem II.2.12 (Deficit–width inequality)**.** *For $n \ge 3$ and $S \in \mathcal{G}_n \setminus \{U_n\}$,*

$$D(S)\,(|K_S| - 1) \;\ge\; 2^{n-1}.$$

Consequently $h(S) \le 2^{n-1} - 2^{(n+1)/2}$ for every $S \in \mathcal{G}_n$, $n \ge 3$.

Proof. As $S \ne U_n$, some $\delta_i > 0$ by (F3). Let $i_0 \le n-2$ be the largest index with $\delta_{i_0} > 0$; then $\delta_{i_0} \ge 2$ by parity. For $i_0 < j \le n-2$ we have $\delta_j = 0$, so $e_j = 2^{n-j-1} = M_j$. By Lemma II.2.5, $T_{i_0} = 2\mathbb{Z} \cap [0, 2^{n-i_0-1}]$, hence $|T_{i_0}| - 1 = 2^{n-i_0-2}$. Level i_0 is not a violation, since $\Delta_{i_0} = 0$. Let v be the number of violations among the levels $1, \dots, i_0 - 1$.

By Lemma II.2.6(a), $|T_{i-1}| - 1 \ge 2(|T_i| - 1)$ at a violation and $|T_{i-1}| \ge |T_i|$ at every level. Hence $|T_0| - 1 \ge 2^v\, 2^{n-i_0-2}$, and by (II.2.1) $|K_S| - 1 \ge 2(|T_0| - 1) \ge 2^{v+n-i_0-1}$.

For the deficits, $\Delta_{i-1} = \Delta_i + \delta_i$. At a non-violating level, $\delta_i \ge \Delta_i$, so $\Delta_{i-1} \ge 2\Delta_i$. Starting from $\Delta_{i_0-1} = \delta_{i_0} \ge 2$ and using the $(i_0 - 1 - v)$ non-violating levels among $1, \dots, i_0 - 1$, we get $D = \Delta_0 \ge 2 \cdot 2^{i_0-1-v}$.

Multiplying, $D(|K_S| - 1) \ge 2^{n-1}$. Finally, for $S \ne U_n$,

$$h(S) = 2^{n-1} - D - (|K_S| - 1) \le 2^{n-1} - 2\sqrt{D(|K_S| - 1)} \le 2^{n-1} - 2^{(n+1)/2},$$

and $h(U_n) = 0$ (note $2^{n-1} \ge 2^{(n+1)/2}$ for $n \ge 3$). □

Remark II.2.13. For $4 \le n \le 12$ the minimum of $D(S)(|K_S| - 1)/2^{n-1}$ over $\mathcal{G}_n \setminus \{U_n\}$ is $4 - 2^{5-n}$, attained at L_n (computation). So the constant in Theorem II.2.12 is probably not optimal; we have not tried to sharpen it.

For lower bounds we use a lifting operation, which also explains the extremal families seen in the data.

Lemma II.2.14 (Deficits as weighted slacks [21])**.** *For $i \ge 1$ and every entry (i, j) of the triangle, $\delta_{i,j} = \sum_{a \ge i} \sum_{b \le j-1} \binom{j-b-1}{a-i} \sigma_{a,b}$, the sum running over the defined slacks. In particular $\delta_i = \delta_{i,n-i} = \sum \binom{n-i-b-1}{a-i} \sigma_{a,b}$ and $D(S) = \sum_{a,b} \binom{n-b-1}{a-1} \sigma_{a,b}$.*

Proof. For $j = 1$ both sides vanish. Assume the formula for column j (all rows). Then $\delta_{i,j+1} = \delta_{i,j} + \delta_{i+1,j} + \sigma_{i,j}$, and Pascal's rule $\binom{j-b-1}{a-i} + \binom{j-b-1}{a-i-1} = \binom{j-b}{a-i}$ gives the formula for column $j+1$; the term $\sigma_{i,j}$ is the summand $b = j$, $a = i$. Summing δ_i over i and using the hockey-stick identity gives the expression for D, which is the deficit identity used in [21, Sect. 2.4]. □

Definition II.2.15 (The lift)**.** For $S = (s_1, \dots, s_N) \in \mathcal{G}_N$ with gaps $g_k = s_{k+1} - s_k$, let $\Lambda(S)$ be the sequence of length $N+1$ with first term 2 and gaps $g'_1 = 1$, $g'_{k+1} = g'_k + g_k$ $(1 \le k \le N-1)$.

Lemma II.2.16. *Let $S \in \mathcal{G}_N$, $N \ge 2$.*

(a) $\Lambda(S) \in \mathcal{G}_{N+1}$. Row 1 of $\Lambda(S)$ is (g'_k), and for $i \ge 2$ row i of $\Lambda(S)$ equals row $i-1$ of S. All slacks of $\Lambda(S)$ in row 1 vanish, and $\sigma_{a+1,b}(\Lambda(S)) = \sigma_{a,b}(S)$.
(b) In bottom indexing the antidiagonal of $\Lambda(S)$ is $(a_1, \dots, a_{N-1}, s_N - 1)$. Hence $\tau_b(\Lambda(S)) = \tau_b(S)$ for $b \le N$, $T_0(\Lambda(S)) = P_{s_N-1}(T_0(S))$ and $|T_0(\Lambda(S))| \le 2|T_0(S)|$.
(c) For $j \ge 0$ and $n = N + j$: $|K_{\Lambda^j(S)}| \le 2^{j+1}|T_0(S)|$ and $D(\Lambda^j(S)) = \sum_{a,b} \binom{N+j-b-1}{N-a-b}\sigma_{a,b}(S)$.

Proof. (a) $|g'_{k+1} - g'_k| = g_k$, so row 2 of $\Lambda(S)$ is row 1 of S, and the later rows follow. The left edge is $g'_1 = 1$ in row 1 and that of S below. The sequence starts $(2, 3)$ and is strictly increasing because $g'_k \ge 1$. In row 1, $\sigma_{1,b} = g'_b + g_b - g'_{b+1} = 0$. (b) The last entry of row 1 is $g'_N = 1 + \sum_{k<N} g_k = s_N - 1$, and the other last entries are those of S. The statements on τ_b follow from (F1), and the size bound from Lemma II.2.6(a). (c) Iterate (b) to get $|T_0(\Lambda^j S)| \le 2^j|T_0(S)|$, and use (II.2.1). Iterating (a), the slacks of $\Lambda^j(S)$ are $\sigma_{a,b}(S)$ placed at $(a+j, b)$. Lemma II.2.14 gives $D = \sum \binom{N+j-b-1}{a+j-1}\sigma_{a,b}$, and $\binom{N+j-b-1}{a+j-1} = \binom{N+j-b-1}{N-a-b}$. □

For example, $\Lambda(L_4) = (2, 3, 5, 9, 15) = V_5$ and $\Lambda(V_n) = V_{n+1}$, so Muney's family is $V_n = \Lambda^{n-4}(L_4)$. The next theorem computes the whole two-parameter family $\Lambda^{n-N}(L_N)$ exactly.

Theorem II.2.17 (Lifted minimal sequences)**.** *Let $N \ge 4$, $j \ge 0$, $n = N + j$, and $S = \Lambda^j(L_N) \in \mathcal{G}_n$. Then*

$$|K_S| = 5 \cdot 2^{n-N}, \qquad \#\mathrm{comp}(K_S) = 2^{n-N}, \qquad D(S) = 2\sum_{k=1}^{N-3} \binom{n-2}{k},$$

and therefore

$$h(S) = 2^{n-1} + 1 - 5 \cdot 2^{n-N} - 2\sum_{k=1}^{N-3} \binom{n-2}{k}.$$

For $N = 4$ this is Muney's family V_n, with $h = 3 \cdot 2^{n-4} - 2n + 5$. For $N = 5$ it is $\alpha_n := \Lambda^{n-5}(L_5) = (U_{n-2}, 2^{n-2} - 1, 2^{n-1} - 2n + 3)$, with $h(\alpha_n) = 11 \cdot 2^{n-5} - n^2 + 3n - 1$.

Proof. The rows $2, \dots, N-2$ of L_N are $(1, 0, \dots, 0)$ and row 1 is $(1, 2, \dots, 2)$. So the only non-zero slacks of L_N are $\sigma_{a,1} = 2$, $2 \le a \le N-2$. By Lemma II.2.16(a) the non-zero slacks of S are $\sigma_{a+j,1} = 2$. Lemma II.2.14 then gives, for the antidiagonal deficit at bottom index $\beta = n - i$ $(2 \le \beta \le n-1$; in this proof δ_β and Δ_β are indexed from the bottom, unlike δ_i, Δ_i in (F3)),

$$\delta_\beta = 2\sum_{a=2}^{N-2} \binom{\beta-2}{a+j-i} = 2\sum_{k=0}^{N-4} \binom{\beta-2}{k}, \qquad \Delta_\beta := \sum_{\beta'<\beta} \delta_{\beta'} = 2\sum_{k=1}^{N-3} \binom{\beta-2}{k},$$

by the hockey-stick identity. Here Δ_β is the sum of the deficits strictly below the level, as in (F3). Taking $\beta = n$ in the second formula gives $D(S)$. Level β is a violation if and only if $\delta_\beta < \Delta_\beta$, i.e. $1 < \binom{\beta-2}{N-3}$, i.e. $\beta \ge N$. By Lemma II.2.16(b) the levels $\beta \le N-1$ are those of L_N, so $\tau_N = T_0(L_N) = \{0, 2, 4\}$. At $\beta = N$ the fold is a violation, so $\tau_{N+1} = 2\mathbb{Z} \cap [a_N - 4, a_N + 4]$ with

$a_N > 4$: five elements, none equal to 0, and two components. Every later level $N < \beta \le n-1$ is again a violation with $0 \notin \tau_\beta$, so both $|\tau_\beta|$ and $c(\tau_\beta)$ double (Lemma II.2.6). For $j \ge 1$ this gives $|T_0| = 5 \cdot 2^{j-1}$, $0 \notin T_0$ and $c(T_0) = 2^j$, hence $|K_S| = 5 \cdot 2^j$ and $\#\mathrm{comp}(K_S) = 2^j$. For $j = 0$ we have $S = L_N$ and $|K_S| = 5$. The identification of α_n follows from $\Lambda(\alpha_n) = \alpha_{n+1}$ and $\Lambda(L_5) = \alpha_6$; both are checked directly from Definition II.2.15. Substituting $N = 4, 5$ gives the two formulas. □

A second family is the lift orbit of a different seed. For $n \ge 7$ put $\beta_n = (U_{n-2},\ 2^{n-2} - 5,\ 2^{n-1} - 6n + 15)$; then $\beta_7 = (2, 3, 5, 9, 17, 27, 37)$ and $\Lambda(\beta_n) = \beta_{n+1}$.

Proposition II.2.18. *For $n \ge 7$: $\beta_n \in \mathcal{G}_n$, $|K_{\beta_n}| = 5 \cdot 2^{n-6}$, $A(\beta_n) = 2^{n-1} - 3n^2 + 17n - 23$, and $h(\beta_n) = 27 \cdot 2^{n-6} - 3n^2 + 17n - 21$.*

Proof. We use bottom indexing and the first part of (F1) (appending k creates the new last entries r_i, whether or not k is valid); the antidiagonal of U_m is $u_b = 2^{b-1}$. The prefix $Q = (U_{n-2}, 2^{n-2} - 5)$ arises from $r = 2^{n-3} - 6 > 0$. Its antidiagonal q' satisfies $q'_{n-2} = r$ and $q'_b = |q'_{b+1} - 2^{b-1}|$. This gives $q'_b = 2^{b-1} - 6$ for $4 \le b \le n-2$, then $q'_3 = |2 - 4| = 2$, $q'_2 = |2 - 2| = 0$ and $q'_1 = 1$; so $Q \in \mathcal{G}_{n-1}$. Appending $x = 2^{n-1} - 6n + 15$ means $r' = x - (2^{n-2} - 5) = 2^{n-2} - 6n + 20 > 0$. The antidiagonal q of β_n satisfies $q_{n-1} = r'$ and $q_b = |q_{b+1} - q'_b|$. For $b \ge 5$ this gives $q_b = 2^{b-1} - 6b + 14$, since $q_{b+1} - q'_b = 2^{b-1} - 6b + 14 \ge 0$. Then $q_4 = |0 - 2| = 2$, $q_3 = |2 - 2| = 0$, $q_2 = 0$ and $q_1 = 1$. Hence $\beta_n \in \mathcal{G}_n$. The reverse tree gives $\tau_2 = \tau_3 = \tau_4 = \{0, 2\}$, $\tau_5 = \tau_6 = \{0, 2, 4\}$ and $\tau_7 = P_{10}(\{0, 2, 4\}) = 2\mathbb{Z} \cap [6, 14]$. For $b \ge 6$, $\max \tau_b = 2^{b-1} - 3b^2 + 17b - 22$ and $q_b - \max \tau_b = 3b^2 - 23b + 36 > 0$, so every fold from $b = 7$ to $n-1$ doubles and keeps 0 out. Hence $|T_0| = 5 \cdot 2^{n-7}$, $|K| = 5 \cdot 2^{n-6}$, and $A = \max \tau_n - 1 = 2^{n-1} - 3n^2 + 17n - 23$. □

Corollary II.2.19. *For all $n \ge N \ge 4$,*

$$2^{(n+1)/2} \ \le\ 2^{n-1} - \max_{S \in \mathcal{G}_n} h(S) \ \le\ 5 \cdot 2^{n-N} + 2 \sum_{k=1}^{N-3} \binom{n-2}{k} - 1.$$

Consequently $\max_{\mathcal{G}_n} h = 2^{n-1}(1 - o(1))$. *More precisely, for all $n \ge 18$,*

$$\tfrac{n+1}{2} \ \le\ \log_2\Bigl(2^{n-1} - \max_{\mathcal{G}_n} h\Bigr) \ \le\ 0.773\, n + 4.$$

Muney's lower bound $3 \cdot 2^{n-4} - 2n + 5$ is the case $N = 4$. Its ratio to the true maximum tends to $3/8$.

Proof. The left inequality is Theorem II.2.12; the right one is Theorem II.2.17. For the asymptotic bound let $H(x) = -x \log_2 x - (1 - x) \log_2(1 - x)$ and take $N = \lfloor 0.227\, n \rfloor$, so $N \ge 4$ for $n \ge 18$. We have $(N - 3)/(n - 2) \le 0.227 < 1/2$, so the standard estimate $\sum_{k \le xm} \binom{m}{k} \le 2^{H(x)m}$ $(0 \le x \le 1/2)$ gives $2 \sum_{k=1}^{N-3} \binom{n-2}{k} \le 2^{H(0.227)n+1} \le 2^{0.773n+1}$, because $H(0.227) = 0.77274\ldots$. Also $5 \cdot 2^{n-N} \le 5 \cdot 2^{0.773n+1}$. The sum is at most $12 \cdot 2^{0.773n} < 2^{0.773n+4}$. □

Remark II.2.20. Optimizing over N in Theorem II.2.17 gives the exponent $1 - \varepsilon^* = 0.77290\ldots$, where $\varepsilon^* \in (0, 1/2)$ solves $H(\varepsilon) = 1 - \varepsilon$. A better exponent would need better seeds for the lift (Lemma II.2.16(c)): sequences of length N with small $|T_0|$ whose slacks lie close to the antidiagonal. The data are compatible with the existence of such seeds. Among the $S \in \mathcal{G}_N$ whose non-zero slacks all satisfy $N - a - b \le 3$, the least width is $5, 7, 9, 10, 11$ for $N = 6, \ldots, 10$ (reported exploratory computation). Seeds with polynomially bounded width and this property, for all N, would prove Conjecture II.2.21 (Theorem II.2.12 gives the other inequality). We have no construction for general N.

Table 5 lists the exact maxima for $n \le 12$ (exhaustive enumeration). For $13 \le n \le 15$ it gives the best values found in the subfamily of sequences with prefix U_{n-2}, next to the families above. For $8 \le n \le 12$ the maximizers are lifts of short seeds: α_n for $n \le 10$ and β_n for $n \ge 10$. For $n = 13$ the best value found is $h(\beta_{13})$. For $n = 14, 15$ other sequences do better, namely $(U_{12}, 4091, 8051)$ and $(U_{13}, 8187, 16237)$, with $|K| = 672$ and 1344. Window searches (Subsection II.2.7) found $h = 28987$, 59279, 120109 for $n = 16, 17, 18$. So the extremal family keeps changing, and a single closed form of the type of [3, Cor. 36] seems unlikely.

n	h^*	status	$h(V_n)$	best $\Lambda^{n-N}L_N$	$h(\beta_n)$	$\lfloor 2^{n-1} - 2^{(n+1)/2} \rfloor$	$\frac{2^{n-1}-h^*}{2^{n/2}}$
5	1	exact max	1	1	–	8	2.65
6	5	exact max	5	5	–	20	3.38
7	15	exact max	15	15	5	48	4.33
8	47	exact max	37	47	31	105	5.06
9	121	exact max	83	121	105	224	5.97
10	281	exact max	177	281	281	466	7.22
11	667	exact max	367	615	667	960	7.89
12	1479	exact max	749	1379	1479	1957	8.89
13	3149	lower bound	1515	2995	3149	3968	10.46
14	6717	lower bound	3049	6317	6541	8010	11.52
15	14095	lower bound	6119	13071	13383	16128	12.65

Table 5: The maximal defect. For $n \le 12$, $h^* = \max_{\mathcal{G}_n} h$ by exhaustive enumeration; the values for $n \le 10$ agree with [3, Table 1], and those for $n = 11, 12$ are new. The maximizer is unique except at $n = 7$ (three maximizers) and $n = 10$ (two: α_{10} and β_{10}). For $13 \le n \le 15$, h^* is the exact defect of an explicit sequence (a lower bound). The column "best $\Lambda^{n-N}L_N$" is the maximum over N of the formula in Theorem II.2.17. The last column is an illustration only.

Conjecture II.2.21. $\log_2(2^{n-1} - \max_{S \in \mathcal{G}_n} h(S)) = \frac{n}{2} + O(\log n)$.

Theorem II.2.12 gives the lower half. Its proof suggests the extremal mechanism: about half of the levels should be non-violating "shifts" (the fold does not enlarge T, while the accumulated deficit doubles) and the other half violations (the set doubles, while the deficit grows only additively). The best sequences found for $n = 14, \dots, 18$ contain one or two such shifts, for example at level $\beta = 10$ for $n = 18$. The conjecture would follow from seeds S with $|T_0(S)| \approx 2^{N/2}$ whose slacks lie close to the antidiagonal, via Lemma II.2.16(c). We have not constructed them.

II.2.6 Interval-completeness frequency: Problem II.2.4 (open)

We do not settle Problem II.2.4. Table 6 extends Muney's data to $n = 12$. The counts N_{11}, N_{12}, N_{13} produced by the same enumeration agree with OEIS A080839 [23] (that entry counts sequences starting $1, 2, \dots$; the shift $S \mapsto S - 1$ identifies them with $\mathcal{G}_n$).

n	N_n	$\#\{h = 0\}$	p_n	mean $\lvert K_S \rvert$	mean $h(S)$	mean $\#\mathrm{comp}(K_S)$
8	1786	1026	0.5745	29.907	3.101	2.027
9	26094	12782	0.4898	43.495	6.410	2.613
10	559127	237073	0.4240	63.359	12.629	3.448
11	17535396	6019863	0.3433	92.273	24.100	4.614
12	804131875	224236288	0.2789	134.414	44.633	6.200

Table 6: Exhaustive data (computation). $p_{11} = 6019863/17535396$ and $p_{12} = 224236288/804131875$ are new; the remaining entries of the first four columns agree with [3, Table 1 and Sect. 16]. The number of increasing extensions summed over $\mathcal{G}_{12}$ is $N_{13} = 53833201737$.

Two observations may help further work. First, by (F1) the sum of widths is tied to the growth of N_n:

$$\sum_{S\in\mathcal{G}_n} |K_S| = 2N_{n+1} + Z_n, \qquad Z_n = \#\{S \in \mathcal{G}_n : \ s_n \in K_S\},$$

because the increasing extensions of S are exactly the $(|K_S| - [\, s_n \in K_S\,])/2$ elements $s_n + d$, $d \in T_0 \setminus \{0\}$. So the mean width is $2N_{n+1}/N_n + O(1)$. Second, interval completeness is not hereditary in either direction (for example, V_5 is defective, while its child $(2, 3, 5, 9, 15, 17)$ is interval-complete). Among the elements of $\mathcal{G}_9, \mathcal{G}_{10}, \mathcal{G}_{11}$ whose parent is complete, the complete fraction is 0.605, 0.571 and 0.507; among those whose parent is defective, it is 0.350, 0.300 and 0.240. The data suggest $p_{n+1}/p_n \approx 0.81$–0.87.

Conjecture II.2.22. *$p_n \to 0$ as $n \to \infty$.*

The mechanism seen in the data is that violations occur mostly at the top levels (largest e_i). Their number grows with n, and by Lemma II.2.6(b) each violation doubles the component count (the count can also grow at non-violating levels). A proof would need control of the typical shape of the right antidiagonal under the uniform measure on $\mathcal{G}_n$, which we do not have.

II.2.7 Exact enumeration and status of numerical data

The preceding theorems have analytic proofs and do not depend on numerical experiments. The large tables through $n = 12$ and the window-search values for $13 \le n \le 18$ are retained as computations reported in the source manuscript. The consolidation independently rechecked every sequence through $n = 9$, including all 26,094 sequences of length 9; it did not rerun the 804,131,875-sequence census at length 12.

Here is a complete reference enumeration. It uses exact integers and no third-party modules. A state records only the right antidiagonal; different paths to an equal antidiagonal remain distinct in the depth-first traversal, so their sequences are counted with the correct multiplicity. Replacing the last argument 9 by 12 requests the larger census, at substantial cost. The code is included as text in this document and is not required to compile it.

```
from collections import Counter

def distances(e):
    t = {1}
    for x in reversed(e):
        t = {x+y for y in t} | {x-y for y in t if y <= x}
    return t

def census(bound):
    rows = [Counter() for _ in range(bound+1)]
    def visit(e):
        n = len(e)+1
        t = distances(e)
        w = 2*len(t) - int(0 in t)
        h = sum(e)+2-w
        z = t | {-x for x in t}
        c = sum(x-2 not in z for x in z)
        r = rows[n]
        r['N'] += 1
        r['complete'] += int(h == 0)
        r['widthsum'] += w
```

```
        r['defectsum'] += h
        r['componentsum'] += c
        r[('width',w)] += 1
        r['maxdefect'] = max(r['maxdefect'],h)
        r['maxcomponents'] = max(r['maxcomponents'],c)
        if n >= 3:
            d = 2**(n-1)-1-sum(e)
            assert not h or w >= 10
            assert (w <= 9) == (sum(e) <= 7)
            assert not d or d*(w-1) >= 2**(n-1)
        if n >= 4:
            assert c <= 2**(n-4)
        if n == bound:
            r['nextN'] += len(t)-int(0 in t)
            return
        for d in t:
            if not d:
                continue
            a, child = d, [d]
            for x in e:
                a = abs(a-x)
                child.append(a)
            assert child[-1] == 1
            visit(tuple(child))
    visit((1,))
    return rows

for n, row in enumerate(census(9)):
    if row:
        print(n, dict(row))
```

The recursion is exactly (F1), and its positive distances are precisely the strictly increasing extensions. Induction on n therefore proves the completeness and absence of duplicate sequence paths in this enumeration. For the families V_n, $\Lambda^j(L_N)$ and β_n, the displayed formulas are established by their proofs; reported finite checks add no logical hypothesis. Search bounds beyond the exhaustive range remain lower bounds, not certificates of maximality.

II.3 Zeros and algebra of Collatz polynomials: questions of Hohertz and of Hohertz–Kalantari

This section treats the open questions on Collatz polynomials posed by M. Hohertz [5, Section 5, "Final remarks"], together with a remark of Hohertz and Kalantari [25] that leaves a converse open. The manuscript [21] whose methods we apply (below: *the manuscript*) already answers, in its Section 3, Hohertz's first question and essentially his fourth; we use its results where needed, always flagged, and we do not count them as new.

Our tools are: (i) a sparse form of $(z-2)P_N(z)$ which turns the zeros of P_N into the solutions of $\varphi_N(u) = 1$ for an explicit probability generating function φ_N (Lemma II.3.9); (ii) the manuscript's realization of parity words by terminating orbits (Lemma II.3.11); (iii) Galois theory of the factors of P_N; (iv) exact computation (integer and rational arithmetic, exact factorization over $\mathbb{Z}$ and over $\mathbb{F}_p$, PARI/GP's exact `polgalois`, and certified ball arithmetic). Table 7 summarizes

the outcome.

Question	Result here	Status
Q1	Analytic bound R_N; source-reported stronger cutoff 0.666 for $N \neq 9$ (Remark II.3.18)	analytic result; stronger cutoff unverified here
Q2	Explicit bound $\lvert z_N\rvert \le 2 - \frac{2d_N^2 N}{3(n+1)^3 2^n}$ off $\lvert z\rvert = 2$ (Thm. II.3.13); explicit N with $d_N = 1$ and $2 - \max \lvert z_N\rvert < 9/(\log_2 N)^2$ (Cor. II.3.15)	partial: closed form in N and $n(N)$; a bound in N alone is not reached
Q3	Decision procedure with exact scope (Thm. II.3.17); source-reported instance at $\lvert z\rvert \le 0.666$ (Remark II.3.18; finite certificates not included)	closed for the stated classes of sets; open for thin sets
Q5	The Hughes–Nikeghbali hypothesis fails for $N = 2^k$ and along every sequence with $n(N) = O(\log N)$; exact criterion; rescaled version (Thm. II.3.20)	disproved as stated; precise partial theory
Q6	Structure for $d_N > 1$, powers of 2, two explicit families (Prop. II.3.22); S_n reported computationally for $N \le 10^4$ with $d_N = 1$ except the six values $x_m = (4^m - 1)/3$, where $P_N = (z+1)H_m$ and H_m has S_{n-1} (Reported computation II.3.23)	partial (computational + structural); conjecture
Q7	Sufficient conditions and an explicit family all of whose zeros are real multiples of roots of unity (Prop. II.3.25); Galois obstruction (Thm. II.3.28); conditional finite-range answer for $N \le 10^4$ (Cor. II.3.29)	partial; finite-data hypothesis explicit
HK	$P_N(-1) = 0$: the family of Hohertz–Kalantari is the only family defined by a parity word; the $4N+1$ chains are due to [25, Lemma 11], sharpened here to an equality (Thm. II.3.30)	partial
Q8	$d_N \mid v_2(N) + 1$; at least $\log_2 N$ zeros in $1 < \lvert z\rvert \le 2$; $\#\{N \le X : d_N > 1\} \le 47\sqrt{X} + \log_2 X$ (Thm. II.3.32)	partial (several orbit-free theorems)
Q4, Q9	Q4 is answered by the manuscript [21, Cor. 3.2]; Q9 asks for a property contradicting a theorem, which would disprove the Collatz conjecture	not treated

Table 7: Targets of this section and their status.

II.3.1 Setting and the questions

Following [5], put $c(N) = N/2$ for even N and $c(N) = (3N+1)/2$ for odd $N \neq 1$, and for $N \ge 2$ whose orbit reaches 1 let

$$n = n(N) = \min\{j \ge 1 : c^j(N) = 1\}, \qquad a_j = c^j(N)\ (0 \le j \le n), \qquad P_N(z) = \sum_{j=0}^{n} a_j z^j.$$

Hohertz assumes that every orbit reaches 1; all our statements concern N whose orbit reaches 1, and every computation below checks this directly. P_N is monic of degree n with $P_N(0) = N$. Its zeros are the *Collatz zeros* z_N. We write

$$O_N = \{0 \le i < n :\ a_i \text{ odd}\}, \qquad R(N) = |O_N|,$$
$$T_N = \{n+1\} \cup \{n - i :\ i \in O_N\}, \qquad d_N = \gcd T_N.$$

T_N coincides with Hohertz's set $\{n+1\} \cup \{j \in [n] : c^{n-j}(N) \text{ odd}\}$. The number of odd terms of the orbit, including the final 1, is $m(N) = R(N) + 1$ in the notation of [25].

We use the following results of [5]; Theorems I–III rest on the Eneström–Kakeya theorem and its sharpness analysis [28], and Theorem IV on a density argument. Theorem I bounds every

zero by $2M/(3M+1) \le |z_N| \le 2$, where M is the least odd iterate of N other than 1 (for N not a power of 2). (The displayed formal definition in [5] writes a maximum, but the introduction and the proof of Theorem II use the least odd iterate. We use only the upper bound $|z_N| \le 2$ of Theorem I, which also follows from Lemma II.3.9 below.) Theorem III states that $|z_N| = 2$ occurs if and only if $d_N > 1$; such zeros are simple and are exactly the points 2ω with $\omega^{d_N} = 1 \neq \omega$; and

$$P_N(2z) = (1 + z + \cdots + z^{d_N - 1})\, Q_N(z^{d_N}), \tag{II.3.1}$$

where Q_N has positive coefficients and, if nonconstant, all its zeros in $|z| < 1$. Theorem IV states that the set of N with a zero on $|z| = 2$ has natural density 0. From the manuscript we use:

(M1) [21, Thm. 3.1] The set of accumulation points of all Collatz zeros is $\{2/3 \le |z| \le 2\}$.
(M2) [21, Thm. 3.1] If $P_N(\xi) = 0$ and $|\xi| \le r < 2/3$, then $N + 1 \le (1 + 3r/2)/((1 - 3r/2)(1 - r))$.
(M3) [21, Section 3.2 and Cor. 3.2] Every nonempty open subset of $\{2/3 < |z| < 2\}$ contains infinitely many distinct Collatz zeros.

Because (M2) enters a certified computation below, we recall its short proof. With $b_j = (a_j + 1)(2/3)^j/(N + 1)$ one has $b_{j+1} = b_j$ after an odd step and $b_{j+1} < b_j$ after an even step, so $1 = b_0 \ge \cdots \ge b_n > 0$. Put $b_{n+1} = 0$, $\delta_j = b_{j-1} - b_j \ge 0$ and $w = 3z/2$; then $(1 - w)\sum_{j \le n} b_j w^j = 1 - \sum_{j=1}^{n+1} \delta_j w^j$ with $\sum \delta_j = 1$. For $|z| \le r < 2/3$ this gives $|\sum_j (a_j + 1) z^j| \ge (N + 1)(1 - 3r/2)/(1 + 3r/2)$, while $|\sum_{j \le n} z^j| \le 1/(1 - r)$; a zero of P_N forces the stated inequality.

The questions treated here are the following (quoted from [5, Section 5]; mathematical notation transcribed).

Problem II.3.1 (Hohertz, Q2)**.** "Strengthen Theorems III and IV by finding an explicit, closed-form upper bound for $|z_N|$ in terms of N."

Problem II.3.2 (Hohertz, Q3)**.** "Figure 1 suggests that certain subsets of $\{z \in \mathbb{C} : |z| \le 2\}$ contain large clusters of Collatz zeros while others are zero-free. Give an algorithm for proving that a subset of $\{z \in \mathbb{C} : |z| \le 2\}$ is free of Collatz zeros."

Problem II.3.3 (Hohertz, Q5)**.** "Under some modest assumptions, the zeros of a polynomial with random coefficients cluster near the unit circle with uniform angular distribution [...]. Prove or disprove that these assumptions hold for the Collatz polynomials." (The omitted citation is to Hughes and Nikeghbali [26].)

Problem II.3.4 (Hohertz, Q6)**.** "Find the Galois groups of P_N for general classes of P_N."

Problem II.3.5 (Hohertz, Q7)**.** "Expanding on Theorem III, find conditions under which P_N has zeros at real multiples (other than 2) of roots of unity."

Problem II.3.6 (Hohertz, Q8)**.** "Prove theorems on z_N which can be applied without calculating the Collatz sequence of N (e.g., if $N \equiv 3 \bmod 4$, then the zeros of P_N lie in ...)."

Problem II.3.7 (Hohertz–Kalantari)**.** Theorem 7 of [25] states that $P_N(-1) = 0$ when N is odd, N is not a power of 2 and $c(N)$ is a power of 2. The authors show that $N = 820569$ satisfies $P_N(-1) = 0$ although $c(N) = 2 \cdot 615427$; by their Lemma 11 (if N is odd and $P_N(-1) = 0$, then $P_{4N+1}(-1) = 0$) they obtain further such N from this one, and they write that "a full converse of Theorem 7 has proved elusive." The question left open is to characterize the N with $P_N(-1) = 0$.

The cited questions motivate the results below. We do not infer that a question remains open merely from an unsuccessful literature search. In particular, the infinite-family Galois results in the later part of this unified paper supersede the earlier computational-only discussion of Q6.

II.3.2 Elementary identities

Lemma II.3.8 (Sparse form). *For $N \geq 2$,*

$$(z-2)P_N(z) = z^{n+1} - \sum_{i \in O_N} (2a_i + 1)\, z^{i+1} - 2N. \tag{II.3.2}$$

Proof. The coefficient of z^j $(1 \leq j \leq n)$ on the left is $a_{j-1} - 2a_j$, which is 0 if a_{j-1} is even and $a_{j-1} - (3a_{j-1}+1) = -(2a_{j-1}+1)$ if a_{j-1} is odd. The coefficients of z^{n+1} and z^0 are $a_n = 1$ and $-2a_0$. □

A closed form of P_N in terms of the odd iterates is Lemma 10 of [25]; (II.3.2) is the form we need.

Lemma II.3.9 (Generating-function form). *Let $N \geq 2$. Put $q_{n+1} = N/2^n$ and $q_{n-i} = 2^{i-n}(2a_i + 1)$ for $i \in O_N$, and $\varphi_N(u) = \sum_{k \in T_N} q_k u^k$.*

1. *$q_k > 0$ for $k \in T_N$, and $\sum_k q_k = 1$.*
2. *$\min_k q_k = q_{n+1} = N/2^n = \prod_{i \in O_N} (3 + 1/a_i)^{-1} \in [4^{-R(N)}, 3^{-R(N)}]$.*
3. *For $u \neq 0$, $(1-u)\, u^n 2^{-n} P_N(2/u) = 1 - \varphi_N(u)$. Hence $\xi \neq 0$ is a zero of P_N if and only if $u = 2/\xi$ satisfies $\varphi_N(u) = 1$ and $u \neq 1$.*

Proof. (3) Substitute $z = 2/u$ in (II.3.2) and multiply by $u^{n+1}/2^{n+1}$. (1) Positivity is clear, and $u = 1$ in (3) gives $\varphi_N(1) = 1$. (2) Since $2a_{i+1}/a_i \in \{1,\ 3+1/a_i\}$, the numbers $2^i a_i$ do not decrease, so $2^i a_i \geq N$ and $q_{n-i} > 2N/2^n$. Finally $1 = a_n = N \prod_{j<n} a_{j+1}/a_j = N 2^{-n} \prod_{i \in O_N} (3 + 1/a_i)$. □

Lemma II.3.9 also gives a short proof of the first part of Theorem III of [5]. If $|u| < 1$ then $|\varphi_N(u)| < 1$. If $|u| = 1$ and $\varphi_N(u) = 1$, then $u^k = 1$ for all $k \in T_N$, that is, $u^{d_N} = 1$.

Lemma II.3.10 (The gcd d_N). *For odd $M \geq 3$ let $g_M = \gcd(\{n(M)\} \cup O_M)$. Then for $k \geq 0$,*

$$d_{2^k M} = \gcd(k+1,\, g_M), \qquad d_{2^k} = k + 1.$$

In particular $d_N \mid v_2(N) + 1$ for every $N \geq 2$, and $d_N = 1$ for every odd $N \geq 3$.

Proof. For $N = 2^k M$ one has $n(N) = k + n(M)$ and $O_N = k + O_M$. Hence $T_N = \{k + n(M) + 1\} \cup \{n(M) - i : i \in O_M\}$. Since $0 \in O_M$, the gcd of the second set is $\gcd(\{n(M)\} \cup O_M) = g_M$, and $g_M \mid n(M)$. For $N = 2^k$ we have $O_N = \emptyset$ and $T_N = \{k+1\}$. □

For odd N, the conclusion $d_N = 1$ is immediate from the definition, since $n, n+1 \in T_N$. It is not stated in [5].

Lemma II.3.11 (Parity words; from the manuscript). *Let $v = (\varepsilon_0, \ldots, \varepsilon_{L-1}) \in \{0,1\}^L$, $R = \sum \varepsilon_j$, and $b_v = \sum_{\varepsilon_j = 1} 2^j 3^{\sum_{i>j} \varepsilon_i}$.*

1. *Let $\tau(x) = x/2$ for even x and $\tau(x) = (3x+1)/2$ for odd x (no stop at 1). If the parities of $x, \tau(x), \ldots, \tau^{L-1}(x)$ are v, then $\tau^L(x) = (3^R x + b_v)/2^L$. Hence each word occurs for at most one residue class of x modulo 2^L.*
2. *If $3^R x + b_v \equiv 0 \pmod{2^L}$, then the first L parities of x are v.*
3. *If $R \geq 1$, the $t \geq 0$ with $2^{L+t} \equiv b_v \pmod{3^R}$ form one residue class modulo $2 \cdot 3^{R-1}$. For every such $t \geq L+1$, the integer $N_t = (2^{L+t} - b_v)/3^R$ has orbit v followed by t halvings $2^t \to \cdots \to 1$, and $n(N_t) = L + t$.*

Proof. (1) is induction on L. (2) Reducing modulo 2 gives $x \equiv \varepsilon_0$. After the first step, $3^{R-\varepsilon_0}\tau(x) + b_{v'} = (3^R x + b_v)/2$ for the shortened word v'; then induct. (3) The last odd letter contributes 2^j to b_v and all others are divisible by 3, so $3 \nmid b_v$. The class of 2 generates $(\mathbb{Z}/3^R)^\times$, which has order $2 \cdot 3^{R-1}$. Here $N_t > 0$, because $b_v/2^L \le (3/2)^R - 1 < 2^t$ for $t \ge L+1$. By (2), the first L parities of N_t are v and $\tau^L(N_t) = 2^t$. Since $\tau(y) \le 2y$, we get $a_i \ge 2^{t-L+i} > 1$ for $i \le L$ when $t \ge L+1$, so the orbit does not stop early. □

This is the construction of [21, Section 3.1], with the explicit condition $t \ge L+1$. Part (1) is the classical fact that the first L parities depend only on $x \bmod 2^L$ and determine it [29, 30].

II.3.3 Q2: an explicit upper bound, and its sharpness

Lemma II.3.12. *Let $S \subseteq \{1, \dots, M\}$ with $M \in S$ and $\gcd S = 1$, and let $x \in \mathbb{R}$ satisfy $\|x\| \ge 1/(2M)$, where $\|\cdot\|$ is the distance to $\mathbb{Z}$. Then $\|kx\| \ge 1/(2M)$ for some $k \in S$.*

Proof. Suppose instead that $kx = m_k + \eta_k$ with $m_k \in \mathbb{Z}$ and $|\eta_k| < 1/(2M)$ for all $k \in S$. For $k, l \in S$, the integer $lm_k - km_l = k\eta_l - l\eta_k$ has absolute value $< (k+l)/(2M) \le 1$, so it vanishes. Thus $y = m_k/k$ does not depend on k, and $ky \in \mathbb{Z}$ for all $k \in S$. As $\gcd S = 1$, we get $y \in \mathbb{Z}$. Taking $k = M$ gives $|x - y| < 1/(2M^2)$, so $\|x\| < 1/(2M)$, a contradiction. □

Theorem II.3.13 (Explicit upper bound)**.** *Let $N \ge 2$, $n = n(N)$, $d = d_N$ and $M = n+1$. Every zero ξ of P_N with $|\xi| \ne 2$ satisfies*

$$|\xi| \;\le\; 2\Big(1 + \frac{2d^2N}{M^2\,2^n}\Big)^{-1/M} \;\le\; 2 - \frac{2d^2N}{3M^3\,2^n}. \tag{II.3.3}$$

In particular, if $d_N = 1$ (for instance if N is odd), then every zero satisfies $|\xi| \le 2 - \frac{2N}{3(n+1)^3 2^n} \le 2 - \frac{2}{3(n+1)^3\,4^{R(N)}}$.

Proof. Let $u = 2/\xi$. By Lemma II.3.9, $\varphi_N(u) = 1$ and $u \ne 1$. All exponents in T_N are multiples of d, so $\varphi_N(u) = \psi(u^d)$ with $\psi(v) = \sum_{k \in S'} q_{dk} v^k$, where $S' = T_N/d \subseteq [1, M']$, $M' = M/d \in S'$ and $\gcd S' = 1$. Put $v = u^d$. If $v = 1$, then $\xi = 2\bar{u}$ with $\bar{u}^d = 1$, so $|\xi| = 2$; this case is excluded. If $M' = 1$ then $\psi(v) = v$, which forces $v = 1$. So $M' \ge 2$, and S' contains an element smaller than M'.

Write $v = \rho e^{2\pi i x}$ with $x \in (-\frac{1}{2}, \frac{1}{2}]$. If $\rho < 1$ then $|\psi(v)| < 1$, so $\rho \ge 1$. If $x = 0$ then $v = \rho > 1$ and $\psi(\rho) > 1$, so $x \ne 0$. Suppose $0 < x \le 1/(2M')$. Then every angle $2\pi kx$ $(k \le M')$ lies in $(0, \pi]$, and the angle lies in $(0, \pi)$ for the element $k < M'$ of S'. Hence $\operatorname{Im} \psi(v) > 0$, which is impossible. The case $-1/(2M') \le x < 0$ is symmetric. Therefore $\|x\| = |x| > 1/(2M')$. By Lemma II.3.12 some $k^* \in S'$ has $\|k^* x\| \ge 1/(2M')$.

Take the real part of $\psi(v) = 1$:

$$\sum_{k \in S'} q_{dk} \rho^k (1 - \cos 2\pi kx) = \psi(\rho) - 1 = \sum_{k} q_{dk}(\rho^k - 1) \le \rho^{M'} - 1.$$

Since $\rho \ge 1$, the left side is at least $q_{dk^*}(1 - \cos(\pi/M')) \ge q_{n+1} \cdot 2/M'^2$, using $1 - \cos t \ge 2t^2/\pi^2$ on $[0, \pi]$ and Lemma II.3.9(2). Hence $|u|^M = \rho^{M'} \ge 1 + 2d^2N/(M^2 2^n) =: 1 + y$, which is the first inequality.

For the second, note that $y \le 2$ and $s := \ln(1+y)/M \le \ln 3/2 < 1$. So $(1+y)^{-1/M} = e^{-s} \le 1 - s/2$. With $\ln(1+y) \ge y/(1+y) \ge y/3$ this gives $2(1+y)^{-1/M} \le 2 - y/(3M)$. The last claim uses Lemma II.3.10 and Lemma II.3.9(2). □

When $d_N > 1$, the zeros of P_N off $|z| = 2$ are the points $2z$ with $Q_N(z^{d_N}) = 0$, by (II.3.1). Theorem II.3.13 therefore also gives an explicit form of the statement in [5, Thm. III] that these zeros lie inside the disk. The bound depends on N and on the stopping time $n(N)$ only. The next result shows that the factor $N2^{-n}$ cannot be dropped.

Proposition II.3.14 (Zeros close to -2)**.** *Let $N \geq 3$ be odd with $n = n(N)$ even and $O_N \subseteq 2\mathbb{Z}$. Let $h \geq 1$ be defined by $a_{n-h} = 2^h$ and a_{n-h-1} odd, and put $p_0 = N/2^n$. Then $d_N = 1$ and $P_N(-2) = N$. If moreover $p_0 \leq 1/18$ and $12p_0(n-h) \leq h+1$, then P_N has a real zero in $(-2,\, -2 + 6p_0/(h+1))$. Hence $2 - \max|\xi| < 6N/((h+1)2^n)$.*

Proof. By Lemma II.3.10, $d_N = 1$. Put $F(w) = P_N(2w)$ and $\pi_i = (2a_i+1)2^{i-n}$, so that $\sum_{i\in O_N} \pi_i = 1 - p_0$. Then (II.3.2) reads $(w-1)F(w) = 2^n(w^{n+1} - \sum_{i\in O_N} \pi_i w^{i+1} - p_0)$. For $w = -s$ with $0 < s \leq 1$, the exponents $n+1$ and $i+1$ ($i \in O_N$) are odd. Hence $F(-s) = 2^n g(s)/(1+s)$ with

$$g(s) = s^{n+1} + p_0 - \sum_{i\in O_N} \pi_i s^{i+1}.$$

At $s = 1$ this gives $F(-1) = 2^n p_0 = N$. Every $i \in O_N$ satisfies $i \leq n-h-1$, so for $0 < s < 1$

$$g(s) \leq p_0 - s^{n-h}(1 - p_0 - s^{h+1}).$$

Let $s = 1-\eta$ with $\eta = 3p_0/(h+1)$. Then $s^{h+1} \leq e^{-3p_0} \leq 1 - 3p_0 + \frac{9}{2}p_0^2$, so $1 - p_0 - s^{h+1} \geq p_0(2 - \frac{9}{2}p_0) \geq \frac{7}{4}p_0$. Also $s^{n-h} \geq 1 - (n-h)\eta \geq \frac{3}{4}$. Hence $g(1-\eta) \leq p_0 - \frac{21}{16}p_0 < 0$. So $F(-(1-\eta)) < 0 < F(-1)$, and F has a zero in $(-1, -1+\eta)$. □

Corollary II.3.15 (An explicit family)**.** *For $m \geq 3$ and $s \geq 1$ put $t = 2\cdot 3^{m-1}s$ and $N_{m,s} = 1 + 4^m(2^t - 1)/3^m$. Then $N_{m,s}$ is an odd integer. Its orbit has parity word $(10)^m$ followed by t halvings $2^t \to \cdots \to 1$, and $n(N_{m,s}) = 2m + t$, $d_{N_{m,s}} = 1$. Moreover*

$$\frac{3^{-m}}{3(n+1)^3} \;<\; 2 - \max_{P_{N_{m,s}}(\xi)=0} |\xi| \;<\; \frac{9^{1-m}}{s}, \qquad \text{and for } s = 1: \quad 2 - \max|\xi| < \frac{9}{(\log_2 N_{m,1})^2}.$$

Proof. For $v = (10)^m$ we have $L = 2m$, $R = m$ and $b_v = \sum_{i<m} 4^i 3^{m-1-i} = 4^m - 3^m$. The congruence $2^{2m+t} \equiv b_v$ modulo 3^m is equivalent to $2^t \equiv 1 \pmod{3^m}$, that is, to $2\cdot 3^{m-1} \mid t$. Since $t \geq 2m+1$, Lemma II.3.11(3) applies and gives $N_{m,s} = (2^{2m+t} - b_v)/3^m = 1 + 4^m(2^t-1)/3^m$, which is odd. Here $O_N = \{0, 2, \ldots, 2m-2\}$, $n = 2m+t$ is even, $h = t+1$ and $n-h = 2m-1$. Also $p_0 = 3^{-m}(1-2^{-t}) + 4^{-m}2^{-t} \in (3^{-m}/2,\, 3^{-m})$, so $p_0 \leq 1/18$ and $12p_0(2m-1) \leq 12\cdot 5/27 < t+2$. Proposition II.3.14 gives $2 - \max|\xi| < 6p_0/(t+2) < 6\cdot 3^{-m}/t \leq 9^{1-m}/s$. The lower bound is Theorem II.3.13. For $s = 1$ we have $\log_2 N_{m,1} < n = 2m + 2\cdot 3^{m-1} \leq 3^m$, so $9^{1-m} = 9/(3^m)^2 < 9/(\log_2 N)^2$. □

Remark II.3.16 (What this says about Q2). Theorem II.3.13 is an explicit closed-form bound in terms of N and $n(N)$. Under Theorem III it strengthens "< 2" (for $d_N = 1$) and "< 2 off the points 2ω" (for $d_N > 1$). Corollary II.3.15 shows three things.

1. No bound $|z_N| \leq 2 - c$ with a constant $c > 0$ holds, even for odd N.
2. Any bound $2 - f(N)$ valid for all N with $d_N = 1$ needs $f(N_{m,1}) < 9/(\log_2 N_{m,1})^2$.
3. The factor $N2^{-n}$ in (II.3.3) is optimal up to the factor $(n+1)^3/(h+1)$.

Obtaining a bound in N alone from this particular estimate would require controlling the stopping time. This observation is not a proof that every possible approach to Q2 requires such a bound. The theorem above does not resolve Q2 in its original input-only form.

II.3.4 Q3: deciding zero-freeness

Let $A = \{2/3 \le |z| \le 2\}$, and let A° be its interior. We call a set $K \subseteq \mathbb{C}$ *semialgebraic over* $\mathbb{Q}$ if it is a finite Boolean combination of sets $\{p(\operatorname{Re} z, \operatorname{Im} z) \ge 0\}$ with $p \in \mathbb{Q}[x, y]$. Examples are finite unions of closed disks and polygons with rational data.

Theorem II.3.17 (Decision procedure)**.** *Let $K \subseteq \{|z| \le 2\}$ be compact.*

1. *If $K \cap A^\circ$ has nonempty interior, then K contains infinitely many Collatz zeros.*
2. *Suppose $K \subseteq \{|z| \le r\} \cup \{|z| = 2\}$ for some $r < 2/3$. A point $\xi \in K$ with $|\xi| \le r$ is a Collatz zero only if $P_N(\xi) = 0$ for some N with $N + 1 \le (1 + 3r/2)/((1 - 3r/2)(1 - r))$. A point $\xi \in K$ with $|\xi| = 2$ is a Collatz zero if and only if $\xi = 2\omega$ with $\omega \ne 1$ a root of unity.*
3. *For K semialgebraic over $\mathbb{Q}$ there is an algorithm that decides whether (1) or (2) applies, and, if one of them applies, whether K is free of Collatz zeros. In case (2) the decision needs the orbits of the finitely many $N \le B(r) := (1 + 3r/2)/((1 - 3r/2)(1 - r)) - 1$ to reach 1; this holds unconditionally whenever $B(r)$ lies below the range in which the Collatz conjecture has been verified. For every other such K there is a procedure that halts exactly when K contains a Collatz zero, and then exhibits one; it needs no assumption on the Collatz conjecture, since the orbit computations for all N can be run in parallel (dovetailing).*

Proof. (1) is (M3). The first part of (2) is (M2). For the circle, Theorem III of [5] shows that every zero of modulus 2 has the form 2ω. Conversely, if ω has order $q \ge 2$, then $N = 2^{q-1}$ gives $P_N(z) = (z^q - 2^q)/(z - 2)$, which vanishes at 2ω.

(3) Each of the following is a first-order sentence over the real field with rational parameters, and so is decidable by Tarski's theorem (see [34]):

- "P_N has a zero in K";
- "$K \cap A^\circ$ has an interior point";
- "$K \subseteq \{|z| \le r\} \cup \{|z| = 2\}$", for a given rational r.

Whether some $r < 2/3$ works is again a first-order sentence ("$\exists \rho < 2/3\ \forall z \in K:\ |z| \le \rho\ \vee\ |z| = 2$"); if it holds, a rational r is obtained from the real algebraic number $\sup\{|z| : z \in K,\ |z| < 2\} < 2/3$. In case (2), finitely many orbits are computed (under the stated hypothesis) and finitely many sentences decided. The set $K \cap \{|z| = 2\}$ is a finite union of arcs and points. An arc of positive length contains some 2ω, and for each of the finitely many isolated points one decides whether it is twice a root of unity. An algebraic number of degree D can only be a root of unity of order q with $\varphi(q) \le D$, so this is a finite check. The semi-decision procedure runs through $N = 2, 3, \ldots$. □

Theorem II.3.17 becomes almost immediate once (M2) and (M3) are available. In practice the test "P_N has a zero in K" is done by certified root isolation. This is valid whenever no zero lies on the boundary of K, and otherwise exact algebraic methods are used. The following instance is certified.

Remark II.3.18 (Source-reported disk computation). The finite computation below is reported by the source manuscript. Its root-isolation certificates are not included here; the present compilation does not independently certify the stronger cutoff 0.666. Subject to that reported computation, The only Collatz zero in the closed disk $|z| \le 0.666$ is the simple real zero $\xi_9 \in [-0.65698976775632, -0.65698976775630]$ of

$$P_9(z) = 9 + 14z + 7z^2 + 11z^3 + 17z^4 + 26z^5 + 13z^6 + 20z^7 + 10z^8 + 5z^9 + 8z^{10} + 4z^{11} + 2z^{12} + z^{13}.$$

Consequently $|z_N| > 0.666$ for every zero of every P_N with $N \ne 9$, and by (M1) and the local finiteness in (M2), $0.666 < \inf_{N \ne 9} \min_{P_N(\xi)=0} |\xi| \le 2/3$.

Reported method, not an included certificate. For $r = 333/500$, the bound in (M2) is $999500/167 = 5985.03\ldots$, so only $N \le 5984$ can contribute. For each $2 \le N \le 5984$, all complex zeros of P_N were isolated in disjoint certified balls (Arb ball arithmetic [38] through python-flint). Each ball was classified rigorously as lying inside or outside $|z| \le r$; no ball met the circle $|z| = r$. The only ball inside is the stated one, for $N = 9$. The original report describes certified Arb ball isolation. □

The source-reported result would give a lower bound uniform in $N \ne 9$. The manuscript's R_N exceeds 0.666 only for $N \ge 5985$ (this is how the cutoff arises), and Hohertz's $2M/(3M+1)$ exceeds 0.666 only when the least odd iterate M exceeds 333. These numerical claims are not used as hypotheses of the analytic results below.

Remark II.3.19 (What remains of Q3). The undecided sets are the compact K that meet $\{2/3 \le |z| < 2\}$ but whose intersection with A° has empty interior. Examples are circles, segments, and the circle $|z| = 2/3$. For circles $|z| = \sqrt{q}$ with $q \in \mathbb{Q}_{>0}$, Theorem II.3.28(2) below excludes every N whose P_N is irreducible with a 2-transitive Galois group; for $q \ne 1$ the same argument applied to $P_N/(z+1)$ covers the N with $P_N(-1) = 0$ (such as 820569 and 6094358), while the unit circle contains -1. So, under Conjecture II.3.24, only the structured N of Proposition II.3.22 and the N with $P_N(-1) = 0$ need to be examined. The unit circle does contain Collatz zeros, for example -1 for $N = 5$.

II.3.5 Q5: the Hughes–Nikeghbali hypotheses

For $F(z) = \sum_{j=0}^{D} f_j z^j$ with $f_0 f_D \ne 0$ put $\mathcal{L}(F) = \log \sum_j |f_j| - \frac{1}{2}\log|f_0| - \frac{1}{2}\log|f_D|$. The deterministic theorem of Hughes and Nikeghbali [26, Thm. 1 and Thm. 3] is as follows. Let (F_k) be polynomials of degrees $D_k \to \infty$ with $\mathcal{L}(F_k) = o(D_k)$. Then, for every $0 < \rho < 1$, the proportion of zeros in $1-\rho \le |z| \le (1-\rho)^{-1}$ tends to 1; quantitatively, $1 - \nu/D \le 2\mathcal{L}(F)/(D\rho)$. Moreover the arguments of the zeros are equidistributed, by the Erdős–Turán inequality [27] $|\frac{1}{D}\#\{\theta \le \arg z < \phi\} - \frac{\phi-\theta}{2\pi}|^2 \le C\mathcal{L}(F)/D$, with an absolute constant C. Their random version [26, Thm. 5] has the same quantity in expectation. For Collatz polynomials,

$$\mathcal{L}(P_N) = \log P_N(1) - \tfrac{1}{2}\log N.$$

Let $A_N = \max_j a_j$ be the largest orbit value.

Theorem II.3.20.

1. *For every $N \ge 2$: $\frac{1}{2}\log A_N \le \mathcal{L}(P_N) \le \log A_N + \log(n+1) - \frac{1}{2}\log N$.*
2. *For every $N \ge 2$, P_N has at least $\lceil \log_2 N \rceil$ zeros (with multiplicity) in $1 < |z| \le 2$. For $0 < \rho < 1$, the number of zeros outside $1-\rho \le |z| \le (1-\rho)^{-1}$ is at least $(\log N + n\log(1-\rho))/\log 2$.*
3. *Let (N_k) satisfy $n(N_k) \to \infty$. The Hughes–Nikeghbali hypothesis $\mathcal{L}(P_{N_k}) = o(n(N_k))$ holds if and only if $\log A_{N_k} = o(n(N_k))$. Suppose instead that $\kappa := \liminf \log N_k / n(N_k) > 0$; this holds for $N_k = 2^k$, for $N_k = 2^k M$ with M fixed, and for every sequence with $n(N_k) = O(\log N_k)$. Then the hypothesis fails, and so does the radial conclusion: for $0 < \rho < 1 - e^{-\kappa}$ the proportion of zeros in $1-\rho \le |z| \le (1-\rho)^{-1}$ is eventually at most $1 - (\kappa - \log\frac{1}{1-\rho})/\log 2 + o(1) < 1$.*
4. *The hypothesis holds along some sequence if and only if $\sup_N n(N)/\log A_N = \infty$. This would imply $\sup_N n(N)/\log N = \infty$, contrary to the conjecture that $\limsup n(N)/\log N = 41.677647\ldots$ [32, Conj. 4.1], [31]. (Infinitely many N with $n(N) > 6.14316\log N$ are known [33].)*
5. *(Rescaling.) Let $\rho_N = N^{1/n}$ and $E(N) = \max_{0\le j\le n}(\log a_j - (1-j/n)\log N) \ge 0$. Then $E(N) \le \mathcal{L}(P_N(\rho_N\,\cdot)) \le E(N) + \log(n+1)$. For every N, the angular discrepancy of the zeros of P_N is at most $(C(E(N)+\log(n+1))/n)^{1/2}$. Along any sequence with $E(N_k) = o(n(N_k))$, the zeros of P_{N_k} cluster at $|z| = \rho_{N_k}$ (in the sense of [26]) with uniform angular distribution.*

For fixed odd M, $E(2^kM)$ is bounded, so along $N = 2^kM$, $k \to \infty$, the zeros cluster uniformly at radius $N^{1/n} \to 2$.

Proof. (1) We have $A_N \le P_N(1) \le (n+1)A_N$ and $N \le A_N$.

(2) P_N is monic with $P_N(0) = N$, so $\prod |z_i| = N$; and $|z_i| \le 2$ (Theorem I, or Lemma II.3.9). Zeros with $|z| \le 1$ contribute $\log|z_i| \le 0$, which gives the first claim. Splitting the zeros into those inside, in, and outside the annulus gives $\log N \le \#\{\text{outside}\}\log 2 + n\log\frac{1}{1-\rho}$.

(3) The equivalence follows from (1), since $\log N \le \log A_N$ and $\log(n+1) = o(n)$. If $\kappa > 0$ then $\mathcal{L} \ge \frac{1}{2}\log N \ge (\kappa/2 + o(1))n$. The radial statement follows from (2). For $N = 2^kM$ we have $n = k + n(M)$ and $\log N/n \to \log 2$.

(4) $\sup_N n(N)/\log A_N = \infty$ forces $n \to \infty$ along a suitable sequence, because n bounded implies $N \le 2^n$ bounded. Then use (3) and $A_N \ge N$.

(5) $\mathcal{L}(P_N(\rho_N\cdot)) = \log\sum_j a_j\rho_N^j - \log N$ and $\log(a_j\rho_N^j/N) = \log a_j - (1-j/n)\log N$; the term $j = 0$ is 0. The angular counts are invariant under $z \mapsto \rho_N z$, so the Erdős–Turán inequality applied to $P_N(\rho_N\cdot)$ gives the discrepancy bound, and [26, Thm. 1] gives the clustering. For $N = 2^kM$ with $j \le k$ the quantity is $j(\log M - n(M)\log 2)/(k+n(M)) \le 0$, because $n(M) \ge \log_2 M$. For $j = k + j'$ it is $\log c^{j'}(M) - \frac{n(M)-j'}{k+n(M)}(k\log 2 + \log M)$, which is bounded in terms of M. □

Remark II.3.21 (Answer to Q5). As stated, with clustering *near the unit circle*, the assumptions fail for the Collatz polynomials: for $N = 2^k$, along every sequence with $n(N) = O(\log N)$, and, under the stochastic-model conjecture cited in (4), along every sequence. The obstruction is the product formula $\prod|z_i| = N$ together with $|z_i| \le 2$: at least $\log_2 N$ zeros lie outside the unit circle. The correct normalization is the radius $N^{1/n(N)}$. After rescaling, the hypothesis becomes $E(N) = o(n)$: the orbit, read on a logarithmic scale, must not rise more than $o(n)$ above the chord from $\log N$ to 0. This holds along explicit families, and the conclusion then holds. Whether it holds for almost all N is out of reach. It would require control of the whole orbit of almost every N, which is stronger than the known almost-all results on the $3x+1$ problem.

II.3.6 Q6: Galois groups

Proposition II.3.22 (Structure)**.**

1. *For $N = 2^k$: $P_N(z) = (z^{k+1} - 2^{k+1})/(z-2) = \prod_{e|k+1,\,e>1} 2^{\varphi(e)}\Phi_e(z/2)$. The splitting field is $\mathbb{Q}(\zeta_{k+1})$, $\mathrm{Gal}(P_N) \cong (\mathbb{Z}/(k+1)\mathbb{Z})^\times$, and P_N is irreducible if and only if $k+1$ is prime.*
2. *Let $d = d_N > 1$ with N not a power of 2, and $r = (n+1)/d - 1 \ge 1$. Then*

$$P_N(z) = \frac{z^d - 2^d}{z-2}\cdot F_N(z^d), \qquad F_N(w) = 2^{-(d-1)}Q_N(w/2^d) \in \mathbb{Z}[w] \text{ monic of degree } r.$$

   The splitting field of P_N is that of $F_N(z^d)$, and it contains ζ_d. Suppose F_N is squarefree (this holds in all computed cases). Fix its roots $\beta_1, \dots, \beta_r$ and δ_i with $\delta_i^d = \beta_i$. Every σ in the Galois group acts by $\zeta_d \mapsto \zeta_d^u$ and $\delta_i \mapsto \zeta_d^{c_i}\delta_{\pi(i)}$. This embeds $\mathrm{Gal}(P_N)$ into $W_{d,r} = (\mathbb{Z}/d)^r \rtimes ((\mathbb{Z}/d)^\times \times S_r)$, of order $d^r\varphi(d)r!$, acting imprimitively on the roots of $F_N(z^d)$ with r blocks of size d.
3. *For $N = x_m := (4^m - 1)/3$ ($m \ge 2$): $(z-2)P_N(z) = z^{2m+1} - (2N+1)z - 2N$ and $P_N(z) = (z+1)H_m(z)$.*
4. *For $N = 2^{2m-1}x_m$ ($m \ge 2$): $d_N = 2m$ and $P_N(z) = (z^{2m} + x_m)\,\dfrac{z^{2m} - 4^m}{z-2}$.*

Proof. (1) is classical. (2) Substitute $z \mapsto z/2$ in (II.3.1). The quotient of the monic integer polynomial P_N by the monic factor $(z^d - 2^d)/(z-2)$ is monic with integer coefficients. The

ratios of the roots of $z^d = \beta_i$ include a primitive d-th root of unity, so ζ_d lies in the splitting field (note $F_N(0) \neq 0$). The map $\sigma \mapsto (u, (c_i), \pi)$ is an injective homomorphism into the group of maps $\zeta_d^c \delta_i \mapsto \zeta_d^{uc+c_i} \delta_{\pi(i)}$, which is $W_{d,r}$.

(3) For $N = x_m$ the orbit is $x_m, 2^{2m-1}, \dots, 1$, so $O_N = \{0\}$ and $n = 2m$; apply (II.3.2). The right-hand side vanishes at $z = -1$ because $-1 + (2N+1) - 2N = 0$.

(4) The orbit is $2^{2m-1}x_m, \dots, x_m, 2^{2m-1}, \dots, 1$, so $n = 4m-1$ and $O_N = \{2m-1\}$. By Lemma II.3.10, $g_{x_m} = 2m$ and $d_N = \gcd(2m, 2m) = 2m$. The numbers $b_j = 2^j a_j$ equal N for $j < 2m$ and 2^n for $j \geq 2m$. So $P_N(2z) = (1 + \cdots + z^{2m-1})(N + 2^n z^{2m})$, which is the stated formula after $z \mapsto z/2$. □

Remark II.3.23 (Source-reported exact computations). The following ranges and group identifications are retained as reports from the source manuscript. Their per-polynomial certificates are not embedded here or independently rerun in this compilation. Any conclusion depending on these finite ranges is conditional on the reported data.

1. For every $2 \leq N \leq 10^4$ with $d_N = 1$, the polynomial P_N is irreducible over $\mathbb{Q}$ with Galois group S_n, except for $N \in \{5, 21, 85, 341, 1365, 5461\} = \{x_m : 2 \leq m \leq 7\}$. For these exceptions, $P_N = (z+1)H_m$ with H_m irreducible and Galois group S_{n-1}.
2. For $2 \leq m \leq 40$, H_m is irreducible with Galois group S_{2m-1}.
3. For every $N \leq 2 \cdot 10^4$ with $d_N > 1$ that is not a power of 2, the factor $F_N(z^d)$ is irreducible. For the 26 values $N \leq 10^4$ with $d_N > 1$ that are not powers of 2, Q_N is irreducible with Galois group S_r, except $N = 170$, where the group is D_4. Among them, the 13 values with $dr \leq 11$ have $|\operatorname{Gal}(F_N(z^d))| = |W_{d,r}|$, except $N = 170$, where the order is $128 = |W_{2,4}|/3$.

Reported method; not an included computational certificate. Factorization over $\mathbb{Z}$ and over $\mathbb{F}_p$ is exact (FLINT [39] through python-flint). The symmetric group was certified by Dedekind's theorem [35]: for a monic $f \in \mathbb{Z}[x]$ squarefree modulo p, the degrees of the irreducible factors of f mod p are the cycle lengths of an element of $\operatorname{Gal}(f)$. For each f we found one prime with pattern $(D-1)+1$, giving a $(D-1)$-cycle, and one prime with exactly one factor of degree 2 and all others of odd degree, giving after an odd power a transposition. A transitive group containing a $(D-1)$-cycle is 2-transitive, hence primitive. A primitive group containing a transposition is symmetric [36]. These certificates cover 9960 values of N and $m \leq 40$. The groups in (3) are from PARI/GP's `polgalois` [37], which is exact for degree ≤ 11. □

Conjecture II.3.24.

1. *If $d_N = 1$, then $P_N/(z+1)^\epsilon$ is irreducible with full symmetric Galois group, where $\epsilon = 1$ if $P_N(-1) = 0$ and $\epsilon = 0$ otherwise.*
2. *If $d_N > 1$ and N is not a power of 2, then $F_N(z^{d_N})$ is irreducible.*

Theorem II.3.23 together with Proposition II.3.22 determines the Galois group for all $N \leq 10^4$ with $d_N = 1$, for all powers of 2, and up to the subgroup question inside $W_{d,r}$ for $d_N > 1$. The case $N = 170$ shows that $W_{d,r}$ is not always attained. The computation does not prove Conjecture II.3.24(1), even for the family (H_m). The later Galois part of this unified paper proves exact groups for another infinite family with $d_N = 4$, hence gives an additional partial answer to Q6 without claiming this conjecture. Modulo 2, the trinomial in (3) reduces to $z(z^m+1)^2$.

II.3.7 Q7: zeros at real multiples of roots of unity

We read Q7 as asking for zeros of the form $\xi = c\,\omega$ with $c > 0$, $c \neq 2$ and ω a root of unity. By Descartes' rule $\omega \neq 1$. The case $\omega = -1$ concerns negative real zeros, and the case $c = 1$ concerns cyclotomic factors.

Proposition II.3.25 (Sufficient conditions)**.**

1. *If $n(N)$ is odd and d_N is odd (for instance N odd with $n(N)$ odd), then P_N has a negative real zero $-c$ with $0 < c < 2$.*
2. *Let $d = d_N > 1$. If Q_N in (II.3.1) has a negative real zero β (for instance if $r = (n+1)/d - 1$ is odd), then P_N has the d zeros $2|\beta|^{1/d}\eta$ with $\eta^d = -1$. These are real multiples, with factor $2|\beta|^{1/d} \in (0, 2)$, of $2d$-th roots of unity.*
3. *For $N = 2^{2m-1}x_m$ ($m \geq 2$), every zero of P_N is a real multiple of a root of unity: the $2m-1$ zeros 2ω with $\omega^{2m} = 1 \neq \omega$, and the $2m$ zeros $x_m^{1/(2m)}\eta$ with $\eta^{2m} = -1$. For $m = 2$ this is $N = 40$, with $P_{40} = (z^4 + 5)(z^3 + 2z^2 + 4z + 8)$.*
4. *$P_N(-1) = 0$ for $N = x_m$ ([25, Thm. 7]), for $N = 820569$, and for the other values listed in Theorem II.3.30(5).*

Proof. (1) A real polynomial of odd degree has a real zero, which is negative by Descartes' rule. It is not -2, because $-2 = 2\omega$ needs $\omega^{d_N} = 1$ with $\omega = -1$, so d_N even. Hence its modulus is < 2. (2) By (II.3.1), zeros z of $Q_N(z^d)$ give zeros $2z$ of P_N. The zeros of Q_N satisfy $|\beta| < 1$, and a real polynomial of odd degree r with positive coefficients has a negative real zero. (3) is Proposition II.3.22(4). (4) is a direct evaluation. □

Lemma II.3.26 (Equimodular zeros)**.** *If all zeros of P_N ($N \geq 2$) have the same modulus, then N is a power of 2.*

Proof. Let ρ be the common modulus. The zeros of the real polynomial P_N are closed under $\beta \mapsto \bar{\beta} = \rho^2/\beta$, so $z^n P_N(\rho^2/z) = \prod_i(\rho^2 - z\beta_i) = \prod_i(-\beta_i)\prod_i(z - \bar{\beta}_i)$. This gives $z^n P_N(\rho^2/z) = P_N(0)P_N(z)$, that is, $a_k\rho^{2k} = Na_{n-k}$ for all k. The case $k = n$ gives $\rho^n = N$. If $n = 1$ then $N = 2$. If $n \geq 2$, the orbit ends with $4, 2, 1$; the cases $k = 1, 2$ give $a_1\rho^2 = 2N$ and $a_2\rho^4 = 4N$, hence $Na_2 = a_1^2$. For odd $N \geq 3$ this reads $4Na_2 = (3N + 1)^2$, which is impossible because $\gcd(N, 3N + 1) = 1$. For even N, $a_1 = N/2$ gives $\rho = 2$, so $N = \rho^n = 2^n$. □

Lemma II.3.27 (2-transitivity)**.** *Let $R \in \mathbb{Q}[z]$ be irreducible of degree $D \geq 3$, with Galois group G acting 2-transitively on its roots, and let α be a nonreal root.*

1. *If $\alpha/\bar{\alpha}$ is a root of unity, then all roots of R have the same modulus.*
2. *$|\alpha|^2 \notin \mathbb{Q}$.*

Proof. Complex conjugation induces an element of G mapping α to $\bar{\alpha} \neq \alpha$. By 2-transitivity, for any two distinct roots β, γ there is $\sigma \in G$ with $\sigma(\alpha) = \beta$ and $\sigma(\bar{\alpha}) = \gamma$. In (1) this gives that β/γ is a root of unity, so $|\beta| = |\gamma|$. In (2), if $|\alpha|^2 = \alpha\bar{\alpha} \in \mathbb{Q}$, then $\beta\gamma = \alpha\bar{\alpha}$ for all distinct roots β, γ. Three distinct roots then give $\beta_1\beta_2 = \beta_1\beta_3$, so $\beta_2 = \beta_3$ (note $R(0) \neq 0$), a contradiction. □

Theorem II.3.28 (Galois obstruction)**.** *Let $N \geq 3$ with $N \neq 4$, so that $n(N) \geq 3$. Suppose P_N is irreducible with 2-transitive Galois group (for example S_n). Then:*

1. *the zeros of P_N at real multiples of roots of unity are exactly its negative real zeros;*
2. *no zero of P_N lies on a circle $|z| = \sqrt{q}$ with $q \in \mathbb{Q}_{>0}$.*

For $N = x_m$ ($m \geq 2$), if H_m has 2-transitive Galois group, the same conclusions hold for the zeros of H_m; the zeros of P_{x_m} at real multiples of roots of unity are then -1 and the negative real zeros of H_m.

Proof. (1) Let $\alpha = c\omega$ be a zero with $\omega \notin \mathbb{R}$. Then $\alpha/\bar{\alpha} = \omega^2$, so by Lemma II.3.27 all zeros are equimodular, and by Lemma II.3.26 $N = 2^k$. Then Proposition II.3.22(1) makes $\mathrm{Gal}(P_N)$ a group of order n acting regularly on $n \geq 3$ roots, which is not 2-transitive.

(2) Nonreal zeros are excluded by Lemma II.3.27(2). A real zero $\pm\sqrt{q}$ would have degree $\leq 2 < n$.

For H_m (monic of degree $D = 2m-1$, $H_m(0) = x_m$), suppose all zeros of H_m have modulus ρ. The same reflection identity gives $h_k\rho^{2k} = x_m h_{D-k}$ for the coefficients h_k of H_m. Comparing coefficients in $P_{x_m} = (z+1)H_m$ gives $h_{D-1} = 1$ and $h_1 = a_1 - x_m = (2^{2m-1}+1)/3$. So $\rho^2 = x_m/h_1 < 2$ while $\rho^D = x_m$. This forces $x_m < 2^{m-1/2}$, which is false for $m \geq 2$. The rest is as in (1) and (2). □

Corollary II.3.29 (Conditional finite-range classification for $N \leq 10^4$)**.** *Assume the finite group and resultant computations reported below are correct. Let $2 \leq N \leq 10^4$. The zeros $c\omega$ of P_N ($c > 0$, $c \neq 2$, ω a root of unity) are as follows.*

1. *If $d_N = 1$: exactly the negative real zeros of P_N. Among them -1 occurs exactly for $N \in \{5, 21, 85, 341, 1365, 5461\}$.*
2. *If $d_N > 1$ and N is not a power of 2: exactly the zeros $2|\beta|^{1/d}\eta$ of Proposition II.3.25(2). They exist exactly for*

$$N \in \{40, 42, 226, 672, 682, 906, 2730, 3616, 3624, 3626\}.$$

3. *If N is a power of 2: none.*

Proof. (1) Combine Theorem II.3.23(1) and (2) with Theorem II.3.28. The list for -1 comes from the exact scan of Theorem II.3.30(5).

(2) A zero z of $F_N(z^d)$ is of the form $c\omega$ if and only if z^d is. So we need the roots of Q_N in $\mathbb{R}_{>0} \cdot \mu_\infty$. There are no positive real roots, since Q_N has positive coefficients. For the nonreal case we computed exactly $S(x) = \mathrm{Res}_y(Q_N(y), Q_N(xy))$, whose roots are the ratios β_i/β_j. In all 26 cases its only cyclotomic factor is $x - 1$, so no ratio of two distinct roots is a root of unity. The source reports that negative real roots were counted by Sturm sequences. This computation is part of the explicit finite-data hypothesis of the corollary.

(3) All zeros are 2ω. □

An even degree alone does not decide the existence of negative real zeros. No general necessary-and-sufficient input-only criterion is established here.

II.3.8 The Hohertz–Kalantari question: when is $P_N(-1) = 0$?

For $N \geq 2$ that is not a power of 2, let $L \geq 1$ be minimal with a_L a power of 2. We call the *core word* of N the parity word $(a_0, \ldots, a_{L-1}) \bmod 2$; it ends with 1. The core word v and t with $a_L = 2^t$ determine $N = (2^{L+t} - b_v)/3^R$ (Lemma II.3.11). Write $\mathcal{Z}_{-1} = \{N : P_N(-1) = 0\}$.

Theorem II.3.30.

1. *(Criterion) $P_N(-1) = 0$ if and only if $\sum_{i \in O_N}(-1)^i(2a_i + 1) = 2N + (-1)^n$.*
2. *(Chains) For odd $N \geq 3$, $P_{4N+1}(-1) = P_N(-1)$. This equality sharpens [25, Lemma 11], which gives the implication from N to $4N + 1$, by adding the converse. Hence the odd elements of $\mathcal{Z}_{-1}$ form chains $N_0 \mapsto 4N_0 + 1 \mapsto \cdots$. The family of [25, Thm. 7] is the chain $\{x_m : m \geq 2\}$ of 5, and it consists exactly of the N with core word (1).*
3. *(Rigidity) For every core word $v \neq (1)$, at most one $N \in \mathcal{Z}_{-1}$ has core word v.*
4. *(Powers of two) $P_{2^k}(-1) \neq 0$. For every odd $M \geq 3$, at most one $k \geq 1$ has $2^k M \in \mathcal{Z}_{-1}$.*
5. *(Source-reported scan, not an included exhaustion certificate) The source manuscript reports that $\mathcal{Z}_{-1} \cap [1, 2 \cdot 10^8]$ consists of 19 numbers: $x_2, \ldots, x_{14}$; the chain $820569 \mapsto 3282277 \mapsto 13129109 \mapsto 52516437$; and the two even numbers $6094358 = 2 \cdot 3047179$ and $46507804 = 4 \cdot 11626951$.*

Proof. (1) Evaluate (II.3.2) at $z=-1$: $-3P_N(-1)=(-1)^{n+1}+\sum_{i\in O_N}(-1)^i(2a_i+1)-2N$.

(2) Write $\mathcal{A}(x)=P_x(-1)$, so that $\mathcal{A}(x)=x-\mathcal{A}(c(x))$ for $x\geq 2$. The orbit of $4N+1$ begins $4N+1,\ 4c(N),\ 2c(N),\ c(N)$. Hence $\mathcal{A}(4N+1)=4N+1-4c(N)+2c(N)-\mathcal{A}(c(N))=N-\mathcal{A}(c(N))=\mathcal{A}(N)$. The core word is (1) exactly when N is odd and $c(N)$ is a power of 2, that is, $N=x_m$.

(3) Let v have length L and R odd letters, with prefix constants $b_{v,i}$, so that $a_i=(3^{o(i)}N+b_{v,i})/2^i$ for $i<L$, where $o(i)$ is the number of odd letters before i. By (1), for every N with core word v and tail 2^t,

$$-3P_N(-1)=2N\Big(\sum_{i\in O(v)}(-1)^i\,3^{o(i)}2^{-i}-1\Big)+(-1)^{L+t+1}+\sum_{i\in O(v)}(-1)^i(2b_{v,i}2^{-i}+1).$$

The admissible t lie in one class modulo the even number $2\cdot 3^{R-1}$, so $t \bmod 2$ is fixed and $-3P_N(-1)$ is an affine function of N with fixed coefficients. If $v\neq(1)$ then $L\geq 2$ and $\max O(v)=L-1\geq 1$. The term with $i=L-1$ has 2-adic valuation $-(L-1)$, and all other terms have larger valuation. So the sum has valuation $-(L-1)<0$, it is not 1, and the slope is nonzero. Distinct t give distinct N, so at most one zero occurs.

(4) $P_{2^k}(-1)=(2^{k+1}+(-1)^k)/3>0$. For odd $M\geq 3$, $P_{2^kM}(-1)=\frac{2}{3}(2^k-(-1)^k)M+(-1)^kP_M(-1)$. The coefficient $\frac{2}{3}(2^k-(-1)^k)$ equals $2,2,6,10,22,\ldots$; it is strictly increasing for $k\geq 2$, and for $k=1,2$ the conditions $P_M(-1)=\pm 2M$ are incompatible.

(5) The source describes a scan using the recursion $\mathcal{A}(x)=x-\mathcal{A}(c(x))$ and subsequent integer rechecks of its listed witnesses. The analytic proof establishes parts (1)–(4); the exhaustiveness of part (5) remains a source-reported computational assertion in this paper. □

Remark II.3.31 (On the "full converse"). Theorem II.3.30(3) is a converse in the following sense. Among all families defined by a common core word, Hohertz–Kalantari's family is the *only* one contained in $\mathcal{Z}_{-1}$; every other core word contributes at most one element. The criterion (1) is a reformulation, not a classification. The scan suggests that $\mathcal{Z}_{-1}$ is very sparse. Of its 19 elements up to $2\cdot 10^8$, 17 lie on the two deterministic chains; the heuristic "$P_N(-1)$ is a random integer of size $\asymp N$" applies only to the starts of chains and to the even elements. (The source additionally reports that a scan to 10^9 adds only 210065749, 840262997, and x_{15}; this larger exhaustion is not certified here.) Whether $\mathcal{Z}_{-1}$ has infinitely many elements outside finitely many chains remains open.

II.3.9 Q8: results that do not use the orbit

Theorem II.3.32. *Let $N\geq 2$.*

1. *$d_N \mid v_2(N)+1$. Hence the zeros of P_N on $|z|=2$ lie among the points 2ω with $\omega^{v_2(N)+1}=1\neq\omega$. If N is odd (in particular if $N\equiv 3 \bmod 4$), all zeros satisfy $|z|<2$, and the stronger cutoff $0.666<|z|<2$ for $N\neq 9$ follows if the source-reported disk computation in Remark II.3.18 is accepted.*
2. *P_N has at least $\lceil\log_2 N\rceil$ zeros in $1<|z|\leq 2$, with equality for powers of 2.*
3. *$\#\{N\leq X:\ P_N \text{ has a zero on } |z|=2\}\leq 47\sqrt{X}+\log_2 X$ for $X\geq 2$.*

Proof. (1) Lemma II.3.10 and Theorem III. (2) Theorem II.3.20(2); for $N=2^k$ all k zeros have modulus 2.

(3) By Lemma II.3.10, $d_N>1$ with $N=2^kM$ forces $M=1$ or $g_M\geq 2$. Let $M\geq 3$ be odd with $g_M\geq 2$, choose a prime $p\mid g_M$, and let $2^L\leq M<2^{L+1}$ ($L\geq 1$). Since $c(y)\geq y/2$, the first L iterates exceed 1, so the first L parities of M are genuine orbit parities. Each odd step among them has index divisible by p, and $\varepsilon_0=1$. There are at most $2^{\lceil L/p\rceil-1}$ such words, and only the

word $10\cdots0$ when $p \geq L$. By Lemma II.3.11(1), each word occurs for at most one residue class modulo 2^L, hence for at most one $M \in [2^L, 2^{L+1})$. Thus

$$\#\{M \in [2^L, 2^{L+1}) \text{ odd}: \ g_M \geq 2\} \leq 1 + \sum_{p<L} 2^{\lceil L/p \rceil - 1} \leq 2^{L/2}(2^{-1/2} + 2^{-L/2} + L\,2^{-L/6-1/3}) \leq 4 \cdot 2^{L/2},$$

since $\max_{L\geq 1} L2^{-L/6} < 3.2$. Summing over $L \leq \log_2 Y$ gives at most $4\sqrt{2}(\sqrt{2}-1)^{-1}\sqrt{Y} < 13.7\sqrt{Y}$ odd $M \leq Y$ with $g_M \geq 2$. Summing over $N = 2^k M \leq X$ gives at most $13.7\sqrt{X}/(1 - 2^{-1/2}) < 47\sqrt{X}$. Powers of 2 add $\lfloor \log_2 X \rfloor$. □

Theorem II.3.32(3) makes Theorem IV of [5], whose proof is qualitative, effective. The source reports counts $23, 39, 74, 134, 228, 410, 726$ for $X = 10^3, \ldots, 10^9$; these exploratory counts are not used in the proof. They suggest growth like $X^{1/4}$, so the exponent $1/2$ is probably not optimal. The lower bound R_N of the manuscript is also a function of N alone.

II.3.10 Computational status

The analytic proofs in this section are included in full. Finite root isolation, exhaustive Galois-group ranges, resultant scans, and large trajectory censuses inherited from the source manuscript are explicitly marked as source-reported computations. They are not silently promoted to independently verified certificates by being included in this compilation. In particular, the unconditional analytic conclusions do not depend on those large reported scans. No external script or archive is required to compile or read this paper.

II.3.11 What remains open

1. (Q2) Is there an explicit $f(N) > 0$ with $|z_N| \leq 2 - f(N)$ for all N with $d_N = 1$? By Corollary II.3.15, necessarily $\liminf f(N)(\log_2 N)^2 \leq 9$.
2. (Q3) Decide zero-freeness for thin sets meeting the annulus, for example the circle $|z| = 2/3$. Determine $\inf_{N\neq 9} \min |z_N|$; the stronger lower cutoff 0.666 is source-reported.
3. (Q5) Do the rescaled hypotheses $E(N) = o(n(N))$ hold for a set of N of density 1?
4. (Q6, Q7) Prove Conjecture II.3.24(1) for the family H_m. By Theorem II.3.28 this would settle Q7 for that family.
5. (HK) Is $\mathcal{Z}_{-1}$ the union of finitely many $4N + 1$ chains together with finitely many even numbers?

II.4 Primes and constructed orbits on Collatz trajectories

This section treats explicitly posed problems of other authors about prime numbers and crossing patterns on trajectories of the standard Collatz map $C(x) = x/2$ (x even), $C(x) = 3x + 1$ (x odd). We also use the shortcut map $T(x) = x/2$, $T(x) = (3x + 1)/2$ and the odd-to-odd map $S(x) = (3x + 1)/2^{v_2(3x+1)}$ of the manuscript [21]. For $n \geq 1$ whose iterates reach 1 we write $\mathrm{Tr}(n) = \{n, C(n), C^2(n), \ldots, 1\}$ (stopped at the first 1, as in the OEIS programs), $\mathcal{O}(n)$ for its set of odd members, and $\mathrm{Orb}(x) = \{C^t(x) : t \geq 0\}$ for the forward orbit. Whenever a statement concerns a trajectory stopped at 1, termination is an explicit hypothesis or is established by construction. Statements about the first prime or about a specified finite orbit segment do not assert termination of the subsequent orbit. Proposition II.4.22 and the last sentence of Theorem II.4.14 retain their stated conditional scope. Large exhaustive scans inherited from the source are identified as source reports rather than independently reproduced certificates.

Source	Problem	Outcome here
OEIS A319227 [9]	at most 2 twin-prime pairs in a trajectory	open; structural reduction and source-reported censuses, explicitly conditional
A319227, comment	bound 2 for every fixed gap $2q$; no gap-14 example	further counterexamples (gaps 8 and 18); maxima modulo a census
OEIS A273216 [40]	every prime except 3 is a first prime	proved (for first primes; see Theorem II.4.14)
OEIS A349325 [42]	every crossing number occurs infinitely often	proved
OEIS A265946 [43]	$a(2n) - a(2n-1) = n - 1$	proved (telescoping)
OEIS A276202 [41]	$a(n)$ exists for every n	proved conditionally on termination
OEIS A233293, A105730, A087256, A033496 [49, 50, 51, 52]	every count $n > 10$ of initial values with a given trajectory maximum occurs	proved (with $M = 4^k$)
OEIS A245691 [53]	step count to 6 equals Collatz step count to 1	proved (conjugacy)
A025586, comment [54]	digital root of the maximum is 7 or a multiple of 4	fails as literally read (an observation, not a stated conjecture)

II.4.1 The twin-prime conjecture of A319227

Problem II.4.1 (Lagneau, OEIS A319227, 2018)**.** The entry [9] is titled "`a(n) is the number of twin primes in the Collatz trajectory of n.`" and its comments state "`Conjecture: a(n) <=2.`" Its Maple program collects the odd members of the trajectory other than 1 into a set (kept sorted by Maple) `lst` and counts the indices j with `isprime(lst[j]) and isprime(lst[j+1]) and lst[j+1]=lst[j]+2`.

Since no odd integer lies strictly between p and $p+2$, $a(n)$ is the number of pairs $\{p, p+2\}$ of primes with $p, p+2 \in \mathrm{Tr}(n)$. Our implementation of this definition reproduces the 87 data terms of the entry.

Definition II.4.2. Let $g \geq 2$ be even. A pair $\{x, x+g\}$ of odd positive integers is *chained* if one member lies in the forward orbit of the other. The member from whose orbit the other is reached is the *head* (unique unless both lie on one cycle, which is impossible on a trajectory reaching 1), the other the *tail*; the pair is *ascending* if the head is x and *descending* if the head is $x+g$. Let E_g be the multiset of heads of chained pairs $\{p, p+g\}$ with p and $p+g$ both prime, and $E = E_2$.

Lemma II.4.3 (Reduction)**.** *Let* $\mathrm{Tr}(n)$ *reach* 1 *and let* $g \geq 2$ *be even.*

1. *Any two distinct members of* $\mathrm{Tr}(n)$ *form a chained pair whose head is the member visited first.*
2. *The number* $U_g(n)$ *of prime pairs* $\{p, p+g\} \subseteq \mathrm{Tr}(n)$ *equals the number of elements of* E_g*, counted with multiplicity, lying in* $\mathrm{Tr}(n)$*. In particular* $a(n) = |\mathrm{Tr}(n) \cap E|$*.*
3. *If* e^* *is the first member of* $\mathrm{Tr}(n)$ *belonging to* E_g*, then* $U_g(n) = U_g(e^*)$*. Consequently* $\sup_n a(n) = \sup\{|\mathrm{Tr}(e) \cap E| : e \in E,\ \mathrm{Tr}(e)\ \textit{reaches}\ 1\}$*.*

Proof. (1) The members $C^s(n)$, s up to the first visit of 1, are distinct: a repetition would make the orbit periodic before it reaches 1, while the only cycle through 1 is $1 \to 4 \to 2 \to 1$, which the trajectory meets only at its end. If $u = C^s(n)$ and $v = C^t(n)$ with $s < t$, then $v = C^{t-s}(u)$. (2) Send a prime pair contained in $\mathrm{Tr}(n)$ to its head, which lies in $\mathrm{Tr}(n)$. Conversely, if a head $h \in \mathrm{Tr}(n)$ has tail $t \in \mathrm{Orb}(h)$, then t precedes the first 1 of $\mathrm{Orb}(h)$, and the part of $\mathrm{Tr}(n)$ from

h on is $\mathrm{Orb}(h)$ up to its first 1; hence $t \in \mathrm{Tr}(n)$. A head is shared by two pairs only when $h-g, h, h+g$ are all prime, which is why E_g is a multiset; for $g=2$ this forces $h=5$, which is not a head because $\mathrm{Orb}(5) = \{5, 16, 8, 4, 2, 1\}$. (3) All heads in $\mathrm{Tr}(n)$ are visited at or after e^*, so they and their tails lie in $\mathrm{Tr}(e^*)$. □

The next lemma is an immediate consequence of the exact parity-word realization of the manuscript ([21, proof of Theorem 5.1]). It explains why the sieve theorems used there (Maynard; Banks–Freiberg–Turnage-Butterbaugh) say nothing about twin pairs on one trajectory: chained pairs do not come in linear families.

Lemma II.4.4 (Rigidity of chained pairs)**.** *Let $w = (e_0, \dots, e_{k-1})$ be a parity word with $u = \sum e_j$ ones and $b_w = \sum_{e_j=1} 2^j 3^{e_{j+1}+\cdots+e_{k-1}}$, and let $\delta \in \mathbb{Z}$. An integer $x \geq 1$ has first k shortcut parities w and satisfies $T^k(x) = x + \delta$ if and only if*

$$3^u x + b_w = 2^k(x+\delta), \qquad \textit{i.e.} \qquad x\,(2^k - 3^u) = b_w - 2^k\delta.$$

Since $2^k \neq 3^u$ for $k \geq 1$, each pair (w, δ) is realised by at most one integer. In particular the chained pairs $\{x, x \pm 2\}$ with a given connecting parity word are isolated solutions, not values of a nonconstant linear form.

Proof. By [21, proof of Theorem 5.1], the integers with parity word w form the class $3^u x + b_w \equiv 0 \pmod{2^k}$, on which $T^k(x) = (3^u x + b_w)/2^k$. If the displayed identity holds, x lies in this class and $T^k(x) = x + \delta$; the converse is immediate. □

For instance the descending pair $\{71, 73\}$ is the solution $3^5 \cdot 73 + 437 = 2^8 \cdot 71$ for the first 8 parities of 73. Table 8 gives the corresponding data for all chained odd pairs $\{x, x+2\}$ with $3 \leq x < 10^{10}$ (Proposition II.4.7); each line was checked against the identity of Lemma II.4.4 with exact integers.

Table 8: All chained odd pairs $\{x, x+2\}$ with $3 \leq x < 10^{10}$: head h, tail t, number k of T-steps from h to t, and number u of odd steps; in each case $3^u h + b_w = 2^k t$ for the parity word w of length k of h.

h	t	k	u	h	t	k	u	h	t	k	u
3	5	1	1	109	107	8	5	913	911	54	34
7	5	7	4	165	167	27	17	1081	1079	54	34
9	7	2	1	231	233	27	17	2049	2051	73	46
9	11	3	2	235	233	27	17	2427	2429	46	29
11	13	3	2	253	251	27	17	2431	2429	73	46
19	17	5	3	265	263	27	17	3075	3077	73	46
55	53	62	39	285	283	27	17	3079	3077	73	46
63	61	54	34	323	325	46	29				
73	71	8	5	327	325	73	46				

Theorem II.4.5 (Conditional source-reported census of chained twin-prime pairs)**.** *Assume the exhaustive search described in the source manuscript is correct. Then the chained pairs $\{p, p+2\}$ of primes with $p < 10^{12}$ are exactly*

$$\{3,5\}, \quad \{5,7\}, \quad \{11,13\}, \quad \{17,19\}, \quad \{71,73\}, \quad \{107,109\},$$

with heads $3, 7, 11, 19, 73, 109$. Hence $E \cap [1, 10^{12}] = E_0 := \{3, 7, 11, 19, 73, 109\}$, and

$$\mathrm{Tr}(e) \cap E_0 = \{3\},\ \{7,11\},\ \{11\},\ \{19,11\},\ \{73\},\ \{109\} \quad \textit{for } e = 3, 7, 11, 19, 73, 109.$$

Proof (exhaustive computation). The connecting paths are $3 \to 10 \to 5$; $7 \to 22 \to 11 \to 34 \to 17 \to 52 \to 26 \to 13 \to 40 \to 20 \to 10 \to 5$; $11 \to 34 \to 17 \to 52 \to 26 \to 13$; $19 \to 58 \to 29 \to 88 \to 44 \to 22 \to 11 \to 34 \to 17$; $73 \to 220 \to 110 \to 55 \to 166 \to 83 \to 250 \to 125 \to 376 \to 188 \to 94 \to 47 \to 142 \to 71$; $109 \to 328 \to 164 \to 82 \to 41 \to 124 \to 62 \to 31 \to 94 \to 47 \to 142 \to 71 \to 214 \to 107$. The odd members of $\mathrm{Tr}(e)$ are $3, 5, 1$ for $e = 3$; $7, 11, 17, 13, 5, 1$ for $e = 7$; $11, 17, 13, 5, 1$ for $e = 11$; $19, 29, 11, 17, 13, 5, 1$ for $e = 19$; and, for $e = 73$ and $e = 109$, the initial segments $73, 55, 83, 125, 47$ and $109, 41, 31, 47$ followed by the common tail 71, 107, 161, 121, 91, 137, 103, 155, 233, 175, 263, 395, 593, 445, 167, 251, 377, 283, 425, 319, 479, 719, 1079, 1619, 2429, 911, 1367, 2051, 3077, 577, 433, 325, 61, 23, 35, 53, 5, 1. Inspection gives the displayed intersections.

Completeness is the additional computational hypothesis in this statement, not a consequence of the six explicit paths. The source reports a segmented twin-prime sieve, exact accelerated orbit tests, and a second independent scan, with 1,870,585,220 twin pairs below 10^{12}. The underlying exhaustive certificates are not supplied here and this compilation did not rerun that search. The six displayed pairs themselves are explicit unconditional examples.

□

Corollary II.4.6. *Part (1) assumes the census in Theorem II.4.5; part (2) additionally assumes the source-reported exhaustive scan through* 10^9.

1. *If* $a(n) \geq 3$, *then* $\mathrm{Tr}(n)$ *contains a twin-prime pair* $\{p, p+2\}$ *with* $p > 10^{12} - 2$, *one member of which lies in the forward orbit of the other; in particular* $\max \mathrm{Tr}(n) > 10^{12}$.
2. $a(n) \leq 2$ *for every* $n \leq 10^9$. *For* $2 \leq n \leq 10^9$ *the values* $0, 1, 2$ *occur* 460,908,552, 168,734,051 *and* 370,357,396 *times, and* $a(n) = 2$ *holds exactly when* 7 *or* 19 *lies in* $\mathrm{Tr}(n)$.

Proof. (1) By Lemma II.4.3, $a(n) \geq 3$ gives three heads on $\mathrm{Tr}(n)$, and the first of them, e^*, has the other two in $\mathrm{Tr}(e^*)$. By Theorem II.4.5 no element of E_0 has two further elements of E_0 in its trajectory, so some head on $\mathrm{Tr}(n)$ exceeds 10^{12}. (2) The source reports a recursive scan of all starts through 10^9, with exact primality and partner-orbit checks for large values. Its numerical counts and exhaustive range are additional computational hypotheses here. Given those checks, the explicit intersections in Theorem II.4.5 show that two heads occur precisely through 7 or 19. □

Proposition II.4.7 (Conditional source-reported census of odd pairs)**.** *Assume the source-reported exhaustive odd-pair scan is correct. Apart from the pair* $\{1, 3\}$, *the odd* $x < 10^{10}$ *for which* $\{x, x+2\}$ *is chained (no primality being required) are exactly the* 25 *values 3, 5, 7, 9, 11, 17, 53, 61, 71, 107, 165, 231, 233, 251, 263, 283, 323, 325, 911, 1079, 2049, 2427, 2429, 3075, 3077; Table 8 lists their heads, tails and connecting words.*

Proof. The source reports exhaustive exact orbit testing through 10^{10}, with smaller independent checks. No exhaustion certificate is included here. The 25 displayed connecting paths can separately be checked by direct integer iteration. □

Remark II.4.8 (Status and heuristics). The conjecture $a(n) \leq 2$ remains open. Subject to their reported finite-data hypotheses, Theorem II.4.5 and Corollary II.4.6 reduce it to the non-existence of three heads on one trajectory and show that a counterexample needs a chained twin-prime pair beyond 10^{12}. By Lemma II.4.4 every chained pair is an isolated solution of an exponential Diophantine equation of the same shape as the cycle equation $x(2^k - 3^u) = b_w$; a proof along these lines would have to control large "near-cycles" $T^k(x) = x \pm 2$, which seems as hard as excluding non-trivial cycles. Chained pairs are, moreover, extremely rare: no odd x with $3077 < x < 10^{10}$ gives one. Even under the generous model in which an odd x is chained with probability c/x independently of primality, with $c \approx 7$ ($25/\sum_{3 \leq x < 3100,\, x \text{ odd}} x^{-1} \approx 6.8$) fitted to the pairs below

3100 (a model that the empty range $(3100, 10^{10})$ already contradicts), the expected number of chained twin-prime pairs beyond $B = 10^{12}$ would be about $\frac{1}{2}\int_B^\infty \frac{c}{x}\cdot\frac{4C_2}{\log^2 x}\,dx = \frac{2cC_2}{\log B}$, which is below 0.4, where $C_2 \approx 0.66$ is the twin-prime constant; three heads on one trajectory are far less likely still. We expect the conjecture to be true.

II.4.2 Fixed even gaps: a second counterexample and maxima below a census bound

Problem II.4.9 (Lagneau, comment in OEIS A319227)**.** The entry [9] continues: "`This sequence is generalizable: let a(n, p, p+2q) be the number of pairs of primes of form (p, p+2q) in the Collatz trajectory of n, q = 1, 2,...`" and "`It is conjectured that a(n, p, p+2q) < =2.`" Its table lists, for each gap $2q \le 12$, the prime pairs of trajectories with $a(n, p, p+2q) = 2$, and for the gap 14 it reports "`No results for n <= 10^6`".

Two counting conventions are possible. Let $\ell_1 < \ell_2 < \dots$ be the elements of $\mathcal{O}(n)$. The convention of the Maple program of A319227 (with 2 replaced by $2q$) is

$$A_q(n) = \#\{j :\ \ell_j,\ \ell_{j+1} \text{ prime},\ \ell_{j+1} - \ell_j = 2q\},$$

while the literal wording suggests $U_q(n) = \#\{p :\ p,\, p+2q \in \mathrm{Tr}(n) \text{ both prime}\}$; $A_1 = U_1 = a$. The table of the entry follows A_q: for instance $\mathcal{O}(7) = \{1, 5, 7, 11, 13, 17\}$ gives $U_3(7) = 3$ (pairs $(5, 11), (7, 13), (11, 17)$) but $A_3(7) = 0$, whereas the table quotes $(41, 47), (47, 53)$ for the gap 6. The manuscript's Proposition 6.2 [21] disproves the case $q = 3$ in both conventions ($A_3(171) = 4$, $U_3(171) = 5$). Lemma II.4.3 applies to every gap; in addition, if e^* is the first head of a chained gap-$2q$ prime pair on $\mathrm{Tr}(n)$, then every pair counted by $A_q(n)$ lies in $\mathrm{Tr}(e^*)$ and $\mathcal{O}(e^*) \subseteq \mathcal{O}(n)$, so $A_q(n) \le A_q(e^*)$.

Proposition II.4.10 (Gap 8)**.** *The trajectory of* 487 *reaches* 1 *after* 141 *standard steps and its odd members are*

$$487, 731, 1097, 823, 1235, 1853, 695, 1043, 1565, 587, 881, 661, 31, 47, 71, 107, 161, 121, 91,$$
$$137, 103, 155, 233, 175, 263, 395, 593, 445, 167, 251, 377, 283, 425, 319, 479, 719, 1079,$$
$$1619, 2429, 911, 1367, 2051, 3077, 577, 433, 325, 61, 23, 35, 53, 5, 1.$$

The pairs $(23, 31)$, $(53, 61)$, $(479, 487)$ *are prime pairs with gap* 8*, each adjacent in the sorted list of odd members. Hence* $A_4(487) = U_4(487) = 3$*, and the conjecture* $a(n, p, p+8) \le 2$ *of Problem II.4.9 is false in either convention. The bound* 2 *is therefore refuted without any exhaustive search.*

Proof. For consecutive entries x, y of the list, $3x + 1 = 2^h y$ with $h \ge 1$, and the sum of the numbers $h + 1$ is 141. The six numbers $23, 31, 53, 61, 479, 487$ are prime (trial division by the primes up to 19), and the odd numbers $25, 27, 29, 55, 57, 59, 481, 483, 485$ do not occur in the list, which gives the adjacency; no other pair $(x, x + 8)$ of primes occurs in the list, so $U_4(487) = 3$. Only this displayed path and six elementary primality checks are needed for the counterexample; no minimality claim is used. □

Proposition II.4.11 (Conditional source-reported census for the tabulated gaps)**.** *Assume the source-reported exhaustive fixed-gap scans and the associated finite maximization checks are correct. For* $g = 2q \in \{4, 6, 8, 10, 12, 14\}$ *the chained pairs* $\{p, p+g\}$ *of primes with* $p < 10^{10}$ *are exactly those listed below (head first).*

g	*chained prime pairs (head → tail)*	$\max A_q$ *(first n)*	$\max U_q$ *(first n)*
4	$7 \to 11$, $17 \to 13$, $107 \to 103$	2 *(7)*	2 *(7)*
6	$11 \to 5$, $7 \to 13$, $11 \to 17$, $19 \to 13$, $43 \to 37$, $41 \to 47$, $47 \to 53$, $97 \to 103$, $109 \to 103$, $257 \to 251$, $257 \to 263$, $587 \to 593$, $599 \to 593$, $1373 \to 1367$	4 *(171)*	5 *(171)*
8	$13 \to 5$, $19 \to 11$, $31 \to 23$, $61 \to 53$, $59 \to 67$, $487 \to 479$, $919 \to 911$	3 *(487)*	3 *(39)*
10	$7 \to 17$, $19 \to 29$, $41 \to 31$, $71 \to 61$, $73 \to 83$, $97 \to 107$, $293 \to 283$, $587 \to 577$	2 *(73)*	3 *(97)*
12	$17 \to 5$, $29 \to 17$, $41 \to 53$, $73 \to 61$, $83 \to 71$, $89 \to 101$, $263 \to 251$, $281 \to 269$, $467 \to 479$, $491 \to 479$, $1607 \to 1619$	2 *(55)*	4 *(1071)*
14	$19 \to 5$, $47 \to 61$, $97 \to 83$, $733 \to 719$, $1381 \to 1367$	1 *(651)*	2 *(97)*

Consequently, for every n such that each prime pair $\{p, p+g\} \subseteq \mathrm{Tr}(n)$ has $p < 10^{10}$, one has $A_q(n) \le \max A_q$ and $U_q(n) \le \max U_q$ with the values in the table, and these maxima are attained first at the indicated n. In particular the conjecture $a(n, p, p+2q) \le 2$ (in the convention A_q of the entry) fails exactly for $q = 3, 4$ among $q = 1, \dots, 7$, unless a prime pair beyond 10^{10} (beyond 10^{12} for $q = 1$) lies in the trajectory; and $A_7(n) \le 1$ for every n whose trajectory contains no gap-14 prime pair beyond 10^{10}, which explains the "No results" entry: a value 2 for the gap 14 cannot occur below that height; the source also reports $A_7(n) \le 1$ for $n \le 10^7$, a range not independently certified here. In the literal convention U_q the bound 2 fails for $q = 3, 4, 5, 6$ (first at $n = 7, 39, 97, 39$) and holds for $q = 1, 2, 7$, under the same proviso.

Proof. The source describes segmented sieving and exact orbit tests. Its completeness and finite maximization checks are hypotheses of this conditional proposition. Given them, Lemma II.4.3(3) and $A_q(n) \le A_q(e^*)$ reduce the asserted maxima to the listed heads. For example, for the gap 14 the heads $19, 47, 97$ are never adjacent to their tails, because $11 \in \mathrm{Tr}(19)$, $53 \in \mathrm{Tr}(47)$, $91 \in \mathrm{Tr}(97)$ lie strictly between; $\mathrm{Tr}(733)$ and $\mathrm{Tr}(1381)$ contain the adjacent pairs $(719, 733)$ and $(1367, 1381)$ respectively, and neither of $733, 1381$ lies in the trajectory of the other. □

Remark II.4.12. Beyond the tabulated gaps the bound also fails for $2q = 18$: the sorted odd members of $\mathrm{Tr}(619)$ contain the adjacent prime pairs $(5, 23)$, $(233, 251)$ and $(911, 929)$, so $A_9(619) = 3$. This explicit path gives a further counterexample; no minimality assertion is needed.

II.4.3 Every prime except 3 is a first prime (A273216)

Problem II.4.13 (McEachen, OEIS A273216, 2016)**.** The entry [40], "`Initial prime numbers encountered by Collatz (3x+1) evaluation at odd, composite, ascending starts.`", lists for $n = 9, 15, 21, 25, \dots$ the first prime in the sequence $n, C(n), C^2(n), \dots$ (its Mathematica program iterates C from n while the current value is not prime). Its comments read "`The sequence is infinite provided the Collatz conjecture is true.`" and "`Conjecture: every prime number is encountered except 3.`"

Theorem II.4.14. *1. No odd composite start encounters the prime 3.*
2. For every prime $p \neq 3$ there is an odd composite n such that p is the first prime among $n, C(n), C^2(n), \dots$.

Hence the set of primes encountered from odd composite starts is exactly the set of primes other than 3. In particular, whenever the entry is defined for all odd composite starts (for instance under the Collatz conjecture, as the entry itself notes), its set of terms is the set of all primes except 3.

Proof. (1) $3x+1 \not\equiv 0 \pmod 3$, and $x/2 \equiv 0 \pmod 3$ forces $x \equiv 0 \pmod 3$. Hence a multiple of 3 in a trajectory is either the start or preceded only by multiples of 3 obtained by halving, so it belongs to the initial segment $n, n/2, \dots$ of multiples of 3. Starting from an odd composite n, this segment is $\{n\}$, and $n \neq 3$.

(2) For $p = 2$ take $n = 21$: $21 \to 64 \to 32 \to 16 \to 8 \to 4 \to 2$. Let $p \geq 5$. Put $\varepsilon = 1$ if $p \equiv 2 \pmod 3$ and $\varepsilon = 2$ if $p \equiv 1 \pmod 3$, so that $2^\varepsilon p \equiv 1 \pmod 3$. For every $e \equiv \varepsilon \pmod 2$, $e \geq 1$, the integer $n_e = (2^e p - 1)/3$ is odd and

$$n_e \;\to\; 3n_e + 1 = 2^e p \;\to\; 2^{e-1} p \;\to \cdots \to\; 2p \;\to\; p$$

is the beginning of its trajectory; the numbers $2^j p$ with $j \geq 1$ are even and larger than 2, hence composite. So it suffices to find such an e with n_e composite. If $q := n_\varepsilon$ is composite we are done. If $p = 5$ then $q = 3$; take $e = 7$, $n_7 = 213 = 3 \cdot 71$. Otherwise q is a prime with $q > 3$ and $q \neq p$. Let d be the order of 2 modulo q and $e = \varepsilon + 2d$. Then $2^e p \equiv 2^\varepsilon p = 3q + 1 \equiv 1 \pmod q$ and $2^e p \equiv 2^\varepsilon p \equiv 1 \pmod 3$, so $3q \mid 2^e p - 1$, i.e. $q \mid n_e$, and $n_e > n_\varepsilon = q$. Thus n_e is composite. □

Remark II.4.15. The witnesses from the order argument can be large, but small ones are usually available: $n = 9, 25, 49, 15, 27, 57$ for $p = 7, 19, 37, 23, 41, 43$ (the case q composite), $n = 117 = 3^2 \cdot 13$ for $p = 11$ ($3 \cdot 117 + 1 = 2^5 \cdot 11$). All witnesses in this proof are verified by the displayed integer identities; their existence does not rely on a finite prime scan.

II.4.4 Every crossing number occurs infinitely often (A349325)

Problem II.4.16 (Xausa, OEIS A349325, 2021)**.** The entry [42] counts the "`Number of times the Collatz plot started at n crosses the y = n line, or -1 if the number of crossings is infinite.`" Its comments specify "`the trajectories from n to 1 of the 3x+1 function`" T and that "`The starting value of the trajectory is considered a crossing.`", and state "`Conjecture: every positive integer appears in the sequence infinitely many times.`" Its programs start a counter at 1 and add 1 whenever an odd step goes from below n to above n or an even step goes from above n to below n.

For $n \geq 2$ whose T-trajectory reaches 1 at time τ, the values $T^j(n)$, $1 \leq j \leq \tau$, differ from n; put $\sigma_j = \operatorname{sign}(T^j(n) - n)$. Odd steps increase and even steps decrease the value, so the programs of the entry compute

$$\mathrm{A349325}(n) = 1 + \#\{1 \leq j < \tau : \; \sigma_j \neq \sigma_{j+1}\}. \tag{II.4.1}$$

Theorem II.4.17. *For every $k \geq 1$ there are infinitely many n with* $\mathrm{A349325}(n) = k$*. More precisely, for each $k \geq 2$ there are a parity word v_k of length L with $R \geq 1$ ones and $3^R < 2^L$, and a residue class $t \equiv t_k \pmod{2 \cdot 3^{R-1}}$, such that*

$$n_t = \frac{2^{L+t} - b_{v_k}}{3^R}$$

satisfies $\mathrm{A349325}(n_t) = k$ *for every sufficiently large t in that class; for $k = 1$ one may take $n = 2^m$, $m \geq 1$.*

Proof. The case $k = 1$ is clear: the trajectory of 2^m decreases. Let $k \geq 2$. For a word $v = (\varepsilon_0, \dots, \varepsilon_{L-1})$ let $r_j = \varepsilon_0 + \cdots + \varepsilon_{j-1}$ and $s_j = \operatorname{sign}(3^{r_j} - 2^j)$, which is never 0 for $j \geq 1$. A letter 1 multiplies $3^{r_j}/2^j$ by $3/2$ and a letter 0 multiplies it by $1/2$; hence from any prefix, appending enough letters 1 (resp. 0) makes the sign $+$ (resp. $-$). Build v_k as follows: begin with the letter 1 if k is even and with 0 if k is odd; then, $k-1$ times, append letters 0 while the current sign is $+$, or letters 1 while it is $-$, stopping as soon as the sign changes. The resulting $s_1, \dots, s_L$ has exactly $k-1$ sign changes, and by parity it ends with $s_L = -$, i.e. $3^R < 2^L$.

By the realization lemma of the manuscript [21, Section 3.1], since $3 \nmid b_{v_k}$ and 2 generates $(\mathbb{Z}/3^R\mathbb{Z})^\times$, the $t \geq 0$ with $2^{L+t} \equiv b_{v_k} \pmod{3^R}$ form a residue class modulo $2 \cdot 3^{R-1}$; for these t the integer n_t has first L shortcut parities v_k and $T^L(n_t) = 2^t$, so its trajectory continues $2^{t-1}, \ldots, 2, 1$ and $\tau = L + t$. For $1 \leq j \leq L$ we have $T^j(n_t) - n_t = ((3^{r_j} - 2^j)n_t + b^{(j)})/2^j$, where $b^{(j)} \geq 0$ is the constant of the prefix of length j. If $s_j = +$ this is positive; if $s_j = -$ it is negative as soon as $n_t > b^{(j)}/(2^j - 3^{r_j})$. For $L \leq j \leq L + t$ we have $T^j(n_t) \leq 2^t < n_t$ as soon as $b_{v_k} < 2^t(2^L - 3^R)$. For all large t in the class these finitely many inequalities hold, so $\sigma_j = s_j$ for $j \leq L$ and $\sigma_j = -$ for $L \leq j \leq \tau$; by (II.4.1), $\mathrm{A349325}(n_t) = 1 + (k - 1) = k$. Distinct t give distinct n_t. □

Remark II.4.18. The construction gives, for instance (in brackets the least $t \geq 1$ of the class; for v_2, v_4 the class also contains $t = 0$, giving $n_0 = 1$): $v_2 = 10$, $t \equiv 0 \pmod 2$, $n = 5$ $(t = 2)$; $v_3 = 0110$, $t \equiv 2 \pmod 6$, $n = 6$ $(t = 2)$; $v_4 = 1010$, $t \equiv 0 \pmod 6$, $n = 113$ $(t = 6)$; $v_5 = 0110110$, $t \equiv 50 \pmod{54}$, $n = 1779199852788342$ $(t = 50)$; $v_6 = 1010110$, $t \equiv 14 \pmod{54}$, $n = 25889$ $(t = 14)$. The parity-word construction and the finitely many explicit inequalities in the proof suffice for every k; no numerical search is needed. For $k = 1, \ldots, 10$, the source gives the illustrative witnesses $1, 3, 6, 7, 14, 19, 54, 109, 110, 73$; their minimality is not used.

II.4.5 Two further entries

Problem II.4.19 (Lagneau, OEIS A265946, 2015)**.** The entry [43] lists, for each n, the pair $(a(2n-1), a(2n)) = (u, v)$, where u is the sum of the positive differences and $-v$ the sum of the negative differences of consecutive members of the trajectory of n; it states "`Conjecture: a(2n) - a(2n-1) = n - 1.`"

Proposition II.4.20. *The conjecture holds for every n whose trajectory reaches 1 (that is, for every n for which the terms are defined).*

Proof. The differences telescope: $u - v = \sum_t (C^{t+1}(n) - C^t(n)) = 1 - n$. □

Problem II.4.21 (Lagneau, OEIS A276202, 2016)**.** For the Collatz function f, A276202 [41] is the "`Smallest integer k such that the number of semiprimes in {k, f(k), f(f(k)), ..., 1} is equal to n`"; the entry states "`Conjecture: a(n) exists for any n.`"

Proposition II.4.22 (conditional)**.** *Let $m \geq 2$. There are infinitely many odd k whose forward orbit contains at least $m - 1$ distinct semiprimes. If, for each m, the trajectory of one such k reaches 1 (in particular, if the Collatz conjecture holds), then A276202(n) exists for every $n \geq 0$.*

Proof. Use the manuscript's array [21, Section 8.1]: $L_i(t) = 2t\,4^i 3^{K-i} + 1$ $(0 \leq i \leq K)$ satisfies $3L_i + 1 = 4L_{i-1}$, so the orbit of $L_K(t)$ passes through $4L_{i-1}, 2L_{i-1}, L_{i-1}$ for $1 \leq i \leq K$. The $K + 1$ linear forms have positive integer coefficients, distinct leading coefficients and the common constant term 1, so they are pairwise non-proportional; they are admissible (for a prime ℓ, take $t \equiv 0 \pmod \ell$; all forms are then $\equiv 1$). Hence Maynard's theorem [8, Thm. 3.1] (its Hypothesis 1 holds for the primes by the Bombieri–Vinogradov theorem) gives, for K large enough in terms of m, infinitely many t for which at least m of the $L_i(t)$ are prime. For such t at least $m - 1$ primes $L_i(t)$ have $i < K$, and then $2L_i(t)$ is a semiprime in the orbit of $k = L_K(t)$. Let $s(x)$ be the number of semiprimes in $\mathrm{Tr}(x)$. If $\mathrm{Tr}(k)$ reaches 1, its members are distinct, so $s(x) = s(C(x)) + [x \text{ is a semiprime}]$ for $x \neq 1$ in $\mathrm{Tr}(k)$, and $s(1) = 0$. Along $\mathrm{Tr}(k)$, s therefore takes every value between 0 and $s(k) \geq m - 1$. As m is arbitrary, every $n \geq 0$ is a value of s. □

We do not know how to remove the termination hypothesis: every $k < 2^{71}$ is known to reach 1 [47, 48], but these finitely many trajectories have bounded length, so for large n every witness lies beyond that range, and the sieve families above have no known termination property.

Remark II.4.23 (Entries examined but not settled). (a) A304524 [44] (Lagneau, 2018) asks "`Is the sequence finite?`" and conjectures "`for all prime p, res(p) < res(993) = 2^61/(3^32*993) = 1.253142...`" where $\mathrm{res}(n) = 2^{E(n)}/(n3^{O(n)})$ with E, O the numbers of halving and tripling steps. Composing the steps gives the exact identity $\mathrm{res}(n) = \prod_{x\in\mathcal{O}(n)\setminus\{1\}}(1 + 1/(3x))$, so res increases along inverse paths; note that $993 = 3 \cdot 331$ has no odd predecessor. Both questions need primes among the inverse images of a fixed vertex, an exponential family to which the sieve theorems do not apply; we did not settle them. (b) A276290 (Ibrishimova, 2016): by a comment in the entry, a counterexample to its conjecture would produce a non-trivial cycle of the Collatz map; we avoided it. (c) A225851 (Lagneau, 2013) concerns n consecutive prime odd members of a trajectory; existence for all n would need prime k-tuples, not Maynard-type results. (d) A233293 (Schoenfield, 2013) and A105730, A087256, A033496 (Hoft, 2016): settled below, in Theorem II.4.28.

II.4.6 Every count above 10 is attained by a power of four (A233293, A105730, A087256, A033496)

For $M \geq 1$ let $S(M)$ be the set of $x \geq 1$ for which M is the largest member of $\mathrm{Tr}(x)$.

Problem II.4.24 (Schoenfield, OEIS A233293, 2013; Hoft, OEIS A105730, A087256, A033496, 2016)**.** A233293 [49] is the "`Smallest number that is the largest value in the Collatz (3x + 1) trajectories of exactly n initial values. (a(n)=0 if no such number exists.)`"; J. E. Schoenfield notes (Dec 14 2013) that no such number exists for $n = 2, 4, 5, 7, 10$ and states "`Conjecture: a(n) > 0 for all n > 10.`" A105730 [50] is the "`Number of different initial values for 3x+1 trajectories in which the largest term appearing in the iteration is 2^(6n+4).`", i.e. $|S(4^{3n+2})|$; H. F. W. Hoft states (Jun 24 2016) "`Conjecture: All numbers greater than 10 appear in the sequence`". On the same date Hoft states in A087256 [51] ("`Number of different initial values for 3x+1 trajectories in which the largest term appearing in the iteration is 2^n.`", i.e. $|S(2^n)|$), "`Conjecture: All numbers m > 10 occur as values in A087256`", and, in the EXAMPLE section of A033496 [52], where $F(n)$ is "`the set of all initial values less than or equal to n whose Collatz trajectories lead to n as their maximum`" (so $F(n) = S(n)$), "`Conjecture: Every number k > 10 occurs as the size of F(n) for some n.`" The entries record finite checks: Schoenfield reports $a(n) > 0$ for all other $n \leq 3000$; Hoft verified the A087256 statement for $10 < m < 133$ and the A033496 statement for fans of every size $10 < m < 355$ with $n \leq 5 \cdot 10^7$.

The proof below uses an explicit inverse-tree construction. Its 18 finite base cases are independently checked by the complete short integer program embedded after the proof; no external data or script is required.

The tree below 4^k. Let $M = 4^k$, $k \geq 2$. Since $\mathrm{Tr}(M)$ decreases, $x \in S(M)$ if and only if $M \in \mathrm{Tr}(x)$ and every member of $\mathrm{Tr}(x)$ before M is smaller than M. Hence $S(M)$ is the vertex set of the tree $\mathcal{T}(M)$ obtained from the root M by repeatedly adjoining to a vertex z its preimages $2z$, if $2z < M$, and $(z-1)/3$, if $z \equiv 4 \pmod 6$ (the root has only the preimage $(M-1)/3$). Label each vertex by the operations on its path from M: e doublings, b odd steps, and

$$v = \frac{2^e M - c}{3^b}, \qquad (e,b,c) = (0,0,0) \text{ at } M, \quad (e,b,c) \mapsto (e{+}1, b, 2c), \quad (e,b,c) \mapsto (e, b{+}1, c{+}3^b),$$

so that $c \geq 0$ depends on the path but not on M. Call b the depth of v. Since $3^b v = 2^e M - c$, the residue $v \bmod 3$ depends only on $M \bmod 3^{b+1}$. The *limit tree* $\mathcal{T}_\infty(M)$ is defined by the same rules except that a doubling $2z$ of a vertex labelled (e,b,c) is adjoined if and only if $2^{e+1} < 3^b$; it records the ratios $v/M \to \theta = 2^e/3^b$ as $M \to \infty$.

Lemma II.4.25 (Stability). *1. $\mathcal{T}_\infty(M) \subseteq \mathcal{T}(M)$ as labelled trees.*
2. Let $M_0 = 4^{k_0}$ and let D be the maximal depth of $\mathcal{T}_\infty(M_0)$ (a finite tree, by (1), since $\mathcal{T}(M_0)$ has vertex set $S(M_0) \subseteq [1, M_0]$). Then $\mathcal{T}_\infty(4^k) = \mathcal{T}_\infty(M_0)$ for every $k \equiv k_0 \pmod{3^D}$, and there is M_* such that $\mathcal{T}(4^k) = \mathcal{T}_\infty(M_0)$ whenever moreover $4^k \geq M_*$.
3. If $|\mathcal{T}(M_0)| = |\mathcal{T}_\infty(M_0)|$, then $|S(4^k)| = |S(M_0)|$ for every $k \geq k_0$ with $k \equiv k_0 \pmod{3^D}$.

Proof. (1) If $2^{e+1} < 3^b$, then $2z = (2^{e+1}M - 2c)/3^b < M$ because $c \geq 0$; odd steps are adjoined by the same rule in both trees. (2) The order of 4 modulo 3^{D+1} is 3^D (as $4 = 1 + 3$), so $k \equiv k_0 \pmod{3^D}$ is equivalent to $4^k \equiv M_0 \pmod{3^{D+1}}$, which fixes the residues modulo 3 of all vertices of depth at most D; these residues decide every odd step, including the absence of odd steps below depth D. Thus the limit trees agree. A doubling with $2^{e+1} > 3^b$ (equality is impossible) is adjoined in $\mathcal{T}(M)$ if and only if $M(2^{e+1} - 3^b) < 2c$; take M_* larger than $2c/(2^{e+1} - 3^b)$ for the finitely many such candidate doublings of vertices of $\mathcal{T}_\infty(M_0)$. By induction on the vertices, $\mathcal{T}(M) = \mathcal{T}_\infty(M)$ for $M \geq M_*$ in the class. (3) By (1) the two trees of M_0 coincide, so no candidate doubling is adjoined at M_0, i.e. $M_0(2^{e+1} - 3^b) \geq 2c$ for each of them; these inequalities persist for $M \geq M_0$, and (2) applies with $M_* = M_0$. □

Write $\theta = 2^e/3^b$ for an odd vertex v of the limit tree and $r = v \bmod 3$. Its doubling chain $v, 2v, \ldots$ has length $\ell(\theta) = \#\{j \geq 0 : 2^j\theta < 1\}$, and its odd children are the vertices $(2^j v - 1)/3$, of ratio $2^j\theta/3$, for the *eligible* j: $1 \leq j < \ell(\theta)$ and $2^j r \equiv 1 \pmod 3$. The root has the single child $y = (M - 1)/3$, of ratio $1/3$, and $y \equiv k \pmod 3$; so for $k \equiv 0, 1 \pmod 3$, $k \geq 2$, the tree is $\{M, y, 2y\}$ and $|S(4^k)| = 3$, and the interesting case is $k \equiv 2 \pmod 3$, i.e. exactly A105730.

Lemma II.4.26 (Free residues). *Let $v_1 = y, v_2, \ldots, v_m$ be a chain of odd vertices, $v_{i+1} = (2^{j_i}v_i - 1)/3$, and let $r_1, \ldots, r_m \in \{0, 1, 2\}$ be such that $2^{j_i}r_i \equiv 1 \pmod 3$ for $i < m$. Then there is a class μ modulo 3^{m+1}, $\mu \equiv 1 \pmod 3$, such that $v_i \equiv r_i \pmod 3$ for all i whenever $M \equiv \mu$; and $M = 4^k$ lies in μ for all k in a residue class modulo 3^m.*

Proof. $v_i \bmod 3$, with $v_i = (2^{E_i}M - c_i)/3^i$, depends on $M \bmod 3^{i+1}$, and replacing M by $M + 3^i t$ changes v_i by $2^{E_i}t$ while leaving $v_1, \ldots, v_{i-1}$ modulo 3 unchanged; choose $t \in \{0, 1, 2\}$ inductively. The powers of 4 run through all classes $\equiv 1 \pmod 3$ modulo 3^{m+1}. □

Lemma II.4.27 (Diamonds and resets). *Let v be an odd vertex of ratio θ with $1/6 < \theta < 1/4$; then $\ell(\theta) = 3$.*

1. *(Diamond) If $\theta > 3/16$, there are two single-child paths from v to an odd vertex of ratio $8\theta/9$: with $r(v) = 1$ through a vertex of ratio $4\theta/3$ with $\ell = 2$ and residue 2, and with $r(v) = 2$ through a vertex of ratio $2\theta/3$ with $\ell = 3$ and residue 1. Their chain lengths add up to $3+2 = 5$ and $3 + 3 = 6$.*
2. *(Reset) If $\theta < 3/16$ and $r(v) = 1$, then v has a single odd child, of ratio $4\theta/3 \in (2/9, 1/4)$.*

In each case the vertex reached again satisfies $1/6 < \theta' < 1/4$, every vertex involved has exactly one eligible doubling, and a vertex of residue 0 has no children.

Proof. Direct inequalities: $4\theta < 1 < 8\theta$ gives $\ell(\theta) = 3$, and the eligible $j < 3$ are $j = 2$ for $r = 1$ and $j = 1$ for $r = 2$. For $3/16 < \theta < 1/4$ the vertex of ratio $4\theta/3 \in (1/4, 1/3)$ has $\ell = 2$ and, for residue 2, only $j = 1$; the vertex of ratio $2\theta/3 \in (1/8, 1/6)$ has $\ell = 3$ (since $8\theta/3 < 1 < 16\theta/3$) and, for residue 1, only $j = 2$; both children have ratio $8\theta/9 \in (1/6, 2/9)$. For $1/6 < \theta < 3/16$, $r = 1$ gives only $j = 2$ and the ratio $4\theta/3 \in (2/9, 1/4)$. The intervals are open because the hypotheses are strict, and $\theta \neq 3/16$ because $\theta = 2^e/3^b$. □

Theorem II.4.28. *For every integer $n \geq 11$ there are infinitely many $k \equiv 2 \pmod 3$ with $|S(4^k)| = n$. Consequently every integer greater than 10 occurs in A105730 and in A087256,*

infinitely often, every $m > 10$ is the size of a fan $F(n)$ as in A033496, and $\mathrm{A233293}(n) > 0$ *for every $n > 10$: the four conjectures of Problem II.4.24 hold.*

Proof. $n \geq 29$. Take $k \equiv 2 \pmod 3$, so y has residue 2 and its only child has ratio $2/9 \in (3/16, 1/4)$; the vertices $M, y, 2y$ contribute 3. From ratio $2/9$ apply, as long as desired, a diamond when $\theta > 3/16$ and a reset when $\theta < 3/16$; this forced sequence begins DDRDDRDDDRDD$\cdots$ and never leaves $(1/6, 1/4)$. Stop at any vertex by giving it residue 0 (its chain contributes 3). With d diamonds, of which s use the second path, and ρ resets, the limit tree is the single path just described, of size

$$n = 3 + 5d + s + 3\rho + 3 = 6 + 5d + 3\rho + s, \qquad 0 \leq s \leq d.$$

By Lemma II.4.26 the residues are realised by $M = 4^k$ for k in a residue class modulo a power of 3, and by Lemma II.4.25(2) all large k in that class give $|S(4^k)| = n$. Along the forced sequence the intervals $[6+5d+3\rho,\ 6+6d+3\rho]$ are consecutive or overlapping as soon as $d \geq 4$ (a diamond raises the lower end by $5 \leq d+1$, a reset by 3), and the first one with $d = 4$ is $[29, 33]$ (after DDRDD); their union is $[29, \infty)$.

$11 \leq n \leq 28$. The following values of k satisfy $|S(4^k)| = n$:

n	11	12	13	14	15	16	17	18	19
k	26	5	14	113	59	68	323	242	203
n	20	21	22	23	24	25	26	27	28
k	80	32	473	140	149	1679	1349	1490	1457

(all $\equiv 2 \pmod 3$). For these 18 listed values, the embedded finite verification checks both $|\mathcal{T}(4^k)| = n$ and $\mathcal{T}(4^k) = \mathcal{T}_\infty(4^k)$ as labelled trees. Lemma II.4.25(3) then gives infinitely many further k for each n. □

Remark II.4.29. Hoft proved in A105730 (by induction) that the sequence has the quasiperiod $6, x, 8, 6, y, 8, 6, 9, z$, together with $a(27n+1) = 12$, $a(27n+4) = 13$, $a(27n+8) = 11$. Lemma II.4.25(3) re-derives these statements: the values $6, 8, 9$ come from $k = 2$, 8 or 17, 23 with $D = 2, 3, 3$, and the values $12, 13, 11$ from $k = 5, 14, 26$ with $D = 4$. The infinite covering argument is analytic; the only finite base cases needed for the theorem are the 18 witnesses above.

II.4.7 Two further minor entries (A245691, A025586)

Problem II.4.30 (Spage, OEIS A245691, 2014)**.** The entry [53] is the "`Irregular triangle of Collatz like iteration, x -> 3x, then repeat (x -> ceiling(x/2) if divisible by 3, otherwise x -> 3x) while x != 6.`"; its comments state "`It is conjectured that the number of steps for the trajectory to arrive at 6 is equal to the number of steps for the Collatz trajectory to arrive at 1 for the same starting value n (n>1), suggesting the length of the n-th row of the irregular array is given by A008908(n).`"

Proposition II.4.31. *Let $g(x) = \lceil x/2 \rceil$ if $3 \mid x$ and $g(x) = 3x$ otherwise, and let $n \geq 2$. The sequence $x_0 = n$, $x_1 = 3n$, $x_{i+1} = g(x_i)$ reaches 6 for the first time at an index $i \geq 1$ equal to the number $\tau(n)$ of Collatz steps from n to 1; if $\tau(n) = \infty$, it never reaches 6. Hence the conjecture holds, and, when $\tau(n) < \infty$, the n-th row has length $\tau(n) + 1 = \mathrm{A008908}(n)$.*

Proof. Put $c_t = C^t(n)$. If c is even, $g(3c) = 3(c/2) = 3C(c)$. If c is odd, $g(3c) = (3c+1)/2$, which is $\equiv 2 \pmod 3$, so $g((3c+1)/2) = 3(3c+1)/2 = 3C^2(c)$. Call t aligned if $x_{1+t} = 3c_t$. Then $t = 0$ is aligned, and by the two cases every aligned t is followed by the aligned $t+1$ (c_t even)

or $t+2$ (c_t odd); at the skipped index $x_{1+t+1} = (3c_t+1)/2$ is not divisible by 3. In particular every t with c_{t-1} even is aligned: if $t-1$ is not aligned, then $t-2$ is aligned with c_{t-2} odd, and $t = (t-2)+2$. If $\tau(n) < \infty$ and $n \geq 3$, then $c_{\tau-2} = 4$ and $c_{\tau-1} = 2$, so $t = \tau - 1$ is aligned and $x_\tau = 6$; for $n = 2$, $x_1 = 6$. Earlier indices give $x_{1+t} = 3c_t$ with $c_t \neq 2$ (the trajectory has distinct members) or a value prime to 3, so 6 is not reached earlier. If $\tau(n) = \infty$, the value $6 = 3 \cdot 2$ would force $c_t = 2$ at an aligned time, which is impossible. □

Problem II.4.32 (Chouraqui, reported by Sloane in OEIS A025586, 2023)**.** For A025586 [54], the "`Largest value in '3x+1' trajectory of n.`", a comment of N. J. A. Sloane (Aug 11 2023) reads "`In an email of Aug 06 2023, Guy Chouraqui observes that the digital root of a(n) appears to be either 7 or a multiple of 4 for all n > 2.`"

Proposition II.4.33. *The observation fails:* A025586(20) = 20 *and* A025586(21) = 64 *have digital roots* 2 *and* 1*, and among* $3 \leq n \leq 200$ *it fails for* 36 *values of* n *(read literally; the comment refers to A006885, and every record maximum other than* $1, 2$ *up to* $3 \cdot 10^6$ *does have digital root* 7*). More precisely, for* $n > 2$ *the maximum* $a(n)$ *is divisible by* 4 *(already noted in a 2019 comment of P. Munn in A025586); if it is attained after an odd step it is* $\equiv 1 \pmod 3$*, so its digital root is* 1*,* 4 *or* 7*, and each of these occurs infinitely often:* $a((4^m-1)/3) = 4^m$ *(*$m \geq 2$*) has digital root* $4, 7, 1$ *for* $m \equiv 1, 2, 0 \pmod 3$*. If the maximum is the starting value, all nine digital roots occur:* $a(2^m) = 2^m$ *gives* $1, 2, 4, 5, 7, 8$*, and* $a(48) = 48$*,* $a(24) = 24$*,* $a(72) = 72$ *give* $3, 6, 9$*.*

Proof. Tr(20) = $20, 10, 5, 16, 8, 4, 2, 1$ and Tr(21) = $21, 64, 32, \ldots, 1$. Divisibility by 4: an odd maximum $x > 1$ would be followed by $3x+1 > x$, and if the maximum is $2x$ with x odd, $x > 1$, then $3x+1$ follows two steps later and exceeds $2x$. A maximum reached by an odd step is $3x+1 \equiv 1 \pmod 3$. For $n = (4^m-1)/3$ the trajectory is $n, 4^m, \ldots, 1$, and $4^m \bmod 9$ cycles through $4, 7, 1$; the digital roots of 2^m cycle through $2, 4, 8, 7, 5, 1$; the trajectories of 24, 48 and 72 never exceed their starting values (their largest later values are 16, 24 and 52). □

Remark II.4.34. The conjugacy proof and the displayed small counterexamples are complete without a numerical scan. These are minor results; the digital-root statement is an observation in a comment, not a named general conjecture.

II.4.8 An embedded certificate for the eighteen fan base cases

The following complete integer computation verifies the only finite base cases needed in the proof of Theorem II.4.28. It uses no imported packages, data files, or auxiliary programs. The integer labels satisfy $3^b z = 2^e M - c$. The ordinary and limit inverse trees are computed separately and compared as labelled trees, not merely by their cardinalities. Running the displayed code gives the final message shown below.

```
cases = [(11,26),(12,5),(13,14),(14,113),(15,59),
 (16,68),(17,323),(18,242),(19,203),(20,80),(21,32),
 (22,473),(23,140),(24,149),(25,1679),(26,1349),
 (27,1490),(28,1457)]

def tree(k, limit):
    M = 4**k
    todo = [(M,0,0,0)]
    seen = {}
    while todo:
        z,e,b,c = todo.pop()
        assert z not in seen
```

```
        assert 3**b*z == 2**e*M-c
        seen[z] = (e,b,c)
        allowed = 2**(e+1)<3**b if limit else 2*z<M
        if allowed:
            todo.append((2*z,e+1,b,2*c))
        if z%6 == 4:
            todo.append(((z-1)//3,e,b+1,c+3**b))
    return seen

for n,k in cases:
    actual = tree(k,False)
    limit = tree(k,True)
    assert len(actual) == n and actual == limit
print("All eighteen finite fan certificates pass.")
```

This is a finite, reproducible certificate. It neither asserts that the listed exponents are the smallest possible nor relies on a scan through all exponents up to 2000.

II.4.9 Every permitted first prime has infinitely many composite preimages

The first-prime conclusion admits a stronger statement than mere existence. It concerns only the finite prefix ending at that prime, and therefore requires no hypothesis about the subsequent orbit.

Theorem II.4.35 (Infinite fibers of the first-prime map)**.** *For every prime $p \neq 3$, infinitely many odd composite integers have p as their first prime under the standard Collatz map. Their first-prime arrival times can be prescribed along an arithmetic progression. More explicitly, choose $e_0 \geq 1$ such that*

$$q = \frac{2^{e_0}p - 1}{3} > 1$$

is an odd integer, and put $D = \operatorname{ord}_{3q}(2)$. Then, for every integer $t \geq 1$,

$$n_t = \frac{2^{e_0+Dt}p - 1}{3}$$

is odd and composite, is divisible by q, and first reaches a prime at p after exactly $e_0 + Dt + 1$ standard Collatz steps. In particular, if $B_p(X)$ counts odd composite starts at most X whose first prime is p, then

$$B_p(X) \geq \frac{\log X}{D \log 2} + O_p(1) \qquad (X \longrightarrow \infty).$$

For $p = 3$ there is no odd composite preimage in this sense.

Proof. For $p \geq 5$, take $e_0 = 1$ when $p \equiv 2 \pmod 3$ and $e_0 = 2$ when $p \equiv 1 \pmod 3$. Then $q \geq 3$. For $p = 2$, one may take $e_0 = 3$, giving $q = 5$. Since $3q$ is odd, the order D exists; it is even because 2 has order 2 modulo 3. The definition of D gives

$$2^{e_0+Dt}p - 1 \equiv 2^{e_0}p - 1 = 3q \equiv 0 \pmod{3q}.$$

Thus n_t is an odd integer divisible by q, and $n_t > q$ for $t \geq 1$. It is therefore composite, whether or not q itself is prime. Its initial orbit is exactly

$$n_t \longmapsto 2^{e_0+Dt}p \longmapsto 2^{e_0+Dt-1}p \longmapsto \cdots \longmapsto 2p \longmapsto p.$$

Every intervening value is even and greater than 2, so p is the first prime. The inequality $n_t \le X$ is equivalent to $t \le (\log_2((3X+1)/p) - e_0)/D$, giving the counting bound. Finally, an odd Collatz step never produces a multiple of 3, while halving preserves divisibility by 3. An odd composite start therefore cannot reach 3. □

This strengthens the surjectivity result associated with OEIS A273216 to infinite fibers and an explicit logarithmic lower bound. It does not say that every odd composite start eventually encounters a prime.

Corollary II.4.36 (Positive frequency among power-of-four exponents)**.** *For each $n \ge 11$ there are integers a_n, D_n and a threshold K_n such that every $k \ge K_n$ with $k \equiv a_n \pmod{3^{D_n}}$ satisfies $|S(4^k)| = n$. Consequently the set of such exponents has lower natural density at least 3^{-D_n}, and*

$$\#\{4^k \le X : |S(4^k)| = n\} \ge \frac{\log X}{3^{D_n} \log 4} + O_n(1).$$

Proof. The diamond/reset construction prescribes a finite inverse tree. The free-residue and stability lemmas give the displayed arithmetic progression of exponents. Counting its members up to $\log_4 X$ proves both assertions. □

II.5 Singularity counting beyond Franel sums: leading symbols, rook walks and G_2 invariants

This section concerns lower bounds for minimal recurrence orders obtained by counting the singularities of the generating function. The method combines the singularity budget of a recurrence with the visibility of Morse critical values of a real moment function [21, 19]. We searched the literature (papers, surveys, OEIS) for questions posed by other authors that ask for the minimal or exact order of a recurrence, or for a proof that no shorter recurrence exists. We then tested the method on each question found. The outcome is as follows.

(F1) *An observation proved and corrected, within the Franel family.* Inside a research question of van Hoeij [55, §7, Question 1] there is an observation on the roots of the characteristic polynomial of the minimal telescoper for $\binom{n}{k}^s$. We determine these roots for every s (Theorem II.5.7). The printed set is correct for odd s and for $s = 2$; for even $s \ge 4$ it omits the negative roots. The methodological part of the question (computing such invariants directly from the module) is not addressed. The result is immediate once the minimal order of the Franel sums is known [19].

(F2) *A limitation, not a closure.* Kauers and Zeilberger found empirically that the recurrences for d-dimensional diagonal rook walks have order d [56]. They pose no lower-bound question; whether shorter recurrences exist is our reading. We show that for $d = 4, 5$ no argument that only counts *distinct* singular points can reach d. The $d = 4, 5$ obstruction is retained conditionally on the operator identities reported in [22, Prop. 2.12]; those full operators are not independently certified in this consolidation (Remark II.5.11). For $d = 2$ the method works (Proposition II.5.10); exactness for $d \le 3$ was proved in [22].

(F3) *An additional theorem, not an explicitly posed problem.* The dimensions of the G_2-invariants in $V_7^{\otimes n}$ (OEIS A059710) have minimal recurrence order exactly 3, so Mihailovs' recurrence is minimal (Theorem II.5.12). The proof needs an extension of the visibility lemma to planar saddles with vanishing amplitudes.

Subsection II.5.6 records the other explicitly posed questions we found and why they are not settled here. Exploratory recurrence searches in the source manuscript are distinguished below from analytic proofs. Finite agreement, and rank tests under a fixed degree bound, do not prove

a recurrence for all indices or rule out all higher coefficient degrees. No external script or log is needed for the proofs retained here.

II.5.1 Width, leading symbol and characteristic polynomial

Definition II.5.1. A *recurrence of width* r for a complex sequence a is a relation $\sum_{j=0}^{r} p_j(n)a(n+j) = 0$, valid for all sufficiently large n, with $p_j \in \mathbb{C}[n]$ and $p_0 p_r \neq 0$. The least such r is the *minimal width* $\mathrm{w}(a)$. In every case below the lower bound is proved over $\mathbb{C}(n)$ and the upper bound is realised over $\mathbb{Q}(n)$, so the two minimal widths agree. For a given recurrence put $D = \max_j \deg p_j$ and $c_j = [n^D]p_j(n)$. Define the *leading symbol* and the *characteristic polynomial* by

$$H(z) = \sum_{j=0}^{r} c_j z^{r-j}, \qquad \chi(x) = \sum_{j=0}^{r} c_j x^j = x^r H(1/x).$$

Lemma II.5.2 (Singularity budget [21, Lemma 2.1])**.** *Let $F(z) = \sum_{n\geq 0} a(n)z^n$ have positive radius of convergence. Suppose a satisfies a recurrence of width r with leading symbol H. Then every nonzero finite point at which some branch of the analytic continuation of the germ of F at 0 is singular is a root of H. In particular there are at most r such points.*

Proof. Multiply the recurrence by z^{n+r} and sum. With $\theta = z\,d/dz$ this gives $\sum_j z^{r-j}p_j(\theta-j)F = R$ for a polynomial R. The coefficient of $(d/dz)^D$ is $z^D H(z)$, and $H \neq 0$. Points with $zH(z) \neq 0$ are ordinary points of this linear equation, so every continued branch is holomorphic there. □

Lemma II.5.3 (Visibility, with nonnegative amplitudes and planar saddles)**.** *Let M be a compact real-analytic manifold of dimension $d \geq 1$ with a positive real-analytic density μ. Let $\Phi, g\colon M \to \mathbb{R}$ be real analytic and $a(n) = \int_M \Phi^n g\,d\mu$. Let $v \neq 0$, and suppose the critical points of Φ on $\Phi^{-1}(v)$ are finite in number and nondegenerate; regular points on the fibre are allowed. The branch of $F(z) = \sum a(n)z^n$ obtained by continuation through the upper half plane in $w = 1/z$ is singular at $z = 1/v$ in each of the following cases.*

(a) The critical points on the fibre are all maxima or all minima. Near each of them $g \geq 0$, and $g \not\equiv 0$ near at least one. The amplitude may vanish at the critical points.

(b) $d = 2$. Near each saddle s on the fibre, choose real-analytic Morse coordinates with $\Phi = v + Y_1Y_2$, and write $g\,\mu = \widehat{g}_s(Y)\,dY_1dY_2$ with $\widehat{g}_s = \sum_{a,b} c_{ab}^{(s)} Y_1^a Y_2^b$. Suppose that $\sum_s c_{aa}^{(s)} \neq 0$ for some $a \geq 0$. The fibre may also contain extrema. (The coefficients $c_{aa}^{(s)}$ do not depend on the choice of Morse chart: by the proof, $c_{aa}^{(s)} = A_{s,a}/(2\pi i^a a!)$, where $A_{s,a}$ is a coefficient of the asymptotic expansion of the localised integral, which depends only on the germs of Φ, g and μ at s.)

Proof. Let $\nu = \Phi_*(g\mu)$, a finite real measure on the compact set $\Phi(M) \subset \mathbb{R}$. Then $G(w) = \int d\nu(u)/(w-u)$ is holomorphic off $\Phi(M)$ and equals $w^{-1}F(1/w)$ near $w = \infty$. Its restriction to the upper half plane is therefore a continuation of the germ of F. Suppose this branch extends holomorphically to a disc B centred at v. By Stieltjes inversion $\nu = -\pi^{-1}\lim_{y\downarrow 0} \operatorname{Im} G(\cdot + iy)$ in the sense of distributions, so ν would have a real-analytic density on $B \cap \mathbb{R}$. It therefore suffices to show that ν has no C^∞ density near v.

Choose a smooth partition of unity subordinate to analytic Morse charts U_p around the critical points p on the fibre, with cutoffs χ_p equal to 1 near p, and to the complement. On the complement, near $\Phi^{-1}(v)$, Φ is a submersion. Its contribution is the pushforward of a smooth compactly supported density under a submersion, which is a C^∞ density near v.

(a) Choose Morse coordinates with $\Phi = v + \varepsilon|y|^2$, where the sign ε is the same for every critical point on the fibre. Write $g\mu = \widehat{g}_p(y)\,dy$ with $\widehat{g}_p$ real analytic and $\widehat{g}_p \geq 0$. Integration over spheres shows that the chart contributes the density $\mathbf{1}_{\varepsilon t>0}\frac{1}{2}|t|^{d/2-1}m_p(|t|)$ at $v+t$, where

$m_p(u) = \int_{\mathbb{S}^{d-1}} (\chi_p \widehat{g}_p)(\sqrt{u}\,\omega)\, d\omega$. For small u we have $\chi_p = 1$. Expand $\widehat{g}_p$ in a power series: odd-degree terms integrate to zero, and a term of degree $2k$ contributes a multiple of u^k. Hence m_p is real analytic, and $m_p \geq 0$. If $\widehat{g}_p \not\equiv 0$ then $\int_{|y|<\delta} \widehat{g}_p = \frac{1}{2}\int_0^{\delta^2} u^{d/2-1} m_p(u)\, du > 0$, so $m_p(u) = \kappa_p u^{k_p} + O(u^{k_p+1})$ with $\kappa_p > 0$. Let k be the least k_p. All charts contribute on the same side $\varepsilon t > 0$, so near v the density has the form

$$\mathbf{1}_{\varepsilon t>0} |t|^{d/2-1} (\kappa |t|^k + O(|t|^{k+1})) + C^\infty, \qquad \kappa > 0.$$

If $\alpha = d/2 - 1 + k$ is not an integer, the derivative of order $\lceil \alpha \rceil$ is unbounded on the side $\varepsilon t > 0$. If α is an integer, the α-th derivative jumps by $\alpha!\kappa$. In either case the density is not C^∞.

(b) Suppose, to the contrary, that ν has a C^∞ density near v. Nondegenerate critical points are isolated, so by compactness Φ has finitely many critical points near the fibre and v is an isolated critical value. Choose $\psi \in C_c^\infty(\mathbb{R})$ with $\psi = 1$ near v, supported in a small interval around v on which ν has a C^∞ density and which contains no other critical value. Then $\psi\nu$ is a C^∞ compactly supported function. Hence, as $\tau \to +\infty$,

$$I(\tau) = \int e^{i\tau u} \psi(u)\, d\nu(u) = \int_M e^{i\tau\Phi} \psi(\Phi)\, g\, d\mu = O(\tau^{-\infty}).$$

On the other hand, localise I to the critical points p on the fibre, using cutoffs equal to 1 near p; the rest of the integral is $O(\tau^{-\infty})$. The phase is exactly quadratic in Morse coordinates, so the method of stationary phase [60, Lemma 7.7.3, Theorem 7.7.5] gives $I(\tau) \sim e^{i\tau v} \sum_{k\geq 0} (\sum_p A_{p,k}) \tau^{-1-k}$. At a saddle, with $\Phi - v = Y_1 Y_2$, the quadratic form has matrix $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, signature 0 and $|\det| = 1$, and

$$A_{s,k} = 2\pi \frac{i^k}{k!} ((\partial_1\partial_2)^k \widehat{g}_s)(0) = 2\pi i^k k!\, c_{kk}^{(s)}.$$

At an extremum, with $\Phi = v + \varepsilon|y|^2$, the same formula gives $A_{p,k} = \pi e^{i\pi\varepsilon/2} (i\varepsilon/4)^k (\Delta^k \widehat{g}_p)(0)/k!$. This equals $i^k \cdot i \cdot$ (a real number). Since g is real,

$$\sum_p A_{p,k} = i^k \Big(2\pi k! \sum_s c_{kk}^{(s)} + i\,\beta_k \Big), \qquad \beta_k \in \mathbb{R}.$$

Choose k with $\sum_s c_{kk}^{(s)} \neq 0$. Then this coefficient is nonzero, which contradicts $I(\tau) = O(\tau^{-\infty})$. The asymptotic expansion of each localised integral, and hence each saddle's contribution $A_{s,k}$, depends only on the germs of Φ, g and μ at s, not on the Morse coordinates. □

Remark II.5.4. In case (b) the hypothesis cannot simply be dropped. For $\Phi = v + Y_1 Y_2$ and $\widehat{g} = Y_1^2 \geq 0$, the local density is $\int_{|t|\leq|Y_1|\leq 1} |Y_1|\, dY_1 = 1 - t^2$, a polynomial, and every coefficient c_{kk} vanishes. A nonnegative amplitude that vanishes along a null line of the Hessian can therefore hide a saddle. With the amplitude 1, part (a) is [21, Lemma 2.2]. Part (a) itself only uses the positivity of the leading term of the spherical mean m_p, not that of the amplitude at p.

Lemma II.5.5 (Rigidity of the leading symbol)**.** *Suppose a satisfies a recurrence of width r. Suppose also that, at r distinct nonzero points $1/v_1, \dots, 1/v_r$, some branch of F is singular. Then $\mathrm{w}(a) = r$. Moreover every recurrence of width r for a has characteristic polynomial $\chi(x) = c\prod_{i=1}^r (x - v_i)$ with $c \neq 0$; in particular $c_0 c_r \neq 0$.*

Proof. By Lemma II.5.2, every recurrence of width r' has a leading symbol of degree at most r' that vanishes at the r points $1/v_i$. Hence $r' \geq r$. If $r' = r$, then $H = \kappa \prod_i (z - 1/v_i)$ with $\kappa \neq 0$, because $H \neq 0$ and $\deg H \leq r$. Therefore $\chi(x) = x^r H(1/x) = \kappa \prod_i (-1/v_i) \cdot \prod_i (x - v_i)$. □

II.5.2 F1. The characteristic polynomial of the minimal telescoper for $\binom{n}{k}^s$

Let $A_s(n) = \sum_k \binom{n}{k}^s$ and $m_s = \lfloor (s+1)/2 \rfloor$.

Problem II.5.6 (van Hoeij [55, §7, Question 1; Remark 6])**.** In [55, Remark 6] the telescoper L of $H = \binom{n}{k}^s$ is shown to have order $r = \lfloor (s+1)/2 \rfloor$, using [2, Theorem 1.1] for the lower bound. Research question 1 of [55, §7] reads, in the arXiv version v2 consulted here: *"The characteristic polynomial for L from Remark 6 appears to have the following roots $\{(\zeta+\zeta^{-1})^s \mid \zeta^s = 1,\ \zeta \neq -1,\ \zeta^2 \neq -1\}$."* The question also states that for the antisymmetric module N_- one should *"just delete the root 2^s"*. The surrounding question asks how to compute such invariant data directly from the module, without computing L.

Theorem II.5.7. *Let $s \geq 2$ and $v_j = (-1)^j \big(2\cos(\pi j/s)\big)^s$ for $0 \leq j < s/2$.*

1. *The numbers v_j are nonzero, real and pairwise distinct, with $|v_0| > |v_1| > \cdots$.*
2. *Every recurrence of width m_s satisfied by A_s has characteristic polynomial $c\prod_{0\leq j<s/2}(x - v_j)$ with $c \neq 0$. The leading symbol is $\kappa\prod_j(z - 1/v_j)$. The nonzero singular points of all branches of $\sum A_s(n)z^n$ are exactly the points $1/v_j$.*
3. *The monic minimal telescoper of $\binom{n}{k}^s$ is the monic minimal recurrence of A_s. Its monic characteristic polynomial is $\prod_j(x - v_j)$, and its root set is*

$$\{v_j\} = \{(1+\omega)^s : \omega^s = 1,\ \omega \neq -1\} = \{\zeta^s(\zeta+\zeta^{-1})^s : \zeta^{2s} = 1,\ \zeta^2 \neq -1\}.$$

4. *For odd s and for $s = 2$ this set equals the set displayed in Problem II.5.6 (for odd s the conditions $\zeta \neq -1$, $\zeta^2 \neq -1$ are vacuous). For even $s \geq 4$ the displayed set, read literally, is the proper subset $\{v_j : j \text{ even}\}$. It omits the negative roots v_j with j odd; for example it gives $\{16\}$ instead of $\{16, -4\}$ for $s = 4$.*

Proof. (1) $0 < \cos(\pi j/s)$ for $0 \leq j < s/2$, and these cosines strictly decrease in j.

(2) Use the torus representation from the proof of [21, Theorem 2.3] (also [19]),

$$A_s(n) = \int_{\mathbb{T}^{s-1}} \Phi_s^n \, \frac{d\theta}{(2\pi)^{s-1}}, \qquad \Phi_s = 2^s \prod_{i<s} \cos(\theta_i/2)\cos\Big(\tfrac{1}{2}\sum_i \theta_i\Big).$$

That proof classifies all critical points on the nonzero fibres. The critical values are exactly the v_j. Each fibre $\Phi_s^{-1}(v_j)$ contains one or two nondegenerate critical points of the same type, a conjugate pair or the single point at $\omega = 1$. Lemma II.5.3(a) with $g = 1$ therefore gives m_s distinct visible singular points $1/v_j$. Stoll's theorem [1] provides a recurrence of width m_s. Lemmas II.5.2 and II.5.5 now give all assertions of (2).

(3) By [55, Remark 6] and [2, Theorem 1.1] the minimal telescoper L has order m_s. By [2, Theorem 2.1] (the coefficient of t^0 in its identity (14)), every telescoper P with $P\binom{n}{k}^s = \Delta_k\big(R\binom{n}{k}^s\big)$, R rational, annihilates $A_s(n)$ for all large n; in particular L does. The monic minimal recurrence of A_s also has order m_s and is unique, so the two monic operators coincide; this is right division with a zero remainder. Part (2) applies to L. The value at $\omega = e^{2\pi ij/s}$ is $(1+\omega)^s = e^{\pi ij}(2\cos(\pi j/s))^s = v_j$, and ω, $\bar\omega$ give the same value. With $\zeta = e^{\pi ij/s}$ one has $\zeta^s(\zeta+\zeta^{-1})^s = (1+\zeta^2)^s$.

(4) First let s be odd and $\zeta^{2s} = 1$. If $\zeta^s = -1$, replace ζ by $-\zeta$. Since s is odd, this replacement leaves $\zeta^s(\zeta+\zeta^{-1})^s$ unchanged. So one may assume $\zeta^s = 1$, and the condition $\zeta \neq -1$ is then automatic; this is the displayed set.

Now let s be even and $\zeta = e^{2\pi ij'/s}$. Then $(\zeta+\zeta^{-1})^s = (2\cos(2\pi j'/s))^s = (2\cos(\pi j/s))^s$, where $j \in [0, s/2]$ is the representative of $\pm 2j'$ modulo s. This j is even, because s is even. The condition $\zeta^2 \neq -1$ excludes exactly $j = s/2$, where the value is 0. The condition $\zeta \neq -1$ removes no value, since $\zeta = 1$ also gives 2^s. So the displayed set is $\{v_j : j \text{ even},\ 0 \leq j < s/2\}$, whose elements are positive. Every v_j with j odd is negative, and $j = 1$ satisfies $1 < s/2$ when $s \geq 4$. □

Remark II.5.8. The characteristic polynomials obtained directly from Theorem II.5.7 are $(x-8)(x+1)$, $(x-16)(x+4)$, $(x-32)(x^2+11x-1)$, $(x-64)(x-1)(x+27)$, $(x-128)(x^3+57x^2-289x-1)$ and $(x-256)(x-16)(x^2+136x+16)$ for $s=3,\dots,8$. These identities can also be checked by the exact product

$$\chi_{\mathrm{monic}}(x)^2=(x-2^s)\prod_{\substack{\omega^s=1\\ \omega\neq -1}}(x-(1+\omega)^s).$$

The displayed set in Problem II.5.6 gives only two roots for $s=6$ and for $s=8$, whereas the correct counts are three and four. The theorem reads these values from the critical-point equations without computing a telescoper. This does not settle the separate methodological question of extracting the data directly from a difference module, nor the statement about the antisymmetric module N_-.

II.5.3 F2. Diagonal rook walks: what singularity counting can and cannot give

Let $R_d(n)$ be the number of rook paths from $(0,\dots,0)$ to $(n,\dots,n)$ in $\mathbb{Z}^d$. Each step moves any positive distance along one coordinate direction. Equivalently, $R_d(n)=[x_1^n\cdots x_d^n]\,\big(1-\sum_i x_i/(1-x_i)\big)^{-1}$.

Problem II.5.9 (Kauers–Zeilberger [56, §2])**.** The recurrences *"turn out (empirically) to have order d"* for $2\le d\le 12$. The authors suggest that one may be able to *"prove this sharp upper bound in general"*; they refer to a priori upper bounds and pose no lower-bound question. A proof of the recurrences for $4\le d\le 12$ is Problem 17 of [57]. The earlier manuscript [22] reports the upper bound d for all d and exactness for $d\le 3$; these additional claims are not proved anew here. See also the published source [58] for $d=3$. The question whether $\mathrm{w}(R_d)\ge d$ for $d\ge 4$ is our reading of the empirical statement.

Proposition II.5.10. *For $n\ge 1$, $R_2(n)=\frac{1}{2\pi}\int_0^{2\pi}(5+4\cos\theta)^{n-1}(2+2\cos\theta)\,d\theta$. Lemma II.5.3(a), applied to $R_2(n+1)=\frac{1}{2\pi}\int\Phi^n\,\frac{\Phi-1}{2}\,d\theta$ with $\Phi=5+4\cos\theta$ (a sequence with the same width and the same nonzero singular points), works at the maximum value 9 (amplitude 4) and at the minimum value 1, where the amplitude $(\Phi-1)/2\ge 0$ vanishes. It shows $\mathrm{w}(R_2)\ge 2$, and R_2 satisfies a recurrence of width 2.*

Proof. Let $u(n)=\sum_k\binom{n}{k}^2 4^k=\mathrm{CT}\big((1+2x)(1+2/x)\big)^n=\frac{1}{2\pi}\int(5+4\cos\theta)^n d\theta$. The generating function of u is $((1-t)(1-9t))^{-1/2}$. Taking the diagonal of $(1-x)(1-y)/(1-2x-2y+3xy)$ gives

$$\sum_{n\ge 0}R_2(n)t^n=\frac{1}{2}\left(1+\sqrt{\frac{1-t}{1-9t}}\right).$$

For example this follows by substituting $y=t/x$ and taking the constant term, using the root of the resulting quadratic that tends to zero with t. Comparison with the generating function of u gives $2R_2(n)=u(n)-u(n-1)$ for $n\ge 1$. Differentiating the displayed algebraic function and comparing coefficients gives the explicit width-2 recurrence

$$(n+1)R_2(n+1)-(10n+4)R_2(n)+9(n-1)R_2(n-1)=0\qquad(n\ge 1).$$

□

Remark II.5.11 (The obstruction for $d=4,5$). The source manuscript reports the following leading symbols for the order-d operators discussed in [22, Proposition 2.12]:

$$H_2\propto(z-1)(9z-1),\quad H_3\propto(z-1)(3z-2)(64z-1),$$

$$H_4 \propto (z-1)^2(4z-1)(625z-1), \qquad H_5 \propto (z-1)^2(4z-3)(125z-8)(7776z-1).$$

The present consolidation does not reproduce or independently certify the full operators for $d = 4, 5$. Accordingly the following obstruction is conditional on those operator identities, rather than a new certified recurrence computation: if the stated symbols are correct, the singularity budget places all nonzero finite singularities among only three points for $d = 4$ and four for $d = 5$. A method that counts only distinct points cannot then prove lower bounds four and five, respectively. The unconditional $d = 2$ result was proved explicitly above. No minimal-order claim for $d \geq 4$ is made here.

The pattern has a heuristic explanation, not used above. The critical points of $x_1 \cdots x_d$ on the smooth part of $\{\sum_i x_i/(1-x_i) = 1\}$ satisfy $y_i(1+y_i) = \text{const}$ with $y_i = x_i/(1-x_i)$ and $\sum y_i = 1$. They are indexed by the unordered split $\{k, d-k\}$ $(k \neq d/2)$ of the coordinates y_i between the two roots, and k and $d-k$ give the same value $t_k = \big((k+1)/(d-k+1)\big)^{d-2k}$. This yields $\lceil d/2 \rceil$ values; together with $t = 1$ there are $\lceil d/2 \rceil + 1$ candidates. In the data for $d \leq 5$, $t = 1$ has multiplicity $\lfloor d/2 \rfloor$ in H_d, which accounts for the remaining width.

A proof of the Kauers–Zeilberger lower bound along these lines needs two ingredients, neither established here. The first is a multiplicity-sensitive budget, such as the Mellin-rank formula used in earlier manuscripts of this series for connections on $\mathbb{G}_m$ [20], which counts vanishing cycles. The second is a proof that the branches of $\sum R_d(n)t^n$ span $\lfloor d/2 \rfloor$ independent singular directions at $t = 1$.

II.5.4 F3. An additional theorem: invariants of G_2 in tensor powers of V_7

Let $T(n) = \dim(V_7^{\otimes n})^{G_2}$ (OEIS A059710 [63]), so $T = 1, 0, 1, 1, 4, 10, 35, 120, 455, \ldots$. Mihailovs conjectured the recurrence

$$(n+5)(n+6)T(n) = 2(n-1)(2n+5)T(n-1)+(n-1)(19n+18)T(n-2)+14(n-1)(n-2)T(n-3),$$

with $n \geq 3$. Bostan, Tirrell, Westbury and Zhang proved it [59, Theorem 1.3]. We found no statement in the literature about its minimality. The following result is therefore *not* a closure of an explicitly posed problem.

Theorem II.5.12. *1. Put*

$$\Phi(\theta) = 1 + 2\cos\theta_1 + 2\cos\theta_2 + 2\cos(\theta_1+\theta_2), \qquad g(\theta) = \frac{1}{12}\prod_{\alpha\in R^+} 4\sin^2\frac{\alpha(\theta)}{2},$$

where R^+ consists of the short forms θ_1, θ_2, $\theta_1+\theta_2$ and the long forms $\theta_1-\theta_2$, $2\theta_1+\theta_2$, $\theta_1+2\theta_2$. Then

$$T(n) = \int_{\mathbb{T}^2} \Phi^n g\,\frac{d\theta}{4\pi^2} = \frac{1}{12}\,\mathrm{CT}\Big[\chi^n \prod_{\alpha\in R}(1-x^\alpha)\Big].$$

Here χ is the sum of the seven monomials x^w over the weights $w \in \{0, \pm e_1, \pm e_2, \pm(e_1+e_2)\}$, $R = R^+ \cup (-R^+)$, and $x^\alpha = e^{i\alpha(\theta)}$.

2. $\mathrm{w}(T) = 3$. *Every recurrence of width 3 for T, in particular Mihailovs', has characteristic polynomial proportional to $(x-7)(x+1)(x+2)$. The nonzero singular points of all branches of $\sum T(n)t^n$ are exactly $1/7$, $-1/2$ and -1.*

Proof. (1) This is the Weyl integration formula for G_2, with $|W| = 12$. The coordinates are those in which the short roots are $\pm e_1$, $\pm e_2$, $\pm(e_1+e_2)$; these generate the root lattice, which for G_2 is the character lattice of the maximal torus. The weights of V_7 are 0 and the short roots, and V_7 is self-dual, so χ is real on the torus. The constant term is 12 for $n = 0$, also forced by the Weyl integration formula applied to the constant function. The recurrence for all n is the published theorem cited above; finite term checks are not needed.

(2) *Classification of critical points.* In $x = e^{i\theta_1}$, $y = e^{i\theta_2}$ the critical-point equations $x\Phi_x = y\Phi_y = 0$ have the lex Gröbner basis

$$(x-y)(x+y), \quad (x-y)(y-1)(y+1), \quad (y-1)(y+1)(y^2+y+1).$$

So there are exactly six critical points in $(\mathbb{C}^*)^2$, all on the torus: $(0,0)$, $(\pi,0)$, $(0,\pi)$, (π,π) and $\pm(2\pi/3, 2\pi/3)$. Their values are $7, -1, -1, -1, -2, -2$. The Hessians are $\left(\begin{smallmatrix} -4 & -2 \\ -2 & -4 \end{smallmatrix}\right)$ (a maximum), $\left(\begin{smallmatrix} 4 & 2 \\ 2 & 0 \end{smallmatrix}\right)$, $\left(\begin{smallmatrix} 0 & 2 \\ 2 & 4 \end{smallmatrix}\right)$, $\left(\begin{smallmatrix} 0 & -2 \\ -2 & 0 \end{smallmatrix}\right)$ (saddles, determinant -4) and $\left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ (minima). All are nondegenerate, and the values $7, -1, -2$ are distinct.

Visibility. The amplitude $g \geq 0$ is real analytic and not identically zero near any point. Lemma II.5.3(a) therefore gives singular points at $1/7$, at the maximum, where g vanishes to order 12, and at $-1/2$, at the two minima, where g vanishes to order 6.

For the saddle value -1, put $u = \theta_1 - \pi$ and $v = \theta_2$. At $(\pi, 0)$ the roots θ_2 and $2\theta_1 + \theta_2$ vanish modulo 2π. The expansion of $12g$ starts in degree 4 with $256v^2(2u+v)^2$, and $\Phi + 1 = 2u(u+v) + O(3)$. In the linear Morse coordinates $Y_1 = 2u$, $Y_2 = u + v$, for which the quadratic part is Y_1Y_2, this form becomes $16Y_1^4 - 128Y_1^2Y_2^2 + 256Y_2^4$. Since g vanishes to order 4, the degree-4 Taylor part of $\widehat{g}$ in analytic Morse coordinates with this linear part equals this form multiplied by $|\det \partial(u,v)/\partial(Y_1,Y_2)|/(48\pi^2) = 1/(96\pi^2)$. The parts of lower degree vanish. Hence $c_{00} = c_{11} = 0$ and $c_{22} = -128/(96\pi^2) \neq 0$. An exact analytic Morse chart confirms this without the argument on higher-order corrections: by the identity $\Phi + 1 = 8\cos\frac{\theta_1}{2}\cos\frac{\theta_2}{2}\cos\frac{\theta_1+\theta_2}{2}$, the functions $Y_1 = 4\sin(u/2)$, $Y_2 = 2\sin((u+v)/2)\cos(v/2)$ satisfy $\Phi + 1 = Y_1Y_2$ exactly near $(\pi, 0)$, and in this chart the degree-4 part of $48\pi^2\widehat{g}$ is $8Y_1^4 - 64Y_1^2Y_2^2 + 128Y_2^4$.

The maps $(\theta_1, \theta_2) \mapsto (\theta_2, \theta_1)$ and $(\theta_1, \theta_2) \mapsto (-\theta_1 - \theta_2, \theta_2)$ preserve Haar measure, Φ and g; because they permute the six short-root weights and the root factors in the squared Weyl denominator. They permute the three saddles transitively. Each such map carries the localised integral at one saddle to the localised integral at another. The contributions $A_{s,2} = 2\pi i^2\, 2!\, c_{22}^{(s)}$ are therefore equal at the three saddles. Composing Morse coordinates with the symmetry also gives equal coefficients $c_{22}^{(s)}$, so $\sum_s c_{22}^{(s)} = 3c_{22} \neq 0$. Lemma II.5.3(b) gives the singular point -1.

Conclusion. There are three distinct visible singular points, and [59, Theorem 1.3] provides width 3. Lemma II.5.5 completes the proof. As a check, the characteristic polynomial of Mihailovs' recurrence is $x^3 - 4x^2 - 19x - 14 = (x-7)(x+1)(x+2)$, by direct expansion. □

Remark II.5.13. The third-order differential operator for $\sum T(n)t^n$ factors [59, §3.4]. This does not contradict Theorem II.5.12, because minimal width and minimal differential order are different invariants. Without Lemma II.5.3(b), visibility of the saddle value would be missed. Remark II.5.4 shows that a vanishing amplitude genuinely requires the computation of c_{22}.

II.5.5 Constant-coefficient filters and characteristic spectra

A natural way to try to simplify a recurrence is to subtract several shifted copies of its solution so that the leading exponential contributions cancel. For the Franel and G_2 moments this never reduces the minimal width. The Franel case recasts the filter result of Part I; the G_2 conclusion is an additional application. This is a stability theorem for the families considered here, not a claim to resolve a separately published conjecture.

Theorem II.5.14 (Rigidity under every nonzero polynomial shift filter)**.** *Let* $Q(x) = \sum_{j=0}^{d} q_j x^j \in \mathbb{C}[x] \setminus \{0\}$.

1. *For* $s \geq 2$ *define*

$$B_{s,Q}(n) = \sum_{j=0}^{d} q_j A_s(n+j).$$

Then $\mathrm{w}(B_{s,Q}) = \lfloor (s+1)/2 \rfloor$. *Every recurrence of this width has characteristic polynomial proportional to*

$$\prod_{0 \le j < s/2} \left(x - (-1)^j (2\cos(\pi j/s))^s \right).$$

2. *For the* G_2 *invariant dimensions define* $B_{G_2,Q}(n) = \sum_{j=0}^{d} q_j T(n+j)$. *Then* $\mathrm{w}(B_{G_2,Q}) = 3$, *and every recurrence of width three has characteristic polynomial proportional to* $(x-7)(x+1)(x+2)$.

In both cases the nonzero finite singular points of continued generating functions are exactly the same as for the unfiltered sequence. This remains true when Q *vanishes at every one of the critical values.*

Proof. Write $a(n) = \int \Phi^n g \, d\mu$ for either moment sequence and $F(z) = \sum_{n \ge 0} a(n) z^n$. The preceding visibility proofs exhibit r distinct singular points $z = 1/v$, where r is the asserted width. We first check that these are not merely poles: the indicated continued branches are not meromorphic there.

Indeed, the measure $\nu = \Phi_*(g\mu)$ has no atom at v. The level set of the nonconstant real-analytic function Φ has volume zero, and $g\mu$ has a continuous density. For its Cauchy transform $G(w) = \int (w-u)^{-1} \, d\nu(u) = w^{-1} F(1/w)$, dominated convergence gives

$$\lim_{y \downarrow 0} iy \, G(v + iy) = \nu(\{v\}) = 0,$$

since $|iy/(v+iy-u)| \le 1$. If G were meromorphic at v, this limit would exclude every pole. It would therefore be holomorphic, contradicting the visibility result. The same conclusion holds for F at $1/v$.

Let $b(n) = \sum_{j=0}^{d} q_j a(n+j)$ and $B(z) = \sum_{n \ge 0} b(n) z^n$. Shifting a power series gives the exact identity

$$B(z) = Q(1/z) F(z) - \sum_{j=0}^{d} q_j z^{-j} \sum_{k=0}^{j-1} a(k) z^k.$$

The second term is a Laurent polynomial and is holomorphic at every nonzero point. Multiplying a nonmeromorphic germ by a nonzero holomorphic function cannot make it meromorphic: otherwise division by that function would make the original germ meromorphic. Thus all r visible singular points persist, even at zeros of $Q(1/z)$. The singularity budget proves $\mathrm{w}(b) \ge r$.

For the reverse inequality, a recurrence of width r for a expresses all sufficiently high forward shifts of a in the $\mathbb{C}(n)$-span of $a(n), \dots, a(n+r-1)$. This span is stable under forward shift, with the coefficients shifted accordingly. Consequently $b(n), \dots, b(n+r)$ are linearly dependent over $\mathbb{C}(n)$. Clearing denominators and removing unused initial or final shifts gives a recurrence of width at most r. Hence the width is exactly r. The rigidity lemma determines the characteristic polynomial, and the budget then excludes any additional nonzero finite singular point. □

Example II.5.15. For $s = 3$, the filtered sequence

$$A_3(n+2) - 7A_3(n+1) - 8A_3(n)$$

still has minimal width two, although the filter $Q(x) = (x-8)(x+1)$ vanishes at both characteristic values. Similarly $T(n+3) - 4T(n+2) - 19T(n+1) - 14T(n)$ still has minimal width three. Cancellation of a leading asymptotic coefficient does not remove the nonmeromorphic singularity that forces the recurrence order.

II.5.6 Record of the other questions found

The following questions were located in the cited sources. None is settled here; this list is not an exhaustive certification of their present status.

1. *Three-dimensional queen paths.* Bostan, Chyzak, van Hoeij and Pech write that their equations for diagonal queen walks are *"very likely"* of minimal order [58]. The generating function is a diagonal of a rational function without a known real moment representation, and Lemma II.5.3 does not apply. We did not attempt this question.
2. *The d-dimensional fcc lattice.* Hassani, Koutschan, Maillard and Zenine confirm for $8 \le d \le 12$ a formula $q = d^2/4 - d/2 + 17/8 - (-1)^d/8$, conjectured in their earlier work on $d = 7$, for the order of the minimal *differential* operator [61]. Singularity counting bounds recurrence width, not differential order, so it gives nothing here.
3. *Regev sums of odd exponent.* Mishna asks whether the annihilating operators of $y_k^{(\beta)}(n) = \sum_{\lambda \vdash n,\ \ell(\lambda) \le k} (f^\lambda)^\beta$ can be usefully bounded for arbitrary β [62, §4]. Even β was treated in an earlier manuscript of this series [20]. For odd β the Weyl-antisymmetric extension of the summand, which gives a torus moment for even β, sums to zero. For $k = 2$, for instance, $y_2^{(\beta)}(n)$ is a half-range sum of the antisymmetric summand $(\binom{n}{j} - \binom{n}{j-1})^\beta$. We found no real moment representation for odd β.
4. *Questions of Chen–Kauers and Straub–Zudilin.* Problem 5 of [57] concerns removable singularities in n of telescopers, and Problem 17 concerns upper bounds. Remarks 2.2 and 4.1 of [2] concern denominators and coefficient degrees. These lie outside a method that controls only the leading symbol.
5. *Other sums.* We found no explicitly posed minimal-order questions for $\sum_k \binom{n}{k}^a \binom{n+k}{k}^b$, for $\sum_k \binom{n}{k}^s \binom{2k}{k}$ with $s \ge 3$, or for sums of powers of trinomial coefficients (OEIS A132303–A132305 [63]). For the first two families we know of no Laurent representation that is real on a torus. For $s = 2$ the second family is a banana-type moment, which was treated earlier. Trinomial power sums do have such representations, $\sum_k ([x^k](1 + x + x^2)^n)^s = \int_{\mathbb{T}^{s-1}} \left[\prod_{i<s}(1 + 2\cos\theta_i) \cdot (1 + 2\cos\sum_i \theta_i)\right]^n \frac{d\theta}{(2\pi)^{s-1}}$, but no recurrence or order question for them is recorded in the sources we found.

Part III

Zero geometry, arithmetic separation, and transcendence

III.1 Conventions, scope, and the structure of the consolidation

Throughout we use the *shortcut* Collatz map

$$T(n) = \begin{cases} n/2, & n \text{ even}, \\ (3n+1)/2, & n \text{ odd}. \end{cases}$$

For a positive integer N whose orbit reaches 1, define

$$\tau(N) = \min\{j \geq 0 : T^j(N) = 1\}, \qquad P_N(z) = \sum_{j=0}^{\tau(N)} T^j(N) z^j.$$

Thus $P_1 = 1$. For $N > 1$ the polynomial is monic with positive integer coefficients, constant term N, and penultimate coefficient 2. The variable z is complex; it is not the integer input N. A zero attached to a prime start is not itself a "prime complex number." Every assertion concerning a full P_N is restricted to terminating starts, unless an explicit family is constructed to terminate. In particular, infinitely many terminating prime starts are not assumed.

Let $\mathcal{Z}$ be the union of all these zero sets, and let $\mathcal{Z}'$ denote its topological derived set: limits of sequences of distinct points of $\mathcal{Z}$. The equality $\mathcal{Z}' = \{2/3 \leq |z| \leq 2\}$ does not assert that every individual zero lies in that annulus, or that every point of the annulus is an exact zero. Indeed, the smallest zero has modulus below 2/3, and no individual zero lies exactly on the inner circle. Pooled accumulation and the limiting distribution of all roots of one polynomial are also different questions.

This edition integrates the three preceding research notes on zero geometry, prime-only loci, and prime radii with Euler's constant, together with the subsequent angle, logarithm, primality, and external-constant investigations. Repeated definitions and arguments are combined where practical. Exploratory questions are retained as questions, and failed proposals are retained as limitations rather than promoted to theorems. The earlier recurrence, difference-triangle and prime-chain results are included in Parts I and II of the present consolidation.

Question	Exact conclusion and scope
What does the annulus mean?	Complete derived-set classification, including composite density; not an all-roots lower bound.
What do the 403 cases prove?	An analytic cutoff reduces a global minimum problem to $2 \leq N \leq 404$; the residual check uses integers only.
Can folds or a line isolate primes?	Fourfold overlaps and axis conditions are algebraic symmetries. Open prime-only regions and certain fixed-prefix rules are excluded.
Do primes have exact fingerprints?	A nonunit prime factor, separated radii, finite uniqueness windows, and sparse exact collisions; composite inputs are not universally excluded.
Can logarithms decide transcendence?	Complete algebraic-versus-transcendental classification for specified linear-form ratios; not for arbitrary external constants.
Why does γ appear?	A harmonic endpoint singularity survives a controlled deformation of a regular polygon.
Does this give instant prime testing?	Exact reformulations and classical certificate families exist, but no new general constant-time or single-step algorithm follows.

Further results proved in this edition. The weighted inverse tree gives a lower growth exponent $s_* \approx 0.1935935685$ for the number of boundary-producing starts (Theorem III.4.9). The joint independence theorem treats all nonunit conjugation classes from pairwise coprime odd starts, including composites (Theorem III.15.3); it yields an exact classification of ratios of algebraic linear forms (Corollary III.15.4). The Euler limit gains its full m^{-1} coefficient and an explicit accelerated estimator in Section III.19. These are additional deductions established here. Their historical novelty has not been certified by an exhaustive literature review.

Established inputs and verification. The polynomial framework and boundary equality classification originate in Hohertz–Kalantari and Hohertz [25, 5]. Gelfond–Schneider, Baker, real quantifier elimination, classical cosecant asymptotics, self-similar dimension results, and standard primality criteria are explicitly credited where used. Finite certificates are distinguished from analytic proofs. The code listings are part of the document, and compilation executes none of them. Some optional exact symbolic checks use SymPy; the 403-case certificate and the displayed primality certificates use only Python's standard library. No external script, image, data file, or bibliography file is needed to compile this `.tex` source.

III.2 The accumulation annulus and prescribed parity patterns

For the shortcut map $T(n) = n/2$ for even n and $T(n) = (3n+1)/2$ for odd n, let

$$\tau(N) = \min\{j : T^j(N) = 1\}, \qquad P_N(z) = \sum_{j=0}^{\tau(N)} T^j(N) z^j.$$

Only starts with $\tau(N) < \infty$ enter statements about P_N; $P_1 = 1$. The polynomial is monic with positive integer coefficients. Let $\mathcal{Z}$ be the union of their zero sets, and $\mathcal{Z}'$ its derived set, consisting of limits of distinct zeros. A derived set need not contain isolated zeros, and it does not specify a probability distribution. Both distinctions will matter below.

Lemma III.2.1 (Universal outer bound and inner finite cutoff)**.** *Every zero of P_N has modulus at most 2. If $0 \le s < 2/3$, then*

$$P_N(\xi) = 0, \quad |\xi| \le s \quad \Longrightarrow \quad N + 1 \le \frac{1 + 3s/2}{(1 - 3s/2)(1 - s)}. \tag{III.2.1}$$

Consequently, every compact subset of $|z| < 2/3$ contains only finitely many zeros pooled over all terminating starts.

Proof. Write $a_j = T^j(N)$ and $d = \tau(N)$. Since $a_j/a_{j+1} \le 2$, the Eneström–Kakeya upper bound gives $|\xi| \le 2$. Here is its elementary argument in this case. The coefficients $c_j = 2^j a_j$ of $H(w) = P_N(2w)$ satisfy $0 < c_0 \le \cdots \le c_d$. Thus

$$(1 - w)H(w) = c_0 + \sum_{j=1}^{d} (c_j - c_{j-1}) w^j - c_d w^{d+1}.$$

If $H(w) = 0$ and $|w| > 1$, taking absolute values gives $c_d |w|^{d+1} \le c_d |w|^d$, a contradiction.

For the inner bound put

$$\widetilde{P}_N(z) = \sum_{j=0}^{d} (a_j + 1) z^j, \qquad b_j = \frac{a_j + 1}{N + 1} (2/3)^j.$$

The inequality $a_{j+1}+1 \le \frac{3}{2}(a_j+1)$, with equality on odd branches, gives $1=b_0 \ge b_1 \ge \cdots \ge b_d > 0$. Set $b_{d+1}=0$ and $e_j = b_{j-1}-b_j$ for $1 \le j \le d+1$. Then $e_j \ge 0$ and $\sum_j e_j = 1$. With $w = 3z/2$ we have the exact identity

$$(1-w)\frac{\widetilde{P}_N(z)}{N+1} = 1 - \sum_{j=1}^{d+1} e_j w^j.$$

When $|z| \le s < 2/3$, its right side has modulus at least $1-3s/2$. Since $|1-w| \le 1+3s/2$, this proves

$$|\widetilde{P}_N(z)| \ge (N+1)\frac{1-3s/2}{1+3s/2}, \qquad |\widetilde{P}_N(z) - P_N(z)| = \left|\sum_{j=0}^{d} z^j\right| \le \frac{1}{1-s}.$$

At a zero of P_N, comparison yields (III.2.1). For each fixed $s < 2/3$ it allows only finitely many integer starts; each terminating polynomial has finitely many zeros. Every compact subset of the open disk lies in such a closed subdisk, proving the final assertion. □

Corollary III.2.2 (A lower bound depending only on the starting value)**.** *Every zero ξ of P_N, for terminating $N \ge 2$, satisfies*

$$|\xi| \ge R_N := \frac{4N}{5N+8+\sqrt{N^2+56N+64}}, \qquad R_N \longrightarrow \frac{2}{3}. \tag{III.2.2}$$

Proof. For $r = |\xi| < 2/3$, rearranging the preceding cutoff gives $3(N+1)r^2 - (5N+8)r + 2N \le 0$. Its smaller root is the displayed rationalized expression. For $r \ge 2/3$ the inequality is automatic. □

Theorem III.2.3 (Every fixed parity prefix has the same accumulation annulus)**.** *Let v be a fixed binary word of length $h \ge 0$. Let $\mathcal{Z}_v$ be the union of zeros of P_N over the starts $N > 1$ whose first h actual parity bits are v and whose orbits reach 1 after at least h steps. Then*

$$\mathcal{Z}_v' = \{z \in \mathbb{C} : 2/3 \le |z| \le 2\}.$$

Equivalently, the same assertion holds after restricting the starting values to any specified residue class modulo any fixed power of 2. In every open set meeting the open annulus, zeros can be obtained from arbitrarily large starting values in the prescribed class.

Proof. First we realize finite words by terminating orbits. If a parity word u has length L and weight $R > 0$, formal composition gives

$$T_u(x) = \frac{3^R x + b_u}{2^L}, \qquad b_u = \sum_{u_j=1} 2^j 3^{\sum_{i>j} u_i}, \qquad 3 \nmid b_u.$$

Since 2 generates the unit group modulo 3^R, infinitely many nonnegative integers t satisfy $2^{L+t} \equiv b_u \pmod{3^R}$. For these t,

$$N_{u,t} = \frac{2^{L+t} - b_u}{3^R}$$

is integral, follows exactly the parity word u, and then reaches 2^t. For sufficiently large t, none of the prescribed prefix values has yet reached 1, and the remaining orbit is the halving tail $2^t, 2^{t-1}, \ldots, 1$. The congruence implying the actual parity word follows by induction modulo 2 through its branches.

Choose an additional word $w = (\varepsilon_0, \ldots, \varepsilon_{k-1})$ of length k and weight $1 \le r \le k$. Put

$$\alpha_j = \frac{3^{\varepsilon_0+\cdots+\varepsilon_{j-1}}}{2^j}, \quad q = \frac{3^r}{2^k}, \quad Q(z) = qz^k, \quad B(z) = \sum_{j=0}^{k-1} \alpha_j z^j,$$

$$E(z) = \sum_{\varepsilon_j=1} \alpha_j z^{j+1}, \qquad A(z) = (1 - z/2)B(z) = 1 - Q(z) + E(z).$$

The nonzero polynomial E records the odd branches. For the fixed prefix $v = (\nu_0, \dots, \nu_{h-1})$, define

$$\gamma_j = \frac{3^{\nu_0+\cdots+\nu_{j-1}}}{2^j}, \qquad C_v(z) = \sum_{j=0}^{h-1} \gamma_j z^j.$$

When $h = 0$, use $C_v = 0$ and $\gamma_h = 1$.

Realize the concatenated word $u = vw^m$, followed by a halving tail. For fixed m and $t \to \infty$ along its permissible progression, the normalized orbit polynomials converge locally uniformly on $|z| < 2$ to

$$g_{v,m}(z) = C_v(z) + \gamma_h z^h f_m(z), \qquad f_m(z) = B(z) \sum_{a=0}^{m-1} Q(z)^a + \frac{Q(z)^m}{1 - z/2}.$$

To justify this convergence explicitly, write the prescribed affine iterates as $T^j(N) = \lambda_j N + \beta_j$ for $0 \le j \le L$, where $L = h + mk$ is fixed and $\lambda_L = 3^R/2^L$. The exact expression is

$$\frac{P_{N_{u,t}}(z)}{N_{u,t}} = \sum_{j=0}^{L-1} \left(\lambda_j + \frac{\beta_j}{N_{u,t}}\right) z^j + \left(\lambda_L + \frac{\beta_L}{N_{u,t}}\right) z^L \sum_{a=0}^{t} (z/2)^a.$$

For $|z| \le s < 2$, its difference from the corresponding limit is at most

$$\frac{C_{u,s}}{N_{u,t}} + \frac{\lambda_L s^L (s/2)^{t+1}}{1 - s/2},$$

with $C_{u,s}$ independent of t. This tends to zero. Grouping the limiting slopes into their initial prefix and m repeated blocks gives the displayed formula for $g_{v,m}$.

The key identity is

$$(1 - z/2)(1 - Q(z))g_{v,m}(z) = F_v(z) - Q(z)^m J_v(z), \tag{III.2.3}$$

where

$$F_v = (1 - z/2)(1 - Q)C_v + \gamma_h z^h A, \qquad J_v = \gamma_h z^h E.$$

Here $F_v(0) = 1$ and J_v is not the zero polynomial.

Set $\rho = 2/3^{r/k} < 2$, so $|Q(z)| = 1$ when $|z| = \rho$. Choose z_0 on this circle with $F_v(z_0)J_v(z_0)(1 - Q(z_0)) \neq 0$; only finitely many points are excluded. Near z_0, choose a holomorphic logarithm $\ell(z) = \log(F_v(z)/J_v(z))$ and write $Q(z_0) = e^{i\theta}$. Choose integers a_m with $\delta_m = 2\pi a_m - m\theta \in [-\pi, \pi]$. For sufficiently large m, Rouché's theorem applied on a radius-one circle centered at $\ell(z_0) + i\delta_m$ gives a bounded solution of

$$u = \ell\big(z_0 e^{u/(km)}\big) + i\delta_m.$$

Indeed, on these circles the perturbation $\ell(z_0 e^{u/(km)}) - \ell(z_0)$ is uniformly $O(1/m)$, whereas $u - \ell(z_0) - i\delta_m$ has modulus 1. Thus $z_m = z_0 e^{u_m/(km)} \to z_0$ satisfies $Q(z_m)^m = F_v(z_m)/J_v(z_m)$. Equation (III.2.3) implies $g_{v,m}(z_m) = 0$. For each such fixed m, a second application of Rouché's theorem transfers a nearby zero to actual normalized polynomials with arbitrarily large permissible t. This application is valid on a sufficiently small circle about z_m: the limit is holomorphic there and is not identically zero, since $g_{v,m}(0) = 1$, and the convergence just proved is uniform on that circle. Every resulting start still has prefix v.

It follows that $\overline{\mathcal{Z}_v}$ contains every circle of radius $2/3^{r/k}$, for rational $r/k \in (0, 1]$. Their radii are dense in $[2/3, 2]$. Density in the open annulus implies that each nonempty open subset of it contains infinitely many distinct points of $\mathcal{Z}_v$, so the entire closed annulus lies in its derived set.

For the reverse inclusion, Lemma III.2.1 places every zero in $|z| \le 2$ and makes the pooled zeros locally finite in $|z| < 2/3$, including after restriction to the prefix. No other derived points are possible.

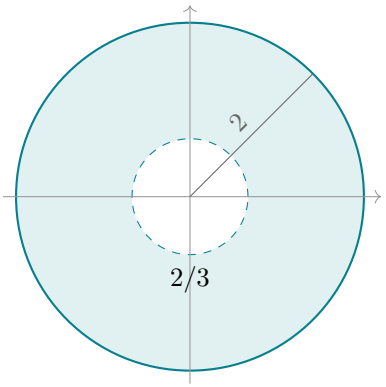
Finally, length-h parity words are in bijection with residue classes modulo 2^h. Inductively, a fixed prefix gives one residue class, and its two lifts modulo 2^{h+1} change the next iterate by the odd integer $3^{\nu_0+\cdots+\nu_{h-1}}$, so precisely one lift has each next parity. Starts reaching 1 before h steps satisfy $N \le 2^{\tau(N)} < 2^h$, since $n \le 2T(n)$. Including these finitely many starts changes the zero set by only finitely many points and does not change its derived set. This proves the equivalent formulation. □

Corollary III.2.4 (Complete classification of the pooled derived set)**.**

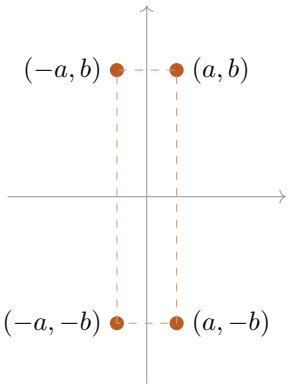
$$\mathcal{Z}' = \{z \in \mathbb{C} : 2/3 \le |z| \le 2\}. \tag{III.2.4}$$

The lower bound (III.2.2) *and the finite cutoff* (III.2.1) *apply to every individual zero.*

Proof. Take the empty prefix in Theorem III.2.3. □



The derived set $\mathcal{Z}'$
not the set of exact zeros



P_{10}: four distinct zeros
fold to (a, b)

Figure 1: Two different geometric objects. Left: the exact accumulation annulus; its inner circle is approached but contains no individual zero. Right: the four zeros of $z^4 + 4z^2 + 5$, a factor of P_{10}, with $a = \sqrt{(\sqrt{5}-2)/2}$ and $b = \sqrt{(\sqrt{5}+2)/2}$. The two axis folds identify this quartet even though the start 10 is composite. Coordinates in the drawing are rounded only for plotting.

Corollary III.2.5 (Even composites alone fill the annulus densely)**.** *Let $\mathcal{Z}_{\text{even,comp}}$ be the union of zeros belonging to terminating even composite starts. Then*

$$\mathcal{Z}'_{\text{even,comp}} = \{z : 2/3 \le |z| \le 2\}.$$

The same conclusion holds if the starts are required to be multiples of any fixed power 2^h, $h \ge 1$, with finitely many small starts discarded.

Proof. Take the prefix $v = (0)$, or $v = (0, \ldots, 0)$ of length h. All sufficiently large starts supplied by the proof are even and greater than 2, hence composite. The proof supplies arbitrarily large starts, so discarding any finite number of them does not remove the constructed accumulation points. The reverse inclusion is unchanged. □

Corollary III.2.6 (Composite density in every prescribed parity class)**.** *In Theorem III.2.3, restricting the starts further to composite values leaves the derived set unchanged. In particular, odd composite starts alone have the same derived annulus, even after any fixed finite initial parity pattern beginning with an odd step is imposed.*

Proof. In each fixed-word family used in the theorem, even starts are eventually composite. For an odd-start family choose a permissible tail length t_0 with $N_0 = N_{u,t_0} > 1$ and let $D = \operatorname{ord}_{3^R N_0}(2)$. Along $t = t_0 + jD$ one has

$$2^{L+t} \equiv 2^{L+t_0} = 3^R N_0 + b_u \equiv b_u \pmod{3^R N_0},$$

so $N_{u,t}$ is divisible by N_0. For $j \geq 1$ it exceeds N_0 and is composite. This subsequence has the same locally uniform polynomial limit as the full tail family. The two Rouché arguments and the density conclusion therefore remain valid with composite starts throughout. □

Corollary III.2.7 (No open geometric region certifies a prime start)**.** *Every open subset of* $\mathbb{C}$ *that meets the closed annulus* $2/3 \leq |z| \leq 2$ *contains a zero from an even composite start. Consequently, no nonempty open region inside that annulus, and no open strip of positive width meeting it, can have the property that every Collatz zero it contains comes only from a prime start.*

Proof. Such an open set contains a nonempty open subset of the open annulus. Apply Corollary III.2.5. □

Corollary III.2.6 gives the same obstruction when inputs have first been restricted to odd values, or to any one fixed residue class modulo a power of 2.

This is a finite-precision obstruction: every neighborhood of a proposed prime zero in the annulus also admits composite-start zeros. It does not settle an exact incidence condition on a zero-width line or curve, a test using algebraic information beyond the location, or a test using the entire collection of roots of one polynomial. The isolated region $|z| < 2/3$ is also outside this particular no-open-region conclusion.

Corollary III.2.8 (Arbitrarily narrow approaches to an interior point)**.** *Let* $2/3 < |\zeta| < 2$*. If* U_j *are nonempty open subsets of the annulus with* $\sup_{z \in U_j} |z - \zeta| \to 0$*, then there are distinct Collatz zeros* $\xi_j \in U_j$*, after passing to a subsequence if necessary. In particular, for every* $p > 0$*, zeros approach* 1 *through the cusp*

$$0 < |\operatorname{Im} z|, \qquad |\operatorname{Re} z - 1| < |\operatorname{Im} z|^p.$$

No nonempty open subset of $2/3 < |z| < 2$ *is free of Collatz zeros.*

Proof. Each nonempty open subset of the annulus contains infinitely many distinct zeros, as proved above. Select a zero avoiding the finitely many earlier selections. For the cusp, also impose $0 < |z - 1| < 1/j$; the resulting open set is nonempty for every sufficiently large j. □

This corollary gives a precise interpretation of arbitrary thin approaches to 1. It also rules out any proposed universal zero-free parabolic region whose interior meets the open annulus.

III.3 The smallest Collatz zero and finite-degree exclusions

The inner circle $|z| = 2/3$ describes an accumulation boundary, not a zero-free boundary. In fact, the orbit of 9 gives

$$\begin{aligned} P_9(z) = 9 + 14z + 7z^2 + 11z^3 + 17z^4 + 26z^5 + 13z^6 \\ + 20z^7 + 10z^8 + 5z^9 + 8z^{10} + 4z^{11} + 2z^{12} + z^{13}. \end{aligned}$$

The following result determines the largest open disk centered at the origin that contains no Collatz zero. It combines an analytic reduction with an exhaustive certificate using integer arithmetic; it is therefore explicitly a computer-assisted result.

Theorem III.3.1 (A unique zero of globally smallest modulus)**.** *Let ρ be the unique zero of $P_9(-x)$ in $(13/20, 23/35)$. Among all zeros of all terminating Collatz polynomials,*

$$\min\{|\xi| : P_N(\xi) = 0,\ N \geq 2,\ \tau(N) < \infty\} = \rho.$$

Equality is attained only by the pair $(N, \xi) = (9, -\rho)$. The zero is simple, and ρ is algebraic of degree 13 over $\mathbb{Q}$. Numerically, for orientation only,

$$\rho = 0.656989767756311659235045725399\ldots.$$

Proof. Take $r = 23/35 < 2/3$. The finite-cutoff bound (III.2.1) gives

$$P_N(\xi) = 0, \quad |\xi| \leq 23/35 \quad \Longrightarrow \quad N + 1 \leq \frac{4865}{12}, \quad \text{hence } N \leq 404.$$

Thus an infinite assertion has reduced to 403 specified starting values. Every one of their orbits reaches 1 in at most 91 steps. The exact certificate below establishes that P_9 has precisely one zero in $|z| < 23/35$, counted with multiplicity, and that every other P_N, $2 \leq N \leq 404$, has none in the closed disk. The same computation establishes that no tested polynomial has a zero on its boundary.

Direct substitution gives

$$P_9(-13/20) = \frac{12282961919783787}{81920000000000000} > 0, \qquad P_9(-23/35) = -\frac{393226451789392713}{118272717781982421875} < 0.$$

The intermediate value theorem therefore supplies a negative real zero in that disk. The count of one proves uniqueness and simplicity. All other Collatz zeros have modulus greater than $23/35 > \rho$, proving the minimum assertion.

For the degree assertion, reduction modulo 2 gives

$$f(x) = x^{13} + x^9 + x^6 + x^4 + x^3 + x^2 + 1.$$

It has no linear factor over $\mathbb{F}_2$, since $f(0) = f(1) = 1$. Repeated squaring gives $x^{2^{13}} \equiv x \pmod{f}$. Because 13 is prime, every irreducible factor of f then has degree 1 or 13; absence of linear factors proves that f is irreducible. Gauss's lemma proves the irreducibility of P_9 over $\mathbb{Q}$. The same holds for $P_9(-x)$, proving $[\mathbb{Q}(\rho) : \mathbb{Q}] = 13$. □

Why the disk-count certificate is exact. For a real polynomial $F(z) = a_0 + \cdots + a_d z^d$ with nonzero constant and leading coefficients, write $F^*(z) = z^d F(1/z)$. Suppose $|a_0| > |a_d|$ and put

$$Q(z) = a_0 F(z) - a_d F^*(z).$$

The leading term cancels, so $\deg Q < d$, while $Q(0) = a_0^2 - a_d^2 \neq 0$. If Q has no unit-circle zero, neither does F: a unit-circle zero of a real polynomial is also a zero of its reciprocal. On the unit circle $|F^*| = |F|$, so Rouché's theorem then shows that F and Q have the same number of zeros in $|z| < 1$. If $|a_0| < |a_d|$, first replace F by F^*; the count for F is d minus the count for F^*. Descending to a nonzero constant proves both the root count and absence of boundary zeros by induction. An equality $|a_0| = |a_d|$ would stop this certificate and is explicitly tested; none occurs in the computation reported here.

To count zeros of P_N in $|z| < 23/35$, apply this descent to the integer polynomial $35^d P_N(23z/35)$. Dividing all coefficients by their positive greatest common divisor after each transformation changes no zeros. Thus no rounding, tolerance, numerical root approximation, or unverified primality test enters the certificate.

Complete in-document verification code. The following standard-library Python code is included as part of the paper; no supplementary source file, package, or data file is required. It bounds every orbit computation in advance, verifies every strict comparison, and checks the resulting exact counts. Counts include multiplicities. The irreducibility check uses binary integers to encode polynomials over $\mathbb{F}_2$, with bit j equal to the coefficient of x^j.

```
from math import gcd
from functools import reduce

def orbit(n):
    a = [n]
    for _ in range(91):
        if a[-1] == 1:
            return a
        x = a[-1]
        a.append(x // 2 if x % 2 == 0 else (3*x+1) // 2)
    assert a[-1] == 1
    return a

def count_disk(a):
    count, sign, steps, reversals = 0, 1, 0, 0
    while len(a) > 1:
        while a[-1] == 0:
            a.pop()
        d = len(a) - 1
        if d == 0:
            break
        u, v = a[0], a[-1]
        assert u != 0 and u*u != v*v
        if u*u < v*v:
            a.reverse()
            u, v = v, u
            count += sign*d
            sign = -sign
            reversals += 1
        b = [u*a[j] - v*a[d-j] for j in range(d)]
        g = reduce(gcd, b)
        assert g > 0
        a = [x // g for x in b]
        steps += 1
    assert a[0] != 0
    return count, steps, reversals

rows = []
for n in range(2, 405):
    a = orbit(n)
    d = len(a)-1
    b = [a[j]*23**j*35**(d-j) for j in range(d+1)]
    count, steps, reversals = count_disk(b)
    assert count == (1 if n == 9 else 0)
    assert steps == d
    assert reversals == (1 if n == 9 else 0)
    rows.append((n, d, count, steps, reversals, max(a)))

assert len(rows) == 403
assert max(x[1] for x in rows) == 91
assert sum(x[1] for x in rows) == 13563
assert max(x[5] for x in rows) == 6560
assert sum(x[2] for x in rows) == 1
assert sum(x[3] for x in rows) == 13563
assert sum(x[4] for x in rows) == 1

f = sum(1 << j for j in [13,9,6,4,3,2,0])
```

```
u, residues = 2, [2]
for _ in range(13):
    u = sum(((u >> j) & 1) << (2*j)
            for j in range(u.bit_length()))
    while u.bit_length() >= 14:
        u ^= f << (u.bit_length()-14)
    residues.append(u)
assert residues == [2,4,16,256,4840,459,1559,
                    4154,7023,2495,3380,6142,3074,2]
assert u == 2
print("403 orbits; 13563 strict descents; only N=9 has one root")
```

The aggregate certificate is:

N range	orbits	$\sum \tau(N)$	$\max \tau(N)$	zeros in $\lvert z\rvert < 23/35$
2–99	98	2119	75	1
100–199	100	3521	79	0
200–299	100	3847	81	0
300–404	105	4076	91	0

Only the descent for $N = 9$ takes the reciprocal branch, and it does so once. All other 13562 comparisons take the direct branch. Every comparison is strict.

Scope of the conclusion. This proves the exact minimum modulus and the complete classification of zeros in $|z| \le 23/35$. It does not claim that P_9 supplies the only zero in the larger disk $|z| < 2/3$. The remaining thin band $23/35 < |z| < 2/3$ is not classified here. In particular, the accumulation annulus and the set of actual zeros must still be distinguished. The result sharpens the universal lower bound 3/5 obtainable from Hohertz's orbit-dependent Eneström–Kakeya bound [5]. No claim of priority is made without a separate literature review.

III.3.1 A finite-degree exclusion sector near the positive axis

Proposition III.3.2 (Degree-dependent separation from 1)**.** *Let P_N have degree $d = \tau(N) \ge 2$. Every zero ξ satisfies*

$$|\arg \xi| > \frac{\pi}{d}, \qquad |\xi - 1| > \sin\left(\frac{\pi}{d}\right),$$

where the argument is taken in $[-\pi, \pi]$. Consequently, if a zero satisfies $|\xi - 1| \le \varepsilon < 1$, then

$$d > \frac{\pi}{\arcsin \varepsilon}.$$

Proof. This is an elementary consequence of strictly positive coefficients, and is not specific to Collatz polynomials. Write $\xi = re^{i\theta}$. If $|\theta| \le \pi/d$, rotate $P_N(\xi)$ by $e^{-id\theta/2}$. Every summand has nonnegative real part, and every interior summand has strictly positive real part. Thus the rotated sum cannot vanish. The Euclidean distance from 1 to the complement of the open sector $|\arg z| < \pi/d$ is $\sin(\pi/d)$; strict angular exclusion gives the strict distance bound. Finally, monotonicity of sine on $[0, \pi/2]$ gives the degree inequality. □

There is no contradiction with zeros approaching 1 or with a dense accumulation annulus: the forbidden sector narrows as the trajectory length increases. A global forbidden sector of a fixed positive angle would contradict the annulus theorem; this finite-degree sector does not. This distinction is particularly important when interpreting a plot formed from only finitely many starting values.

III.3.2 The extremal zero has full symmetric Galois group

Proposition III.3.3 (An explicit S_{13} example)**.** *The Galois group of P_9 over $\mathbb{Q}$ is S_{13}. Consequently, the extremal radius ρ cannot be expressed by radicals over $\mathbb{Q}$.*

Proof. The preceding irreducibility proof shows that the Galois group G acts transitively on 13 roots. A transitive group of prime degree is primitive: all blocks in a block system have equal size dividing 13. Modulo 17, direct multiplication gives

$$\begin{aligned} P_9(x) \equiv{}& (x+4)(x+5)(x^2+5x-3) \\ &\cdot (x^3+x^2-6x+3)(x^3+8x^2+x-3)(x^3-4x^2+7x+2). \end{aligned}$$

The quadratic and the three cubics have no roots in $\mathbb{F}_{17}$, as checked by the complete finite calculation below. They are therefore irreducible over that field. All six monic factors are distinct, so the reduction is squarefree. By the standard Frobenius–Dedekind theorem [75] relating a squarefree factorization modulo a prime to a permutation in the Galois group, G contains an element of cycle type $(1,1,2,3,3,3)$. Its cube is a transposition.

For completeness, a primitive permutation group containing a transposition is the full symmetric group. Form a graph whose edges are the pairs swapped by all G-conjugates of that transposition. Its connected components form a G-invariant block system. Since the graph has an edge, primitivity makes it connected. Transpositions along the edges of a connected graph generate the full symmetric group, and all those transpositions belong to G. Thus $G = S_{13}$. Since S_{13} is not solvable, the irreducible polynomial $P_9(-x)$ is not solvable by radicals; in particular its root ρ is not expressible by radicals over $\mathbb{Q}$. □

Here is the complete factorization check, with coefficient lists in ascending order. It is independent of any computer algebra package.

```
A = [9,14,7,11,17,26,13,20,10,5,8,4,2,1]
factors = [[4,1], [5,1], [-3,5,1], [3,-6,1,1],
           [-3,1,8,1], [2,7,-4,1]]
product = [1]
for a in factors:
    b = [0]*(len(product)+len(a)-1)
    for j, u in enumerate(product):
        for k, v in enumerate(a):
            b[j+k] = (b[j+k]+u*v) % 17
    product = b
assert product == [a % 17 for a in A]
for a in factors[2:]:
    assert all(sum(c*t**j for j,c in enumerate(a)) % 17
               for t in range(17))
assert len({tuple(c % 17 for c in a)
            for a in factors}) == 6
```

This determines the Galois group of one explicit extremal polynomial; it does not purport to classify Galois groups of general Collatz polynomials, the broader direction raised in [5, Section 5, question 6].

III.4 Exact boundary zeros and the arithmetic clock

Throughout this section, $T(n) = n/2$ for even n and $T(n) = (3n+1)/2$ for odd n, and an orbit polynomial stops at its first visit to 1. Write $a_j = T^j(N)$, $n = \tau(N)$, and $P_N(z) = \sum_{j=0}^{n} a_j z^j$, initially assuming $n < \infty$ and $N > 1$.

III.4.1 The established boundary classification, with a direct proof

The following classification is Hohertz's Theorem III [5]. We include a short proof using a coefficient identity, both to make the mechanism explicit and to distinguish this known theorem from the quantitative estimate proved below.

Theorem III.4.1 (Hohertz's boundary classification)**.** *Define*

$$d_N = \gcd(\{n+1\} \cup \{j+1 : 0 \le j < n,\ a_j \text{ odd}\}).$$

The zeros of P_N *on* $|z| = 2$ *are precisely*

$$\{2\omega : \omega^{d_N} = 1,\ \omega \neq 1\}.$$

There are $d_N - 1$ *such zeros, all simple. Every other zero has modulus strictly less than* 2*.*

Proof. Direct subtraction of coefficients gives the defect identity

$$(z-2)P_N(z) = z^{n+1} - 2N - \sum_{\substack{0 \le j < n \\ a_j \text{ odd}}} (2a_j + 1) z^{j+1}. \tag{III.4.1}$$

Indeed, the coefficient of z^{j+1} in the left side is $a_j - 2a_{j+1}$, which is 0 for even a_j and $-(2a_j+1)$ for odd a_j. Evaluating at $z = 2$ yields

$$2^{n+1} = 2N + \sum_{\substack{0 \le j < n \\ a_j \text{ odd}}} (2a_j + 1) 2^{j+1}.$$

For a zero $z = 2\omega$, $|\omega| = 1$, equality in the triangle inequality in (III.4.1) forces

$$\omega^{n+1} = 1, \qquad \omega^{j+1} = 1 \quad \text{for every odd } a_j, j < n,$$

because the constant term $2N$ on the right is positive. These conditions are equivalent to $\omega^{d_N} = 1$ and conversely make the right side of (III.4.1) vanish. The value $z = 2$ must be discarded because $P_N(2) > 0$.

Put $G(z) = (z-2)P_N(z)$. At any of the displayed roots of G,

$$zG'(z) = (n+1)2^{n+1} - \sum_{\substack{0 \le j < n \\ a_j \text{ odd}}} (j+1)(2a_j+1)2^{j+1} > 0.$$

The last inequality follows from the preceding identity and $j + 1 < n + 1$. Thus these roots are simple. Finally, the positive-coefficient Eneström–Kakeya bound gives $|z| \le 2$ for all zeros, since $a_j/a_{j+1} \le 2$. Equivalently, the same upper bound follows from (III.4.1): for $r > 2$, dividing by r^{n+1} makes every term on its right smaller than its value at $r = 2$. □

Hohertz uses the equivalent gcd of $n + 1$ and the integers $n - j$ at odd positions. The two forms agree because $(n + 1) - (j + 1) = n - j$.

III.4.2 A valuation formula and prime starting values

Write $e = v_2(N)$. Remove the initial divisions by 2, and denote the subsequent odd values by

$$u_0 = N/2^e,\ u_1, \ldots, u_\ell = 1, \qquad k_i = v_2(3u_i + 1), \quad u_{i+1} = (3u_i + 1)/2^{k_i}.$$

When N is a power of 2, take $\ell = 0$ and $u_0 = 1$.

Proposition III.4.2 (The arithmetic clock)**.** *With the preceding convention,*

$$\boxed{d_N = \gcd(e+1, k_0, \ldots, k_{\ell-1}).}$$

In particular $d_N \mid e+1$, and the number of boundary zeros is at most $v_2(N)$.

Proof. The odd positions are $j_0 = e$ and $j_{i+1} = j_i + k_i$, with $j_\ell = n$. The gcd in Theorem III.4.1 is the gcd of $j_0 + 1, \ldots, j_\ell + 1$, hence the gcd of $e+1$ and their successive differences k_i. In the power-of-2 case it is simply $e+1$. □

Corollary III.4.3 (Prime starts cannot hit the outer circle)**.** *Every terminating odd starting value, prime or composite, has $d_N = 1$. Consequently all zeros of P_p have $|z| < 2$ for every odd prime p whose orbit terminates. The sole prime exception is $p = 2$, for which $P_2(z) = 2 + z$ has the zero -2.*

This is a parity restriction, not a primality criterion. In particular, the absence of outer-boundary zeros cannot distinguish odd primes from odd composite integers.

Proposition III.4.4 (An unconditional terminating decision procedure)**.** *There is a finite procedure, independent of the general Collatz conjecture, which decides whether a positive integer belongs to the set of terminating starts having an outer-boundary zero.*

Proof. For $N > 1$, start with $g = v_2(N) + 1$ and $u = N/2^{v_2(N)}$. If $g = 1$, answer no. While $u > 1$ and $g > 1$, set $k = v_2(3u+1)$ and replace g by $\gcd(g, k)$. If this makes $g = 1$, answer no immediately; otherwise replace u by $(3u+1)/2^k$ and continue. Every such continuation has $k \geq 2$, and therefore $1 \leq (3u+1)/2^k < u$ for $u > 1$. Thus the procedure stops. If it reaches $u = 1$ with $g > 1$, its explicitly constructed orbit terminates and Proposition III.4.2 gives $d_N = g > 1$. Conversely, if a gcd ever becomes 1, every later gcd stays 1, so a terminating orbit cannot have a boundary zero. For $N = 1$, the polynomial is constant and the answer is no. □

III.4.3 Integral factorization and exact two-circle examples

The block factorization below is the integral-variable version of Hohertz's factorization, not a new boundary classification.

Proposition III.4.5 (Block factorization)**.** *Set $d = d_N$ and*

$$B_d(z) = \frac{z^d - 2^d}{z - 2} = \sum_{j=0}^{d-1} 2^{d-1-j} z^j.$$

Then

$$P_N(z) = B_d(z) Q_N(z^d), \qquad Q_N \in \mathbb{Z}_{>0}[w],$$

where Q_N is monic. If $m = (n+1)/d - 1$, explicitly

$$Q_N(w) = \sum_{t=0}^{m} a_{dt+d-1} w^t.$$

All roots of Q_N have modulus strictly less than 2^d. The largest order of rotational invariance of the polynomial $(z-2)P_N(z)$ is exactly d.

Proof. All odd positions satisfy $j \equiv d - 1 \pmod{d}$. Thus within each block of d coefficients, $a_{dt+j} = 2^{d-1-j} a_{dt+d-1}$ for $0 \leq j < d$. This gives the factorization and positive integral coefficients; the last block ends in $a_n = 1$, so Q_N is monic. Its root bound follows from Theorem III.4.1, since B_d already accounts for all boundary zeros. Finally, the nonconstant support exponents in (III.4.1) have gcd d. Hence $G(\omega z) = G(z)$ holds for all z exactly when $\omega^d = 1$. □

Proposition III.4.6 (An explicit family of two regular polygons)**.** *For every even integer $d \geq 4$, set*

$$q = \frac{2^d - 1}{3}, \qquad N = 2^{d-1}q.$$

Then

$$\boxed{P_N(z) = \frac{z^d - 2^d}{z - 2}(z^d + q).}$$

Its zeros consist of $d-1$ points on the outer circle and d points on an inner circle:

$$2e^{2\pi ij/d} \ (1 \leq j < d), \qquad q^{1/d}e^{(2j+1)\pi i/d} \ (0 \leq j < d).$$

The inner radii satisfy $q^{1/d} < 2$ and tend to 2 as $d \to \infty$. Within this family, q is prime if and only if $d = 4$.

Proof. The orbit begins with $2^{d-1}q, \ldots, q$, and because $3q + 1 = 2^d$ it continues with $2^{d-1}, \ldots, 1$. These are two blocks of d coefficients, giving the formula. The root locations follow by solving the two factors. For the final assertion write $d = 2h$. Then

$$q = \frac{(2^h - 1)(2^h + 1)}{3}.$$

Exactly one factor is divisible by 3. If $h \geq 3$, both factors remain greater than 1 after this division; when $h = 2$, the result is $q = 5$. □

For example,

$$P_{40}(z) = (z^3 + 2z^2 + 4z + 8)(z^4 + 5).$$

The three outer zeros are $-2, 2i, -2i$; the other four form a square of radius $5^{1/4}$. The inner factor is irreducible over $\mathbb{Q}$ by Eisenstein's criterion at 5. This is an exact arithmetic explanation of one visually distinctive pattern in a root plot.

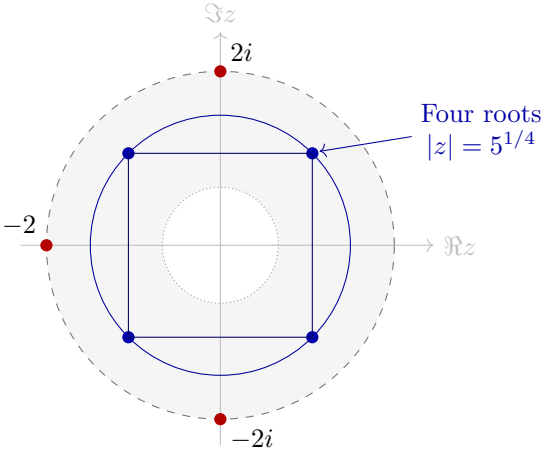


Figure 2: Exact zeros of P_{40}. The outer geometric factor contributes three points; the irreducible factor $z^4 + 5$ contributes a rotated square. The dotted inner circle $|z| = 2/3$ is shown only as a reference.

III.4.4 A quantitative bound for boundary starts

Hohertz proves that the set of starting integers with outer-boundary zeros has natural density zero. The next argument gives a stronger explicit bound. We make no claim here that the bound has not appeared elsewhere.

Lemma III.4.7 (Finite parity prefixes)**.** *For the shortcut map, each prescribed parity word of length k corresponds to exactly one residue class of starting integers modulo 2^k.*

Proof. After a fixed prefix with r odd branches, the iterate is $(3^r N + b)/2^k$ for an integer b. Inductively, a length-k prefix specifies one class modulo 2^k. Its two lifts modulo 2^{k+1} differ by 2^k in the input and by the odd integer 3^r in the kth iterate, so exactly one lift has either prescribed next parity. The length-zero case starts the induction. □

Theorem III.4.8 (A square-root upper bound)**.** *For $X \geq 4$, put $k = \lfloor \log_2 X \rfloor$ and*

$$A_d(X) = \#\{2 \leq N \leq X : \tau(N) < \infty,\ d \mid d_N\}, \qquad d \geq 2.$$

Then

$$A_d(X) \leq 3\, 2^{\lfloor k/d \rfloor} - 1.$$

Consequently, if $E(X)$ counts the starts $2 \leq N \leq X$ whose terminating orbit polynomials have a zero on $|z| = 2$, then

$$\boxed{E(X) \leq 3\, 2^{\lfloor k/2 \rfloor} + 3(k-1)2^{\lfloor k/3 \rfloor} = O(\sqrt{X}).}$$

In particular, $E(X) \leq (3 + o(1))\sqrt{X}$. No global convergence assumption is used.

Proof. If $d \mid d_N$, odd positions before the terminal 1 must lie at $j \equiv d - 1 \pmod{d}$, and $n + 1$ must be divisible by d. First count starts with $n \geq k$. Among their first k parity bits, only $\lfloor k/d \rfloor$ positions can be odd. There are at most $2^{\lfloor k/d \rfloor}$ such words. By Lemma III.4.7, each is one residue class modulo 2^k; because $2^k \leq X < 2^{k+1}$, each class supplies at most two starts. This part contributes at most $2 \cdot 2^{\lfloor k/d \rfloor}$.

For the starts with $n < k$, write $n = dq - 1$, where $1 \leq q \leq \lfloor k/d \rfloor$. Only the $q - 1$ positions $d - 1, 2d - 1, \ldots, (q-1)d - 1$ can be odd before time n. There are at most 2^{q-1} words. A complete parity word and the terminal value 1 determine at most one starting value, because each branch is an injective affine map. Summing over q gives at most $2^{\lfloor k/d \rfloor} - 1$, proving the first estimate.

If $d_N > 1$, then $2 \leq d_N \leq v_2(N) + 1 \leq k + 1$. Take the union over $d = 2, \ldots, k + 1$, separate the term $d = 2$, and use $\lfloor k/d \rfloor \leq \lfloor k/3 \rfloor$ for the other $k - 1$ terms. This gives the displayed bound. Finally, $kX^{1/3} = o(X^{1/2})$. □

The exponent $1/2$ is an upper bound, not an established optimal exponent or an asymptotic formula. Determining the actual growth of $E(X)$ remains a concrete counting problem. These facts explain algebraic patterns and rarity on the outer circle; they do not identify Collatz zeros with zeta zeros or imply a statement about the Riemann hypothesis.

III.4.5 A positive power lower bound from an inverse tree

The previous upper bound leaves open how many starts actually produce boundary zeros. A complementary lower bound follows from an explicit terminating inverse tree. This is an additional deduction of the present compilation; no claim of historical priority is made.

Theorem III.4.9 (Polynomially many starts with the boundary zero -2)**.** *For $B \geq 2$, let $s_B > 0$ be the unique solution of*

$$\sum_{j=1}^{B} \left(\frac{3}{64^j} \right)^{s_B} = 1.$$

Let $E(X)$ count terminating starts $2 \le N \le X$ whose polynomials have a zero on $|z| = 2$. For every $X \ge 170$,

$$E(X) > \frac{(X/170)^{s_B} - 1}{B - 1}.$$

Every start constructed in this proof has $d_N = 2$, so its only outer-boundary zero is -2. In particular,

$$s_* \le \liminf_{X\to\infty} \frac{\log E(X)}{\log X} \le \limsup_{X\to\infty} \frac{\log E(X)}{\log X} \le \frac{1}{2}, \qquad 64^{s_*} = 1 + 3^{s_*}, \tag{III.4.2}$$

where $s_* = 0.1935935685\ldots$. For example, $s_2 = 0.1433841276\ldots$.

Proof. Use the accelerated odd map $U(v) = (3v+1)/2^{v_2(3v+1)}$. Start the inverse tree at $v = 85$; then $U(85) = 1$ and $v_2(3 \cdot 85 + 1) = 8$. For every odd $v \equiv 1 \pmod 3$, exactly one $a_0 \in \{1, 2, 3\}$ satisfies

$$4^{a_0} v \equiv 4 \pmod 9.$$

Indeed, $v \bmod 9$ belongs to $\{1, 4, 7\}$, and multiplication by 4 cycles these three residues. For $j = 1, \ldots, B$, define the children

$$v_j = \frac{4^{a_0+3(j-1)} v - 1}{3}.$$

Each child is an odd integer congruent to 1 modulo 3, and

$$U(v_j) = v, \qquad v_2(3v_j + 1) = 2a_0 + 6(j-1), \qquad v < v_j < c_j v, \quad c_j = 64^j/3.$$

The strict growth uses $v \ge 85$. Every node therefore has a terminating orbit, with all accelerated valuation exponents even. Distinct children have different valuation exponents, so are different; a node has its unique parent under U. Strict growth rules out cycles. If two nodes at different depths were equal, applying their common unique successive parents would make a positive iterate of 85 return to 85, whereas its first successor is 1. Thus every word in the child indices gives a distinct node.

For a word $w = (j_1, \ldots, j_k)$ put $C(w) = \prod_{i=1}^{k} c_{j_i}$, with $C(\varnothing) = 1$. Its node obeys $v(w) \le 85C(w)$, so the even start $N(w) = 2v(w)$ obeys $N(w) \le 170C(w)$. Its initial 2-adic valuation is 1, and every following odd-step valuation is even; hence the arithmetic clock gives

$$d_{N(w)} = \gcd(2, \text{the following even valuations}) = 2.$$

It remains to count words with $C(w) \le Y = X/170$.

Stop a word at the first time $C(w) > Y$. Since every $c_j > 1$, this produces a finite full B-ary tree. Its internal nodes are exactly the words with $C(w) \le Y$; all of them give counted starts. Assign a child of index j the relative mass $c_j^{-s_B}$. These masses sum to 1, so the sum of the leaf masses $C(w)^{-s_B}$ is 1. Every leaf has $C(w) > Y$, and hence has mass less than Y^{-s_B}. If L is the number of leaves, then $L > Y^{s_B}$. A finite full B-ary tree has $(L-1)/(B-1)$ internal nodes. Consequently

$$E(X) \ge \frac{L-1}{B-1} > \frac{Y^{s_B} - 1}{B-1}.$$

As B increases, s_B increases to the unique positive solution of

$$\sum_{j=1}^{\infty} (3/64^j)^s = \frac{3^s}{64^s - 1} = 1,$$

which is s_*. For every fixed B the lower bound implies $\liminf \log E(X)/\log X \ge s_B$; taking the supremum over B gives the left side of (III.4.2). The square-root upper bound gives the right side. □

The interval of possible growth exponents is narrower, but neither an exact exponent nor a full asymptotic for $E(X)$ is proved.

III.5 Zero-cloud measures and the frequency of odd Collatz steps

Topological accumulation describes where a zero can be found arbitrarily nearby. It does not describe the proportion of all zeros in a region. The distinction is substantial for Collatz polynomials. The results in this section are elementary deductions from their coefficient structure and the finite-word construction; no assertion of historical priority is made.

Let $N > 1$ reach 1 under the shortcut map T, set $a_j = T^j(N)$, $\tau = \min\{j : a_j = 1\}$, and write

$$P_N(z) = \sum_{j=0}^{\tau} a_j z^j = \prod_{\nu=1}^{\tau} (z - \zeta_\nu).$$

Zeros are counted with algebraic multiplicity. Define

$$r_N = \#\{0 \le j < \tau : a_j \text{ is odd}\}, \qquad \mu_N = \frac{1}{\tau} \sum_{\nu=1}^{\tau} \delta_{\zeta_\nu}.$$

Theorem III.5.1 (Exact logarithmic radius and a small error term)**.** *Put*

$$E_N = \sum_{\substack{0 \le j < \tau \\ a_j \ odd}} \log\left(1 + \frac{1}{3a_j}\right).$$

Then

$$\log N = \tau \log 2 - r_N \log 3 - E_N, \tag{III.5.1}$$

$$0 \le E_N \le \frac{1}{6} \log(2r_N + 1), \tag{III.5.2}$$

$$\left(\prod_{\nu=1}^{\tau} |\zeta_\nu|\right)^{1/\tau} = N^{1/\tau} = 2\, 3^{-r_N/\tau} e^{-E_N/\tau}. \tag{III.5.3}$$

In particular, $r_N/\tau < \log 2/\log 3$.

Proof. On an even step $a_{j+1}/a_j = 1/2$, and on an odd step $a_{j+1}/a_j = (3/2)(1 + 1/(3a_j))$. Multiplying these identities from $j = 0$ to $\tau - 1$, with $a_\tau = 1$, gives (III.5.1). A terminating orbit cannot repeat a value before its first 1, since a repeat would trap it in a cycle. Its odd values before that time are therefore distinct integers at least 3. If ordered increasingly as $u_1 < \cdots < u_R$, they satisfy $u_i \ge 2i + 1$. For $R \ge 1$, using $\log(1 + x) \le x$ gives

$$E_N \le \frac{1}{3} \sum_{i=1}^{R} \frac{1}{2i+1} \le \frac{1}{3} \int_0^R \frac{dx}{2x+1} = \frac{1}{6} \log(2R + 1).$$

For $R = 0$, both sides are zero. The monic polynomial has constant term N, so Vieta's formula gives $\prod |\zeta_\nu| = N$, proving (III.5.3). Finally, $\log N > 0$ and $E_N \ge 0$ in (III.5.1) give the strict bound on r_N/τ. □

This identifies an arithmetic feature of the orbit inside the zero cloud: the average logarithmic radius determines the proportion of odd steps up to an error $O(\log \tau/\tau)$. It concerns ordinary odd steps of the shortcut map, not the number of prime values visited.

Proposition III.5.2 (Exact terminal moments and universal moment bounds)**.** *For every integer* $k \ge 1$*, the real power sums satisfy*

$$-2^k \le \sum_{\nu=1}^{\tau} \zeta_\nu^k \le (k-1)2^k.$$

If $\tau \geq 3$, then

$$\sum_{\nu} \zeta_\nu = -2, \qquad \sum_{\nu} \zeta_\nu^2 = -4, \qquad \sum_{\nu} \zeta_\nu^3 = -8.$$

If $\tau \geq 4$, then

$$\sum_{\nu} \zeta_\nu^4 = 48 - 4a_{\tau-4} = \begin{cases} -16, & a_{\tau-4} = 16, \\ 28, & a_{\tau-4} = 5. \end{cases}$$

Proof. Every inverse step satisfies $a_j \leq 2a_{j+1}$. Therefore $c_j = 2^{-j} a_{\tau-j}$ decreases from $c_0 = 1$ to $c_\tau > 0$. For $w_\nu = \zeta_\nu/2$, the reversed normalized polynomial is

$$F(u) = \sum_{j=0}^{\tau} c_j u^j = \prod_{\nu=1}^{\tau} (1 - w_\nu u).$$

Put $d_j = c_{j-1} - c_j$ for $1 \leq j \leq \tau$ and $d_{\tau+1} = c_\tau$. Then $d_j \geq 0$, $\sum d_j = 1$, and

$$(1-u)F(u) = 1 - D(u), \qquad D(u) = \sum_{j=1}^{\tau+1} d_j u^j.$$

Comparing coefficients in the formal logarithms gives

$$S_k := \sum_{\nu} w_\nu^k = -1 + k \sum_{m \geq 1} \frac{1}{m} [u^k] D(u)^m.$$

Every coefficient on the right is nonnegative, proving $S_k \geq -1$. Write $(1 - D(u))^{-1} = \sum_{n \geq 0} v_n u^n$. The recurrence $v_0 = 1$ and $v_n = \sum_{j=1}^{n} d_j v_{n-j}$, with $d_j = 0$ beyond its defined range, shows by induction that $0 \leq v_n \leq 1$. Consequently, for $k \geq 1$,

$$\sum_{m \geq 1} \frac{1}{m} [u^k] D(u)^m \leq \sum_{m \geq 1} [u^k] D(u)^m = v_k \leq 1.$$

Thus $S_k \leq k - 1$, and multiplication by 2^k proves the bound. This argument also shows that the sums are real; alternatively, nonreal roots occur in conjugate pairs.

The final three predecessors of a first visit to 1 are necessarily $8, 4, 2$ whenever they exist. The next predecessor is either 16 or 5. Substitution into Newton's identities in the original variable gives the displayed exact sums. The lower bound is sharp, and the upper bound is sharp for $k \geq 2$, across the family: for $N = 2^t$, $S_k = -1$ when $(t+1) \nmid k$, and $S_{t+1} = t$. □

Theorem III.5.3 (Constraints on every limiting full zero cloud)**.** *Let $N_j \to \infty$ be terminating starting values, and suppose μ_{N_j} converges weakly to a probability measure μ. Then*

1. *μ is supported on $2/3 \leq |z| \leq 2$;*
2. *$\int z^k \, d\mu(z) = 0$ for every integer $k \geq 1$;*
3. *$0 \leq \int \log|z| \, d\mu(z) \leq \log 2$, and*

$$\frac{r_{N_j}}{\tau(N_j)} \longrightarrow \frac{\log 2 - \int \log|z| \, d\mu(z)}{\log 3};$$

4. *for every point z_0 in the support,*

$$\mu(\{z_0\}) \leq \frac{2 - |z_0|}{2 + |z_0|}.$$

In particular, there are no atoms on $|z| = 2$, and the possible mass at $z = 1$ is at most $1/3$.

Proof. The coefficient bound $N \le 2^\tau$ implies $\tau(N_j) \to \infty$. The outer radius bound $|\zeta| \le 2$ and the initial-value lower bound tending to $2/3$ give the asserted support. Proposition III.5.2, divided by $\tau(N_j)$, proves the vanishing analytic moments. Since all sufficiently late zero clouds stay away from zero, weak convergence also gives

$$\int \log|z|\, d\mu(z) = \lim_j \frac{\log N_j}{\tau(N_j)} \in [0, \log 2].$$

Equation (III.5.1) and $E_{N_j}/\tau(N_j) = O(\log\tau(N_j)/\tau(N_j))$ then prove the odd-step frequency formula.

For the atom bound, scale $w = z/2$ and write $w_0 = re^{i\theta}$. For $0 < t < 1$, the function

$$H_t(w) = \operatorname{Re}\frac{1 + te^{-i\theta}w}{1 - te^{-i\theta}w}$$

is positive on the closed unit disk. Its uniformly convergent power series and the vanishing analytic moments give $\int H_t\, d\mu = 1$ (with the scaling understood). At w_0 its value is $(1+tr)/(1-tr)$. Hence $\mu(\{z_0\}) \le (1-tr)/(1+tr)$; let $t \uparrow 1$. □

One consequence is that the full zero cloud cannot converge to a probability measure carried by a circle of radius strictly smaller than 1. Such a measure would have negative average $\log|z|$. Nevertheless, individual zeros accumulate on every such circle between $2/3$ and 1. The topological annulus theorem and this statistical restriction are compatible because individual exceptional zeros may carry vanishing mass in μ_N.

Theorem III.5.4 (Outer-circle equidistribution for sparse odd steps)**.** *If $N_j \to \infty$ are terminating starts and $r_{N_j} = o(\tau(N_j))$, then μ_{N_j} converges to normalized arclength on $|z| = 2$. For every $0 < r < 2$, the following quantitative radial bound holds:*

$$\#\{\nu : |\zeta_\nu| \le r\} \le \frac{\tau\log 2 - \log N}{\log(2/r)} = \frac{r_N \log 3 + E_N}{\log(2/r)}.$$

Proof. Every zero has modulus at most 2, and Vieta's formula gives

$$\sum_\nu \log\frac{2}{|\zeta_\nu|} = \tau\log 2 - \log N.$$

Each zero in $|z| \le r$ contributes at least $\log(2/r)$, proving the bound. Its right side is $o(\tau)$ under the stated hypothesis. Every weakly convergent subsequence is therefore supported on $|z| = 2$. Its analytic moments vanish by Proposition III.5.2. After scaling to the unit circle, all positive Fourier coefficients vanish; the negative coefficients are their complex conjugates. Uniqueness of a probability measure from its Fourier coefficients identifies the limit as normalized arclength. Compactness of probability measures on $|z| \le 2$ then proves convergence of the entire sequence. □

For example, $N = 2^t$ has $r_N = 0$ and

$$P_{2^t}(z) = 2^t \sum_{j=0}^{t} (z/2)^j.$$

Its roots are exactly $2\exp(2\pi ik/(t+1))$, $1 \le k \le t$, which visibly become uniformly spaced on the outer circle. The theorem allows an unbounded number of odd steps as long as their proportion tends to zero.

There is also a concrete family with an exact interior zero. For odd $t \geq 3$, put $N_t = (2^{t+1} - 1)/3$. Its orbit is $N_t, 2^t, 2^{t-1}, \ldots, 1$, so it has exactly one odd step and degree $t + 1$. An alternating geometric sum gives

$$P_{N_t}(-1) = \frac{2^{t+1} - 1}{3} + \sum_{j=1}^{t+1} 2^{t+1-j}(-1)^j = 0.$$

Moreover, $P'_{N_t}(-1) = (2N_t - t - 1)/3 > 0$, so this zero is simple. Thus the starts $5, 21, 85, 341, \ldots$ retain the exact zero -1, while their full zero-cloud measures converge to uniform arclength on $|z| = 2$. The one interior root has normalized mass $1/(t + 1)$, which tends to zero.

Corollary III.5.5 (Dense selected zeros with all statistical mass at the rim)**.** *Let $U_1, U_2, \ldots$ be nonempty open sets, each meeting the open annulus $2/3 < |z| < 2$. There are terminating starts $N_j \to \infty$ and zeros ξ_j of P_{N_j} such that*

$$\xi_j \in U_j, \qquad \mu_{N_j} \Longrightarrow \textit{normalized arclength on } |z| = 2.$$

In particular, the selected zeros may be dense throughout the entire annulus while the normalized full zero clouds converge to its outer rim.

Proof. The repeated-word proof of the annulus theorem has a useful feature: for each open set U_j one can first fix a finite parity prefix so that its power-of-two-tail limit has a zero in U_j, and then let the tail length t tend to infinity through an arithmetic progression. Rouché's theorem supplies a zero in U_j for all sufficiently large permitted t. The prefix length and its odd-step count are fixed during this last limit, whereas the total length is the prefix length plus t. Choose t large enough that $r_{N_j}/\tau(N_j) < 1/j$ and $N_j > j$. The preceding theorem gives the claimed convergence. To obtain density, choose the U_j from a countable base of open disks meeting the annulus, repeating each such disk infinitely often. □

The distribution results do not assume that all positive integers, or all primes, terminate. They do not assert a universal statistical distribution when the odd-step proportion stays positive.

III.6 Exact vertical lines and a composite imaginary-axis family

All statements below concern positive starting integers $N > 1$ whose shortcut Collatz orbit reaches 1. Write

$$P_N(z) = \sum_{j=0}^{L} a_j z^j, \qquad a_0 = N, \quad a_L = 1,$$

where L is the first hitting time of 1. Thus P_N is monic with integer coefficients. We use the established bound $|\alpha| \leq 2$ for its zeros.

Proposition III.6.1 (Rational vertical lines)**.** *If a nonreal zero α of P_N has rational real part, then $k = 2\operatorname{Re}\alpha$ is an integer. Its monic minimal polynomial has even degree and has the form*

$$f(z) = g(z^2 - kz), \qquad g \in \mathbb{Z}[z].$$

In particular, the only rational-real-part vertical lines on which a nonreal zero can lie are

$$\operatorname{Re} z \in \left\{ -\frac{3}{2}, -1, -\frac{1}{2}, 0, \frac{1}{2}, 1, \frac{3}{2} \right\}.$$

If N is odd and $c = f(0)$, then $\gcd(c, k) = 1$.

Proof. The sum $\alpha + \overline{\alpha}$ is a rational algebraic integer, so it is an integer k. Complex conjugation and minimality give $f(k-z) = (-1)^d f(z)$, where $d = \deg f$. If d were odd, substituting $z = k/2$ would give a rational zero of f, contradicting irreducibility and the nonreality of α. Hence d is even and $f(k-z) = f(z)$. Every integral polynomial invariant under $z \mapsto k - z$ belongs to $\mathbb{Z}[z^2 - kz]$: subtract its leading coefficient times $(z^2 - kz)^{d/2}$ and proceed by induction on the degree. The radius bound and nonreality imply $|k| < 4$, proving the displayed list.

Since f is monic, division in $\mathbb{Z}[z]$ gives $P_N = fh$ for some $h \in \mathbb{Z}[z]$. A prime dividing both $f(0)$ and $f'(0)$ would divide both $P_N(0)$ and $P'_N(0)$. For odd N these two integers are N and $(3N+1)/2$, which are coprime. Thus $\gcd(f(0), f'(0)) = 1$. Since $f'(0) = -kg'(0)$, we obtain $\gcd(c, k) = 1$. □

Corollary III.6.2 (Imaginary roots of odd starts are units)**.** *For every odd starting integer N, any purely imaginary zero of P_N is an algebraic unit. More generally, every even monic integral factor of P_N has constant coefficient ± 1.*

Proof. An even factor f satisfies $f'(0) = 0$. If $P_N = fh$, then $f(0)$ divides both $P_N(0) = N$ and $P'_N(0) = (3N+1)/2$, whose gcd is 1. For a purely imaginary zero, its minimal polynomial is even by the preceding proposition with $k = 0$. □

This is a restriction on all odd starts, including odd composites; it is not a primality criterion. In particular, if p is an odd prime and G_p is the unique irreducible factor of P_p with constant coefficient of absolute value p, then G_p has no imaginary-axis zero. Possible zeros of unit-constant factors are not excluded by this argument.

Proposition III.6.3 (Infinitely many composite starts on the imaginary axis)**.** *For every integer $r \geq 0$, put $N_r = 42 \cdot 16^r$. The polynomial P_{N_r} has exactly one conjugate pair of imaginary-axis zeros. They are*

$$\alpha_r^{\pm} = \pm i\sqrt{x_r},$$

where x_r is the unique positive solution of

$$x_r^{2r+1}(43 + x_r^3) = 84 \cdot 16^r.$$

Moreover $0 < x_r < 4$, x_r increases strictly with r, and $x_r \longrightarrow 4$. Consequently $\alpha_r^{\pm} \longrightarrow \pm 2i$ from the interior of the radius-two disk. Every N_r is composite.

Proof. More generally let $N = 42 \cdot 4^u = 21 \cdot 2^{2u+1}$, with $u \geq 0$. After $2u+1$ halvings the orbit reaches 21, followed by $32, 16, 8, 4, 2, 1$. Pairing consecutive coefficients therefore gives

$$P_N(z) = (z+2)R_u(z^2),$$

where

$$R_u(t) = 21\sum_{j=0}^{u} 4^{u-j}t^j + 16t^{u+1} + 4t^{u+2} + t^{u+3}.$$

A finite geometric sum yields the polynomial identity

$$(4-t)R_u(t) = 84 \cdot 4^u + (43 - t^3)t^{u+1}.$$

For $u = 2r$ and $x > 0$, it follows that

$$R_{2r}(-x) = \frac{84 \cdot 16^r - x^{2r+1}(43 + x^3)}{4 + x}.$$

The numerator decreases strictly from a positive value to negative infinity. It is negative at $x = 4$, since its value there is $(84 - 428)16^r < 0$. Hence it has exactly one positive zero, and that zero lies below 4. This proves the claim about imaginary-axis zeros.

Divide the defining equation by 16^r and write

$$(x_r/4)^{2r} x_r(43 + x_r^3) = 84.$$

For a fixed $x \in (0, 4)$, the left-hand side decreases strictly when r increases, while it increases strictly with x. Thus $x_{r+1} > x_r$. The increasing bounded sequence has a limit $\ell \le 4$. If $\ell < 4$, the left-hand side would tend to 0, contradicting its constant value 84. Therefore $\ell = 4$. □

Corollary III.6.4 (Exact first-order approach to the boundary)**.** *For the imaginary-axis family above,*

$$\lim_{r\to\infty} r(2 - \sqrt{x_r}) = \frac{1}{2} \log \frac{107}{21}.$$

Proof. The defining equation gives

$$2r \log(x_r/4) + \log \frac{x_r(43 + x_r^3)}{84} = 0.$$

Since $x_r \to 4$, it follows that $r \log(4/x_r) \to \frac{1}{2} \log(107/21)$. Moreover $(2 - \sqrt{x})/\log(4/x) \to 1$ as $x \to 4$, which proves the assertion. □

For example, $N_0 = 42$ gives

$$P_{42}(z) = (z + 2)(z^6 + 4z^4 + 16z^2 + 21).$$

At $N_1 = 672$, the factorization becomes

$$P_{672}(z) = (z + 2)(z^6 + 21)(z^2 - 2z + 4)(z^2 + 2z + 4),$$

so its interior imaginary zeros are exactly $\pm i\, 21^{1/6}$. The last two quadratic factors also give composite-start zeros on the vertical lines $\operatorname{Re} z = \pm 1$, on the outer boundary.

Scope. These propositions establish exact obstructions to several proposed prime-only geometric loci. They do not prove that every line contains composite-start zeros, and they do not prove that imaginary-axis zeros are impossible for all odd starts. In particular, a finite computation finding no such odd-start zero cannot replace a proof of its absence.

III.7 Two axis folds and exact four-point overlaps

Define the folding map by

$$\mathcal{F}(x + iy) = (|x|, |y|).$$

For $a, b > 0$, the four possible preimages of (a, b) are $a + ib, a - ib, -a + ib, -a - ib$. We call their simultaneous occurrence among the distinct zeros of one polynomial a genuine four-point overlap. This definition does not combine zeros from different starting values.

Write $P_N(z) = E_N(z^2) + zO_N(z^2)$ and set $H_N(t) = \gcd(E_N(t), O_N(t))$, chosen monic. A genuine four-point overlap exists if and only if H_N has a nonreal zero. Indeed, a pair of opposite zeros $\alpha, -\alpha$, with $\alpha \neq 0$, gives $E_N(\alpha^2) = O_N(\alpha^2) = 0$. The number α^2 is nonreal exactly when α lies off both axes. Conversely, a nonreal zero of H_N, together with its complex conjugate, produces the four distinct square roots in the displayed orbit of the folding map.

For an odd starting value, every zero in such a quartet is an algebraic unit. Indeed, the monic even polynomial $H_N(z^2)$ divides P_N, and its constant term divides both $P_N(0) = N$ and $P_N'(0) = (3N + 1)/2$. These integers are coprime, so that constant term has absolute value 1.

All its monic irreducible factors consequently have constant term ± 1. In particular, for an odd prime start p, the unique factor G_p of constant term of absolute value p cannot contribute a zero to a genuine quartet. This argument does not assume that the minimal polynomial of each individual quartet member is even.

The exact gcds for $1 \le N \le 10$ are

$$H_N(t) = \begin{cases} t+4, & N = 8, \\ t^2+4t+5, & N = 10, \\ 1, & N \in \{1,2,3,4,5,6,7,9\}. \end{cases}$$

Since $t+4$ has only a real zero, $N = 10$ is the smallest positive starting value giving a genuine four-point overlap. Its factorization is

$$P_{10}(z) = (z+2)(z^4+4z^2+5).$$

The quartic zeros are $\pm a \pm ib$, with independent signs, where

$$a = \sqrt{\frac{\sqrt{5}-2}{2}}, \qquad b = \sqrt{\frac{\sqrt{5}+2}{2}}.$$

All four fold to the point

$$(a,b) \approx (0.3435607497225124641,\ 1.4553466902253548081).$$

For comparison,

$$P_5(z) = (z+1)(z^3+z^2+3z+5).$$

Its nonreal pair is approximately $0.2013139705930618974 \pm 1.8772879069162397302i$, which gives a two-point overlap under $\mathcal{F}$. The other two zeros are -1 and approximately -1.4026279411861237947; they fold to different points on the horizontal axis.

The exact odd-start search in Section III.10 proves that no individual P_N with odd $3 \le N \le 10^6$ has a genuine four-point overlap. This is a finite conclusion. It also does not exclude an apparent overlap assembled from zeros of several different polynomials in a pooled plot.

Proposition III.7.1 (Arbitrarily many four-point overlaps from composites)**.** *For every integer $u \ge 0$, let $N_u = 10 \cdot 4^u$. Then P_{N_u} has exactly $\lfloor u/2 \rfloor + 1$ distinct genuine four-point overlaps. If u is even it has no imaginary-axis zero; if u is odd its only imaginary-axis zeros are $\pm 2i$. Every N_u is composite.*

Proof. Pairing consecutive orbit coefficients gives

$$P_{N_u}(z) = (z+2)R_u(z^2), \qquad R_u(t) = 5\sum_{j=0}^{u} 4^{u-j}t^j + 4t^{u+1} + t^{u+2}.$$

All coefficients of R_u are positive, so it has no nonnegative real zero. Moreover

$$(4-t)R_u(t) = 20 \cdot 4^u + (11-t^2)t^{u+1}.$$

For u even and $x \ge \sqrt{11}$, the numerator of $(4+x)R_u(-x)$ is positive. For $0 < x < \sqrt{11}$, use

$$x^{u+1}(11-x^2) = 4^u(x/4)^u x(11-x^2) \le 4^u \frac{22}{3}\sqrt{\frac{11}{3}} < 20 \cdot 4^u.$$

Thus R_u has no real zero when u is even.

For u odd, a negative zero is equivalent to

$$x^{u+1}(x^2 - 11) = 20 \cdot 4^u, \qquad x > \sqrt{11}.$$

The left side increases strictly on that interval, and $x = 4$ satisfies the equation. Hence -4 is the unique real zero of R_u.

All zeros of R_u are simple. A repeated zero of R_u would also annul the derivative of $(4-t)R_u(t)$, namely

$$t^u(11(u+1) - (u+3)t^2).$$

It cannot be 0, and the remaining derivative zeros are real with squared value $11(u+1)/(u+3) < 11$. None is a real zero of R_u. There are therefore $u+2$ nonreal zeros of R_u for even u, and $u+1$ for odd u. Each conjugate pair gives one distinct four-point overlap of P_{N_u}. This proves the count and the axis assertions. □

For $u = 1$, the particularly simple factorization

$$P_{40}(z) = (z+2)(z^2+4)(z^4+5)$$

gives a four-point overlap exactly on the diagonal:

$$\mathcal{F}(\text{each zero of } z^4+5) = \left(\frac{5^{1/4}}{\sqrt{2}}, \frac{5^{1/4}}{\sqrt{2}}\right).$$

Thus neither four-point overlap itself nor its occurrence on this diagonal isolates prime starting values.

III.8 A rigorous obstruction for bounded prefixes

Fix $L \geq 1$, and define a different polynomial

$$Q_{N,L}(z) = \sum_{j=0}^{L} T^j(N) z^j.$$

Here T continues to be iterated even if the orbit has already reached 1. This definition exists for every positive integer N and does not assume the Collatz conjecture. A *fixed semialgebraic root rule* means a fixed first-order formula over the ordered real field, using the real and imaginary coordinates of roots and the coefficients of $Q_{N,L}$. Examples include existence of a root on any prescribed line, circle, or real algebraic curve; membership of all roots in a prescribed semialgebraic region; prescribed root counts or multiplicities; and finite Boolean combinations of these conditions. The formula and L must be independent of N.

Theorem III.8.1 (Eventual periodicity of bounded-prefix geometric rules)**.** *For a fixed L and a fixed semialgebraic root rule, the set of positive integers N accepted by the rule is eventually a union of residue classes modulo 2^L. In particular, if infinitely many starts are accepted, then infinitely many composite starts are accepted. Such a rule cannot identify an infinite set consisting only of primes.*

Proof. If $N \equiv M \pmod{2^L}$, their first L parity decisions coincide. Indeed, at each step their difference is multiplied by either $1/2$ or $3/2$, so after $j < L$ steps it is still divisible by 2^{L-j}. Fixing a residue class r modulo 2^L therefore fixes integers s_j, b_j such that, on that class,

$$T^j(N) = \frac{3^{s_j} N + b_j}{2^j} \qquad (0 \leq j \leq L).$$

Thus all coefficients of $Q_{N,L}$ are affine functions of N on the class. Temporarily regard N as a real variable and substitute these affine functions in the fixed rule. Real and imaginary parts of polynomial equations in roots are real polynomial equations in their coordinates. The Tarski–Seidenberg quantifier-elimination theorem [65] therefore says that the accepted real parameter set S_r is semialgebraic.

A semialgebraic subset of the real line is a finite union of intervals and points. It follows that membership in S_r is constant for all sufficiently large positive real parameters. In particular, acceptance is eventually constant on the integer residue class r. Taking a maximum of the finitely many thresholds proves eventual periodicity.

If the accepted integer set is infinite, one residue class is eventually entirely accepted. Choose an integer $a > 1$ in this class. The numbers

$$a(1 + 2^L k) \equiv a \pmod{2^L} \qquad (k \geq 1)$$

are distinct composites and eventually exceed the threshold. They are therefore accepted. □

Scope. This is a consequence of classical real algebraic geometry; no priority claim is made. It concerns a uniformly bounded orbit prefix and a fixed semialgebraic rule. It does *not* rule out a criterion using full P_N, whose degree $\tau(N)$ is unbounded, nor arbitrary transcendental curves, rules of unbounded complexity, or arithmetic/integer quantifiers. Restricting the input domain in advance to starts known to terminate is also not part of the theorem. These restrictions are essential.

III.8.1 Extension to exponential and logarithmic rules

The semialgebraic hypothesis can be replaced by the substantially broader hypothesis of definability in a fixed o-minimal expansion of the real ordered field. An expansion is called *o-minimal* if every definable subset of the real line, allowing real parameters, is a finite union of points and intervals. The real field with the exponential function is o-minimal by Wilkie's theorem [104]; logarithms on positive arguments are definable by taking the inverse graph. These are established external results. The following application requires only the defining one-dimensional finiteness property.

Theorem III.8.2 (Fixed-prefix definable rules have asymptotically composite outputs)**.** *Fix $L \geq 1$, an o-minimal expansion $\mathcal{R}$ of the real ordered field, and a first-order rule definable in $\mathcal{R}$, with fixed real parameters, whose input is N and the coefficients and root coordinates of*

$$Q_{N,L}(z) = \sum_{j=0}^{L} T^j(N) z^j.$$

The rule and L do not depend on N; quantifiers range over real variables. Let $A \subseteq \mathbb{N}$ be its accepted positive integer starts. Then A differs by a finite set from a union of residue classes modulo 2^L. If A is infinite, there is an integer $r \in \{1, \ldots, 2^L\}$ with

$$\#(A \cap [1, X]) = \frac{r}{2^L} X + O(1), \qquad \frac{\#\{n \leq X : n \in A,\ n \text{ composite}\}}{\#(A \cap [1, X])} \longrightarrow 1.$$

In particular, no such rule has an infinite prime-only output.

Proof. For each residue class c modulo 2^L, the first L parity decisions are fixed. Consequently there are fixed integers s_j, b_j such that $T^j(N) = (3^{s_j} N + b_j)/2^j$ on that class. Substitute these affine coefficient expressions into the rule, temporarily treating N as real. The resulting accepted

parameter set $S_c \subseteq \mathbb{R}$ is definable in $\mathcal{R}$. Quantified root coordinates cause no difficulty: real and imaginary parts of the polynomial root equations are polynomial equations, and definable sets are closed under real quantification.

By o-minimality, S_c is a finite union of intervals and points. Its membership is therefore constant for all sufficiently large positive parameters. Taking the maximum of the finitely many thresholds proves eventual constancy on every integer residue class. If r classes are eventually accepted, counting integers in these classes gives the first formula. If A is infinite then $r \geq 1$. Finally, the classical estimate $\pi(X) = o(X)$ shows that the number of accepted primes is $o(X)$, whereas the number of accepted integers is asymptotic to $rX/2^L > 0$. Removing the possible accepted value 1 proves the second formula. □

This applies, for example, to fixed rules made from polynomial root coordinates, real arithmetic, inequalities, exponential functions, and logarithms on positive arguments. It supplies no effective numerical bound on the final exceptional start. It excludes neither full-orbit rules of unbounded degree nor integer quantifiers, unrestricted periodic trigonometric functions, or functions not known to be definable in the chosen o-minimal structure. In particular, the theorem makes no blanket claim about rules involving the Gamma function. This is a direct application of classical o-minimality, not a claimed new general impossibility theorem for primality testing.

III.9 Inverse moments and the comparison with zeta

Throughout this section $N > 1$ is a terminating shortcut-Collatz start, $d = \tau(N)$, and

$$a_j = T^j(N), \qquad P_N(z) = \sum_{j=0}^{d} a_j z^j, \qquad a_0 = N, \quad a_d = 1.$$

Write the roots, with multiplicity, as $\xi_1, \ldots, \xi_d$. All are nonzero. The following identities are elementary consequences of polynomial factorization; their use here is to specify exactly what orbit information the zero configuration contains. No historical novelty is claimed for these identities.

Proposition III.9.1 (An exact zero-to-orbit formula)**.** *Put $c_j = a_j/N$ for $0 \leq j \leq d$, put $c_j = 0$ for $j > d$, and define*

$$S_m(N) = \sum_{k=1}^{d} \xi_k^{-m} \qquad (m \geq 1).$$

Then, for $|z| < \min_k |\xi_k|$,

$$-\frac{zP_N'(z)}{P_N(z)} = \sum_{m \geq 1} S_m(N) z^m. \tag{III.9.1}$$

For every $m \geq 1$ the coefficients satisfy

$$S_m = -mc_m - \sum_{j=1}^{m-1} c_j S_{m-j}, \qquad mc_m = -\sum_{j=1}^{m} S_j c_{m-j}. \tag{III.9.2}$$

In particular,

$$S_1(N) = \begin{cases} -\dfrac{1}{2}, & N \text{ even}, \\ -\dfrac{3}{2} - \dfrac{1}{2N}, & N \text{ odd}. \end{cases} \tag{III.9.3}$$

For odd $N > 1$, the first inverse-root moment alone recovers the start:

$$N = -\frac{1}{2S_1(N) + 3}.$$

Proof. The normalization at zero gives

$$\frac{P_N(z)}{N} = \prod_{k=1}^{d} \left(1 - \frac{z}{\xi_k}\right).$$

Logarithmic differentiation and the convergent geometric-series expansions of $z/(\xi_k - z)$ prove (III.9.1). Comparing coefficients in

$$\left(\sum_{j\geq 0} c_j z^j\right)\left(\sum_{m\geq 1} S_m z^m\right) = -\sum_{m\geq 1} m c_m z^m$$

gives (III.9.2). Taking $m = 1$ gives $S_1 = -T(N)/N$ and hence (III.9.3). □

For example, $3 \mapsto 5 \mapsto 8 \mapsto 4 \mapsto 2 \mapsto 1$ gives $P_3(z) = 3 + 5z + 8z^2 + 4z^3 + 2z^4 + z^5$. Its roots therefore satisfy $S_1 = -5/3$ and $S_2 = (5/3)^2 - 2(8/3) = -23/9$, without numerical root finding. The same relations reconstruct every normalized orbit coefficient from the inverse-root moments. This is an encoding identity, not an efficient primality test: obtaining exact roots or moments does not remove the need to determine the underlying integer.

III.9.1 A strip representation and an obstruction to identifying zeta zeros

The change of variables

$$z = 2\,3^{-s} = 2\exp(-s\log 3), \qquad F_N(s) = P_N(2\,3^{-s})$$

turns each orbit polynomial into an entire exponential polynomial, periodic with imaginary period $2\pi i/\log 3$. It sends the annulus $2/3 \leq |z| \leq 2$ to the strip $0 \leq \operatorname{Re} s \leq 1$. More precisely, each root ξ lifts to

$$s = \frac{\log(2/|\xi|) - i(\arg\xi + 2\pi k)}{\log 3}, \qquad k \in \mathbb{Z}. \tag{III.9.4}$$

Multiplicity is preserved. Choose one lift of each of the d polynomial roots, counted with multiplicity. By Theorem III.5.1, their average real part equals the proportion of odd steps up to $O(\log d/d)$. Here the pooled zeros come from many polynomials, whereas the zeta zeros belong to one fixed meromorphic function. The apparent resemblance to the critical strip of the Riemann zeta function is a coordinate normalization; every annulus of positive inner radius can similarly be normalized to a strip.

Corollary III.9.2 (The lifted accumulation set)**.** *By Theorem III.2.4, the pooled set $\mathcal{Z}$ of terminating Collatz-polynomial roots satisfies:*

$$\mathcal{Z}' = \{z : 2/3 \leq |z| \leq 2\}.$$

If $\mathcal{W} = \{s : F_N(s) = 0$ for some terminating $N > 1\}$, then

$$\mathcal{W}' = \{s : 0 \leq \operatorname{Re} s \leq 1\}.$$

In particular, $\mathcal{W}$ is dense throughout the open strip. The isolated root $-\rho$ from Theorem III.3.1 lifts to $\operatorname{Re} s = \log(2/\rho)/\log 3 > 1$: the strip describes accumulation, not every individual zero. The assertion that all these zeros lie on $\operatorname{Re} s = 1/2$ is false.

Proof. The exponential map $s \mapsto 2\,3^{-s}$ has nonzero derivative and is locally one-to-one. On a sufficiently small neighborhood of any finite s, distinct points and their images correspond under a holomorphic inverse. Therefore s is a distinct-point accumulation point of $\mathcal{W}$ if and only if $2\,3^{-s}$ is a distinct-point accumulation point of $\mathcal{Z}$. The claimed equality follows from $|2\,3^{-s}| = 2\,3^{-\operatorname{Re} s}$. □

Proposition III.9.3 (No common nonconstant holomorphic zero identification)**.** *Let U be a nonempty connected open subset of $2/3 < |z| < 2$. Suppose $f : U \to \mathbb{C}$ is holomorphic and $\zeta(f(\xi)) = 0$ for every $\xi \in \mathcal{Z} \cap U$. Then f is constant, with value a zero of the Riemann zeta function. Also, no nonzero meromorphic function on U can vanish at every point of $\mathcal{Z} \cap U$.*

Proof. The zeta function is meromorphic on $\mathbb{C}$, has a single pole at 1, and is not identically zero. Its zero set is therefore closed and discrete in $\mathbb{C}$; a sufficiently small neighborhood of its pole is zero-free. By continuity and density, $f(U)$ is contained in that zero set. Since the continuous image of a connected set is connected, $f(U)$ consists of a single zero. The last assertion is the identity theorem for meromorphic functions, applied on any disk avoiding their isolated poles. □

This obstruction concerns a single holomorphic coordinate map applied to *all* pooled Collatz roots in an open region. It does not exclude relations involving a selected subfamily, arithmetic transforms, or other functions constructed from Collatz dynamics. It does rule out treating the two zero sets as the same analytic object merely because both can be drawn inside a strip.

The established zeta–prime connection has additional arithmetic structure: for $\operatorname{Re} s > 1$,

$$\zeta(s) = \prod_{p \text{ prime}} (1 - p^{-s})^{-1}, \qquad -\frac{\zeta'(s)}{\zeta(s)} = \sum_{n \geq 1} \frac{\Lambda(n)}{n^s},$$

where $\Lambda(p^r) = \log p$ and $\Lambda(n) = 0$ otherwise. These are the Euler product and its logarithmic derivative [71, 72]. The nontrivial zeros of ζ are discrete in $0 < \operatorname{Re} s < 1$; the Riemann hypothesis places them on $\operatorname{Re} s = 1/2$ [73]. Classical explicit formulas connect those zeros to prime-power counting [74]. In contrast, (III.9.1) encodes Collatz orbit ratios. No Euler product over rational primes for P_N or F_N, and no implication for the Riemann hypothesis or the distribution of primes, is established here.

III.10 Exact finite searches for vertical-line zeros

The results in this section are exhaustive finite computations, not claims about unbounded starting values. All the tested orbits reached 1. Only integer and rational polynomial arithmetic was used; a floating-point distance from a line was never used to certify absence.

Imaginary axis. Write

$$P_N(z) = E_N(z^2) + zO_N(z^2).$$

For a nonzero real b, the equality $P_N(ib) = 0$ implies $E_N(-b^2) = O_N(-b^2) = 0$. An exhaustive computation gave

$$\gcd_{\mathbb{Q}[t]}(E_N, O_N) = 1 \qquad (3 \leq N \leq 10^6, \quad N \text{ odd}).$$

Thus none of these 499 999 odd starts has a purely imaginary zero. The stronger finite conclusion is that none has a nonconstant even polynomial factor. The constant polynomial $P_1 = 1$ also has no zeros. This does not settle the same question for every odd starting value.

Other rational vertical lines. For each $2 \leq N \leq 3000$ and $k \in \{-3, -2, -1, 1, 2, 3\}$, we tested the line $\operatorname{Re} z = k/2$. The complete list of nonreal intersections was as follows; an entry lists starting values, not distinct zeros.

Real part	Starting values N	Factor producing the zeros
-1	$4, 32, 84, 256, 672, 2048$	$z^2 + 2z + 4$
1	$32, 672, 2048$	$z^2 - 2z + 4$
$-3/2, -1/2, 1/2, 3/2$	none	—

All the listed starts are composite. The roots are $-1 \pm i\sqrt{3}$ or $1 \pm i\sqrt{3}$, respectively, and lie on $|z| = 2$. Real zeros, such as -1 for $N = 5$, are outside the scope of this table.

Together with the preceding rational-real-part proposition, these finite computations imply that no nonreal zero of P_p, for a prime $p \leq 3000$, has rational real part. The case $p = 2$ follows directly from $P_2 = z + 2$. This observation is not a proposed prime-only line: the searches found no prime-start nonreal zero on any rational vertical line in this range.

Exact certificates and modular filters. The first search computes the gcd of the even and odd coefficient parts. Reducing first modulo 2, then modulo 101, left respectively 167 521 and 1 661 candidates from the 499 999 odd starts. An exact rational gcd eliminated every remaining candidate. One of E_N, O_N is monic, so their monic rational gcd has integer coefficients. A nonconstant such gcd retains its degree under reduction modulo every prime. Hence a modular gcd equal to 1 is a valid certificate of absence, rather than a heuristic filter.

For the second search, put

$$F_{N,k}(w) = P_N(w + k/2) = E_{N,k}(w^2) + wO_{N,k}(w^2).$$

A nonreal zero on the specified line forces a nonconstant gcd of $E_{N,k}$ and $O_{N,k}$. Their coefficients belong to $\mathbb{Z}[1/2]$, and one is monic; reduction modulo the odd primes 101 and 103 is therefore sound. Every survivor was checked by the exact rational gcd of $P_N(z)$ and $P_N(k - z)$. The only nonconstant gcds obtained were the quadratic factors in the table, whose roots are explicit. Thus a possible symmetric factor with no actual zero on the line was not mistaken for an intersection.

Self-contained reproduction. The following complete Python listing uses SymPy for the final exact polynomial gcds. It generates its own orbit data and requires no companion program or data file. The output records the limits, candidate counts, and all surviving factors. The two computations may be run separately by omitting the other final function call.

```
import sympy as sp

def orbit(n):
    a = [n]
    while a[-1] != 1:
        v = a[-1]
        a.append(v // 2 if v % 2 == 0 else (3*v + 1) // 2)
    return a

def binary_gcd(a, b):
    # Bit j is the coefficient of t^j over F_2.
    while b:
        while a and a.bit_length() >= b.bit_length():
            a ^= b << (a.bit_length() - b.bit_length())
        a, b = b, a
    return a

def modular_gcd(a, b, p):
    a = [v % p for v in a]
    b = [v % p for v in b]
    while a and a[-1] == 0:
        a.pop()
```

```
    while b and b[-1] == 0:
        b.pop()
    while b:
        inv = pow(b[-1], -1, p)
        while a and len(a) >= len(b):
            q = a[-1] * inv % p
            shift = len(a) - len(b)
            for j, v in enumerate(b):
                a[shift+j] = (a[shift+j] - q*v) % p
            while a and a[-1] == 0:
                a.pop()
        a, b = b, a
    return a

def shifted_coefficients(a, k, p):
    # Ascending coefficients of P(w + k/2) modulo p.
    c = k * pow(2, -1, p) % p
    b = []
    for v in reversed(a):
        out = [0] * (len(b) + 1)
        for j, u in enumerate(b):
            out[j] = (out[j] + c*u) % p
            out[j+1] = (out[j+1] + u) % p
        out[0] = (out[0] + v) % p
        b = out
    return b

def imaginary_axis_search(limit=1000000):
    t = sp.Symbol('t')
    counts = [0, 0, 0]
    hits = []
    for n in range(3, limit + 1, 2):
        a = orbit(n)
        E, O = a[::2], a[1::2]
        e = sum((v & 1) << j for j, v in enumerate(E))
        o = sum((v & 1) << j for j, v in enumerate(O))
        counts[0] += 1
        if binary_gcd(e, o) <= 1:
            continue
        counts[1] += 1
        if len(modular_gcd(E, O, 101)) <= 1:
            continue
        counts[2] += 1
        f = sp.Poly.from_list(E[::-1], t)
        g = sp.Poly.from_list(O[::-1], t)
        h = sp.gcd(f, g)
        if h.degree() > 0:
            hits.append((n, str(h.as_expr())))
    print('imaginary', limit, counts, hits)

def vertical_line_search(limit=3000):
    z = sp.Symbol('z')
    ks = [-3, -2, -1, 1, 2, 3]
    counts = {k: [0, 0] for k in ks}
    hits = {k: [] for k in ks}
    for n in range(2, limit + 1):
        a = orbit(n)
        for k in ks:
            b = shifted_coefficients(a, k, 101)
            if len(modular_gcd(b[::2], b[1::2], 101)) <= 1:
                continue
            counts[k][0] += 1
            b = shifted_coefficients(a, k, 103)
```

```
            if len(modular_gcd(b[::2], b[1::2], 103)) <= 1:
                continue
            counts[k][1] += 1
            f = sp.Poly.from_list(a[::-1], z)
            g = sp.Poly(f.as_expr().subs(z, k-z), z)
            h = sp.gcd(f, g)
            if h.degree() > 0:
                hits[k].append((n, str(h.as_expr())))
    print('vertical', limit, counts, hits)

imaginary_axis_search()
vertical_line_search()
```

The first output is `imaginary 1000000 [499999, 167521, 1661] []`. The second gives the table above; its successive modular-survivor counts, in the order $k = -3, -2, -1, 1, 2, 3$, are $(25, 2), (36, 7), (30, 0), (35, 0), (28, 4), (36, 0)$. These finite data motivate further study of odd-start even factors and exact vertical lines, but do not establish their absence at all scales.

III.11 Prime starting values and arithmetic information in the zeros

Throughout this section, $T(n) = n/2$ for even n and $T(n) = (3n + 1)/2$ for odd n. The starting value is assumed to reach 1, and

$$P_N(z) = \sum_{j=0}^{d} T^j(N) z^j, \qquad d = \tau(N).$$

Thus P_N is monic, belongs to $\mathbb{Z}[z]$, and has constant term N. The universal bound $|\alpha| \leq 2$ for its complex zeros is used below. The arithmetic statements in this section do not prove that infinitely many prime starting values terminate. Neither do they identify the accumulation set obtained by restricting the starting values to primes.

III.11.1 Norms, degrees, and rational radii

Theorem III.11.1 (Prime factor and degree bound)**.** *Let p be a prime whose orbit terminates, and factor P_p into monic irreducible polynomials in $\mathbb{Z}[z]$, counting multiplicity. Exactly one factor, denoted G_p, has constant term of absolute value p. Every other factor has constant term 1 or -1. Moreover,*

$$\deg G_p \;\geq\; \lceil \log_2 p \rceil.$$

Every zero of G_p has algebraic degree $\deg G_p$ and algebraic norm of absolute value p. No zero of G_p lies on the complex unit circle. If two different prime starting values have a common zero, that zero is an algebraic unit.

Proof. Gauss's lemma gives the asserted monic integral factorization. The product of the nonzero integer constant terms is p, so exactly one factor has constant term of absolute value p, and all remaining factors have constant term of absolute value 1. Its occurrence has multiplicity one. If $r = \deg G_p$ and its complex roots are $\alpha_1, \ldots, \alpha_r$, then

$$p = \prod_{j=1}^{r} |\alpha_j| \leq 2^r,$$

which proves the degree estimate. Irreducibility identifies the algebraic degree and norm of each of its roots.

For the unit-circle assertion, let f be the monic minimal polynomial of an algebraic integer α satisfying $|\alpha| = 1$. Its coefficients are real, so $\alpha^{-1} = \overline{\alpha}$ is also a root. Consequently f and $f^*(z) = z^{\deg f} f(1/z)$ have a common root. Irreducibility implies $f^* = f(0)f$; comparison of constant terms gives $f(0)^2 = 1$. Thus every algebraic integer on the unit circle is an algebraic unit, whether or not it is a root of unity. This excludes zeros of G_p. Finally, the minimal polynomial of a zero common to P_p and P_q has constant term dividing both distinct primes p and q, hence has constant term 1 or -1. □

Theorem III.11.2 (Arithmetic restrictions on exact radii)**.** *If a zero α of a Collatz polynomial has rational squared modulus, then $|\alpha|^2$ is a positive integer. In particular, no zero lies exactly on $|z| = 2/3$, even though that circle belongs to the accumulation set. More strongly, if $p \geq 5$ is prime and α is a zero of P_p, then*

$$|\alpha|^2 \in \mathbb{Q} \quad \Longrightarrow \quad |\alpha| = 1.$$

Consequently every zero of G_p has irrational squared modulus.

Proof. Both α and $\overline{\alpha}$ are algebraic integers, so $m = \alpha\overline{\alpha}$ is an algebraic integer. If rational, it is an integer. It is positive because the constant term of P_N is nonzero. This immediately excludes $m = 4/9$.

Now let f be the monic minimal polynomial of a zero of P_p, with degree r and constant term c. Its constant term divides p, so $|c|$ is 1 or p. If $m = |\alpha|^2$ is an integer, then $m/\alpha = \overline{\alpha}$ is a root of f. Thus f divides $z^r f(m/z)$, and comparison of leading coefficients gives $z^r f(m/z) = cf(z)$. Comparison of constant terms then gives

$$c^2 = m^r.$$

If $|c| = 1$, this forces $m = 1$. If $|c| = p$, unique factorization forces either $r = 1, m = p^2$ or $r = 2, m = p$. For $p \geq 5$, both possibilities contradict $m \leq 4$, the universal complex root bound. Finally, the roots of G_p cannot have modulus one by Theorem III.11.1. □

Remark III.11.3. The theorem concerns the minimal polynomial and all algebraic conjugates of a zero. A complex zero merely lying at a particular location in the annulus does not provide a primality test. The irreducibility conclusion is also deliberately limited: prime p need not make P_p irreducible. For example,

$$P_5(z) = z^4 + 2z^3 + 4z^2 + 8z + 5 = (z + 1)(z^3 + z^2 + 3z + 5).$$

The cubic is irreducible by the rational-root test; it is G_5. In particular, a prime starting value can give an exact unit-circle zero.

Corollary III.11.4 (A sufficient irreducibility criterion)**.** *If p is prime, its orbit terminates, and all zeros of P_p have modulus strictly greater than 1, then P_p is irreducible over $\mathbb{Q}$.*

Proof. A proper factorization would have a nonconstant monic factor with constant term of absolute value 1. The product of the moduli of its roots would then be 1, contradicting the strict lower bound for every root. □

Proposition III.11.5 (An orbit edge has no shared zero)**.** *If $N > 1$ terminates and $d = \tau(N)$, then*

$$\gcd(P_N, P_{T(N)}) = 1, \qquad \operatorname{Res}(P_N, P_{T(N)}) = N^{d-1}.$$

For a prime starting value $p > 2$, the only possible rational zero of P_p is -1.

Proof. The recurrence $P_N(z) = N + zP_{T(N)}(z)$ rules out any common zero. At each of the $d-1$ roots of the monic polynomial $P_{T(N)}$, the value of P_N is N. Hence the resultant in that order is N^{d-1}. Reversing the order introduces the sign $(-1)^{d(d-1)} = 1$. The formula also holds for $d = 1$, when $P_{T(N)} = 1$. For the rational-root assertion, a rational root is an integer divisor of p; positivity of the coefficients excludes positive roots, and the bound $|\alpha| \le 2$ excludes $-p$ when $p > 2$. □

III.11.2 An exact invariant in an inverse-orbit family

Hohertz and Kalantari proved that $P_N(-1) = 0$ when the only nonterminal odd term of the orbit is N, and that a zero at -1 propagates through an inverse-orbit construction. They also gave the example $N = 820569$, which is not of the first type [25, Theorem 7 and Lemma 11]. The following polynomial identity records that mechanism at every complex argument and completely classifies common zeros of adjacent members.

Theorem III.11.6 (Inverse-family identity and common-zero classification)**.** *Let $n > 1$ be odd and terminating. Then $4n+1$ terminates and*

$$P_{4n+1}(z) = z^2 P_n(z) + (z+1)\big((2n+1)z + 4n + 1\big). \tag{III.11.1}$$

Consequently,

$$P_{4n+1}(-1) = P_n(-1),$$

and, with the gcd chosen monic in $\mathbb{Q}[z]$,

$$\gcd(P_n, P_{4n+1}) = \begin{cases} z+1, & P_n(-1) = 0, \\ 1, & P_n(-1) \neq 0. \end{cases}$$

In particular, the only possible common zero is -1, and its multiplicity in the gcd is exactly one.

Proof. Write $t = T(n) = (3n+1)/2$. The initial orbit segment of $4n+1$ is $4n+1, 4t, 2t, t$, after which the orbit of t is followed. Hence

$$P_{4n+1} = 4n + 1 + 4tz + 2tz^2 + z^3 P_t, \qquad P_n = n + zP_t.$$

Subtracting $z^2 P_n$ gives $(4n+1) + (6n+2)z + (2n+1)z^2$, which factors as in (III.11.1). Evaluating at -1 proves the invariant. Any common zero must be -1 or $-(4n+1)/(2n+1)$. The latter is a noninteger rational number, since $(4n+1)/(2n+1) = 2 - 1/(2n+1)$. It cannot be a zero of a monic integer polynomial. At $z = -1$ the second linear factor in the right side of (III.11.1) equals $2n \neq 0$, so the difference has a simple zero there. The gcd therefore cannot contain $(z+1)^2$. □

Corollary III.11.7 (Invariant along the entire family)**.** *For $n_a = 4^a n + (4^a - 1)/3$, $a \ge 0$, with odd terminating $n > 1$,*

$$\tau(n_a) = \tau(n) + 2a, \qquad P_{n_a}(-1) = P_n(-1).$$

If $P_n(-1) = 0$, then

$$P'_{n_a}(-1) = P'_n(-1) + 2\sum_{j=0}^{a-1} n_j.$$

Thus at most one member of such a family has a multiple zero at -1.

Proof. The orbit calculation in the theorem adds two steps to the stopping time; iteration gives the first two formulas. Differentiating (III.11.1) at -1 gives

$$P'_{4n+1}(-1) = P'_n(-1) - 2P_n(-1) + 2n.$$

When the invariant vanishes, iteration proves the displayed derivative formula. The derivative values strictly increase because each $n_j > 0$. □

III.11.3 A certified prime with a prescribed unit-circle zero

The initial example $p = 5$ is not the only prime example. One explicit, moderately sized example is

$$p = 860429309269 = 4^{10} \cdot 820569 + \frac{4^{10} - 1}{3}. \tag{III.11.2}$$

Its stopping time is 127, and $P_p(-1) = 0$. These are exact arithmetic assertions, not conclusions from a numerical root plot.

For completeness, the finite orbit checks can be reproduced entirely by the following integer procedure. Start with $(x, j, E, O) = (N, 0, 0, 0)$; add x to E if j is even and to O if j is odd. If $x = 1$, stop; otherwise replace x by $T(x)$, increase j by one, and repeat. The output is $(\tau(N), E, O)$ and $P_N(-1) = E - O$. The exact outputs are

N	$\tau(N)$	E	O
820569	107	5125979	5125979
860429309269	127	1720862103379	1720862103379.

The first row also appears as a zero example in the cited literature; the second follows from the invariant, independently of rerunning the larger orbit. No external program is needed to specify the verification.

Here is a compact primality certificate for (III.11.2). Use the elementary form of Pocklington's criterion: if a known prime q divides $m - 1$, $q > \sqrt{m}$, and some integer a satisfies

$$a^{m-1} \equiv 1 \pmod{m}, \qquad \gcd(a^{(m-1)/q} - 1, m) = 1,$$

then m is prime. Indeed, every prime divisor r of m must have $q \mid r - 1$, so every such divisor exceeds $\sqrt{m}$, which precludes compositeness.

The following table supplies a chain of certificates; in every row $a = 2$, $2^{m-1} \bmod m = 1$, and the gcd of the last entry minus one with m is 1.

m	q	$(m-1)/q$	$2^{(m-1)/q} \bmod m$
17827	2971	6	64
32052947	17827	1798	20474078
860429309269	32052947	26844	585577902723

Each displayed q exceeds $\sqrt{m}$. The seed 2971 is prime by trial division: the primes at most $\sqrt{2971} < 55$ are $2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53$, and the corresponding nonzero remainders are $1, 1, 1, 3, 1, 7, 13, 7, 4, 13, 26, 11, 19, 4, 10, 3$. All modular powers in the table can be checked by repeated squaring. This proves the primality of the stated starting value with a short, self-contained certificate. It makes no claim of a new largest prime or of infinitely many prime values in this fixed inverse family.

III.11.4 A distinguished zero in the p-adic metric

The geometry of a complex annulus and divisibility by a prime measure different properties. A direct link to divisibility does become exact when zeros are considered p-adically. We normalize the p-adic valuation by $v_p(p) = 1$; its associated absolute value is $|x|_p = p^{-v_p(x)}$.

Theorem III.11.8 (The distinguished p-adic zero)**.** *Let p be an odd prime with a terminating shortcut-Collatz orbit. In an algebraic closure of $\mathbb{Q}_p$, the valuations of the zeros of P_p, counted with multiplicity, are exactly*

$$1, 0, 0, \ldots, 0.$$

The unique zero ρ_p of positive valuation belongs to $\mathbb{Q}_p$, is simple, and satisfies

$$\rho_p \equiv -2p \pmod{p^2}.$$

More precisely,

$$\rho_p \equiv \begin{cases} -2p + 4p^2 \pmod{p^3}, & p \equiv 1 \pmod 4, \\ -2p - 4p^2 \pmod{p^3}, & p \equiv 3 \pmod 4. \end{cases}$$

It is a zero of the unique factor G_p in Theorem III.11.1.

Proof. The constant and linear coefficients of P_p are p and $a_1 = (3p+1)/2$, respectively. The reduction modulo p has a simple zero at 0, since $a_1 \not\equiv 0 \pmod p$. One can lift this zero without any assumed factorization: given x_k with $P_p(x_k) \equiv 0 \pmod{p^k}$ and $x_k \equiv 0 \pmod p$, there is a unique $t \pmod p$ such that

$$P_p(x_k + tp^k) \equiv P_p(x_k) + tp^k P_p'(x_k) \equiv 0 \pmod{p^{k+1}}.$$

The resulting compatible residues define a unique $\rho_p \in p\mathbb{Z}_p$. Its derivative is a unit, proving simplicity. Substitution modulo p^2 gives $\rho_p \equiv -p/a_1 \equiv -2p \pmod{p^2}$, so $v_p(\rho_p) = 1$.

Every zero of a monic polynomial over $\mathbb{Z}_p$ has nonnegative valuation: at a putative zero of negative valuation, the leading term would have strictly smaller valuation than every other term and could not cancel. The product of all zeros has valuation $v_p(p) = 1$. The one zero already found contributes 1, so all others contribute 0. A monic factor with constant term 1 or -1 has only zeros of valuation zero. Therefore ρ_p belongs to G_p.

For the refinement, the quadratic coefficient is

$$a_2 = \begin{cases} (3p+1)/4, & p \equiv 1 \pmod 4, \\ (9p+5)/4, & p \equiv 3 \pmod 4. \end{cases}$$

Put $\rho_p = -2p + cp^2 \pmod{p^3}$. Terms of degree at least three vanish modulo p^3, and substitution yields $c/2 - 3 + 4a_2 \equiv 0 \pmod p$. The two residues of a_2 give $c \equiv 4$ or -4, as claimed. □

Remark III.11.9 (What the p-adic theorem does not assert). The prime is an input to the theorem. The lifted zero does not certify that an unknown integer is prime: for an odd composite N, the same simple-root lifting works separately at every odd prime dividing N, because $T(N) = (3N+1)/2$ is coprime to those primes. The new information is an exact description of divisibility in the algebraic roots, not a shortcut to primality testing. Nor is ρ_p an extra point in the complex annulus; it is a zero interpreted in a different absolute value.

III.11.5 Limits and provenance

The factor-norm argument and the p-adic lifting method are elementary consequences of standard algebra; their specialization here should not be advertised as the solution of a famous open problem. The invariant at -1 extends a mechanism already present in Hohertz–Kalantari. The exact polynomial identity, the adjacent-family gcd classification, and the explicit certified prime example provide concrete refinements suitable for independent checking. Historical novelty for these refinements is not asserted. None establishes a connection to the nontrivial zeros of the Riemann zeta function, whose analytic structure is substantially different.

III.12 Prime starting values: radial separation and algebraic rotations

Let $T(N) = N/2$ for even N and $T(N) = (3N+1)/2$ for odd N. Only starting values that reach 1 are considered, and

$$P_N(z) = \sum_{j=0}^{\tau(N)} T^j(N) z^j.$$

This is a monic integer polynomial with constant term N. An algebraic integer is called a unit if its monic minimal polynomial has constant term 1 or -1. The results below concern exact algebraic zeros and exact moduli. They do not assume that all positive integers, or infinitely many primes, have terminating orbits.

III.12.1 Which irreducible factor carries the prime?

Proposition III.12.1 (Constant terms of irreducible factors)**.** *If $N > 1$ is odd and terminating, the constant terms of distinct nonunit irreducible factors of P_N are pairwise coprime, and every such factor occurs with multiplicity one. Each entire prime power dividing N belongs to the constant term of exactly one factor.*

Proof. Gauss's lemma factors P_N into monic irreducible integer polynomials. Their nonzero constant terms, with multiplicities, multiply to N. For prime $N = p$, exactly one has absolute value p, and the others have absolute value 1.

For odd N, the constant and linear coefficients satisfy

$$\gcd(P_N(0), P_N'(0)) = \gcd\left(N, \frac{3N+1}{2}\right) = 1.$$

If a prime ℓ divided the constant terms of two factors, or the constant term of a factor occurring twice, differentiation of the factorization would show that ℓ divides $P_N'(0)$ as well as N. This is impossible. The final assertion follows by taking the product of the constant terms. □

III.12.2 One zero with positive valuation

Lemma III.12.2 (A local separation property)**.** *Let $N > 1$ be odd and terminating, let ℓ be a prime dividing N, and put $e = v_\ell(N)$. Among the zeros of P_N in $\overline{\mathbb{Q}}_\ell$, precisely one has positive valuation. It belongs to $\mathbb{Q}_\ell$, is simple, and has valuation e. Every other zero has valuation zero.*

Proof. Write $P_N(z) = N + a_1 z + \cdots + z^d$. Here $a_1 = (3N+1)/2$ is a unit modulo ℓ, so 0 is a simple root modulo ℓ. To lift it, given a root x_k modulo ℓ^k with $x_k \equiv 0 \pmod{\ell}$, choose the unique $t \pmod{\ell}$ solving

$$P_N(x_k) + t\ell^k P_N'(x_k) \equiv 0 \pmod{\ell^{k+1}}.$$

The compatible residues define a unique simple root $\rho \in \ell\mathbb{Z}_\ell$. In the equation $N + a_1\rho + \sum_{j\geq 2} a_j \rho^j = 0$, a unique term would have smallest valuation unless $v_\ell(\rho) = e$.

Every zero of a monic polynomial over $\mathbb{Z}_\ell$ has nonnegative valuation: at a zero of negative valuation, the leading term alone would have smallest valuation. The product of all zeros has valuation e, already contributed by ρ. All the remaining valuations must therefore be zero. □

Lemma III.12.3 (Distinct radii inside an odd-start polynomial)**.** *Let $N > 1$ be odd and terminating. If α is a nonunit zero of P_N and β is any zero of P_N with $|\beta| = |\alpha|$, then*

$$\beta = \alpha \quad \text{or} \quad \beta = \overline{\alpha}.$$

Proof. Choose a prime ℓ dividing the constant term of the minimal polynomial of α. At least one of its conjugates has positive ℓ-adic valuation, and by Lemma III.12.2 that conjugate must be the unique positive-valuation zero ρ of P_N. Choose an embedding σ of a splitting field into $\overline{\mathbb{Q}}_\ell$ sending α to ρ. Put $e = v_\ell(\rho) > 0$.

The equality of complex moduli is the algebraic identity

$$\alpha\overline{\alpha} = \beta\overline{\beta}.$$

If α is nonreal, its conjugate is distinct, so the valuation of the left side after applying σ is exactly e. The right side can have positive valuation only if $\sigma(\beta) = \rho$ or $\sigma(\overline{\beta}) = \rho$. Injectivity gives the conclusion. If α is real, the left side has valuation $2e$. Two distinct zeros cannot both have positive valuation, so β must be real and satisfy $\sigma(\beta) = \rho$, again giving $\beta = \alpha$. □

III.12.3 Circles separating different prime starts

Theorem III.12.4 (Radial separation of the prime factors)**.** *Let p be an odd prime and q any prime, both with terminating orbits. Let α be a zero of G_p, and let β be any zero of P_q. Then*

$$|\alpha| = |\beta| \quad \Longrightarrow \quad q = p \quad \textit{and} \quad \beta \in \{\alpha, \overline{\alpha}\}.$$

Thus the circle centered at 0 through a zero of G_p contains no zero from any other prime starting value. Within P_p, it contains exactly that conjugate pair, or just that one zero if it is real. The statement excludes the unit factors of P_p.

Proof. Let $d = [\mathbb{Q}(\alpha) : \mathbb{Q}]$, let $h = [\mathbb{Q}(\beta) : \mathbb{Q}]$, and let c be the absolute value of the constant term of the minimal polynomial of β. Proposition III.12.1 gives $c \in \{1, q\}$. Choose a finite number field L containing $\alpha, \overline{\alpha}, \beta, \overline{\beta}$, and put $D = [L : \mathbb{Q}]$. Taking field norms of $\alpha\overline{\alpha} = \beta\overline{\beta}$ gives

$$p^{2D/d} = c^{2D/h}.$$

The exponents are positive integers. Thus $c \neq 1$, and unique prime factorization forces $q = p$. The zeros now belong to the same polynomial P_p, so Lemma III.12.3 completes the proof. □

Corollary III.12.5 (Multiplicative independence of selected radii)**.** *Choose a zero α_i of G_{p_i} for each of finitely many distinct terminating odd primes $p_1, \ldots, p_s$, and put $r_i = |\alpha_i|$. Then the r_i are multiplicatively independent:*

$$\prod_{i=1}^{s} r_i^{n_i} = 1, \quad n_i \in \mathbb{Z} \quad \Longrightarrow \quad n_1 = \cdots = n_s = 0.$$

Equivalently, the real numbers $\log r_i$ are linearly independent over $\mathbb{Q}$.

Proof. Square the proposed relation and choose a finite number field L containing all α_i and their complex conjugates. With $D = [L : \mathbb{Q}]$ and $d_i = \deg G_{p_i}$, its norm is

$$\prod_{i=1}^{s} p_i^{2Dn_i/d_i} = 1.$$

Every exponent is an integer. Unique prime factorization, allowing negative exponents in $\mathbb{Q}^\times$, forces each exponent to vanish. Clearing denominators proves the equivalent statement about rational linear relations between logarithms. □

Corollary III.12.6 (Transcendental logarithmic ratios)**.** *Let $p \neq q$ be terminating odd primes, let α be a zero of G_p, and let β be a zero of G_q. Then*

$$\frac{\log|\alpha|}{\log|\beta|} \quad \textit{is transcendental,}$$

where both logarithms are the real natural logarithms of positive numbers.

Proof. Put $r_p = |\alpha|$ and $r_q = |\beta|$. These are positive algebraic numbers: their squares are respectively $\alpha\overline{\alpha}$ and $\beta\overline{\beta}$. Corollary III.12.5 implies that neither equals 1, and that $t = \log r_p / \log r_q$ is not rational. If t were algebraic irrational, the classical Gelfond–Schneider theorem [66, Theorem 3.2] would make r_q^t transcendental. That theorem states that every value of a^b is transcendental when a is algebraic, $a \neq 0, 1$, and b is algebraic irrational. In the present positive real branch, however,

$$r_q^t = \exp(t \log r_q) = r_p,$$

which is algebraic. This contradiction proves the assertion. This is a direct application of the classical theorem, with no claim of historical novelty. □

III.12.4 Rotations and irrational arguments

Theorem III.12.7 (No nontrivial root-of-unity rotations)**.** *Let α be a zero of G_p, where p is an odd terminating prime. If q is another terminating prime, $P_q(\beta) = 0$, and ζ is a root of unity such that*

$$\beta = \zeta\alpha \quad \textit{or} \quad \beta = \zeta\overline{\alpha},$$

then $q = p$ and $\zeta = 1$. Consequently every nonreal zero α of G_p satisfies

$$\frac{\arg\alpha}{\pi} \notin \mathbb{Q}.$$

Real zeros are negative and have argument π modulo 2π; they are expressly excluded from this irrational-angle assertion.

Proof. The case $q = 2$ is impossible because its only zero is -2 and norms in a common number field would equate a power of p with a power of 2. For odd q, equality of moduli and Theorem III.12.4 first give $q = p$.

To exclude the rotation, suppose first that $\beta = \zeta\alpha$. Embed a splitting field containing ζ into $\overline{\mathbb{Q}}_p$ so that α maps to the unique positive-valuation zero ρ of P_p. A root of unity has valuation zero, because its finite power is 1. Thus the image of β also has positive valuation and must equal ρ. Injectivity implies $\beta = \alpha$ and $\zeta = 1$. Replace α by its conjugate to handle the other case.

If a nonreal α had $\arg\alpha/\pi \in \mathbb{Q}$, then $\overline{\alpha}/\alpha$ would be a root of unity. The result just proved would give $\overline{\alpha} = \alpha$, a contradiction. Positivity of all coefficients of P_p rules out nonnegative real zeros, proving the final assertion. □

Remark III.12.8. The same valuation argument proves the within-polynomial rotation restriction for any nonunit zero of an odd-start P_N. Thus irrational arguments and the absence of opposite nonunit pairs are not, individually, properties exclusive to primes. What separates different prime starts in Theorem III.12.4 is the field norm together with unique prime factorization.

III.12.5 What a composite radius collision would require

Proposition III.12.9 (Necessary conditions for a composite collision)**.** *Let α be a zero of G_p, with $d = \deg G_p$, and let β be a zero of any terminating P_N. If $|\alpha| = |\beta|$, then the minimal polynomial of β has constant term of absolute value p^e and degree ed for an integer $e \geq 1$.*

If N is odd, then $v_p(N) = e$, and the possibilities are further restricted as follows:

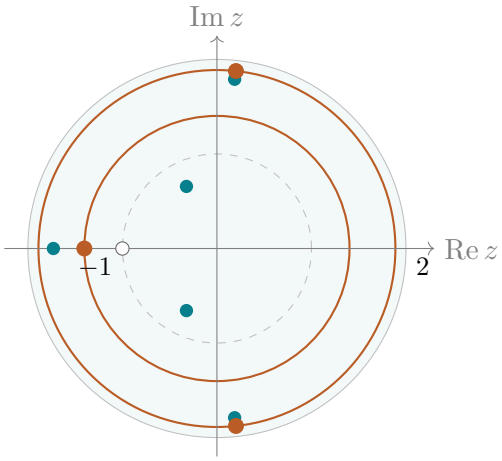
- *If α is nonreal, then β is nonreal and $e = 1$.*
- *If α is real and β is real, then $e = 1$.*
- *If α is real and β is nonreal, then $e = 2$.*

These are necessary conditions, not existence statements.

Proof. Use the common-field norm argument in the proof of Theorem III.12.4, now with arbitrary nonzero integer constant term c and degree h for the minimal polynomial of β. It yields $p^{2D/d} = |c|^{2D/h}$, or equivalently $p^h = |c|^d$. Hence $|c| = p^e$ and $h = ed$ for an integer $e \geq 1$.

When N is odd, Proposition III.12.1 implies $e = v_p(N)$. Embed a field containing the relevant roots into $\overline{\mathbb{Q}}_p$ so that α maps to the unique positive-valuation root of P_p. The image of $\alpha\overline{\alpha}$ has valuation 1 when α is nonreal, and 2 when it is real. By Lemma III.12.2, the image of $\beta\overline{\beta}$ has valuation either 0 or e if β is nonreal, and either 0 or $2e$ if β is real. Equating these valuations gives exactly the listed possibilities. □

III.12.6 Examples and limitations



filled teal: zeros of P_3
filled orange: zeros of G_5

Figure 3: The two circles determined by G_5. No zero belonging to any other terminating prime start lies on either circle, by Theorem III.12.4. The additional zero -1 of P_5 is a unit zero and is excluded. The plotted coordinates are numerical illustrations; the separation is proved exactly.

For $p = 5$,

$$P_5(z) = (z+1)G_5(z), \qquad G_5(z) = z^3 + z^2 + 3z + 5.$$

The cubic is irreducible by the rational-root test. Its unique real root is $a \approx -1.402627941186$, and its nonreal roots form a conjugate pair of common modulus $\sqrt{-5/a}$. The two radii supplied by its nonunit factor are therefore

$$-a \approx 1.402627941186, \qquad \sqrt{-5/a} \approx 1.888051164617.$$

Theorem III.12.4 says that neither circle contains a zero from any other terminating prime start. The additional zero -1 belongs to a unit factor and is outside the theorem's scope. Composite starts are not excluded from either circle by this theorem. In particular, a finite check excluding the exact factor G_5 from other P_N does not exclude a different minimal polynomial whose roots have the same modulus.

Having one nonunit Galois orbit is not a primality criterion. The composite start 9 has orbit

$$9, 14, 7, 11, 17, 26, 13, 20, 10, 5, 8, 4, 2, 1.$$

Its degree-13 polynomial is irreducible over $\mathbb{Q}$. Here is a short exact certificate: its reduction modulo 2 is

$$f(z) = z^{13} + z^9 + z^6 + z^4 + z^3 + z^2 + 1,$$

and repeated squaring gives $z^{8192} \equiv z \pmod f$, while $f(0) = f(1) = 1$ (see the integer-only certificate accompanying Theorem III.3.1). Over $\mathbb{F}_2$, every irreducible factor of $z^{2^{13}} - z$ has degree dividing 13. Since 13 is prime and f has no linear factor, this proves that f is irreducible. Thus P_9 has one nonunit Galois orbit of zeros, with algebraic norm of absolute value 9, despite its composite starting value. Its nonreal zeros also satisfy the irrational-angle property described above.

These results describe exact arithmetic restrictions on locations in the annulus. They provide no efficient algorithm for testing an unknown starting value for primality, and no theorem that a fixed visible curve contains zeros from all prime starts and none from composite starts. The proofs are applications of standard algebraic-number methods; historical novelty is not asserted.

III.13 Exact collisions with a prime root are quantitatively sparse

For real $X > 1$, define

$$A_p(X) = \#\{N \in \mathbb{Z} : 1 < N \le X,\ \tau(N) < \infty,\ G_p \mid P_N\}.$$

Equivalently, choose any root α of G_p and count the terminating starts whose polynomial vanishes at that exact algebraic number. The count is independent of which conjugate root is chosen, because $P_N(\alpha) = 0$ is equivalent to $G_p \mid P_N$.

Theorem III.13.1 (Sparse exact collisions with a prime root)**.** *For every odd terminating prime p and every real $X > 1$,*

$$\boxed{A_p(X) \le 3X^{\beta_p}, \qquad \beta_p = \frac{\log 2}{\log(2p)} < 1.}$$

In particular, the set of starts sharing a root of G_p has natural density zero. Every such start other than p is composite.

Proof. We first obtain a prime-adic conjugate of the chosen root. Normalize the p-adic valuation by $v_p(p) = 1$. The polynomial P_p satisfies

$$P_p(0) = p, \qquad P_p'(0) = \frac{3p+1}{2} \not\equiv 0 \pmod p.$$

Thus 0 is a simple root modulo p. Hensel's lemma gives $\rho \in p\mathbb{Z}_p$ with $P_p(\rho) = 0$. In the equation

$$0 = p + \frac{3p+1}{2}\rho + \sum_{j=2}^{\tau(p)} T^j(p)\rho^j,$$

valuation $v_p(\rho) > 1$ would leave the constant term as the unique term of smallest valuation. Hence $v_p(\rho) = 1$. Write $P_p = G_p H_p$, where $H_p(0) = \pm 1$. Since $H_p(\rho) \equiv H_p(0) \not\equiv 0 \pmod p$, the root ρ belongs to G_p. Consequently every start counted by $A_p(X)$ satisfies $P_N(\rho) = 0$.

Fix an integer $r \geq 1$. For a start with $\tau(N) \geq r - 1$, terms of degree at least r vanish modulo p^r at ρ, so

$$0 = P_N(\rho) \equiv \sum_{j=0}^{r-1} T^j(N)\rho^j \pmod{p^r}.$$

Fixing N modulo 2^{r-1} fixes its first $r-1$ parity decisions. Indeed, along common initial decisions, a difference initially divisible by 2^{r-1} is at each step multiplied by either $1/2$ or $3/2$; it remains even until the last required decision. Thus on each of these 2^{r-1} residue classes there are fixed integers s_j, b_j such that

$$T^j(N) = \frac{3^{s_j} N + b_j}{2^j} \qquad (0 \leq j \leq r-1),$$

with $s_0 = b_0 = 0$. Because p is odd, all these rational coefficients are p-adic integers. Substituting them into the truncated evaluation gives a congruence

$$U_v N + V_v \equiv 0 \pmod{p^r}, \qquad U_v = \sum_{j=0}^{r-1} \frac{3^{s_j}}{2^j}\rho^j \equiv 1 \pmod p.$$

Here v labels the chosen residue class modulo 2^{r-1}. The coefficient U_v is a p-adic unit. Therefore this equation specifies exactly one residue class for N modulo p^r. By the Chinese remainder theorem, together with the chosen class modulo 2^{r-1} it specifies one class modulo $2^{r-1}p^r$. The number of counted starts with $\tau(N) \geq r - 1$ is consequently at most

$$2^{r-1}\left(\frac{X}{2^{r-1}p^r} + 1\right) = \frac{X}{p^r} + 2^{r-1}.$$

The shorter orbits contribute at most 2^{r-1} further starts. To see this, use $n \leq 2T(n)$ at every step to obtain $N \leq 2^{\tau(N)}$. If $\tau(N) < r - 1$, then $N < 2^{r-1}$; for $r = 1$ there are no such orbits. We have proved, for every $r \geq 1$,

$$A_p(X) \leq \frac{X}{p^r} + 2^r.$$

Take

$$r = \left\lceil \frac{\log X}{\log(2p)} \right\rceil.$$

Then $r \geq 1$, $p^r \geq X^{1-\beta_p}$ and $2^r \leq 2X^{\beta_p}$. Substitution gives the claimed bound. Since $\beta_p < 1$, dividing it by X proves density zero. Finally, divisibility $G_p \mid P_N$ implies $p \mid N$ by constant terms; hence any positive such $N \neq p$ is composite. □

Example: the prime 5. Here

$$P_5(z) = (z+1)(z^3 + z^2 + 3z + 5), \qquad \beta_5 = \frac{\log 2}{\log 10} = 0.3010299956\ldots.$$

For any exact root α of $z^3 + z^2 + 3z + 5$,

$$\#\{N \leq X : \tau(N) < \infty,\ P_N(\alpha) = 0\} \leq 3X^{0.3010299956\ldots}.$$

For instance, at $X = 10^6$ the exact exponent gives the upper bound $3 \cdot 2^6 = 192$, including the known start $N = 5$. This is an upper bound, not an assertion that any additional start exists. The exponent in the theorem is the exact logarithmic ratio, not its displayed decimal approximation.

Meaning and limitations. The result controls exact equality of algebraic zeros, not numerical closeness. It is compatible with composite-start zeros being dense in every open part of the accumulation annulus. It does not prove that $A_p(X) = 1$ for all $X \geq p$, and it does not prove that there is even one composite collision. It also does not assume that all Collatz orbits terminate: only the starts whose finite polynomials exist are counted. No historical priority claim is made for this deduction.

This sparsity mechanism is not an exclusive property of prime inputs. The proof only needs a fixed algebraic root with a conjugate $\rho \in \mathbb{Q}_q$ of positive valuation $v_q(\rho) = e$ at an odd prime q. In that situation the same argument uses q^{er} in place of p^r and gives an exponent $\log 2/\log(2q^e)$. Suitable nonunit factors from composite starts can therefore satisfy analogous bounds. The theorem supplies a quantitative restriction on possible imitators of a prime's exact root; it is not a prime-only geometric locus or a new primality test.

III.14 Shared zeros and a finite uniqueness window for prime starts

Let $T(n) = n/2$ for even n and $T(n) = (3n+1)/2$ for odd n, and let $P_N(z) = \sum_{j=0}^{\tau(N)} T^j(N) z^j$ for a starting value whose orbit reaches 1. For a prime p, write G_p for the unique monic irreducible factor of P_p whose constant term has absolute value p. All other irreducible factors have constant term 1 or -1.

III.14.1 A prime norm does not identify a prime start

The exact factorization

$$P_{10}(z) = (z+2)(z^4 + 4z^2 + 5)$$

shows that a composite starting value can have an irreducible factor with prime constant term. Indeed, translation by 1 changes the quartic to

$$z^4 + 4z^3 + 10z^2 + 12z + 10,$$

which is Eisenstein at 2. Thus the quartic is irreducible, and each of its zeros has algebraic norm 5, although the starting value 10 is composite. Those zeros satisfy $z^2 = -2 \pm i$, so their squared moduli are $\sqrt{5}$. Prime algebraic norm, irrational squared modulus, and location strictly inside the outer circle therefore do not, together, certify prime input.

Even an exact zero can be shared between a prime and infinitely many composite starts. For $p = 2$, $G_2(z) = z + 2$, and

$$G_2 \mid P_{2^{2r+1}} \qquad (r \geq 0).$$

This follows by substituting -2 in the geometric polynomial for an orbit consisting entirely of powers of 2. Thus the same nonunit zero -2 belongs to the prime start 2 and the composite starts $8, 32, 128, \ldots$. The unit zero -1 is also shared: it belongs to P_5 and to $P_{21}, P_{85}, \ldots$. These facts do not decide whether a zero of G_p for an odd prime can be shared with a composite start.

III.14.2 A necessary divisibility pattern

Theorem III.14.1 (Restriction on a shared prime factor)**.** *Let p be an odd prime with a terminating orbit. Suppose N terminates and $G_p \mid P_N$. Put $k = v_2(N)$. Then*

$$N = 2^k p^{k+1} u, \qquad u \text{ odd}, \qquad u \equiv (-1)^k \pmod{p}. \tag{III.14.1}$$

In particular, $v_p(N) = v_2(N) + 1$. For odd N, the stronger-looking integer congruence is

$$N \equiv p \pmod{2p^2}.$$

Proof. The divisibility of constant terms first gives $p \mid N$. The polynomial P_p has constant term p and linear coefficient $(3p+1)/2$, a unit modulo p. Its simple zero 0 modulo p lifts to a zero $\rho \in \mathbb{Q}_p$ satisfying

$$v_p(\rho) = 1, \qquad \rho/p \equiv -2 \pmod{p}.$$

Every root of a monic integral polynomial has nonnegative p-adic valuation. A monic factor with constant term 1 or -1 can therefore have no root of positive valuation. Consequently ρ is a root of G_p and hence of P_N.

Write $N = 2^k m$ with m odd, and put $e = v_p(N) \geq 1$. Because $p \mid m$, we have $m > 1$, so the coefficient following the initial halving segment is present. The coefficients $a_j = T^j(N)$ satisfy

$$a_j = N/2^j \quad (0 \leq j \leq k), \qquad a_{k+1} = (3m+1)/2.$$

The terms $a_j \rho^j$ for $0 \leq j \leq k$ have valuations $e + j$; the term of index $k+1$ has valuation $k+1$; and every later term has valuation at least $k+2$. Unless $e = k+1$, a unique term has smallest valuation, making cancellation impossible. Hence $e = k+1$.

Set $N = 2^k p^{k+1} u$. Divide $P_N(\rho) = 0$ by p^{k+1} and reduce modulo p. Only the constant term and the term of index $k+1$ remain:

$$2^k u + \frac{1}{2}(-2)^{k+1} \equiv 0 \pmod{p}.$$

This is exactly $u \equiv (-1)^k \pmod{p}$. Since $k = v_2(N)$, u is odd. When $k = 0$, N/p is odd and congruent to 1 modulo the odd number p; therefore $N/p \equiv 1 \pmod{2p}$, proving the last assertion. □

Corollary III.14.2 (A finite uniqueness window)**.** *Let p be an odd terminating prime and let α be a zero of G_p. If $P_N(\alpha) = 0$ for a terminating $N \neq p$, then*

$$N \geq p(2p+1).$$

More precisely, an odd such N is $p + 2p^2 h$ for an integer $h \geq 1$, whereas an even such N satisfies

$$N \geq 2p^2(2p-1).$$

Thus, among all terminating starts $1 \leq N < p(2p+1)$, the algebraic zero α identifies the starting value p uniquely.

Proof. The minimal polynomial of α is G_p, so $P_N(\alpha) = 0$ implies $G_p \mid P_N$. For odd N, Theorem III.14.1 gives $N = p + 2p^2 h$. Positivity implies $h \geq 0$, and $N \neq p$ excludes $h = 0$. For even N, consider $k = v_2(N) \geq 1$. If $k = 1$, the positive odd number u is -1 modulo p, and its smallest possible value is $2p - 1$. Thus $N \geq 2p^2(2p-1)$. If $k \geq 2$, then $N \geq 2^k p^{k+1} \geq 4p^3 > 2p^2(2p-1)$. Finally $2p^2(2p-1) > p(2p+1)$ for every odd prime. □

Remark III.14.3. The congruence is necessary, not sufficient. For example $p = 3$, $N = 21$ satisfies it, but direct polynomial subtraction gives

$$P_{21}(z) = zP_3(z) + 11z^2 + 29z + 21.$$

In fact the two polynomials are coprime: modulo 5, the quadratic remainder becomes $z^2 - z + 1$, and reducing P_3 modulo that polynomial gives the nonzero constant 2. Thus no root of G_3 is a root of P_{21}. The finite uniqueness result concerns an exact algebraic zero with its minimal polynomial known. It is not a criterion based on a plotted point or an approximate distance to a line, and it does not establish uniqueness over unbounded starting values.

III.14.3 An exact line selecting one known prime in a finite range

Theorem III.14.4 (A line through one algebraic zero)**.** *Let p be an odd terminating prime, let α be a zero of G_p, and let t be any transcendental real number. Define the real line*

$$L_{\alpha,t} = \{x + iy : \ y - \operatorname{Im}\alpha = t(x - \operatorname{Re}\alpha)\}.$$

The only algebraic complex number on this line is α. Consequently, for every terminating starting value $1 \le N < p(2p+1)$,

$$P_N \text{ has a zero on } L_{\alpha,t} \quad \Longleftrightarrow \quad N = p.$$

Proof. The real and imaginary parts of an algebraic complex number are algebraic, since its complex conjugate and i are algebraic. If an algebraic number $\beta \neq \alpha$ lay on the line and had the same real part as α, the line equation would force the same imaginary part, a contradiction. Otherwise that equation would give

$$t = \frac{\operatorname{Im}\beta - \operatorname{Im}\alpha}{\operatorname{Re}\beta - \operatorname{Re}\alpha},$$

which is algebraic, again a contradiction. All zeros of P_N are algebraic, so a zero on the line must equal α. Corollary III.14.2 now gives $N = p$ in the stated range; conversely $P_p(\alpha) = 0$ and α is on the line. □

Worked example. Let α be the unique real zero of

$$G_5(z) = z^3 + z^2 + 3z + 5.$$

The polynomial is irreducible by the rational-root test, and its derivative $3z^2 + 2z + 3$ is positive for real z, proving uniqueness of the real zero. Numerically $\alpha \approx -1.402627941186$, and its exact definition is the specified cubic root. The concrete line

$$\boxed{y = \pi(x - \alpha)}$$

has a zero of P_N on it, among all starts $1 \le N \le 54$ whose orbits terminate, exactly when $N = 5$. Its unique algebraic point is $\alpha + i0$, an original zero of the Collatz polynomial P_5, with $2/3 < |\alpha| < 2$. The bounded exact computation below extends this verified selection property to $1 \le N \le 1000000$ for this particular example. The displayed decimal approximation is not the definition of the intercept: replacing α by a rounded rational number changes the algebraic point selected by the line.

Remark III.14.5 (Scope of the construction). This line is deliberately constructed through a zero of a known prime start. It is a selector for that known start within the proved or checked range, not an efficient primality algorithm for unknown integers and not an observed universal arrangement of prime zeros. An unbounded claim would require proving that no composite starting value shares the exact zero α, which is not established here for odd p. The same construction works with any transcendental real slope. Such a line has only one algebraic point, so it cannot contain infinitely many distinct Collatz-zero locations.

III.14.4 An exact finite exclusion check

For

$$p \in \{3, 5, 7, 11, 13, 17, 19, 23, 29\}$$

and every $1 \le N \le 100000$, exact modular polynomial arithmetic found

$$G_p \mid P_N \quad \Longleftrightarrow \quad N = p.$$

The degrees of G_p, in that order, are $5, 3, 11, 10, 7, 9, 14, 11, 13$. For these starts, $G_p = P_p$ except that $G_5 = z^3 + z^2 + 3z + 5$. This is a bounded computation, not a proof of the corresponding statement for all positive integers.

Here is the complete verification procedure without a companion program. Let $M = 1000000007$. For each displayed p, compute P_p from its finite orbit and factor it over $\mathbb{Q}$ to obtain G_p. Maintain remainders $R_N = P_N \bmod (G_p, M)$, with $R_1 = 1$. For each $N = 2, \ldots, 100000$, follow its integer orbit until reaching a previously processed value; then process that path backwards using

$$R_x = x + zR_{T(x)} \pmod{G_p, M}.$$

This recurrence uses integer arithmetic only. Equivalently, if $G_p = z^d + g_{d-1}z^{d-1} + \cdots + g_0$ and $R_{T(x)} = \sum_{j=0}^{d-1} r_j z^j$, the new coefficient vector is

$$(x - r_{d-1}g_0,\ r_0 - r_{d-1}g_1, \ldots, r_{d-2} - r_{d-1}g_{d-1}) \pmod{M}.$$

The only zero remainder with $p \mid N \le 100000$ was $N = p$ in every case; starts not divisible by p are already excluded by constant terms. A nonzero modular remainder rigorously excludes divisibility over $\mathbb{Z}$, regardless of whether the reduction of G_p modulo M is irreducible. This check does not require probabilistic primality tests or floating-point approximations to zeros.

For $p = 5$ alone, a separate exact integer computation extends the exclusion through $N = 1000000$. It tested all 180000 multiples of 5 in $100001 \le N \le 1000000$, with no zero remainders. To reproduce it, read each finite orbit's coefficients backwards and use Horner reduction in $\mathbb{Z}[z]/(z^3 + z^2 + 3z + 5)$. If the current remainder is $a + bz + cz^2$ and the next coefficient is v, the next remainder vector is

$$(a, b, c) \longmapsto (v - 5c,\ a - 3c,\ b - c).$$

Initialize with $(0, 0, 0)$; after the final coefficient, divisibility holds exactly when all three entries vanish. This extension used exact integers, without modular filtering or numerical root comparison. The larger range is asserted here only for G_5, not for the other prime starts listed above.

III.15 Joint independence of radii, arguments, and logarithmic forms

The local separation lemma (Lemma III.12.2) and the factor-constant partition (Proposition III.12.1) hold for odd composite starts as well as primes. The following extension isolates every nonunit root, not just one selected root per prime. The same argument applies to monic integer polynomials with coprime constant and linear coefficients and pairwise coprime constant terms.

Theorem III.15.1 (All nonunit zeros are independent modulo units)**.** *Let $N_1, \ldots, N_t > 1$ be pairwise coprime odd terminating starts. List all the distinct nonunit zeros of their polynomials as $\alpha_1, \ldots, \alpha_m$, including both members of every nonreal conjugate pair. If ε is an algebraic unit and $n_1, \ldots, n_m \in \mathbb{Z}$ satisfy*

$$\prod_{j=1}^{m} \alpha_j^{n_j} = \varepsilon,$$

then $n_1 = \cdots = n_m = 0$ and $\varepsilon = 1$. In particular, these m complex algebraic numbers are multiplicatively independent.

Proof. Fix i, and let $f_i \in \mathbb{Z}[z]$ be the monic minimal polynomial of α_i. Since α_i is a nonunit, choose a prime $\ell \mid f_i(0)$. The factor f_i divides one of the P_{N_a}, and $f_i(0)$ divides N_a. The ℓ-adic roots of f_i have nonnegative valuations with positive sum. One therefore has positive valuation, and Lemma III.12.2 identifies it with the unique root ρ of P_{N_a} having positive valuation $e = v_\ell(N_a)$. An embedding of $\mathbb{Q}(\alpha_i)$ sending α_i to ρ extends to an embedding σ of a number field containing all the α_j and ε into $\overline{\mathbb{Q}}_\ell$.

Any other zero from P_{N_a} has an image different from ρ, by injectivity of σ, and hence has valuation zero. For $b \neq a$, coprimality gives $\ell \nmid N_b$. The roots of P_{N_b} have nonnegative valuations whose sum is $v_\ell(N_b) = 0$, so all those valuations are zero as well. Every image of an algebraic unit has valuation zero, since both the unit and its inverse are integral. Taking the valuation of the proposed identity after applying σ consequently gives

$$n_i e = 0.$$

As $e > 0$, this proves $n_i = 0$. Applying the argument to every index proves the theorem. It also shows that a nonunit zero cannot belong to two of the polynomials in this pairwise-coprime collection. □

Corollary III.15.2 (Independent radii modulo unit radii)**.** *Choose one representative from each conjugation class of nonunit zeros in Theorem III.15.1, and call their moduli $r_1, \dots, r_k$. For algebraic units $\varepsilon_1, \dots, \varepsilon_s$,*

$$\prod_{j=1}^{k} r_j^{u_j} \prod_{h=1}^{s} |\varepsilon_h|^{v_h} = 1, \qquad u_j, v_h \in \mathbb{Z},$$

implies $u_1 = \cdots = u_k = 0$. Thus all selected radii are different from 1 and from one another.

Proof. Square the identity and use $|\alpha|^2 = \alpha\overline{\alpha}$. A nonreal conjugate pair then has exponent u_j on each of its two members; a real root has exponent $2u_j$. The remaining factor is an algebraic unit. Theorem III.15.1 makes every such exponent zero. □

III.15.1 Baker's theorem gives joint linear independence

We use the classical, inhomogeneous form of Baker's theorem: if complex numbers $\lambda_1, \dots, \lambda_s$ are linearly independent over $\mathbb{Q}$ and every e^{λ_j} is algebraic, then $1, \lambda_1, \dots, \lambda_s$ are linearly independent over $\overline{\mathbb{Q}}$. The independence and nonzero-linear-form conclusions are due to Baker; they are external theorems, not proved or claimed as new here [92, 93].

Theorem III.15.3 (Joint radius–argument independence)**.** *For the starts in Theorem III.15.1, select one representative of each nonunit conjugation class, choosing the upper-half-plane root for every nonreal class. Let $r_1, \dots, r_k$ be all the selected moduli, and let $\theta_1, \dots, \theta_h \in (0, \pi)$ be the principal arguments of the nonreal representatives. Then*

$$\boxed{1,\ \pi,\ \log r_1, \dots, \log r_k,\ \theta_1, \dots, \theta_h \quad \text{are linearly independent over } \overline{\mathbb{Q}}.}$$

All radius logarithms are real logarithms. Arguments of real roots are omitted: all real roots are negative, so their principal arguments equal π.

Proof. All coefficients of P_N are positive, so it has no nonnegative real root. List its nonunit real roots as $\eta_a < 0$, and its upper-half-plane nonunit roots as ξ_b, with their conjugates. For the combined collection of starts use the following fixed logarithms:

$$L_0 = i\pi, \qquad L_{\eta_a} = \log|\eta_a| + i\pi, \qquad L_{\xi_b} = \log|\xi_b| + i\theta_b, \qquad L_{\overline{\xi_b}} = \log|\xi_b| - i\theta_b.$$

Their exponentials are respectively $-1, \eta_a, \xi_b, \overline{\xi_b}$, all algebraic. Suppose there is a rational linear relation between these logarithms. After multiplying by a common denominator and exponentiating, it becomes

$$(-1)^{n_0} \prod_a \eta_a^{n_a} \prod_b \xi_b^{u_b} \overline{\xi_b}^{v_b} = 1$$

with integer exponents. The right-hand unit and Theorem III.15.1 force all exponents on nonunit roots to vanish. The original additive relation then reduces to $n_0 i\pi = 0$, so $n_0 = 0$ too. These logarithms are therefore linearly independent over $\mathbb{Q}$. Baker's theorem makes the same logarithms, together with 1, linearly independent over $\overline{\mathbb{Q}}$.

Finally perform the following invertible change of coordinates, whose coefficients are algebraic:

$$\pi = -iL_0, \qquad \log|\eta_a| = L_{\eta_a} - L_0,$$

$$\log|\xi_b| = \frac{L_{\xi_b} + L_{\overline{\xi_b}}}{2}, \qquad \theta_b = \frac{L_{\xi_b} - L_{\overline{\xi_b}}}{2i}.$$

Its inverse is exactly the displayed definition of the L's. Linear independence over $\overline{\mathbb{Q}}$ is preserved, which proves the claim. □

Corollary III.15.4 (Exact classification of ratios of linear forms)**.** *Let*

$$\mathbf{v} = (1, \pi, \log r_1, \ldots, \log r_k, \theta_1, \ldots, \theta_h).$$

For algebraic coefficient vectors $\mathbf{a}, \mathbf{b}$ *of the same length, with* $\mathbf{b} \neq 0$*, the denominator* $\mathbf{b} \cdot \mathbf{v}$ *is nonzero. Moreover,*

$$\frac{\mathbf{a} \cdot \mathbf{v}}{\mathbf{b} \cdot \mathbf{v}} \in \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad \mathbf{a} = c\mathbf{b} \text{ for some } c \in \overline{\mathbb{Q}}.$$

In the proportional case the ratio equals c*. In every other case it is transcendental.*

Proof. The denominator assertion is Theorem III.15.3. If the ratio equals an algebraic number c, then $(\mathbf{a} - c\mathbf{b}) \cdot \mathbf{v} = 0$. The same theorem implies $\mathbf{a} = c\mathbf{b}$. The converse follows by direct substitution. □

Corollary III.15.5 (Consequences for individual angles and radii)**.** *Each* $\log r_i$ *and each* θ_j *is transcendental. Each of the ratios*

$$\frac{\theta_j}{\pi}, \qquad \frac{\log r_i}{\pi}, \qquad \frac{\theta_j}{\log r_i}$$

is transcendental. For distinct selected conjugation classes, the ratios of their radius logarithms are transcendental; the same is true for the arguments of any two distinct nonreal selected classes. In particular, no nonreal nonunit zero has argument a rational multiple of π*.*

Proof. Compare distinct coordinate vectors, or a nonconstant coordinate vector with the coordinate vector of 1, in Corollary III.15.4. □

An explicit check against an apparent norm relation. For the start 5,

$$P_5(z) = 5 + 8z + 4z^2 + 2z^3 + z^4 = (z + 1)(z^3 + z^2 + 3z + 5).$$

The cubic has no rational root, hence is irreducible. Its derivative $3z^2 + 2z + 3$ is positive on the real line. It has one real root $-r$ and a nonreal pair $Re^{\pm i\theta}$, where $r, R > 0$ and $0 < \theta < \pi$. The theorem says that

$$1, \ \pi, \ \log r, \ \log R, \ \theta$$

are linearly independent over $\overline{\mathbb{Q}}$. Vieta's identity gives

$$rR^2 = 5, \qquad \log r + 2\log R = \log 5.$$

There is no contradiction: $\log 5$ is an additional number, not an algebraic coefficient or one of the listed coordinates. The identity becomes a dependence only after $\log 5$ is adjoined. For example, it also yields the exact classification

$$\frac{a\log r + b\log R}{\log 5} \in \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad b = 2a \qquad (a, b \in \overline{\mathbb{Q}}),$$

with value a in the algebraic case.

Limits of the conclusion. Linear independence over $\overline{\mathbb{Q}}$ is not algebraic independence: no conclusion about arbitrary polynomial relations between these logarithms and angles is asserted. Nor may the theorem automatically adjoin logarithms of primes dividing the starting integers; Vieta's identity above explains why. The theorem concerns a finite collection of exact algebraic zeros. It supplies neither a primality criterion nor a theorem about an uncontrolled limit of such logarithmic expressions.

III.15.2 When can a nonunit radius have a rational power?

Lemma III.15.6 (A root bound and the linear coefficient of a factor)**.** *Every zero of a terminating P_N has modulus at most 2. If N is odd and a monic integer factor $f(z) = c + bz + \cdots$ divides P_N, then $\gcd(c, b) = 1$. Consequently an odd-start polynomial has no nonunit rational zero.*

Proof. Write $a_j = T^j(N)$ and $d = \tau(N)$. Since $a_{j+1} \geq a_j/2$, the coefficients $b_j = 2^j a_j$ of $Q(w) = P_N(2w)$ are positive and nondecreasing. The identity

$$(1-w)Q(w) = b_0 + \sum_{j=1}^{d}(b_j - b_{j-1})w^j - b_d w^{d+1}$$

precludes a zero w with $|w| > 1$: the modulus of the last term would exceed the sum of the moduli of all the others, which is at most $b_d|w|^d$. This proves the root bound.

Write $P_N = fg$ with $g \in \mathbb{Z}[z]$. A prime dividing c and b would divide both $P_N(0)$ and $P_N'(0) = b\,g(0) + c\,g'(0)$, which is impossible for odd N. Finally a rational zero of a monic integer polynomial is an integer. Positivity of the coefficients and the root bound leave only -1 and -2. The latter cannot occur when N is odd, by the rational root theorem, and -1 is a unit. □

Proposition III.15.7 (Four precise quadratic exceptions)**.** *Let $N > 1$ be odd and terminating, and let α be a nonunit zero of P_N, with $r = |\alpha|$. The following are equivalent:*

1. *$r^b \in \mathbb{Q}$ for some integer $b \geq 1$;*
2. *the monic minimal polynomial of α is one of*

$$z^2 + z + 3, \qquad z^2 - z + 3, \qquad z^2 + 2z + 3, \qquad z^2 - 2z + 3.$$

In every such case $r = \sqrt{3}$ and $v_3(N) = 1$. Condition (1) is equivalent to saying that r is a rational power of a positive rational number: clearing the denominator of the exponent gives one direction, and $r = (r^b)^{1/b}$ gives the other. The proposition is a conditional classification: it does not assert that any of these four factors occurs in an odd-start Collatz polynomial.

Proof. Assume $r^b \in \mathbb{Q}$ and set $u = \alpha\overline{\alpha} = r^2$. Then $u^b \in \mathbb{Q}$. Let f be the minimal polynomial of α, $d = \deg f$, and choose a prime $\ell \mid f(0)$. Let L be a splitting field of P_N inside $\mathbb{C}$ and $D = [L : \mathbb{Q}]$. As in the proof of Theorem III.15.1, there is an embedding into $\overline{\mathbb{Q}}_\ell$ taking α to the unique positive-valuation root ρ of P_N. For that embedding, u has positive valuation, since both factors are integral and one has positive valuation. Because u^b is rational, every embedding $\sigma : L \longrightarrow \overline{\mathbb{Q}}_\ell$ must give $\sigma(u)$ the same positive valuation. Thus for every such σ,

$$\sigma(\alpha) = \rho \quad \text{or} \quad \sigma(\overline{\alpha}) = \rho.$$

There are D embeddings in total. Exactly D/d send α to ρ, and exactly D/d send $\overline{\alpha}$ to ρ. If α is nonreal these events are disjoint, so $D = 2D/d$ and $d = 2$. If α is real they are the same event, so $D = D/d$ and $d = 1$; this is impossible for a nonunit zero by Lemma III.15.6. Hence α is nonreal and quadratic.

Its minimal polynomial is $z^2 + az + c$, where $c = \alpha\overline{\alpha} = r^2$ is a positive integer. Since $c \mid N$, it is odd; since the root is a nonunit and $r \leq 2$, one has $1 < c \leq 4$, hence $c = 3$. The discriminant is negative, so $a^2 < 12$. Moreover $\gcd(a, 3) = 1$ by Lemma III.15.6. The only possibilities are $a = \pm 1, \pm 2$. Conversely, every zero of any of the displayed quadratics has modulus $\sqrt{3}$, proving the equivalence. The constant term 3 of the factor contains the entire power of 3 dividing N, by Proposition III.12.1; therefore $v_3(N) = 1$. □

Corollary III.15.8 (Rational-base logarithmic probes)**.** *Under the preceding assumptions, suppose α does not belong to one of the four exceptional quadratic factors. For every rational $q > 0$ with $q \neq 1$,*

$$\frac{\log |\alpha|}{\log q} \quad \textit{is transcendental.}$$

If an exceptional factor does occur, its radius is $\sqrt{3}$: the same ratio is rational exactly when $q = 3^a$ for a nonzero integer a, and its value is then $1/(2a)$. For all other positive rational $q \neq 1$ the ratio is transcendental.

Proof. A rational value of the ratio implies that some positive integral power of $|\alpha|$ is rational, excluded in the first case by Proposition III.15.7. An algebraic irrational value would contradict the Gelfond–Schneider theorem, since q and $|\alpha|$ are algebraic.

In the exceptional case the ratio is $\log 3/(2 \log q)$. Unique prime factorization of positive rational numbers shows that it is rational exactly when q is a nontrivial integer power of 3. Gelfond–Schneider again excludes an algebraic irrational value in every remaining case. □

Corollary III.15.9 (No quadratic exception for an odd prime start)**.** *For any terminating odd prime p and any nonunit zero α of P_p, no positive integral power of $|\alpha|$ is rational. Consequently every rational-base ratio in Corollary III.15.8 is transcendental.*

Proof. An exceptional factor has constant term 3, so it could divide P_p only if $p = 3$. The orbit $3, 5, 8, 4, 2, 1$ gives

$$P_3(z) = z^5 + 2z^4 + 4z^3 + 8z^2 + 5z + 3.$$

The remainders on division by $z^2 + az + 3$, for $a = -2, -1, 1, 2$ respectively, are

$$6z - 39, \qquad -4z - 6, \qquad -12, \qquad -10z - 15.$$

None is zero, excluding all four factors. □

The remaining quadratic question. No proof here excludes the four displayed quadratic factors from every odd composite start, and no occurrence is asserted. Eliminating them globally would strengthen the rational-base corollary, but would not strengthen the already proved joint independence of the selected radii and arguments. The latter holds even if a quadratic exception exists. The results are applications of elementary local algebra together with the classical theorems of Baker and Gelfond–Schneider; no claim of historical priority or of a newly resolved named transcendence conjecture is made.

III.16 Explicit transcendental quantities and what they cannot classify

III.16.1 An elementary logarithm test detects prime powers

Proposition III.16.1. *For a fixed prime p and an integer $n \geq 2$,*

$$\frac{\log n}{\log p} \in \mathbb{Q} \quad \Longleftrightarrow \quad n = p^k \quad (k \geq 1).$$

In every other case the ratio is transcendental. The same dichotomy holds for its reciprocal.

Proof. A rational ratio a/b, with positive integers a, b, implies $n^b = p^a$, so unique factorization makes n a power of p. Conversely a power gives the integer ratio k. If the ratio were algebraic irrational, Gelfond–Schneider applied to $p^{\log n/\log p} = n$ would contradict the algebraicity of n. Taking reciprocals preserves both rationality and transcendence for nonzero numbers. □

Thus a prime input $n \neq p$ and a composite input not a power of p give the same transcendence answer. For example $\log 7/\log 5$ and $\log 21/\log 5$ are both transcendental. Replacing n by n^k changes the ratio by an integer factor and does not change its arithmetic type.

III.16.2 A prime and a composite with the same independence pattern

The polynomials for $N = 9$ and $N = 29$ both have degree 13, are irreducible over $\mathbb{Q}$, and have exactly one real root. Their coefficient lists, in ascending powers, are

$$P_9 : (9, 14, 7, 11, 17, 26, 13, 20, 10, 5, 8, 4, 2, 1),$$
$$P_{29} : (29, 44, 22, 11, 17, 26, 13, 20, 10, 5, 8, 4, 2, 1).$$

These finite statements admit exact checking. Reduction modulo 2 certifies irreducibility for P_9, as detailed later; reduction modulo 19 does so for P_{29}. In the latter case the degree-13 polynomial f satisfies $x^{19^{13}} \equiv x \pmod f$ and $\gcd(f, x^{19} - x) = 1$ in $\mathbb{F}_{19}[x]$. The standard finite-field factor criterion proves irreducibility because 13 is prime. Sturm's theorem gives one real root for each. A complete symbolic check, using only exact polynomial operations, is:

```
import sympy as S
x = S.symbols('x')
tails = [11,17,26,13,20,10,5,8,4,2,1]
for n, head, ell in [(9,[9,14,7],2),(29,[29,44,22],19)]:
    a = head + tails
    f = S.Poly(sum(v*x**j for j,v in enumerate(a)), x)
    g = S.Poly(f, x, modulus=ell)
    h = S.Poly(x, x, modulus=ell)
    for j in range(13):
        h = (h**ell).rem(g)
        if j == 0:
```

```
            assert S.gcd(g,h-S.Poly(x,x,modulus=ell)).degree()==0
    assert h == S.Poly(x,x,modulus=ell)
    assert f.count_roots(-S.oo,S.oo) == 1
```

Consequently each polynomial has seven distinct nonunit radii, one for the real root and six for the nonreal pairs. The joint independence theorem applies to both. All ratios of logarithms of different radii are transcendental in both examples. Adding all nonunit radii from P_5 gives nine independent radii in either case, since 5 is coprime to both 9 and 29. This is an exact counterexample to distinguishing prime from composite by this independence pattern or its rank alone. The numeric values themselves differ; equality of arithmetic type is not equality of the numbers.

III.16.3 A transcendental angle, pitch, and fractal dimension

Write

$$G_5(z) = z^3 + z^2 + 3z + 5, \qquad Q(r) = r^3 - r^2 + 3r - 5.$$

The derivative $Q'(r) = 3r^2 - 2r + 3$ is positive. Since $Q(1) < 0 < Q(3/2)$, there is a unique root $r \in (1, 3/2)$. The real zero of G_5 is $-r$. Let α be its upper-half-plane zero and $R = |\alpha|$. Then

$$R^2 = \frac{5}{r}, \qquad r \approx 1.402627941186, \quad \alpha \approx 0.201313970593 + 1.877287906916i.$$

The normalized angle is

$$\frac{\arg \alpha}{\pi} \approx 0.465995480153,$$

and is transcendental by the joint logarithm theorem. This is stronger than merely saying that the angle is not a rational multiple of π. It does not say that $\arg \alpha$ alone is algebraically independent of every familiar constant.

There is also a very general elementary observation. For any nonreal algebraic β, put $s = |\beta| \neq 1$ and choose an argument θ. Then

$$\frac{\theta}{\log s} \quad \text{is transcendental.}$$

Indeed, if this nonzero real number t were algebraic, the algebraic irrational exponent $1+it$ would give $s^{1+it} = \beta$, contradicting Gelfond–Schneider. In the G_5 example the value is approximately 2.303483768698. This pitch result requires neither a prime input nor a Collatz polynomial.

Proposition III.16.2 (An explicit Weierstrass graph dimension)**.** *Let r be the root of Q above and define*

$$W_r(x) = \sum_{j=0}^{\infty} r^{-j} \cos(2\pi 2^j x).$$

Its graph on $[0, 1]$ has transcendental Hausdorff dimension

$$D = 2 - \frac{\log r}{\log 2} \approx 1.51186762724349.$$

Proof. The series converges uniformly because $r > 1$. Shen's theorem for the classical cosine Weierstrass function [94] applies with $b = 2$ and $\lambda = 1/r \in (1/2, 1)$ and gives the displayed dimension. The cubic Q is irreducible by the rational-root test, and $\mathrm{Norm}_{\mathbb{Q}(r)/\mathbb{Q}}(r) = 5$. If $\log r / \log 2 = a/b$ with integers $b > 0$, then $r^b = 2^a$; taking norms gives $5^b = 2^{3a}$, which is impossible. Gelfond–Schneider excludes an algebraic irrational ratio. Subtracting it from 2 therefore gives a transcendental number. □

The dimension formula is an established external theorem, not a consequence of a scale-counting heuristic. The explicit application constructs a provably transcendental dimension; it does not settle a previously named open dimension problem.

III.16.4 A self-similar set from the same cubic

There is a second geometric realization with only rational contractions. Let $\rho = R^2 = 5/r$. Substituting in $Q(r) = 0$ gives

$$\rho^3 - 3\rho^2 + 5\rho - 25 = 0,$$

and the useful quartic identity is

$$\rho^4 = \rho^3 + \rho^2 + 15\rho + 50.$$

The quartic factors as $(x + 2)(x^3 - 3x^2 + 5x - 25)$. Use the following prefix-free base-4 words, interpreted as the usual subinterval maps of $[0, 1]$: one word of length one, 0; one of length two, 10; fifteen of length three,

$$\{11a, 12a, 13a : a = 0, 1, 2, 3\} \cup \{200, 201, 202\};$$

and fifty of length four,

$$\{21ab, 22ab, 23ab : a, b = 0, 1, 2, 3\} \cup \{2030, 2031\}.$$

Their open interval images are disjoint, so the open set condition holds. The classical self-similar dimension theorem [95] gives the unique d with

$$4^{-d} + 4^{-2d} + 15\,4^{-3d} + 50\,4^{-4d} = 1.$$

Thus $4^d = \rho$ and

$$d = \frac{\log R}{\log 2} \approx 0.9168978610654274.$$

The cubic for ρ is irreducible modulo 3 and has norm 25. If $\log \rho / \log 4$ were rational, taking norms of a power relation would equate a nonzero power of 25 with a power of 4. As before, Gelfond–Schneider proves transcendence. This is an explicit constructed example using a classical dimension theorem.

III.16.5 Euler's constant is not the Gamma function

The Euler–Mascheroni constant γ and the function $\Gamma(z)$ are different objects. To use a theorem about logarithms of algebraic numbers to prove transcendence of an independently given constant c, one must first prove an *exact* suitable identity, for example $c = \log a / \log b$ for specified positive multiplicatively independent algebraic a, b. A decimal match is not such an identity.

In particular, proposed identities $\log r_p = \gamma^2$ and $\log r_q = \gamma$ would give a ratio equal to γ, but no such identities have been established. The second alone, if true with $r_q \neq 1$ algebraic, would already prove γ transcendental by Hermite–Lindemann. If instead the proposal is $r_p = \gamma^2$ and $r_q = \gamma$, the logarithmic ratio is 2, contrary to the independent prime-radii theorem. These are two different substitutions.

Finally, a sequence of transcendental quantities can converge to a rational number, for example $\pi/n \to 0$. Neither a boundary limit equal to γ nor increasingly accurate numerical approximation proves the irrationality or transcendence of γ.

III.17 Boundary distances, Euler's constant, and the zeta function

For the shortcut Collatz map, pure halving gives

$$A_m(z) := P_{2^{m-1}}(z) = \sum_{j=0}^{m-1} 2^{m-1-j} z^j = \frac{z^m - 2^m}{z-2}, \qquad m \geq 2.$$

Its $m-1$ zeros are $\alpha_k = 2e^{2\pi ik/m}$, $1 \leq k < m$. Thus their reciprocal distances from the boundary point 2 give

$$S_m := \sum_{k=1}^{m-1} \frac{1}{|2-\alpha_k|} = \frac{1}{4} \sum_{k=1}^{m-1} \csc \frac{\pi k}{m}, \qquad E_m := \frac{2\pi}{m} S_m - \log \frac{2m}{\pi}.$$

Proposition III.17.1 (A classical cosecant asymptotic in zero coordinates)**.** *We have $E_m \to \gamma$, where γ is the Euler–Mascheroni constant [70]. More precisely,*

$$E_m = \gamma - \frac{\pi^2}{72m^2} + \frac{7\pi^4}{43200m^4} + O(m^{-6}).$$

For every $m \geq 2$, the following rigorous bounds hold:

$$-\frac{\pi^2}{72m^2} < E_m - \gamma < -\frac{\pi^2}{72m^2} + \frac{7\pi^4}{43200m^4}. \tag{III.17.1}$$

Proof. A self-contained proof of the limit removes the endpoint singularities:

$$f(x) = \csc(\pi x) - \frac{1}{\pi x} - \frac{1}{\pi(1-x)}, \qquad \int_0^1 f(x)\,dx = \frac{2}{\pi} \log \frac{2}{\pi}.$$

The function extends smoothly to $[0,1]$. Writing H_j for the jth harmonic number gives the exact identity

$$E_m = H_{m-1} - \log m + \frac{\pi}{2m} \sum_{k=1}^{m-1} f(k/m) - \log \frac{2}{\pi}.$$

The definition of γ and the Riemann-sum limit prove convergence. For the asymptotic expansion, Euler–Maclaurin summation uses

$$f(0) = f(1) = -1/\pi, \quad f'(1) - f'(0) = -\pi/3 + 2/\pi,$$

$$f'''(1) - f'''(0) = -7\pi^3/30 + 12/\pi.$$

Combining this with the harmonic-number expansion gives the displayed coefficients. The all-m inequality (III.17.1) is exactly the cosecant bound of Blagouchine–Moreau [69], Theorem 7, equation (74), after multiplication by $\pi/(2m)$. □

Attribution and interpretation. The complete cosecant asymptotic is classical [68, 69], appearing in Watson's 1916 work; this is a geometric application, not a new asymptotic formula. See G. N. Watson, "The sum of a series of cosecants," *Philosophical Magazine* 31 (1916), 111–118, doi:10.1080/14786440208635481; and I. V. Blagouchine and E. Moreau, "On a finite sum of cosecants appearing in various problems," arXiv:2312.16657, Theorems 4b and 7. The constant comes from the harmonic growth of inverse chord lengths. The same mechanism works for regular polygons on any circle, and the starts 2^{m-1} are composite for $m \geq 3$.

Proposition III.17.2 (A finite inverse-distance zeta function)**.** *Define, using real logarithms of the positive bases,*

$$Z_m(s) = \frac{1}{2}\sum_{k=1}^{m-1}\left(\frac{4\pi}{m|2-\alpha_k|}\right)^s.$$

Then $Z_m(s) \to \zeta(s)$ locally uniformly on $\operatorname{Re} s > 1$. At the endpoint $s = 1$, its logarithmic finite part is

$$Z_m(1) - \log(2m/\pi) = E_m \longrightarrow \gamma.$$

Proof. Pair the terms k and $m-k$. For $1 \le k < m/2$ their combined contribution is

$$\left(\frac{\pi}{m\sin(\pi k/m)}\right)^s \longrightarrow k^{-s}.$$

For a compact set with $1 < \sigma_0 \le \operatorname{Re} s \le \sigma_1$, the inequality $\sin(\pi k/m) \ge 2k/m$ bounds these terms uniformly by $(\pi/2)^{\sigma_1}k^{-\sigma_0}$. Dominated convergence for the series is therefore uniform on that compact set. When m is even, the unpaired middle term $\frac{1}{2}(\pi/m)^s$ tends uniformly to zero. □

This convergence is confined to $\operatorname{Re} s > 1$. It gives no conclusion about zeta zeros in the critical strip, the Riemann hypothesis, or arithmetic properties of γ.

An exact extension beyond pure halving. For even $m \ge 4$, set $q_m = (2^m - 1)/3$ and $N_m = 2^{m-1}q_m$. The orbit has the two blocks $q_m(2^{m-1}, \dots, 1)$ and $(2^{m-1}, \dots, 1)$, hence

$$P_{N_m}(z) = (z^m + q_m)A_m(z).$$

Its $m-1$ boundary zeros yield exactly the same S_m and Z_m; the other m zeros have modulus $q_m^{1/m} < 2$. These are all composite starts. For this boundary-only statistic, m is the boundary-zero count plus one, not the full degree $2m-1$.

III.18 An Euler–Mascheroni limit from odd terminating starts

This construction uses a specific infinite family of terminating odd starts. Every member beyond $N_4 = 5$ is composite, so the limit below is not a prime-detection phenomenon.

Theorem III.18.1 (Odd-start reciprocal-distance limit)**.** *For even integers $m \ge 4$, let*

$$N_m = \frac{2^m - 1}{3}, \qquad P_{N_m}(z) = N_m + \sum_{j=1}^{m} 2^{m-j}z^j.$$

The displayed polynomial is the shortcut Collatz polynomial of N_m. If α ranges over its zeros, counted with multiplicity, then

$$\frac{2\pi}{m}\sum_{P_{N_m}(\alpha)=0}\frac{1}{|2-\alpha|} = \log\frac{2m}{\pi} + \gamma + O\left(\frac{\log m}{m}\right) \qquad (m \longrightarrow \infty,\ m \text{ even}),$$

where γ is the Euler–Mascheroni constant [70].

Proof. For even $m \ge 4$, N_m is odd and exceeds 1, and $T(N_m) = 2^{m-1}$. Hence its remaining orbit consists of successive halvings down to 1, proving the stated polynomial formula. Write $\epsilon_m = 2^{-m}$,

$$b_m = \frac{1-\epsilon_m}{3}, \qquad a_m = 1 - b_m = \frac{2+\epsilon_m}{3}, \qquad D_m(w) = 2^{-m}P_{N_m}(2w) = b_m + \sum_{j=1}^{m} w^j.$$

Then

$$(w-1)D_m(w) = w^{m+1} - a_m w - b_m. \tag{III.18.1}$$

The right side has roots 1 and $-1/2$; the latter uses that m is even.

We first locate every other root with an error that remains uniform near the singular point $w = 1$. On $|w| > 3/4$ define the principal logarithm

$$L_m(w) = \operatorname{Log}(a_m + b_m/w).$$

It is holomorphic there, since the argument of the logarithm has positive real part. Uniformly for $|w| \geq 3/4$, its modulus lies between $2/9$ and $10/9$, and the absolute value of its argument is at most $\arcsin(2/3)$. Consequently $|L_m(w)| < 2$.

Put $\zeta_j = e^{2\pi i j/m}$, $0 \leq j < m$. For $m \geq 8$, the map

$$\Phi_{m,j}(u) = L_m(\zeta_j e^{u/m})$$

maps the closed disk $|u| \leq 2$ to itself, because $|\zeta_j e^{u/m}| \geq e^{-1/4} > 3/4$. Its derivative satisfies

$$|\Phi'_{m,j}(u)| = \left| \frac{-b_m}{m(a_m \zeta_j e^{u/m} + b_m)} \right| \leq \frac{2}{m} \leq \frac{1}{4}.$$

Thus it has a unique fixed point u_j in that disk. The numbers

$$w_j = \zeta_j e^{u_j/m}$$

satisfy $w_j^m = a_m + b_m/w_j$, hence are roots of (III.18.1). Because $|\operatorname{Im} u_j| < \pi/2$, they occupy disjoint angular sectors centered at the mth roots of unity and are distinct. Also $u_0 = 0$ and $w_0 = 1$. The additional root $-1/2$ is outside their annulus. These $m+1$ distinct roots exhaust the trinomial, so the roots of D_m are exactly $-1/2, w_1, \ldots, w_{m-1}$.

The equality $L_m(1) = 0$ gives a relative, rather than merely absolute, root-location estimate. Along the unit circle,

$$\left| \frac{d}{dt} L_m(e^{it}) \right| = \frac{b_m}{|a_m e^{it} + b_m|} \leq \frac{b_m}{a_m - b_m} \leq 1.$$

Integrating along the shorter arc from 1 to ζ_j gives $|L_m(\zeta_j)| \leq (\pi/2)|1 - \zeta_j|$. The contraction estimate then implies

$$|u_j| \leq \frac{|L_m(\zeta_j)|}{1 - 2/m} \leq \frac{2\pi}{3}|1 - \zeta_j|, \qquad |w_j - \zeta_j| \leq \frac{e^{1/4}}{m}|u_j| \leq \frac{3}{m}|1 - \zeta_j|.$$

For $j \neq 0$, the triangle inequality therefore yields

$$\left| \frac{1}{|1 - w_j|} - \frac{1}{|1 - \zeta_j|} \right| \leq \frac{5}{m} \frac{1}{|1 - \zeta_j|} \qquad (m \geq 8). \tag{III.18.2}$$

For completeness, the corresponding roots-of-unity sum has the asymptotic

$$\frac{\pi}{m} \sum_{j=1}^{m-1} \frac{1}{|1 - \zeta_j|} = \frac{\pi}{2m} \sum_{j=1}^{m-1} \csc \frac{\pi j}{m} = \log \frac{2m}{\pi} + \gamma + O(1/m). \tag{III.18.3}$$

To see this without a singular Riemann-sum argument, subtract the two endpoint poles and put

$$h(x) = \csc(\pi x) - \frac{1}{\pi x} - \frac{1}{\pi(1-x)}.$$

The function h extends to a continuously differentiable function on $[0,1]$, with $h(0) = h(1) = -1/\pi$, and direct integration gives

$$\int_0^1 h(x)\,dx = \frac{2}{\pi}\log\frac{2}{\pi}.$$

Therefore the left side of (III.18.3) is

$$H_{m-1} + \frac{\pi}{2m}\sum_{j=1}^{m-1} h(j/m) = \log m + \gamma + \log\frac{2}{\pi} + O(1/m).$$

In particular the unnormalized reciprocal-distance sum is $O(m\log m)$. Summing (III.18.2) and multiplying by π/m changes (III.18.3) by $O((\log m)/m)$. The extra root $-1/2$ contributes only $2\pi/(3m)$. Finally $\alpha = 2w$ converts $\pi/(m|1-w|)$ to $2\pi/(m|2-\alpha|)$, completing the proof. □

Corollary III.18.2 (An explicit error bound)**.** *Define*

$$C_m = \frac{2\pi}{m}\sum_{P_{N_m}(\alpha)=0}\frac{1}{|2-\alpha|} - \log\frac{2m}{\pi}.$$

For every even $m \geq 8$,

$$\boxed{|C_m - \gamma| \leq \frac{5\log m + 21}{m}.}$$

Proof. We make the elementary Riemann-sum error above explicit. The function h used there satisfies $|h'(x)| < 8$ on $[0,1]$. Indeed, for $0 < x \leq 1/2$ put $t = \pi x$. The inequalities

$$0 \leq t - \sin t \leq t^3/6, \qquad 0 \leq 1 - \cos t \leq t^2/2, \qquad \sin t \geq 2t/\pi$$

give

$$|\sin^2 t - t^2\cos t| \leq \frac{5}{6}t^4, \qquad |h'(x)| \leq \frac{5\pi^3}{24} + \frac{4}{\pi} < 8.$$

Symmetry $h(1-x) = h(x)$ covers the other half of the interval. The right-endpoint Riemann sum therefore has error at most $4/m$; removing its last endpoint, where $h(1) = -1/\pi$, gives

$$\left|\frac{1}{m}\sum_{j=1}^{m-1} h(j/m) - \int_0^1 h(x)\,dx\right| \leq \frac{4 + 1/\pi}{m}.$$

Together with $|H_{m-1} - \log m - \gamma| \leq 1/m$, this yields

$$\left|B_m - \log\frac{2m}{\pi} - \gamma\right| \leq \frac{2\pi + 3/2}{m} < \frac{8}{m}, \qquad B_m = \frac{\pi}{m}\sum_{j=1}^{m-1}\frac{1}{|1-\zeta_j|}.$$

In particular $B_m \leq \log m + 2$ for $m \geq 8$, using $\gamma < 1$. The relative bound (III.18.2) and the contribution $2\pi/(3m)$ of the outlier give

$$|C_m - \gamma| \leq \frac{5B_m + 8 + 2\pi/3}{m} \leq \frac{5\log m + 21}{m},$$

as claimed. □

Corollary III.18.3 (First logarithmic correction)**.** *With the notation of Theorem III.18.1,*

$$\frac{2\pi}{m}\sum_{P_{N_m}(\alpha)=0}\frac{1}{|2-\alpha|} - \log\frac{2m}{\pi} = \gamma + \frac{\log m}{3m} + O(1/m).$$

In particular the normalized quantities eventually exceed γ.

Proof. The fixed-point estimates in the proof give, uniformly for $1 \le j < m$,

$$w_j = \zeta_j + \frac{\zeta_j L_m(\zeta_j)}{m} + O\Big(\frac{|1-\zeta_j|}{m^2}\Big).$$

Define

$$A_m(\zeta) = \frac{\zeta L_m(\zeta)}{1-\zeta}, \qquad A_m(1) = b_m.$$

The apparent singularity at 1 is removable, since $L_m'(1) = -b_m$. On the unit circle these functions are uniformly bounded and satisfy $A_m(\zeta) - b_m = O(|1-\zeta|)$; this follows from uniform bounds for the derivatives of L_m near 1, and compactness away from 1. Consequently

$$\frac{1}{|1-w_j|} = \frac{1}{|1-\zeta_j|}\left(1 + \frac{\operatorname{Re} A_m(\zeta_j)}{m} + O(m^{-2})\right).$$

If B_m denotes the roots-of-unity sum on the left side of (III.18.3), summation and the bounded remainder $(\operatorname{Re} A_m(\zeta) - b_m)/|1-\zeta|$ give

$$\frac{\pi}{m} \sum_{D_m(w)=0} \frac{1}{|1-w|} = B_m + \frac{b_m}{m} B_m + O(1/m).$$

The root $-1/2$ is included in the $O(1/m)$ term. Substituting $B_m = \log(2m/\pi) + \gamma + O(1/m)$ and $b_m = 1/3 + O(2^{-m})$ proves the formula. Its positive logarithmic term eventually dominates the $O(1/m)$ remainder. □

The uniform relative bound (III.18.2) is essential: weak convergence of root-counting measures alone would not justify this limit, because the reciprocal-distance kernel is unbounded at $w = 1$.

To verify the compositeness claim, write $m = 2r \ge 6$. Then

$$N_m = \frac{(2^r - 1)(2^r + 1)}{3}.$$

Exactly one of the two factors is divisible by 3. After dividing that factor by 3, both factors exceed 1 because $r \ge 3$. Thus the same Euler–Mascheroni limit holds along this explicit family of odd composites.

Numerical illustration and reproduction. The following values illustrate the convergence. They are ordinary floating-point approximations, not interval certificates or substitutes for the proof. The reference value is $\gamma \approx 0.577215664901533$.

m	Approximate C_m
128	0.607570723152
2048	0.579564141044
32768	0.577390648950
131072	0.577262936443

The complete Python code below uses only its standard library and implements the contraction proved above. Conjugate pairs are counted together, and the exact scaled outlier $-1/2$ is included separately. No external script or polynomial root-finding package is required. For large m, floating-point underflow replaces 2^{-m} by zero; the analytic proof uses its exact value throughout.

```
import cmath
import math
```

```
def gamma_approx(m):
    assert m >= 8 and m % 2 == 0
    b = (1 - 2.0**(-m)) / 3
    a = 1 - b
    weights = [2/3]
    for j in range(1, m//2 + 1):
        zeta = cmath.exp(2j * math.pi * j / m)
        u = 0j
        for step in range(30):
            v = cmath.log(a + b / (zeta * cmath.exp(u/m)))
            if abs(v-u) < 2e-16:
                u = v
                break
            u = v
        w = zeta * cmath.exp(u/m)
        weight = 1 / abs(1-w)
        weights.append(weight if 2*j == m else 2*weight)
    total = math.pi/m * math.fsum(weights)
    return total - math.log(2*m/math.pi)

for m in (128, 2048, 32768, 131072):
    print(m, format(gamma_approx(m), '.12f'))
```

III.19 Stability of the Euler limit and its complete first correction

The appearance of Euler's constant in the preceding distance sums is an endpoint phenomenon. The first two statements below isolate elementary forms of that mechanism; they are not claims of a new general theory of Euler's constant. The final theorem evaluates the full coefficient of m^{-1} for the specific odd Collatz family, refining the earlier $O(m^{-1})$ remainder after the logarithmic correction.

Lemma III.19.1 (Endpoint subtraction)**.** *Suppose that*

$$k(x) = \frac{a_0}{x} + \frac{a_1}{1-x} + g(x), \qquad 0 < x < 1,$$

where a_0, a_1 are fixed real numbers and g is Lipschitz continuous on $[0,1]$. Then

$$\frac{1}{m}\sum_{j=1}^{m-1} k(j/m) = (a_0 + a_1)(\log m + \gamma) + \int_0^1 g(x)\,dx + O(m^{-1}).$$

Proof. The two pole sums are exactly $(a_0 + a_1)H_{m-1}$. The remaining sum is a Riemann sum for g with error $O(m^{-1})$, and $H_{m-1} = \log m + \gamma + O(m^{-1})$. □

For the inverse-chord kernel $k(x) = \pi/(2\sin \pi x)$ one has $a_0 = a_1 = 1/2$ and $\int_0^1 g = \log(2/\pi)$. Thus the two sides of the singular point jointly produce one copy of the harmonic sum. The additional logarithm in $\log(2m/\pi)$ removes a geometric constant. A different grid or kernel need not have the same finite part, and the mere presence of logarithmic divergence does not force the constant to be γ.

Proposition III.19.2 (Relative perturbation stability)**.** *Fix $R > 0$ and write $\zeta_{m,j} = e^{2\pi i j/m}$ for $1 \le j < m$. Suppose a multiset of points contains $m-1$ labelled points $\alpha_{m,j}$ satisfying*

$$|\alpha_{m,j} - R\zeta_{m,j}| \le \delta_m R|1 - \zeta_{m,j}|, \qquad 0 \le \delta_m < 1,$$

and additional points $\beta_{m,\ell} \neq R$, counted with multiplicity. Set

$$B_m = \frac{\pi}{m} \sum_{j=1}^{m-1} \frac{1}{|1-\zeta_{m,j}|}, \qquad U_m = \frac{\pi R}{m} \left(\sum_{j=1}^{m-1} \frac{1}{|R-\alpha_{m,j}|} + \sum_{\ell} \frac{1}{|R-\beta_{m,\ell}|} \right).$$

Then

$$\left| U_m - \log \frac{2m}{\pi} - \gamma \right| \leq \frac{\pi^2}{72m^2} + \frac{\delta_m}{1-\delta_m} B_m + \frac{\pi R}{m} \sum_{\ell} \frac{1}{|R-\beta_{m,\ell}|}.$$

In particular, if $\delta_m \log m \to 0$ and $\sum_\ell |R-\beta_{m,\ell}|^{-1} = o(m)$, then

$$U_m - \log \frac{2m}{\pi} \longrightarrow \gamma.$$

Proof. Put $d_j = R|1-\zeta_{m,j}|$. The triangle inequality gives $\big||R-\alpha_{m,j}| - d_j\big| \leq \delta_m d_j$, so

$$\left| \frac{1}{|R-\alpha_{m,j}|} - \frac{1}{d_j} \right| \leq \frac{\delta_m}{1-\delta_m} \frac{1}{d_j}.$$

Sum this estimate and use the classical cosecant bound (III.17.1), which implies $|B_m - \log(2m/\pi) - \gamma| \leq \pi^2/(72m^2)$. Finally $B_m = O(\log m)$. □

This proposition explains why an absolute error estimate or weak convergence of zero-counting measures is insufficient near the singular point. It also shows that no Collatz-specific or prime-specific mechanism is needed once the required relative perturbation estimate has been established.

Theorem III.19.3 (The full coefficient of the first correction)**.** *For the odd family of Theorem III.18.1, define*

$$C_m = \frac{2\pi}{m} \sum_{P_{N_m}(\alpha)=0} \frac{1}{|2-\alpha|} - \log \frac{2m}{\pi}, \qquad u = \arcsin \frac{1}{\sqrt{3}}.$$

As $m \to \infty$ through even integers,

$$C_m = \gamma + \frac{\log m}{3m} + \frac{K}{m} + O\Big(\frac{\log m}{m^2} \Big),$$

where

$$\boxed{K = \frac{\gamma + \log(2/\pi)}{3} + u^2 - \frac{2\sqrt{2}}{3} u + \frac{1}{3} + \frac{2\pi}{3}.}$$

Numerically, $K \approx 2.26814152647080423359$.

Proof. Keep the notation $b = b_m = (1-2^{-m})/3$, $a = 1-b$, and $\zeta_j = e^{2\pi i j/m}$ from the odd-family proof. For $0 \leq b \leq 1/3$ put

$$L_b(\zeta) = \operatorname{Log}(1 - b + b/\zeta), \qquad A_b(\zeta) = \frac{\zeta L_b(\zeta)}{1-\zeta}, \qquad A_b(1) = b.$$

The logarithm is the principal branch. The expression is analytic in a fixed annulus around the unit circle, and the apparent singularity of A_b at 1 is removable. All bounds below are uniform in $b \in [0, 1/3]$.

The contraction proof gives $|u_j| = O(|1-\zeta_j|)$ and $u_j - L_b(\zeta_j) = O(|1-\zeta_j|/m)$. Expanding $w_j = \zeta_j e^{u_j/m}$ therefore yields

$$w_j = \zeta_j + \frac{\zeta_j L_b(\zeta_j)}{m} + O\Big(\frac{|1-\zeta_j|}{m^2} \Big).$$

It follows, uniformly for $1 \le j < m$, that

$$\frac{1}{|1-w_j|} = \frac{1}{|1-\zeta_j|}\left(1 + \frac{\operatorname{Re} A_b(\zeta_j)}{m} + O(m^{-2})\right).$$

Define

$$f_b(x) = \frac{\operatorname{Re} A_b(e^{2\pi i x}) - b}{|1-e^{2\pi i x}|}, \qquad 0 < x < 1.$$

These functions extend to $f_b(0) = f_b(1) = 0$ and are uniformly Lipschitz. To check the endpoint assertion explicitly, $A_b(\overline{\zeta}) = \overline{A_b(\zeta)}$ makes $\operatorname{Re} A_b(e^{it})$ an even smooth function near $t = 0$. Its difference from b is therefore $O(t^2)$, with derivative $O(|t|)$, whereas $|1 - e^{it}|$ is comparable to $|t|$. Compactness handles the rest of the circle. Consequently

$$\frac{1}{m}\sum_{j=1}^{m-1} f_b(j/m) = \int_0^1 f_b(x)\,dx + O(m^{-1}).$$

Writing $J(b) = \pi \int_0^1 f_b(x)\,dx$, including the exact outlier $w = -1/2$, and using $B_m = O(\log m)$ now gives

$$\frac{\pi}{m}\sum_{D_m(w)=0} \frac{1}{|1-w|} = B_m + \frac{b}{m}B_m + \frac{J(b)}{m} + \frac{2\pi}{3m} + O\!\left(\frac{\log m}{m^2}\right). \tag{III.19.1}$$

It remains to evaluate J. Symmetry around the real axis gives

$$J(b) = \int_0^\pi \frac{\operatorname{Re} A_b(e^{it}) - b}{2\sin(t/2)}\,dt.$$

The uniform endpoint estimates justify differentiation under the integral sign. Since

$$\frac{\partial A_b(\zeta)}{\partial b} = \frac{1}{1 - b + b/\zeta},$$

one obtains

$$J'(b) = \int_0^\pi \frac{b(1-2b)\sin(t/2)}{1 - 4b(1-b)\sin^2(t/2)}\,dt.$$

The substitution $v = \cos(t/2)$ evaluates this as

$$J'(b) = 2\sqrt{\frac{b}{1-b}}\arcsin\sqrt{b} \qquad (0 \le b \le 1/3).$$

Using $J(0) = 0$ and then the substitution $s = \arcsin\sqrt{b}$ gives

$$J(b) = (\arcsin\sqrt{b})^2 - 2\sqrt{b(1-b)}\arcsin\sqrt{b} + b.$$

Finally, the earlier classical expansion supplies $B_m = \log(2m/\pi) + \gamma + O(m^{-2})$, while $b_m = 1/3 + O(2^{-m})$. Insert these expressions into (III.19.1) to obtain the claimed K. □

Corollary III.19.4 (An accelerated expression without an unknown input)**.** *Put*

$$J_0 = u^2 - \frac{2\sqrt{2}}{3}u + \frac{1}{3}, \qquad \widehat{\gamma}_m = \frac{mC_m - \frac{1}{3}\log(2m/\pi) - J_0 - 2\pi/3}{m + 1/3}.$$

Then

$$\widehat{\gamma}_m = \gamma + O\!\left(\frac{\log m}{m^2}\right).$$

Every quantity in the definition of $\widehat{\gamma}_m$ is determined by the finite Collatz polynomial and elementary constants; the unknown target γ is not supplied as an input.

Proof. In the preceding theorem, collect the terms containing γ and multiply by m. Division by $m + 1/3$ gives the result. □

The acceleration is a consequence of an asymptotic expansion, not a transcendence criterion. In particular, the formulas do not establish the irrationality of γ or identify e^{γ} with an algebraic zero.

III.20 What a geometric primality criterion would need to detect

Write $T(n) = n/2$ for even n, and $T(n) = (3n + 1)/2$ for odd n. For a start N which reaches 1, let $\tau(N)$ be the first such time and put

$$P_N(z) = \sum_{j=0}^{\tau(N)} T^j(N) z^j.$$

Its leading coefficient is 1 and its constant coefficient is N. Consequently its complete root multiset determines N by Vieta's formula. Recovering N this way and applying an ordinary primality test is an exact reformulation, but it supplies neither a new primality test nor a prime-only line in the complex plane.

III.20.1 An exact polynomial certificate and the first failing degree

The following constructive certificate applies to the reciprocal Collatz polynomial, but the proof is a general binomial argument. It belongs to the same classical polynomial-congruence principle as Lemma 2.1 of Agrawal–Kayal–Saxena, *PRIMES is in P* (2004); no claim of a new primality-testing principle or efficiency improvement is made.

Theorem III.20.1 (First failure of a Frobenius congruence)**.** *Let $N \geq 2$, and let $F(z) \in \mathbb{Z}[z]$ satisfy $F(0) = 1$ and $\gcd(F'(0), N) = 1$. Set*

$$D_{N,F}(z) = F(z)^N - F(z^N).$$

Then

$$N \text{ is prime} \quad \Longleftrightarrow \quad D_{N,F}(z) \equiv 0 \pmod{N}.$$

If N is composite and q is its least prime divisor, then the least degree having a nonzero coefficient in $D_{N,F}$ modulo N is exactly q. More precisely, if $e = v_q(N)$, then

$$v_q([z^q] D_{N,F}(z)) = e - 1.$$

In particular, it is enough to check that the coefficients in degrees $1, \ldots, \lfloor \sqrt{N} \rfloor$ all vanish modulo N.

Proof. For prime N, the Frobenius identity in $\mathbb{F}_N[z]$ and the congruence $a^N \equiv a \pmod{N}$ for integer coefficients a give $F(z)^N \equiv F(z^N) \pmod{N}$.

Suppose N is composite, let q be its least prime divisor, and write $F = 1 + U$, where $U(0) = 0$ and $a = U'(0)$ is coprime to N. For $1 \leq j < N$, the term $F(z^N)$ has zero coefficient in degree j, and the binomial theorem gives

$$[z^j] D_{N,F} = \sum_{k=1}^{j} \binom{N}{k} [z^j] U(z)^k.$$

Terms $k > j$ contribute nothing because $U(0) = 0$. When $j < q$, each $1 \le k \le j$ is coprime to N; from $k\binom{N}{k} = N\binom{N-1}{k-1}$ we obtain $N \mid \binom{N}{k}$. Thus all degrees below q, including degree zero, vanish modulo N.

Write $e = v_q(N)$. For $k < q$ the same identity implies $v_q\binom{N}{k} \ge e$. However,

$$\binom{N}{q} = \frac{N}{q}\binom{N-1}{q-1}, \qquad \binom{N-1}{q-1} \equiv (-1)^{q-1} \not\equiv 0 \pmod{q},$$

so $v_q\binom{N}{q} = e - 1$. Since $[z^q]U^q = a^q$ is a q-adic unit,

$$[z^q]D_{N,F} \equiv \binom{N}{q}a^q \pmod{q^e}$$

has valuation exactly $e-1$. This coefficient does not vanish modulo N, proving the characterization and the first-failure claim. Finally, every composite N has a least prime divisor $q \le \sqrt{N}$, proving the truncated coefficient criterion. □

Corollary III.20.2 (Reciprocal Collatz polynomial certificate)**.** *For odd $N > 1$ whose orbit reaches 1, let $d = \tau(N)$ and*

$$R_N(z) = z^d P_N(1/z) = 1 + 2z + \cdots + Nz^d.$$

Then N is prime if and only if

$$R_N(z)^N \equiv R_N(z^N) \pmod{N}.$$

For composite odd N, the first nonzero coefficient of the difference modulo N occurs precisely at its least prime divisor.

Proof. The last state before the first occurrence of 1 is 2: the even equation $n/2 = 1$ has solution $n = 2$, whereas $(3n + 1)/2 = 1$ has no positive integer solution. Hence $R_N(0) = 1$, $R'_N(0) = 2$, and the theorem applies because N is odd. □

An exact algorithm. Handle $N \le 1$, $N = 2$, and even $N > 2$ immediately. For odd N, choose an admissible F (for example R_N if a terminating orbit is supplied), and let $B = \lfloor\sqrt{N}\rfloor$. Compute

$$H(z) = F(z)^N \pmod{(N, z^{B+1})}$$

by repeated squaring with truncated polynomial multiplication. If $H(z) = 1$, return PRIME. Otherwise, the least positive degree with nonzero coefficient is the least prime divisor of N, and return COMPOSITE with that divisor. The proof above verifies this algorithm since $B < N$ and $F(z^N) \equiv 1 \pmod{z^{B+1}}$.

What is and is not gained. This supplies a coefficient-level if-and-only-if test and even an exact factor location for composite inputs. It is not a prime-only line in the original complex zero annulus. The exponent and truncation length depend on N, and the $\sqrt{N}$ bound is exponential in the bit length of N. More fundamentally, $F(z) = 1 + z$ already satisfies all hypotheses for every $N \ge 2$, so the Collatz orbit is unnecessary for this test. Taking that choice also removes any dependence on an unproved universal termination claim. The later, much more substantial reductions in the AKS algorithm are not established merely by this congruence.

III.21 Exact primality encodings by boundary power maps

Here an integer power map $u \mapsto u^a$ is called a fold in the earlier construction. It is distinct from the two axis reflections considered in Section III.7.

III.21.1 A factorial encoding using one boundary zero

The construction in this section is an exact reformulation of classical primality arithmetic. It uses a known boundary zero of a very simple Collatz polynomial and the Gelfond–Schneider theorem. It is not a new primality algorithm, and it does not classify the starting value of an arbitrary Collatz polynomial from its unmodified zeros.

Lemma III.21.1 (A root-of-unity logarithm dichotomy)**.** *Fix a prime $p \geq 3$ and let u be any root of unity. Put*

$$R = p + |1-u|^2 = p + 2 - u - u^{-1}, \qquad L = \frac{\log R}{\log p},$$

where the logarithms are real. If $u = 1$, then $L = 1$. If $u \neq 1$, then $L > 1$ is transcendental.

Proof. The number R is an algebraic integer. Every conjugate of R is of the form $p+2-v-v^{-1} = p + |1-v|^2$, where v is a root of unity. Thus every conjugate is positive and lies in $[p, p+4]$.

Suppose first that $L = a/b \in \mathbb{Q}$, with $b > 0$. Then $R^b = p^a$. Each conjugate of R must therefore be the unique positive real bth root of p^a. All conjugates are equal, so separability in characteristic zero implies $R \in \mathbb{Q}$. A rational algebraic integer is an integer. Unique factorization in $\mathbb{Z}$ now gives $R = p^k$ for an integer $k \geq 0$. Since

$$p \leq R \leq p + 4 < p^2 \qquad (p \geq 3),$$

we must have $R = p$, hence $u = 1$ and $L = 1$.

If $u \neq 1$, we have therefore excluded rational L. Were L algebraic irrational, the Gelfond–Schneider theorem would make $p^L = R$ transcendental, contrary to the algebraicity of R. Consequently L is transcendental. Finally $u \neq 1$ gives $R > p$, so $L > 1$. □

Theorem III.21.2 (Prime versus composite after a factorial fold)**.** *Fix any prime reference $p \geq 3$. For every integer $n \geq 2$, define*

$$A_n = ((n-1)!)^2, \qquad \zeta_n = e^{2\pi i/n},$$

$$R_n = p + |1 - \zeta_n^{A_n}|^2, \qquad L_n = \frac{\log R_n}{\log p}.$$

Then

n *prime*	$\iff$	L_n *transcendental,*
n *composite*	$\iff$	$L_n = 1$.

In the prime case, $R_n = p + 2 - 2\cos(2\pi/n)$.

Proof. If n is prime, Wilson's congruence gives $(n-1)! \equiv -1 \pmod{n}$, and hence $A_n \equiv 1 \pmod{n}$. For completeness, the congruence follows by pairing every nonzero residue modulo n with its inverse; the unpaired residues are $1, -1$ when n is odd. The case $n = 2$ is immediate. Thus $\zeta_n^{A_n} = \zeta_n \neq 1$, and Lemma III.21.1 gives the transcendence assertion.

If n is composite, write $n = ab$ with $1 < a, b < n$. Both a and b divide $(n-1)!$. Writing $(n-1)! = ar = bs$ shows

$$((n-1)!)^2 = abrs = nrs.$$

This argument also covers $a = b$ and $n = 4$. Consequently $A_n \equiv 0 \pmod{n}$, $\zeta_n^{A_n} = 1$, and $L_n = 1$. Every integer $n \geq 2$ is either prime or composite, so both equivalences follow. □

The exact Collatz connection. For the shortcut Collatz map, start at $N = 2^{n-1}$. Its orbit is the halving chain $2^{n-1}, 2^{n-2}, \ldots, 1$, and its polynomial is

$$P_{2^{n-1}}(z) = \sum_{j=0}^{n-1} 2^{n-1-j} z^j = 2^{n-1} \sum_{j=0}^{n-1} (z/2)^j.$$

Thus $\alpha_n = 2\zeta_n$ is an actual zero on the boundary $|z| = 2$. The theorem may equivalently be written using

$$R_n = p + \left|1 - (\alpha_n/2)^{A_n}\right|^2.$$

This identifies the primality of the index n, not the primality of the Collatz starting value 2^{n-1}. The latter is composite for every $n \geq 3$. The prime/composite information enters through the factorial congruence; the boundary zero supplies a geometric representation of that congruence, and the logarithm changes its output to 1 versus a transcendental number.

The same conclusion holds simultaneously for every nontrivial nth root ζ_n^k, $1 \leq k < n$: prime n leaves all these folded roots nontrivial, while composite n sends every one to 1. Therefore all corresponding logarithms are transcendental in the prime case, and all are 1 in the composite case. The single primitive root already suffices for the stated criterion.

Exact examples with $p = 5$.

n	$A_n \bmod n$	R_n	L_n
2	1	9	transcendental
3	1	8	transcendental
4	0	5	1
5	1	$(15 - \sqrt{5})/2$	transcendental
6	0	5	1
7	1	$7 - 2\cos(2\pi/7)$	transcendental
9	0	5	1

The transcendent entries are proved by the lemma, not inferred from decimal approximations. In actual computation one would calculate $((n-1)!)^2 \bmod n$ directly; numerical testing of transcendence offers no shortcut. No claim of historical novelty is made for this repackaging of classical results.

III.21.2 A factorial-free product using all boundary zeros

The following construction gives an exact prime/composite equivalence using products of squared chord lengths. Its arithmetic content is the classical absence of zero divisors modulo a prime. The construction requires no primality label as input and no factorial. No claim of a new efficient primality algorithm or of historical novelty is made.

Lemma III.21.3 (The folded chord product)**.** *For each integer $n \geq 2$, let $\zeta_n = e^{2\pi i/n}$ and define*

$$D_n = \prod_{j=1}^{n-1} \prod_{k=1}^{n-1} |1 - \zeta_n^{jk}|^2.$$

Then

$$\boxed{D_n = \begin{cases} n^{2(n-1)}, & n \text{ prime}, \\ 0, & n \text{ composite}. \end{cases}}$$

Proof. Suppose first that $n = q$ is prime. For each $1 \le j < q$, multiplication by j permutes the nonzero residue classes modulo q. Moreover

$$\prod_{r=1}^{q-1}(x - \zeta_q^r) = 1 + x + \cdots + x^{q-1}, \qquad \prod_{r=1}^{q-1}(1 - \zeta_q^r) = q.$$

Consequently each inner product in D_q equals

$$\prod_{k=1}^{q-1}|1 - \zeta_q^{jk}|^2 = \left|\prod_{r=1}^{q-1}(1 - \zeta_q^r)\right|^2 = q^2.$$

There are $q - 1$ such products, proving $D_q = q^{2(q-1)}$.

If n is composite, write $n = ab$ with $1 < a, b < n$. The indices $j = a$, $k = b$ occur in the product and give $|1 - \zeta_n^{ab}|^2 = |1 - 1|^2 = 0$. Hence $D_n = 0$. □

Theorem III.21.4 (An exact transcendence criterion with a fixed prime base)**.** *Fix any prime $p \ge 2$. For $n \ge 2$, define*

$$F_p(n) = \frac{\log(p + D_n)}{\log p}.$$

Then

$$\boxed{\begin{array}{rcl} n \text{ prime} & \iff & F_p(n) \text{ transcendental,} \\ n \text{ composite} & \iff & F_p(n) = 1. \end{array}}$$

In the prime case $F_p(n) > 1$. The reference prime p is fixed independently of the tested integer n.

Proof. For a composite n, Lemma III.21.3 gives $D_n = 0$ and hence $F_p(n) = 1$.

Now let $n = q$ be prime and put $M = p + q^{2(q-1)}$. We first show that M is not an integer power of p. If $q \ne p$, then $p \nmid M$, since $M \equiv q^{2(q-1)} \not\equiv 0 \pmod{p}$. If $q = p$, then

$$M = p(1 + p^{2p-3}).$$

Here $2p - 3 \ge 1$, so the exponent of p in M is exactly 1, whereas $M > p$. This also excludes a power of p. In particular the case $n = p = 2$ is covered: $M = 6$.

If $\log M / \log p = a/b \in \mathbb{Q}$, with $b > 0$, then $M^b = p^a$. Unique factorization forces the positive integer M to be an integer power of p, a contradiction. Thus $F_p(q)$ is irrational. If it were algebraic irrational, the Gelfond–Schneider theorem would make $p^{F_p(q)} = M$ transcendental, contradicting the fact that M is an integer. Therefore $F_p(q)$ is transcendental. Finally $M > p$ proves $F_p(q) > 1$, and the two exhaustive cases give both stated equivalences. □

The exact connection with all Collatz boundary zeros. Use the shortcut Collatz map $T(N) = N/2$ for even N and $T(N) = (3N + 1)/2$ for odd N, stopping at the first 1. The orbit of 2^{n-1} is its halving chain, so

$$P_{2^{n-1}}(z) = \sum_{r=0}^{n-1} 2^{n-1-r} z^r = 2^{n-1} \sum_{r=0}^{n-1} (z/2)^r.$$

Its zeros are precisely

$$\alpha_j = 2\zeta_n^j, \qquad 1 \le j < n.$$

Every one lies on the outer circle $|z| = 2$. Therefore the chord product can be written entirely in terms of these zeros and their integer folds:

$$\boxed{D_n = \prod_{P_{2^{n-1}}(\alpha)=0} \prod_{k=1}^{n-1} \left|1 - (\alpha/2)^k\right|^2.}$$

The product uses every zero. If n is prime, no listed fold lands at 1, and the product has the exact positive value $n^{2(n-1)}$. If n is composite, at least one listed fold lands at 1, and one zero chord makes the entire product vanish.

Exact examples with $p = 5$.

n	D_n	$5 + D_n$	$F_5(n)$
2	4	9	$\log 9/\log 5$ (transcendental)
3	81	86	$\log 86/\log 5$ (transcendental)
4	0	5	1
5	390625	390630	transcendental
6	0	5	1

Scope of the criterion. The tested integer is $n = \deg P_{2^{n-1}} + 1$. This does not test the primality of the Collatz starting value 2^{n-1}, which is composite for $n \geq 3$. It also makes no primality assertion about the transformed integer $p + D_n$: for example, $p = 5$, $n = 2$ gives $p + D_n = 9$.

The result expresses a classical divisibility criterion through the geometry of known Collatz boundary zeros, then uses a standard transcendence theorem to give its output a different form. Numerical testing of transcendence supplies no computational shortcut. Directly evaluating a product with $(n-1)^2$ factors is not an asserted improvement over established primality algorithms. Nothing here proves a prime-only locus among the unmodified zeros of arbitrary P_N.

An entirely integral version. Let $A_n(x) = (x^n - 1)/(x - 1)$. The same chord product is

$$D_n = \prod_{k=1}^{n-1} \operatorname{Res}(A_n(x), 1 - x^k)^2.$$

Indeed, the resultant is the product of $1 - \zeta_n^{jk}$ over $1 \leq j < n$; it is n if $\gcd(k, n) = 1$ and zero otherwise. Thus exact resultants remove numerical root coordinates, but expose the same ordinary modular divisibility test. They do not create a faster transcendence oracle.

III.22 A known prime as a certificate for a larger integer

This appendix records the exact primality statements investigated after the zero-geometry questions. The results belong to classical large-factor primality theory. They do not constitute a new general primality algorithm, a consequence of the Collatz conjecture, or a method for deciding arbitrary integers from one previously known prime. In particular, a single displayed congruence is not a constant-time computation. Its evaluation may require substantial arithmetic.

III.22.1 An exact one-base test on a specified family

Theorem III.22.1 (A classical large-prime-factor criterion)**.** *Let P be a proven odd prime, let k be a positive even integer, and put $N = kP + 1$. If*

$$2^k < N < (2P + 1)^2, \tag{III.22.1}$$

then

$$N \text{ is prime} \iff 2^{N-1} \equiv 1 \pmod{N}. \tag{III.22.2}$$

Proof. If N is prime, the congruence follows from Fermat's little theorem, since N is odd. Conversely, assume the congruence and write $d = \operatorname{ord}_N(2)$. Then $d \mid kP$. The inequality $1 < 2^k < N$ shows that $2^k \not\equiv 1 \pmod{N}$, so $d \nmid k$. If $P \nmid d$, coprimality of d and P would imply $d \mid k$; therefore $P \mid d$. Lagrange's theorem gives $d \mid \varphi(N)$, and hence $P \mid \varphi(N)$.

Write $N = \prod_j q_j^{e_j}$. Because $N \equiv 1 \pmod{P}$, no q_j equals P. The product formula

$$\varphi(N) = \prod_j q_j^{e_j-1}(q_j - 1)$$

therefore implies $P \mid q - 1$ for some prime divisor q of N. Suppose N is composite and set $h = N/q > 1$. Reduction modulo P gives $q \equiv h \equiv 1 \pmod{P}$. Both q and h are odd. Since P is odd, an odd integer greater than 1 and congruent to 1 modulo P is at least $2P + 1$. Consequently

$$N = qh \geq (2P + 1)^2,$$

contrary to (III.22.1). Thus N is prime. □

The order argument is part of the classical $N - 1$ theory of Pocklington and Brillhart–Lehmer–Selfridge [76, 77]; the explicit $kP + 1$ specialization also appears in [78, Theorem 2.1]. The hypotheses matter: without them, a base-2 Fermat congruence alone is only a probable-prime condition. For example, $341 = 11 \cdot 31$ satisfies $2^{340} \equiv 1 \pmod{341}$.

Corollary III.22.2 (Mersenne seeds)**.** *Suppose $P = 2^m - 1$ is prime and $m \geq 3$. For every even integer k with $2 \leq k \leq m$, the number $N_k = kP + 1$ satisfies* (III.22.1)*. Thus* (III.22.2) *is an exact primality test for each such N_k.*

Proof. We have $2^k \leq 2^m = P + 1 < 2P + 1 \leq N_k$. Also $m \leq P$, so $N_k \leq P^2 + 1 < (2P + 1)^2$. □

This corollary neither predicts that one of the candidates is prime nor proves that infinitely many of them are prime. It specifies a finite family on which one congruence is conclusive. Failure of that congruence proves compositeness; success proves primality under the stated hypotheses.

III.22.2 Exact examples and reproducible certificates

For $P = 127 = 2^7 - 1$, exact modular exponentiation gives

$$2^{508} \bmod 509 = 1, \qquad 2^{762} \bmod 763 = 64.$$

Since $509 = 4P + 1$ and $763 = 6P + 1$ satisfy the range conditions, Theorem III.22.1 proves that 509 is prime and 763 is composite. Indeed, $763 = 7 \cdot 109$.

For the larger seed $P = 2^{127} - 1$, the following two candidates also have residue 1 and are therefore proven primes:

k	$N = k(2^{127} - 1) + 1$	$2^{N-1} \bmod N$
114	193960949144934924174123526236107880528 79	1
124	210975067490981847347292256607696291101 49	1

Both integers have 41 decimal digits. These computations certify their primality; they do not assert that these integers were previously unknown primes.

For completeness, the seed certificates use the classical Lucas–Lehmer theorem [79]: for a prime exponent $m \geq 3$, define $s_0 = 4$ and $s_{j+1} = s_j^2 - 2 \pmod{2^m - 1}$. Then $2^m - 1$ is prime if and only if $s_{m-2} = 0$. The exponents 7 and 127 are themselves verified by trial division in the complete Python code below. The code uses only exact integers and the Python standard library. It contains all data needed for these examples and the following small-divisor table; no separate source file or numerical package is required.

```
from math import isqrt

def small_prime(n):
    return n >= 2 and all(n % d for d in range(2, isqrt(n)+1))

def mersenne_prime(m):
    assert small_prime(m)
    p = (1 << m) - 1
    if m == 2:
        return p
    s = 4
    for _ in range(m-2):
        s = (s*s - 2) % p
    assert s == 0                 # Lucas--Lehmer certificate
    return p

def classify(p, k):
    # Precondition: p has already been PROVEN prime.
    n = k*p + 1
    assert p > 2 and k > 0 and k % 2 == 0
    assert (1 << k) < n < (2*p + 1)**2
    residue = pow(2, n-1, n)
    return n, residue, ('prime' if residue == 1 else 'composite')

p7 = mersenne_prime(7)
assert classify(p7, 4) == (509, 1, 'prime')
assert classify(p7, 6) == (763, 64, 'composite')

p127 = mersenne_prime(127)
expected = {
    114: 193960949144934924174123526236107880528 79,
    124: 210975067490981847347292256607696291101 49,
}
for k, value in expected.items():
    assert classify(p127, k) == (value, 1, 'prime')

m = 136279841
factors = (3, 5, 607, 3, 11, 281, 3, 7, 13, 3, 103, 5, 3)
for k, q in zip(range(2, 28, 2), factors):
    assert small_prime(q)
    assert (k*(pow(2, m, q)-1)+1) % q == 0
print('All certificates and small-divisor identities verified.')
```

III.22.3 What can be checked cheaply at the record-sized seed

Set $M = 2^{136279841} - 1$. For the candidates $N_k = kM + 1$, the following exact proper divisors already prove compositeness:

k	2	4	6	8	10	12	14	16	18	20	22	24	26
$q \mid N_k$	3	5	607	3	11	281	3	7	13	3	103	5	3

Each entry is checked by the small computation

$$k((2^{136279841} \bmod q) - 1) + 1 \equiv 0 \pmod{q}.$$

It is unnecessary to construct the decimal expansion of M or evaluate a modular power with modulus N_k. These divisibility certificates do not even require an assumption that M is prime. In contrast, applying Corollary III.22.2 to a candidate that survives such sieving requires a certified prime seed and a potentially very expensive congruence computation. No prime larger than M is certified by this table, and failure to find a small divisor is not a primality proof.

III.22.4 Related certificates and the boundary of the method

Proposition III.22.3 (Pocklington certificate)**.** *Let $N > 1$, and let $F \mid N-1$ have a complete factorization into proven primes, with $F > \sqrt{N}$. Suppose that for every prime $q \mid F$ there is an integer a_q satisfying*

$$a_q^{N-1} \equiv 1 \pmod{N}, \qquad \gcd(a_q^{(N-1)/q} - 1, N) = 1.$$

Then N is prime.

Proof. For each prime divisor r of N, the order of a_q modulo r divides $N-1$ but does not divide $(N-1)/q$. Its q-adic valuation is therefore $v_q(N-1)$. As the order divides $r-1$, this implies $F \mid r-1$. Hence every prime divisor of N exceeds $\sqrt{N}$, which is impossible for a composite N. □

With a single known prime P, one may extract $F = P^{v_P(N-1)}$ by repeated division, without factoring the remaining cofactor. If $F^2 > N$, the proposition supplies a useful certificate. Its failure for a particular base does not by itself prove compositeness: a greatest common divisor equal to N can simply mean that another base is needed. A proper greatest common divisor, or a failed Fermat congruence, does prove compositeness.

There is a precise $N+1$ analogue using Lucas sequences. Let N be odd, P a proven odd prime dividing $N+1$, and

$$U_0 = 0, \quad U_1 = 1, \quad U_{j+2} = AU_{j+1} - BU_j, \qquad \Delta = A^2 - 4B.$$

If $\gcd(N, 2B\Delta) = 1$, $U_{N+1} \equiv 0 \pmod{N}$, $\gcd(U_{(N+1)/P}, N) = 1$, and $2P - 1 > \sqrt{N}$, then N is prime. Indeed, for each prime $r \mid N$, the rank of apparition of r in this Lucas sequence divides $r - (\Delta/r)$ and, by the two displayed conditions on U, is divisible by P. Thus $r \equiv \pm 1 \pmod{P}$. Oddness gives $r \geq 2P - 1 > \sqrt{N}$. This is a certificate, not an unconditional binary test for an arbitrary fixed choice of A, B; see the classical $N+1$ theory in [77].

An elliptic-curve certificate provides a different route [80]. Let E be an elliptic curve modulo N with discriminant coprime to N. Suppose a point R reduces to a nonidentity point modulo every prime divisor of N, and

$$[Q]R = O, \qquad Q \text{ is a proven prime}, \qquad Q > (N^{1/4} + 1)^2.$$

Then N is prime: if a prime $r \leq \sqrt{N}$ divided N, reduction would give a point of order Q over $\mathbb{F}_r$, contrary to Hasse's bound $Q \leq \#E(\mathbb{F}_r) \leq (\sqrt{r}+1)^2 \leq (N^{1/4}+1)^2$. The nonidentity condition must hold at every prime divisor, not merely as a formal inequality $R \neq O$ over a composite modulus. Finding a suitable curve, point, and certified Q is a separate computational problem. None of these certificates guarantees a short proof for an arbitrary prescribed N from one fixed seed prime.

III.23 External constants: established applications and unresolved extensions

The arithmetic arguments for Collatz zero moduli invite comparison with constants that were defined independently of this investigation. Such a comparison must distinguish an application of an established theorem from the solution of a previously open problem. This section supplies several exact applications, then records the hypotheses that fail for harder targets. No previously open named constant is claimed to be classified here.

III.23.1 A reusable criterion for logarithms of algebraic units

We use real logarithms of positive real numbers. The Gelfond–Schneider theorem implies the familiar dichotomy

$$\alpha, \beta \in \overline{\mathbb{Q}} \cap \mathbb{R}_{>0}, \quad \alpha \neq 1, \quad \beta \neq 1 \quad \Longrightarrow \quad \frac{\log \alpha}{\log \beta} \in \mathbb{Q} \text{ or } \frac{\log \alpha}{\log \beta} \text{ is transcendental.} \tag{III.23.1}$$

Indeed, an algebraic irrational value t would make $\beta^t = \alpha$ transcendental, a contradiction. This is classical transcendence theory, not a property specific to Collatz trajectories; see [67].

An algebraic unit is an algebraic number u for which both u and u^{-1} are algebraic integers. Its norm in any number field containing it is 1 or -1.

Proposition III.23.1 (Unit logarithms)**.** *If $u > 1$ is an algebraic unit and $b \geq 2$ is an integer, then $\log u / \log b$ is transcendental.*

Proof. If the ratio were rational, write it as m/n with positive integers m, n. Then $u^n = b^m$. In a number field K containing u, taking norms gives $(\pm 1)^n = b^{m[K:\mathbb{Q}]}$, which is impossible. Apply (III.23.1). □

Thus the same norm calculation can certify transcendence in a setting with no distinguished prime divisor at all. It distinguishes an algebraic unit from an integer greater than one; it does not automatically distinguish two different units from each other.

III.23.2 Hard-square strips: a complete finite-width calculation

In the hard-square model, occupied sites of a square lattice may not be adjacent. At activity 1, its partition function counts independent sets of the grid graph. Transfer matrices encode compatibility between consecutive rows; their standard definition is given in [81, Section 2].

Fix a width $w \geq 1$ with open boundary conditions across a row. Let $\mathcal{F}_w$ be the family of independent subsets of the path with w vertices. Put

$$A_{ST} = \mathbf{1}_{S \cap T = \varnothing} \qquad (S, T \in \mathcal{F}_w). \tag{III.23.2}$$

The family $\mathcal{F}_w$ is downward closed: every subset of an allowed row is allowed. This simple fact determines the determinant of A exactly.

Proposition III.23.2 (Compatibility factorization)**.** *Let $\mathcal{F}$ be a finite nonempty downward-closed family of subsets of a finite set. Define A by (III.23.2), with indices in $\mathcal{F}$, and set*

$$Z_{SU} = \mathbf{1}_{U \subseteq S}, \qquad D_{UU} = (-1)^{|U|}.$$

Then

$$A = ZDZ^{\mathsf{T}}, \qquad \det A = (-1)^{\sum_{U \in \mathcal{F}} |U|}. \tag{III.23.3}$$

Every eigenvalue of A is an algebraic unit.

Proof. For $S, T \in \mathcal{F}$, downward closure gives

$$(ZDZ^{\mathsf{T}})_{ST} = \sum_{U \subseteq S \cap T} (-1)^{|U|} = \begin{cases} 1, & S \cap T = \varnothing, \\ 0, & S \cap T \neq \varnothing. \end{cases}$$

Order $\mathcal{F}$ by nondecreasing cardinality. Then Z is lower triangular with all diagonal entries equal to 1. This proves the determinant formula. In particular, A is invertible over $\mathbb{Z}$. An eigenvalue λ is an algebraic integer because A is integral; λ^{-1} is an algebraic integer because A^{-1} is integral. □

The factorization is the elementary subset-poset form of the classical Möbius factorization for meet matrices. The proof is included to avoid any dependence on a numerical determinant calculation. It also applies to independent subsets of a cycle and, more generally, of any finite graph.

Let λ_w be the spectral radius of the open-strip matrix A. Since the empty row is compatible with every row, A^2 is entrywise positive. Perron–Frobenius theory therefore gives a simple positive dominant eigenvalue. The principal submatrix on the empty row and any one-site row is

$$\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix},$$

so $\lambda_w \geq (1+\sqrt{5})/2 > 1$. Proposition III.23.2 and Proposition III.23.1 prove

$$\boxed{\frac{\log \lambda_w}{\log b} \text{ is transcendental for every } w \geq 1 \text{ and integer } b \geq 2.} \tag{III.23.4}$$

The same is true after division by w, which expresses strip entropy per site in base b. For example, at width 1 it gives the transcendence of $\log((1+\sqrt{5})/2)/\log 2$.

At every finite prime of every number field containing an eigenvalue, the valuation of that eigenvalue is zero. Thus a method requiring a nonzero valuation at a distinguished prime cannot be transferred unchanged to this model. Formula (III.23.4) succeeds for a different reason: the comparison number b is not a unit.

III.23.3 Why the infinite-width limit remains a different problem

If $I(n,n)$ counts independent sets of an n by n square grid, the hard-square growth constant is

$$\kappa_2 = \lim_{n\to\infty} I(n,n)^{1/n^2} = 1.503048\ldots.$$

It can also be approached through increasing strip widths. The 2026 preprint [82] explicitly treats the algebraicity of this limit as unresolved. Finite transfer matrices have algebraic eigenvalues, but their dimensions and the associated number fields grow with the width. Neither their unit property nor (III.23.4) classifies the limit.

The following exact example shows that this is a logical obstruction, not merely a missing error estimate.

Proposition III.23.3 (Transcendental logarithm ratios can converge to a rational)**.** *For $n \geq 2$, let $u_n > 2$ be the real root of*

$$x^n - 2x^{n-1} - 1 = 0.$$

Then u_n is an algebraic unit, $\log u_n/\log 2$ is transcendental, and

$$\lim_{n\to\infty} \frac{\log u_n}{\log 2} = 1.$$

Proof. The function $x^{n-1}(x-2)$ increases strictly from 0 to infinity on $(2,\infty)$, so the stated root exists and is unique. Its monic polynomial has constant term -1; the reciprocal also satisfies a monic integral polynomial. Hence u_n is a unit. Moreover,

$$0 < u_n - 2 = u_n^{-(n-1)} < 2^{-(n-1)}.$$

This proves $u_n \to 2$, while Proposition III.23.1 proves transcendence at each finite index. □

Consequently, even a convergent sequence consisting entirely of units, with explicitly transcendental normalized logarithms, supplies no such classification of its limit without an additional uniform argument.

III.23.4 Gamma values: a transcendental combination need not identify a factor

Euler's gamma function Γ must be distinguished from the Euler–Mascheroni constant γ. A classical theorem of Schneider on abelian integrals proves the transcendence of

$$B\left(\frac{1}{5},\frac{1}{5}\right)=\frac{\Gamma(1/5)^2}{\Gamma(2/5)}. \tag{III.23.5}$$

See [83, 84]. It follows that $\Gamma(1/5)$ and $\Gamma(2/5)$ cannot both be algebraic. It does not identify which individual value is transcendental. Algebraicity of $\Gamma(1/5)$ alone would be compatible with (III.23.5) if $\Gamma(2/5)$ were transcendental. Grinspan's stronger result states that at least one of these two gamma values is transcendental over $\mathbb{Q}(\pi)$ [85]; the conclusion still does not isolate a specified one of them.

Nor does taking logarithms turn these numbers into an instance of (III.23.1). That theorem requires its logarithm arguments to be algebraic. Substituting gamma values whose required arithmetic nature has not been established assumes the missing hypothesis. Similarly, for a positive $x\neq 1$,

$$\frac{\log(x^2)}{\log x}=2$$

is an identity for every such x, and cannot classify x.

III.23.5 Somos's constant: the finite versus infinite logarithm issue

Somos's quadratic recurrence constant has the convergent product and series

$$\sigma=\prod_{n=1}^{\infty} n^{2^{-n}},\qquad \log\sigma=\sum_{n=1}^{\infty}2^{-n}\log n. \tag{III.23.6}$$

Sondow and Hadjicostas [86] connect it to the generalized Euler-constant function

$$\gamma_{\mathrm{gen}}(z)=\sum_{n\geq 1}z^{n-1}\left(\frac{1}{n}-\log\frac{n+1}{n}\right),\qquad \gamma_{\mathrm{gen}}(1/2)=2\log(2/\sigma).$$

This is a useful exact identity, but it transfers the same unknown arithmetic information between two expressions.

Every finite product $\sigma_N=\prod_{n=1}^{N}n^{2^{-n}}$ is algebraic; for $N\geq 2$ it is greater than one. Its real logarithm is transcendental by the Hermite–Lindemann theorem. The limit in (III.23.6) is an infinite logarithmic sum, whereas Baker's linear-independence results apply to finitely many logarithms of algebraic numbers.

One can make the failure of a naive prime-by-prime reduction explicit. All summands are nonnegative, so rearranging the real series gives

$$\log\sigma=\sum_{p\ \text{prime}}c_p\log p,\qquad c_p=\sum_{j\geq 1}\frac{1}{2^{p^j}-1}. \tag{III.23.7}$$

Indeed, $\sum_{n\geq 1}2^{-n}v_p(n)=\sum_{j\geq 1}\sum_{k\geq 1}2^{-kp^j}$. Formula (III.23.7) has infinitely many nonzero coefficients, and it does not establish that those coefficients are algebraic. It therefore meets neither the finiteness condition nor the coefficient hypothesis of the usual finite logarithm theorem. Furthermore, real convergence alone does not justify taking a p-adic valuation of an infinite product. These observations leave the arithmetic classification of σ unresolved by the present methods.

III.23.6 The sophomore's dream: rapid convergence is not enough

Another independently established target is

$$S = \sum_{n=1}^{\infty} \frac{1}{n^n} = \int_0^1 x^{-x}\, dx. \tag{III.23.8}$$

The integral identity follows by expanding $e^{-x\log x}$ into its nonnegative power series and integrating: $\int_0^1 x^k(-\log x)^k\, dx = k!/(k+1)^{k+1}$. Hančl and Kristensen discuss its unresolved irrationality in their 2018 work [87]; no resolution is claimed here.

Let $S_N = \sum_{n=1}^{N} n^{-n}$ and use the natural common denominator

$$D_N = \operatorname{lcm}(1^1, 2^2, \ldots, N^N).$$

The tail $R_N = S - S_N$ obeys the elementary bounds

$$(N+1)^{-(N+1)} \le R_N \le \frac{N+2}{N+1}(N+1)^{-(N+1)}. \tag{III.23.9}$$

For the upper bound, successive terms in the tail have ratio at most $1/(N+2)$. On the other hand, every prime $p \in (N/2, N]$ contributes the factor p^p to D_N. Writing $\vartheta(x) = \sum_{p\le x} \log p$, the prime number theorem yields

$$\log D_N \ge \sum_{N/2<p\le N} p\log p \ge \frac{N}{2}(\vartheta(N) - \vartheta(N/2)) \ge cN^2$$

for some $c > 0$ and all sufficiently large N. Together with (III.23.9), this proves

$$D_N R_N \longrightarrow \infty. \tag{III.23.10}$$

If $S = a/b$ were rational, then bD_NR_N would be an integer. The usual elementary contradiction would require that positive integer to lie below 1; (III.23.10) shows why this particular denominator-and-tail argument cannot do so. This does not rule out better approximants or cancellation in the reduced denominator of S_N. It identifies the failure of the displayed method, not an impossibility theorem for proving irrationality.

III.23.7 Established modular examples really do admit stronger conclusions

There are external infinite series for which a deeper theorem supplies the missing structure. Define

$$L_k = \sum_{n\ge1} \frac{n^k}{2^n - 1}, \qquad k = 1, 3, 5.$$

These three constants are algebraically independent, hence each is transcendental. This is an established consequence of Nesterenko's modular theorem [88], not a new application of a Collatz argument. For completeness, put

$$E_2(q) = 1 - 24\sum_{n\ge1} \frac{nq^n}{1-q^n},$$

$$E_4(q) = 1 + 240\sum_{n\ge1} \frac{n^3q^n}{1-q^n},$$

$$E_6(q) = 1 - 504\sum_{n\ge1} \frac{n^5q^n}{1-q^n}.$$

Nesterenko proves that, for $0 < |q| < 1$, the field $\mathbb{Q}(q, E_2(q), E_4(q), E_6(q))$ has transcendence degree at least 3. At the algebraic point $q = 1/2$, this forces the three series values to be algebraically independent. The displayed rational affine changes of variables give the assertion for L_1, L_3, L_5.

Related established consequences include the transcendence of

$$\theta_3(1/2) = \sum_{n\in\mathbb{Z}} 2^{-n^2} \qquad \text{and} \qquad \sum_{n\geq 1} \frac{1}{F_n^2},$$

where $F_1 = F_2 = 1$ are the Fibonacci numbers; see [89, Theorem 46 and Corollaries 51–52]. The reciprocal-square series must not be confused with $\sum_{n\geq 1} 1/F_n$. These examples show what an effective external connection requires: an exact identification with functions covered by a substantial theorem, rather than numerical resemblance or a shared use of the word "zero."

III.23.8 The Moser–de Bruijn decimal constant: an explicit classical proof

Let E consist of the nonnegative integers whose base-4 digits belong to $\{0, 1\}$:

$$E = \{0, 1, 4, 5, 16, 17, 20, 21, \ldots\}.$$

The established decimal construction

$$\chi = 3 \sum_{n\in E} 10^{-n} = 3.300330000000000330033\ldots \tag{III.23.11}$$

appears in van der Poorten's discussion of an observation of Blanchard and Mendès France [90]. Its transcendence is already accessible to Roth's theorem. The proof below is recorded as a transparent classical application, not as a newly solved open problem.

Proposition III.23.4. *The number χ in (III.23.11), and its reciprocal, are transcendental.*

Proof. Put $M_k = (4^k - 1)/3$ and truncate at the last position representable with k base-4 digits:

$$\chi_k = 3 \sum_{\substack{n\in E \\ n\leq M_k}} 10^{-n} = \frac{A_k}{q_k}, \qquad q_k = 10^{M_k}.$$

Since $M_k \in E$, the integer A_k ends in the decimal digit 3. Thus $\gcd(A_k, q_k) = 1$. The next nonzero digit occurs at position 4^k, and therefore

$$0 < \chi - \frac{A_k}{q_k} \leq 3 \sum_{j=4^k}^{\infty} 10^{-j} = \frac{10}{3} 10^{-4^k} = \frac{1}{3q_k^3}. \tag{III.23.12}$$

The decimal expansion has infinitely many nonzero digits and arbitrarily long zero blocks. It is neither terminating nor ultimately periodic, so χ is irrational. If it were algebraic, Roth's theorem [91], for example with exponent $2 + 1/2$, would allow only finitely many reduced rational approximations satisfying (III.23.12). The denominators q_k tend to infinity, a contradiction. A nonzero number and its reciprocal have the same algebraic-versus-transcendental classification. □

The examples in this section therefore have three different outcomes: finite-strip logarithms are certified by a direct classical norm argument; modular series and the Moser–de Bruijn construction have established transcendence proofs; the proposed routes to the hard-square limit, individual fifth-argument gamma values, Somos's constant, and the sophomore's dream do not supply the missing classification.

III.24 Remaining questions and exact limits of the results

The results above close specific classification, separation, and asymptotic questions, while leaving several natural extensions open within this investigation. For context, Hohertz's 2022 list [5, Section 5] asks for an explicit lower bound in terms of N and for zeros approaching 1. The starting-value bound in Section III.2 supplies the former; the annulus and cusp results prove the latter and specify arbitrarily thin approaches. They do not validate an ambiguous global claim about all zeros being bounded by a parabola. The S_{13} computation is a concrete instance. Part IV additionally determines full splitting-field groups in an infinite family and the associated radical towers; arbitrary starting values remain unclassified. These comparisons assert mathematical coverage, not that the corresponding questions remained unsolved elsewhere until this manuscript.

1. Can an exact zero of G_p, for an odd prime p, occur in some P_N with $N \neq p$? The congruence restrictions, finite uniqueness window, sparse counting bound, and bounded searches do not give global uniqueness.
2. Do any of $z^2 \pm z + 3$ and $z^2 \pm 2z + 3$ divide an odd-start Collatz polynomial? The rational-power classification reduces the exceptional case exactly to these four factors; it does not exclude them from every odd composite start.
3. What is the true growth rate of boundary-producing starts? The lower exponent s_* and the upper exponent $1/2$ leave a genuine gap. The inverse tree proves a lower bound, not an asymptotic formula.
4. What can be proved about full zero clouds when the frequency of odd steps does not tend to zero? Part IV realizes every radial probability law in an explicit outer band, but a classification of all possible limiting measures remains open.
5. Which exact zero locations, or limiting zero sets, occur when only prime starts are allowed? Composite density proves an obstruction to open prime-only regions; it does not determine the prime-restricted derived set.
6. Can a proven arithmetic identity link a previously unresolved constant to one of the independent logarithmic forms here? The Gamma, Somos, hard-square, and related investigations supplied no such identity resolving a named open transcendence problem.

The scope restrictions are substantive. A statement about every terminating P_N does not prove that every positive integer terminates. An algebraic-coefficient *linear* independence theorem does not establish algebraic independence. A finite verification is not an unbounded conjecture. A limit equal to Euler's constant does not decide its arithmetic nature. A geometric reformulation of a primality congruence does not automatically improve its complexity.

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Part IV
Full Galois groups and prescribed zero distributions

IV.1 Definitions, results, and the published questions

We use the shortcut Collatz map

$$T(n) = \begin{cases} n/2, & n \text{ even}, \\ (3n+1)/2, & n \text{ odd}. \end{cases}$$

For a positive integer $N > 1$ whose orbit reaches 1, let $\tau(N)$ be its first hitting time of 1, and define

$$a_j = T^j(N), \qquad P_N(z) = \sum_{j=0}^{\tau(N)} a_j z^j, \qquad \mu_N = \frac{1}{\tau(N)} \sum_{P_N(\alpha)=0} \delta_\alpha.$$

Multiplicities are included. Every starting integer constructed below has an explicit finite path to $5, 8, 4, 2, 1$. No assumption that all integers reach 1 is used.

The starting framework is due to Hohertz and Kalantari [25]. Hohertz's subsequent research questions include finding Galois groups for general classes of Collatz polynomials, and investigating whether assumptions from random-polynomial zero distribution results apply [5, Section 5, questions 5–6]. Sections IV.2–IV.3 determine an exact symmetric quotient and then the full group for an explicit infinite family, providing a family-specific answer to the Galois research direction. Section IV.4 proves a deterministic realization theorem; it is not presented as a resolution of the specific random-coefficient assumptions in question 5.

For broader context, Bergweiler and Eremenko [98] characterize limit zero measures for arbitrary positive-coefficient polynomials. The additional constraint here is that coefficients must be successive values of an actual terminating Collatz orbit. Related methods for symmetric Galois groups of trinomial factors appear in Garmi [99]; no new general Galois method is claimed here.

The conclusions can be read concretely as follows.

- For infinitely many prime indices k, a Collatz polynomial of degree $4k+3$ has a degree-$4k$ irreducible factor whose roots cannot be expressed by radicals. For every prime $k \geq 23$ with $11 \nmid k+1$, the full group is $(C_4)^k \rtimes (S_k \times C_2)$, and the associated radical tower is determined at every level.
- Given finitely many radii in $[2\,3^{-1/4}, 2]$ and proportions totaling one, entire zero clouds can be made to converge to those circles in those proportions, uniformly in angle. Arbitrary probability distributions over these radii follow by approximation.

The second statement concerns a limit as the degree grows. It does not say that finite-degree zeros lie exactly on the prescribed circles. Proofs of the stated results and references to the classical tools are included in this single `.tex` file; compilation requires no external code, data, figures, or bibliography file.

IV.2 An infinite family with a symmetric Galois quotient

The problem of describing Galois groups for general classes of Collatz polynomials, raised by Hohertz [5, Section 5, question 6], motivates the following exact infinite-family result. We first

determine a symmetric Galois quotient of unbounded degree. This is the foundation for the full-group theorem in Section IV.3; the quotient calculation alone does not determine the entire group.

For $k \geq 1$ define

$$F_k(x) = x^k + 5\sum_{j=0}^{k-1} 16^{k-1-j}x^j, \qquad N_k = 10 \cdot 4^{2k-1}.$$

Every N_k has an explicitly terminating shortcut Collatz orbit: it halves repeatedly to 5, then follows $5, 8, 4, 2, 1$.

Theorem IV.2.1 (An unbounded symmetric quotient)**.** *For every integer $k \geq 1$,*

$$P_{N_k}(z) = (z+2)(z^2+4)F_k(z^4),$$

and $F_k(z^4)$ is irreducible over $\mathbb{Q}$ of degree $4k$. If $k \geq 23$ is prime and $11 \nmid k+1$, then

$$\mathrm{Gal}(F_k/\mathbb{Q}) \cong S_k.$$

Consequently the splitting-field Galois group of P_{N_k} admits a surjective homomorphism onto S_k. None of the $4k$ zeros of $F_k(z^4)$ is expressible by radicals over $\mathbb{Q}$.

Proof. The orbit gives the explicit formula

$$P_{N_k}(z) = 5\sum_{j=0}^{4k-1} 2^{4k-1-j}z^j + 8z^{4k} + 4z^{4k+1} + 2z^{4k+2} + z^{4k+3}.$$

Multiplying the displayed factors recovers these coefficients and proves $\deg P_{N_k} = 4k+3$ as well as the factorization. Both $F_k(x)$ and $F_k(z^4)$ are Eisenstein at 5: their nonleading coefficients are divisible by 5, whereas their constant coefficient $5 \cdot 16^{k-1}$ is not divisible by 25. They are therefore irreducible.

Assume henceforth that $k \geq 23$ is prime and $11 \nmid k+1$. Let K be the splitting field of F_k, choose a root β, and put $E = \mathbb{Q}(\beta)$. Thus $[E : \mathbb{Q}] = k$. We first examine the ramification in these fields.

The prime 2 is unramified in K. The integral polynomial

$$g_k(y) := 16^{1-k}F_k(16y) = 16y^k + 5\sum_{j=0}^{k-1} y^j$$

reduces modulo 2 to $(y^k-1)/(y-1)$. Since k is odd, this reduction has degree $k-1$ and is squarefree. All its roots lie in a finite extension of $\mathbb{F}_2$, and Hensel's lemma lifts these $k-1$ distinct roots to the corresponding unramified extension of $\mathbb{Q}_2$. Multiplying them by 16 gives $k-1$ roots of F_k. The remaining root lifts the simple root $x = 1$ of

$$F_k(x) \bmod 2 = x^{k-1}(x+1),$$

so it lies in $\mathbb{Q}_2$. Thus the entire local splitting field is unramified at 2.

The prime 11 is unramified in K. The geometric-sum identity gives

$$T_k(x) := (x-16)F_k(x) = x^{k+1} - 11x^k - 5 \cdot 16^k, \qquad T_k'(x) = x^{k-1}((k+1)x - 11k).$$

Modulo 11, the derivative is $(k+1)x^k$. Under our congruence hypothesis it has no common root with T_k, whose constant coefficient is nonzero modulo 11. Hence T_k, and therefore F_k, is squarefree modulo 11.

Every other ramified prime except 5 *supplies a transposition.* Let $q \notin \{2, 5, 11\}$ be prime. If $q \mid k$ or $q \mid k+1$, the displayed derivative shows that T_k is squarefree modulo q. Otherwise its only possible repeated root is

$$c = \frac{11k}{k+1} \pmod{q}.$$

This is nonzero and

$$T_k''(c) = (k+1)c^{k-1} \neq 0 \pmod{q}.$$

Thus any repeated root of T_k has multiplicity exactly two, and there is at most one such root. Dividing by $x - 16$ cannot increase multiplicities, so the same statement holds for F_k.

All roots of F_k are integral at q. Inertia fixes their reductions; every simple residual root lifts in an unramified extension and is fixed by inertia. There are at most two roots remaining to be permuted. Consequently, if q actually ramifies in K, its nontrivial inertia acts on the roots as a transposition.

A prime of this last kind must exist. Suppose otherwise that E were unramified outside 5. The Eisenstein condition at 5 makes $E/\mathbb{Q}$ totally ramified there with local degree k. Since $k \neq 5$, this ramification is tame and its discriminant exponent is $k - 1$. We would obtain

$$|D_E| = 5^{k-1}.$$

The Minkowski discriminant bound [97, Corollary 14.20] instead gives, since the number r_2 of complex places satisfies $r_2 \leq (k-1)/2$,

$$|D_E| \geq \left(\frac{\pi}{4}\right)^{2r_2} \left(\frac{k^k}{k!}\right)^2 \geq \left(\frac{\pi}{4}\right)^{k-1} \left(\frac{k^k}{k!}\right)^2 > 5^{k-1}.$$

Here is a rational certificate for the last strict inequality. Put

$$B_k = \left(\frac{\pi}{20}\right)^{k-1} \left(\frac{k^k}{k!}\right)^2.$$

Using $\pi > 157/50$, the two integer inequalities

$$157^{22} 23^{46} > 1000^{22} (23!)^2, \qquad 157 \cdot 24^{46} > 1000 \cdot 23^{46}$$

prove respectively $B_{23} > 1$ and $(\pi/20)(24/23)^{46} > 1$. Moreover

$$\frac{B_{k+1}}{B_k} = \frac{\pi}{20} \left(1 + \frac{1}{k}\right)^{2k} > 1 \qquad (k \geq 23),$$

because $(1+1/k)^k$ increases with k. Hence $B_k > 1$ for every $k \geq 23$, as required. The contradiction proves that some prime $q \notin \{2, 5, 11\}$ ramifies in E, and therefore in K.

It follows that $G = \mathrm{Gal}(F_k/\mathbb{Q})$ contains a transposition. Irreducibility makes G transitive, and its prime degree makes it primitive. For completeness, take the graph whose edges are all the conjugates in G of that transposition. Its connected components form a G-invariant block system. Primitivity and the existence of an edge force this graph to be connected; its edge transpositions generate S_k. Thus $G = S_k$.

Taking fourth powers of the roots of $F_k(z^4)$ produces all roots of F_k. The splitting field of F_k is therefore contained in the splitting field of P_{N_k}, and restriction of automorphisms gives a surjection onto S_k. Since $k \geq 5$, this quotient is not solvable. In particular no root of the irreducible polynomial F_k is expressible by radicals. If a root of $F_k(z^4)$ were expressible by radicals, its fourth power would be such a root of F_k, a contradiction. □

Corollary IV.2.2 (Infinitely many explicitly terminating examples)**.** *There are infinitely many indices satisfying the hypotheses of Theorem IV.2.1. The irreducible factors $F_k(z^4)$ have unbounded degree, and their splitting fields have Galois groups of order at least $k!$.*

Proof. There are infinitely many primes $k \equiv 1 \pmod{11}$; in this special case an elementary cyclotomic argument suffices. If $p_1, \dots, p_s$ were the complete list of such primes, put $A = 11p_1 \cdots p_s$ and choose a prime divisor q of

$$\Phi_{11}(A) = 1 + A + \cdots + A^{10} > 1.$$

This integer is 1 modulo every prime dividing A, so q is neither 11 nor any p_i. Since $A^{11} \equiv 1 \pmod{q}$, the order of A modulo q divides 11. It cannot be 1, for otherwise $\Phi_{11}(A) \equiv 11 \pmod{q}$, forcing $q = 11$. Thus the order is 11, and $q \equiv 1 \pmod{11}$, a contradiction. Every prime in this progression is at least 23 and satisfies $11 \nmid k + 1$. □

For the first eligible index $k = 23$, this gives

$$N = 10 \cdot 4^{45}, \qquad \deg P_N = 95, \qquad \deg F_{23}(z^4) = 92, \qquad \mathrm{Gal}(F_{23}/\mathbb{Q}) = S_{23}.$$

The 92 off-axis zeros belong to one irreducible factor and are not expressible by radicals. The remaining zeros are the explicit axis zeros -2 and $\pm 2i$.

There is also a much smaller concrete example outside the threshold used in the uniform proof. For $k = 5$, modulo 3 one has

$$F_5(x) \equiv (x^2 + x - 1)(x^3 + x^2 - x + 1).$$

Neither factor has a root in $\mathbb{F}_3$, so both are irreducible and the factorization is squarefree. The Galois group therefore contains a permutation of cycle type (2)(3); its cube is a transposition. Eisenstein irreducibility and prime degree again give $\mathrm{Gal}(F_5/\mathbb{Q}) = S_5$. Consequently

$$N = 10 \cdot 4^9 = 2\,621\,440$$

already yields an irreducible degree-20 off-axis factor whose zeros are not expressible by radicals. Here $\deg P_N = 23$.

Precise scope and attribution. The quotient argument alone does not determine the full group. Section IV.3 closes that gap for every odd k coprime to 5, and in particular for every index covered by the symmetric-quotient theorem. The full group is not S_{4k}: for $k > 1$, its fourth-power blocks make the action imprimitive. The result gives an exact S_k quotient in an explicit infinite family and uses classical Eisenstein, Hensel, tame-ramification, Minkowski, and permutation- group arguments. Historical priority of this particular family and deduction has not been established.

IV.3 Full Galois groups and the entire power-of-two radical tower

The symmetric quotient can be upgraded to the full group for a much broader class than one congruence progression. The proof combines local square classes at 2 with tame inertia at 5; no condition on the parity of $\mathrm{ord}_k(2)$ is needed. For odd $k \geq 3$, write $G_k = \mathrm{Gal}(F_k/\mathbb{Q})$ and let K_k be its splitting field. For $a \geq 1$, put $q = 2^a$, let C_q denote the cyclic group of order q, and let $L_{k,a}$ be the splitting field of $F_k(X^q)$ over $\mathbb{Q}$.

Theorem IV.3.1 (Full lifting without an order-of-2 restriction)**.** *For every odd $k \geq 3$ with $5 \nmid k$, and every $a \geq 1$,*

$$\mathrm{Gal}(L_{k,a}/\mathbb{Q}) \cong (C_{2^a})^k \rtimes \big(G_k \times (\mathbb{Z}/2^a\mathbb{Z})^\times\big), \qquad [L_{k,a} : \mathbb{Q}] = 2^{ak}\varphi(2^a)|G_k|.$$

Here G_k permutes the k coordinates, and a unit modulo 2^a multiplies every coordinate by that unit. In particular, for every prime $k \geq 23$ with $11 \nmid k+1$,

$$\boxed{\mathrm{Gal}(P_{10\cdot 4^{2k-1}}/\mathbb{Q}) \cong (C_4)^k \rtimes (S_k \times C_2).}$$

There are infinitely many such prime indices, and the displayed formula determines the full splitting-field Galois group of the original Collatz polynomial.

Proof. Write $K = K_k$ and label the roots of F_k as $\beta_1, \ldots, \beta_k$. We first prove the local statements, then check that adjoining roots of unity cannot create a relation among their square classes.

Step 1: independent square classes over K for every odd k. Let $f = \mathrm{ord}_k(2)$ and let $U/\mathbb{Q}_2$ be the unramified extension of degree f. It contains all kth roots of unity. The simple root 1 of F_k mod 2 lifts to a root $\beta_0 \in \mathbb{Q}_2$. Since

$$F_k(X) \equiv X^{k-1}(X+5) \pmod{16},$$

this unit root satisfies $\beta_0 \equiv -5 \pmod{16}$. Hence $\beta_0 = -5s^2$ for some $s \in \mathbb{Q}_2^\times$. The field $\mathbb{Q}_2(\sqrt{-5})$ is ramified quadratic, so β_0 is not a square in *any* unramified extension of $\mathbb{Q}_2$. For an elementary verification of ramification, the unique unramified quadratic extension is $\mathbb{Q}_2(\sqrt{5})$, obtained from $X^2 - X - 1$; it differs from $\mathbb{Q}_2(\sqrt{-5})$ because -1 is not a square in $\mathbb{Q}_2$.

For a nontrivial kth root of unity $t \in U$, put

$$H(Y) = 16^{1-k}F_k(16Y) = 16Y^k + 5\sum_{j=0}^{k-1} Y^j.$$

One has $H(t) = 16$, and $H'(t)$ is a unit: the derivative of the geometric sum at t is $kt^{k-1}/(t-1)$, whose reduction is nonzero. Hensel's lemma gives a root y_t with $y_t - t \in 16\mathcal{O}_U$. The quotient $y_t/t \in 1 + 16\mathcal{O}_U$ is a square by strong Hensel, and t is a square in the odd-order group of kth roots of unity. Thus $\beta_t = 16y_t$ is a square in U. These $k-1$ roots and β_0 account for all roots. In particular, K is unramified at 2. The forms of Hensel's lemma used here are recalled in [102].

Suppose $\prod_j \beta_j^{e_j}$ is a square in K, with $e_j \in \{0,1\}$. Eisenstein irreducibility makes G_k transitive. For each j, choose a 2-adic embedding taking β_j to the unique unit root β_0. Every other root then becomes a square in U, while β_j does not. Thus $e_j = 0$. We have proved that

$$\iota : \mathbb{F}_2^k \longrightarrow K^\times/K^{\times 2}, \qquad (e_j) \longmapsto \left[\prod_j \beta_j^{e_j}\right]$$

is injective. It is also G_k-equivariant for the permutation action.

Step 2: the cyclotomic square-class kernel is fixed pointwise. Put $B = K(\zeta_{2^a})$. Since K is unramified at 2 and every nontrivial subfield of $\mathbb{Q}(\zeta_{2^a})$ is totally ramified there, these two Galois fields have trivial intersection. Hence

$$\mathrm{Gal}(B/\mathbb{Q}) = G_k \times (\mathbb{Z}/2^a\mathbb{Z})^\times.$$

Every quadratic intermediate field of B/K is the compositum of K with a quadratic subfield of $\mathbb{Q}(\zeta_{2^a})$. The latter, when present, are among $\mathbb{Q}(i)$, $\mathbb{Q}(\sqrt{2})$, and $\mathbb{Q}(\sqrt{-2})$. Consequently every element of

$$A = \ker(K^\times/K^{\times 2} \longrightarrow B^\times/B^{\times 2})$$

has a rational representative $d \in \{1, -1, 2, -2\}$. This also covers the smaller levels $a = 1, 2$, where fewer classes occur. The group G_k therefore fixes A pointwise.

Let $R = \iota^{-1}(A)$ be the space of relations among the root square classes over B. For $e \in R$ and $\sigma \in G_k$, $\iota(\sigma e) = \sigma\iota(e) = \iota(e)$. Injectivity of ι implies $\sigma e = e$. Transitivity forces every coordinate of e to be the same. Thus the only possible nonzero relation is the product of all roots. Since k is odd, that product is

$$\prod_{j=1}^{k} \beta_j = -5\,16^{k-1}.$$

Step 3: tame inertia at 5 excludes the last relation. Here we use $5 \nmid k$. The 5-Eisenstein polynomial F_k defines a totally tamely ramified extension of $\mathbb{Q}_5$ of degree k. A totally tame extension of degree k is generated by a kth root of a suitable uniformizer of the ground field; see [103, Theorem 11.10]. Its normal closure is obtained by also adjoining the kth roots of unity, an unramified extension because $5 \nmid k$. Therefore the inertia group in its local normal closure is cyclic of order k, which is odd. This is the inertia group at 5 in $K/\mathbb{Q}$. Every quadratic subfield of K is consequently unramified at 5.

If the product of all roots represented an element of A, then $-5/d$ would be a square in K for some $d \in \{1, -1, 2, -2\}$. The field $\mathbb{Q}(\sqrt{-5/d})$ would be a quadratic subfield of K ramified at 5, since $v_5(-5/d) = 1$. This is impossible. Thus $R = 0$: all k root square classes remain independent over B.

Step 4: the full Kummer kernel and the split action. Write $q = 2^a$. The root classes generate a copy of $(\mathbb{Z}/q\mathbb{Z})^k$ in $B^\times/B^{\times q}$. Indeed, if $\prod \beta_j^{e_j} = c^q$ with $0 \le e_j < q$ and not all $e_j = 0$, put $r = \min_{e_j \neq 0} v_2(e_j) < a$ and $e_j = 2^r d_j$. Then

$$\prod \beta_j^{d_j} = c^{2^{a-r}}\xi, \qquad \xi \in \mu_{2^r}.$$

The right side is a square in B: every 2^rth root of unity is a square there because $r + 1 \le a$. Some d_j is odd, contradicting the square-class independence. Kummer theory [101] now gives

$$L_{k,a} = B(\alpha_1, \ldots, \alpha_k), \qquad \alpha_j^q = \beta_j, \qquad \mathrm{Gal}(L_{k,a}/B) \cong (C_q)^k.$$

The splitting field contains B, since ratios of roots lying over one β_j give all qth roots of unity.

The monomials $\alpha_1^{j_1} \cdots \alpha_k^{j_k}$ with $0 \le j_i < q$ form a B-basis, by the degree q^k just proved. For $(\sigma, u) \in G_k \times (\mathbb{Z}/q\mathbb{Z})^\times$, extend its action on B by $\alpha_j \mapsto \alpha_{\sigma(j)}$. These are well-defined automorphisms, since the only defining algebra relations over B are $\alpha_j^q = \beta_j$, as follows from the basis. The lifts form a section of the restriction homomorphism. Conjugation permutes Kummer coordinates by σ and multiplies them by u. This proves the semidirect product.

For prime $k \ge 23$ with $11 \nmid k + 1$, the preceding symmetric-quotient theorem gives $G_k = S_k$, and $5 \nmid k$ automatically. At $a = 2$, the splitting field of $F_k(z^4)$ already contains i, so the remaining roots $-2, \pm 2i$ of P_{N_k} add no extension. Infinitely many eligible indices were established by the elementary argument with primes $k \equiv 1 \pmod{11}$. □

Proposition IV.3.2 (The only possible quadratic obstruction)**.** *Let $k \ge 3$ be any odd integer; divisibility by 5 is now allowed. The group formula in Theorem IV.3.1 always holds for $a = 1$. For any $a \ge 2$, it holds if and only if $\sqrt{5} \notin K_k$. In particular, the same single condition decides maximality simultaneously at every level $a \ge 2$.*

Proof. Step 1 did not use $5 \nmid k$, and proves the assertion for $a = 1$. Step 2 also applies to every odd k and leaves only the all-roots relation. For $a \ge 2$, $i \in B$, so that relation is present exactly when $\sqrt{5} \in B$. If $\sqrt{5} \notin K$ but $\sqrt{5} \in B$, the quadratic extension $K(\sqrt{5})/K$ must equal one of $K(i)$, $K(\sqrt{2})$, or $K(\sqrt{-2})$. Equality of quadratic extensions would put one of $\sqrt{-5}, \sqrt{10}, \sqrt{-10}$

in K, contradicting its unramifiedness at 2. Hence $\sqrt{5} \in B$ if and only if $\sqrt{5} \in K$. If it is absent, Step 4 proves the formula. If it is present, the nonzero square-class relation prevents Kummer degree 2^{ak}, so the asserted full-order group is impossible. □

Corollary IV.3.3 (All levels of the infinite radical tower)**.** *For every odd $k \geq 3$ with $5 \nmid k$, choose compatible roots $\alpha_{j,a+1}^2 = \alpha_{j,a}$ and roots of unity. Then*

$$\mathrm{Gal}\left(\bigcup_{a\geq 1} L_{k,a}/\mathbb{Q}\right) \cong \mathbb{Z}_2^k \rtimes (G_k \times \mathbb{Z}_2^\times).$$

For every prime $k \geq 23$ with $11 \nmid k+1$, replace G_k by S_k. The conclusion also holds for any odd k satisfying $\sqrt{5} \notin K_k$.

Proof. The splitting fields are nested. Use the compatible radicals in the sections of the restriction maps constructed above. Restriction reduces the translation and unit coordinates modulo 2^a and retains the same action on K. Taking inverse limits proves the formula. □

For $k = 23$,

$$N = 10 \cdot 4^{45} = 12\,379\,400\,392\,853\,802\,748\,991\,242\,240.$$

The full Galois group of its degree-95 Collatz polynomial has order

$$2 \cdot 4^{23} \cdot 23! = 3\,638\,347\,904\,750\,568\,937\,046\,801\,299\,537\,920\,000.$$

This group consists of independent quarter-turns within the 23 root quartets, arbitrary permutations of those quartets, and a common inversion of the root-of-unity coordinate. These are algebraic permutations, not arbitrary Euclidean symmetries of a plotted zero set.

What has been strengthened. The order-of-2 hypothesis in the earlier local argument has been removed for every odd index coprime to 5. The new ingredient is that the cyclotomic square-class kernel has rational representatives and hence is fixed pointwise by the transitive root action. This leaves just one possible relation, which odd tame inertia at 5 excludes. The proposition also identifies the exact remaining obstruction for indices divisible by 5; it does not assert that this obstruction actually occurs in this family.

Scope of the tower statement. The case $a = 2$ is the actual Collatz polynomial factor. Other levels $F_k(X^{2^a})$ form an associated radical tower, not successive Collatz orbit polynomials. This determines the full group allowed by the power-map structure in the stated family. It does not determine the Galois group of every Collatz polynomial or solve the general inverse Galois problem.

IV.4 Realization of prescribed radial zero-cloud distributions

Put $R_* = 2\,3^{-1/4}$. For $R > 0$, let σ_R denote normalized arclength on the circle $|z| = R$.

Theorem IV.4.1 (Prescribed rotationally invariant limits)**.** *For every probability measure ν on $[R_*, 2]$, there is a sequence of terminating positive integers $N_j \to \infty$ such that*

$$\frac{1}{\tau(N_j)} \sum_{P_{N_j}(\alpha)=0} \delta_\alpha \implies \int_{[R_*,2]} \sigma_R \, d\nu(R).$$

Zeros are counted with multiplicity. In particular, every circle $R_* \le R \le 2$ supports a realizable limiting full zero cloud. The result concerns the full empirical distribution, not a selected zero or a pooled accumulation set. It makes no assertion about historical priority or an exact classification of all possible limits.

Proof. We give a construction for finite mixtures, followed by approximation.

Admissible inverse blocks. Start with the odd integer $m_0 = 5$, whose shortcut orbit reaches 1. If $m \ge 5$ is odd and $3 \nmid m$, an exponent $a \ge 1$ is admissible when

$$n = \frac{2^a m - 1}{3}$$

is integral and $3 \nmid n$. Then n is odd, $n \ge 5$, and its shortcut orbit reaches m after exactly a steps, with precisely one odd step in that block. The intermediate values are $2^{a-1}m, 2^{a-2}m, \dots, m$.

The required parity of a is even if $m \equiv 1 \pmod 3$, and odd if $m \equiv 2 \pmod 3$. Among three consecutive exponents of this parity, exactly one gives $3 \mid n$: the corresponding residues $2^a m \pmod 9$ run through $1, 4, 7$. Thus at most one of any two consecutive such exponents is forbidden. Consequently an admissible small exponent $a_- \le 4$ always exists: use $2, 4$ in the even case and $1, 3$ in the odd case. For every real $h \ge 4$, an admissible large exponent satisfies $h \le a_+ < h + 4$.

Choose $a_k = a_+$ whenever $D_{k-1} \le 0$, and $a_k = a_-$ otherwise, where

$$A_k = \sum_{j=1}^{k} a_j, \qquad D_k = A_k - hk, \qquad D_0 = 0.$$

Induction gives $1 - h \le D_k < 4$. All exponents are bounded in terms of h. The exact iterated inverse identity is

$$n_k = \frac{2^{A_k}}{3^k} \left(m - \sum_{j=1}^{k} \frac{3^{j-1}}{2^{A_j}} \right).$$

The quantity subtracted from m is at most

$$\frac{1}{2(1 - 3/2^h)} \le \frac{8}{13}.$$

Since $m \ge 5$, it follows uniformly in the block length and its starting node that

$$\log n_k = \log m + A_k \log 2 - k \log 3 + O(1).$$

All values remain positive and the inverse construction supplies a finite terminating orbit; no Collatz termination conjecture is used.

Let b_ℓ be its orbit coefficients read backwards from the first visit to 1. Within this block, at backward time ℓ measured from its initial node, the preceding estimate and the bounded gap between successive A_k imply

$$\log b_\ell - \log m = \ell \left(\log 2 - \frac{\log 3}{h} \right) + O_h(1).$$

The error is uniform throughout the block and independent of its length or initial node. In particular, a block with this rule has asymptotic logarithmic coefficient slope $\log R$, where $R = 2\,3^{-1/h}$. The limiting slope $\log 2$ can also be realized: take a single admissible inverse exponent of arbitrarily large size. All but its last inverse step multiply by 2, and the final division by 3 changes the logarithm by only a bounded amount.

Prescribing a concave coefficient profile. Fix $2 \ge R_1 > \cdots > R_s \ge R_*$ and positive weights $\lambda_1 + \cdots + \lambda_s = 1$. For $R_i < 2$ use $h_i = \log 3 / \log(2/R_i) \ge 4$. For an integer parameter $L \to \infty$,

concatenate inverse blocks having backward lengths $\lambda_i L + O(1)$, in the displayed order. This is accomplished with $\lfloor \lambda_i L/h_i \rfloor$ odd inverse steps for finite h_i, and one long inverse step if $R_i = 2$. The fixed orbit of 5 adds only a bounded initial segment. Denote the resulting start by N_L and its degree by $d_L = L + O(1)$.

Write the reversed orbit polynomial as

$$F_L(w) = w^{d_L} P_{N_L}(1/w) = \sum_{j=0}^{d_L} b_{L,j} w^j.$$

Let $f : [0,1] \to \mathbb{R}$ be continuous, start at $f(0) = 0$, and have slope $\log R_i$ on consecutive intervals of lengths λ_i. It is concave and piecewise linear. The block estimates give the uniform coefficient-profile limit

$$\max_{0 \le j \le d_L} \left| \frac{\log b_{L,j}}{d_L} - f(j/d_L) \right| \longrightarrow 0.$$

Also $N_L \to \infty$, since $f(1) \ge \log R_* > 0$.

The zero measure of a piecewise linear profile. For $w \neq 0$ set

$$U(w) = \max_{0 \le t \le 1} \{ f(t) + t \log |w| \}, \qquad u_L(w) = \frac{1}{d_L} \log |F_L(w)|.$$

The triangle inequality and the coefficient-profile limit yield $u_L(w) \le U(w) + o(1)$ locally uniformly away from 0. On the positive real axis all coefficients are positive, so comparison with the largest summand also yields $u_L(r) \to U(r)$ for every $r > 0$. These subharmonic functions are locally bounded above and do not converge uniformly to $-\infty$; the standard subharmonic compactness alternative (see, for example, [100, p. 15]) therefore provides L^1_{loc} convergent subsequences.

For any such limit v, one has $v \le U$. On each annulus between the finitely many circles $|w| = 1/R_i$, the function U is harmonic. The submean inequality gives $\limsup_L u_L(r) \le v(r)$: apply it on a small disk centered at r, pass to the L^1 limit in its area average, and then shrink the disk. Convergence on the positive real axis therefore gives $v(r) = U(r)$ at an interior point of that annulus. The maximum principle applied to $v - U \le 0$ then gives $v = U$ throughout the annulus. These annuli cover the punctured plane except for finitely many circles, which have planar measure zero. Thus every subsequential L^1 limit is U, and $u_L \to U$ in L^1_{loc} away from 0.

Taking distributional Laplacians shows that the empirical zero measures of F_L tend to

$$\frac{1}{2\pi} \Delta U = \sum_{i=1}^{s} \lambda_i \sigma_{1/R_i}.$$

Indeed, the slope of U as a function of $\log |w|$ jumps by λ_i at $1/R_i$. There is no escaped mass: every orbit coefficient ratio a_j/a_{j+1} lies between $3/5$ and 2, so Eneström–Kakeya confines all zeros of P_{N_L} to $3/5 \le |z| \le 2$, and hence all zeros of F_L to a fixed compact annulus. Inversion $z = 1/w$ is continuous on that annulus and sends σ_{1/R_i} to σ_{R_i}. We obtain the asserted finite mixture for the original orbit polynomials.

Finally approximate an arbitrary probability measure ν on $[R_*, 2]$ weakly by finite atomic probability measures. For the jth approximation choose a sufficiently far member of its constructed sequence so that its empirical zero measure is within $1/j$ in a metric for weak convergence of the corresponding finite mixture, and its starting value exceeds j. A diagonal sequence proves the general statement. □

Scope. This is a realization theorem on an outer subannulus, with an explicit inverse-tree construction for each prescribed finite mixture. For arbitrary real parameters or an arbitrary measure ν, the statement is mathematical existence; no uniform algorithm for noncomputable

input measures is asserted. It neither classifies all possible nonradial limits nor decides which radial laws involving radii below $2\,3^{-1/4}$ can occur. It does, however, show that full zero clouds do not have one universal limiting distribution, even among explicitly terminating starts.

Corollary IV.4.2 (A fixed inverse path and quantitative radial concentration)**.** *For every real $h \geq 4$ there is one infinite inverse path of odd terminating starts n_k, all eventually reaching 5, such that, with $R_h = 2\,3^{-1/h}$,*

$$\mu_{n_k} \Longrightarrow \sigma_{R_h}.$$

For every fixed $\varepsilon > 0$, the number of roots satisfying $\big||\alpha| - R_h\big| \geq \varepsilon$ is $O_{h,\varepsilon}(1)$, uniformly in k.

Proof. Use the single balanced inverse block indefinitely, with initial node 5. The reversed orbit coefficients b_j then satisfy

$$C_h^{-1} R_h^j \leq b_j \leq C_h R_h^j \qquad (j \geq 0)$$

for one fixed $C_h \geq 1$; the finite initial tail is absorbed into the same constant. This follows from the bounded discrepancy $A_k - hk$ and the uniform within-block estimate proved above. The single-segment coefficient profile is $f(t) = t \log R_h$, so the preceding proof gives weak convergence to σ_{R_h}.

For the quantitative statement write $d = \tau(n_k)$, $Q_k(w) = w^d P_{n_k}(1/w)$, and

$$J_k(v) = Q_k(v/R_h) = \sum_{j=0}^{d} c_j v^j, \qquad C_h^{-1} \leq c_j \leq C_h, \quad c_0 = 1.$$

For fixed $0 < r < \rho < 1$, the geometric-series bound gives $|J_k(v)| \leq C_h/(1-\rho)$ on $|v| = \rho$, independently of d. Jensen's formula therefore bounds the number of zeros $|v| \leq r$ by

$$\frac{\log(C_h/(1-\rho))}{\log(\rho/r)}.$$

Apply the same argument to $v^d J_k(1/v)$, whose constant coefficient is at least C_h^{-1} and whose other coefficients obey the same bound. The number of zeros of J_k with $|v| \geq 1/r$ is at most

$$\frac{\log(C_h^2/(1-\rho))}{\log(\rho/r)}.$$

At an exceptional radius containing zeros, Jensen's formula is understood in its limiting-radius form. Both bounds are independent of k. Since $v = R_h/\alpha$, suitable fixed choices of r and ρ give the stated radial bound for every fixed ε. □

IV.5 A concrete distribution and the scope of the extension

For example, put

$$R_1 = 2\,3^{-1/8} = 1.74337\ldots, \qquad R_2 = 2\,3^{-1/4} = 1.51967\ldots.$$

Theorem IV.4.1 provides terminating starts N_j such that

$$\mu_{N_j} \Longrightarrow \frac{2}{5}\sigma_{R_1} + \frac{3}{5}\sigma_{R_2}.$$

For every sufficiently thin fixed annular neighborhood of the first circle, the proportion of all zeros in that neighborhood tends to 2/5; for the second it tends to 3/5. Each circle also has uniform angular distribution. The construction uses a backward block with average inverse exponent 8, followed by one with average inverse exponent 4, in the corresponding ordinary-step proportions.

More generally, choosing ν to be normalized Lebesgue measure on $[R_*, 2]$ realizes a rotationally invariant limit whose radius is uniformly distributed in that interval. This is different from uniform planar area: the latter is obtained by choosing

$$d\nu(R) = \frac{2R}{4 - R_*^2}\, dR \qquad (R_* \le R \le 2).$$

Thus even normalized area measure on this annulus is a limit of full zero clouds of terminating Collatz polynomials.

What is established. The full-group theorem determines the exact splitting-field Galois group for every prime index $k \equiv 23 \pmod{88}$ and the entire associated power-of-two radical tower. The measure theorem proves every prescribed radial law in the displayed outer band is attainable. The one-circle corollary adds a uniform, degree-independent bound on the number of radial outliers. These claims follow from the proofs above and do not depend on numerical root finding or an external verification program.

Proposition IV.5.1 (A necessary stopping-time condition for smaller circles)**.** *If $N_j \to \infty$ are terminating starts and $\mu_{N_j} \Longrightarrow \sigma_R$, then*

$$\frac{\log N_j}{\tau(N_j)} \longrightarrow \log R.$$

In particular $R \ge 1$. For $R = 1$ this requires $\tau(N_j)/\log N_j \to \infty$.

Proof. Since P_N is monic with constant coefficient N, the product of the moduli of all its roots is N. Hence

$$\int \log|z|\, d\mu_N(z) = \frac{\log N}{\tau(N)}.$$

All its zeros lie in the fixed annulus $3/5 \le |z| \le 2$, as shown above. The function $\log|z|$ is continuous and bounded there, so weak convergence gives the asserted limit. Its nonnegativity forces $R \ge 1$. If $R = 1$, the positive ratios $\log N_j/\tau(N_j)$ tend to zero; taking reciprocals proves the last assertion. □

This proposition is an elementary consequence of the root-product identity, not a claimed new stopping-time estimate. It identifies a genuine obstacle to simply extending the one-circle construction down to radius 1. A uniform bound $\tau(N) \le C \log N$ over terminating starts would instead force every attainable one-circle limit to have $R \ge e^{1/C} > 1$. Neither such a bound nor a unit-circle realization is proved here.

What remains open here. For odd indices divisible by 5, full lifting has been reduced to the single obstruction $\sqrt{5} \in K_k$; we have not classified its occurrence. We have not removed all restrictions on the symmetric quotient, determined which zero-cloud limits occur below R_*, or classified possible nonrotational limits. Neither theorem establishes universal Collatz termination, a primality test, a prime-only starting family, or transcendence of the Euler–Mascheroni constant. The cutoff R_* comes from the uniform inverse-block construction; no optimality is claimed. The arbitrary-measure diagonalization is an existence result and is not a computability claim for noncomputable input measures. These statements are research deductions with complete proofs, not a claim that a named famous conjecture has been resolved or that their historical priority has been verified.

IV.6 A sharper radial band from paired inverse steps

Pairing odd inverse steps improves the preceding radial endpoint. The optimal pair cost is 7; optimality of the resulting radial endpoint remains open.

Lemma IV.6.1 (Exact uniform cost of two odd inverse steps)**.** *For every odd integer $m \geq 5$ with $3 \nmid m$, there are positive integers a, b with $a + b \leq 7$ such that*

$$m_1 = \frac{2^a m - 1}{3}, \qquad m_2 = \frac{2^b m_1 - 1}{3}$$

are odd integers not divisible by 3. The constant 7 is the best possible uniform bound, even if m is restricted to terminating starts.

Proof. Integrality and nondivisibility by 3 of both intermediate nodes depend only on m mod 27. The following table gives one valid choice in each of the eighteen unit classes:

m mod 27	a	b	m mod 27	a	b	m mod 27	a	b
1	2	2	2	1	2	4	2	3
5	3	2	7	4	2	8	1	3
10	2	2	11	3	1	13	2	1
14	3	2	16	4	2	17	1	1
19	4	1	20	1	2	22	2	1
23	5	1	25	6	1	26	1	1

Each entry is checked by the two displayed integer formulas modulo 27 and 9, respectively; every row has $a + b \leq 7$. All nodes are at least 5: an odd positive integral predecessor of an odd $m \geq 5$ is at least 3, and the value 3 is excluded by nondivisibility by 3.

For optimality take $m \equiv 25 \pmod{27}$. The first exponent must be even. The choice $a = 2$ gives $3 \mid m_1$, whereas $a = 4$ gives $m_1 \equiv 7 \pmod 9$ and forces the second admissible exponent to be at least 4. Thus $a + b \leq 6$ is impossible; $a \geq 6$ already forces $a + b \geq 7$. This obstruction occurs at the terminating start $m = 25$, whose shortcut trajectory is

$$25, 38, 19, 29, 44, 22, 11, 17, 26, 13, 20, 10, 5, 8, 4, 2, 1.$$

The pair $(a, b) = (6, 1)$ attains cost 7. □

Theorem IV.6.2 (Improved full zero-cloud realization band)**.** *Put*

$$R_{**} = 2\,3^{-2/7} = 1.4611999\ldots.$$

For every probability measure ν on $[R_{**}, 2]$ there are terminating starts $N_j \to \infty$ whose empirical zero measures converge to

$$\int_{[R_{**},2]} \sigma_R \, d\nu(R).$$

Moreover, for every real $h \geq 7/2$ there is a single infinite inverse path of odd starts, all eventually reaching 5, for which

$$\mu_{N_j} \Longrightarrow \sigma_{2\,3^{-1/h}}.$$

For every fixed $\varepsilon > 0$, this path has only $O_{h,\varepsilon}(1)$ zeros outside the ε-neighborhood in radius of its limiting circle, independently of its degree.

Proof. Fix $h \geq 7/2$. A *small pair* is the pair in Lemma IV.6.1, and has total exponent at most 7. For a *large pair*, first choose an admissible exponent

$$2h - 1 \leq a < 2h + 3,$$

then choose an admissible exponent $b \le 4$ at the resulting node. The existence of the first choice follows from the same elementary modulo-9 rule used above: among consecutive exponents of the required parity at most one is forbidden. The total $S = a + b$ then satisfies

$$2h \le S < 2h + 7.$$

All exponents in either type of pair are bounded in terms of h.

Starting at an odd node $m \ge 5$ not divisible by 3, write A_j for the sum of the first j inverse exponents, and put

$$D_r = A_{2r} - 2hr, \qquad D_0 = 0.$$

Choose a large pair when $D_{r-1} \le 0$ and a small pair otherwise. Since a small pair has total exponent between 2 and $7 \le 2h$, induction gives

$$2 - 2h \le D_r < 7.$$

At an odd prefix, a small pair has $D_{r-1} > 0$ and $a \ge 1$, whereas a large pair has $D_{r-1} \ge 2 - 2h$ and $a \ge 2h - 1$. In either case,

$$A_{2r-1} \ge h(2r - 1) + 1 - h.$$

Thus $A_j - hj = O_h(1)$ at every prefix, not just at pair endpoints.

To verify that the additive correction remains uniformly harmless, use the exact inverse identity

$$n_j = \frac{2^{A_j}}{3^j}\left(m - \sum_{\ell=1}^{j} \frac{3^{\ell-1}}{2^{A_\ell}}\right).$$

With $q = 9/2^{2h}$, the odd and even terms in pair r are bounded by $\frac{1}{2}q^{r-1}$ and $\frac{3}{4}q^{r-1}$, respectively. Hence

$$0 \le \sum_{\ell=1}^{j} \frac{3^{\ell-1}}{2^{A_\ell}} \le \frac{5}{4(1 - 9/2^{2h})} \le \frac{160}{119} < 5 \le m.$$

The correction is therefore bounded above and bounded away from m, uniformly in the block length and starting node. Consequently

$$\log n_j = \log m + A_j \log 2 - j \log 3 + O(1).$$

Between odd inverse nodes there are only $O_h(1)$ ordinary shortcut steps. The reversed orbit coefficients in such a block therefore obey

$$\log b_\ell - \log m = \ell\left(\log 2 - \frac{\log 3}{h}\right) + O_h(1),$$

with an error independent of its length and initial node.

For finitely many requested radii, concatenate these balanced paired blocks in decreasing order of radius, using $\lfloor \lambda_i L/(2h_i) \rfloor$ pairs in the block of weight λ_i, where $h_i = \log 3/\log(2/R_i) \ge 7/2$. Its ordinary-step length is $\lambda_i L + O(1)$ by the discrepancy bound. The endpoint $R_i = 2$ is handled by one long inverse step as before. Thus the same concave coefficient profile and subharmonic argument in the preceding realization proof give every finite mixture of circle measures in $[R_{**}, 2]$; finite atomic approximation and diagonalization give every probability measure on that interval.

For one fixed h, continue the paired construction indefinitely from 5. The coefficient estimate becomes $C_h^{-1} R^\ell \le b_\ell \le C_h R^\ell$, where $R = 2\,3^{-1/h}$. The same Jensen bounds for the scaled reversed polynomial and its reciprocal give the asserted degree-independent number of radial outliers. □

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